III

VOLUME THREE OF THREE

The Question Bank

Whether it has stuck

1,016 single best answer questions in thirty blocks,
with answers, explanations and references
A complete answer-key grid for rapid marking
A mark sheet: one row per block, three attempts
207 viva questions with model answers
A viva tracker

Every block names the Volume II section it tests and the part of Notes In Orthodontics *the material came from, so a poor score tells you exactly what to re-read.*

For candidates sitting the examination of 5 November 2026

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The set

Three volumes, in the order you should use them. Each is cross-referenced to the other two, so you can enter from whichever you happen to have open.

I	**Notes In Orthodontics** Revised edition · 275 pp	The book that teaches the material. A crash notebook of forty-two schematics and four number cards; the twenty essay topics with their skeletons; the sixteen chapters; model viva answers; and four new units on ethics and consent, radiation protection, infection control, and governance. **Reach for it** when you are learning something for the first time, planning an essay, or want the detail behind a fact.
II	**A Candidate's Revision Handbook** 161 pp	The book that maps the material onto the examination. Thirty sections against the sixty-five curriculum items, the numbers on six cards, the diagnostic cases, the case presentation, the oral, the literature by theme, and the dated countdown. **Reach for it** when you are revising against the syllabus, preparing the components that are not written, or covering what Volume I does not.
III	**The Question Bank** 300 pp	The book that tells you whether it has stuck. 1,016 single best answers in thirty blocks with explanations and references, a mark sheet, 207 viva questions with model answers, and a viva tracker. **Reach for it** at the end of every study session, and let the score decide what you re-read.

LEARN, REVISE, TEST

Volume I teaches the material. Volume II checks it against the sixty-five curriculum items and covers the four assessed components that are not knowledge — the diagnostic cases, the case presentation, the oral and examination technique. Volume III tells you whether any of it has stuck. Work in that order, and let a poor score in Volume III send you back to the section Volume II names.

The plan

Seven working weeks and a weekend, ending **Thursday 5 November 2026**. The written paper is the engine — revising for it is simultaneously revising for the oral and the diagnostic cases, because all three test the same knowledge. The case submissions depend on other people, so their *requests* go out first.

Before Monday — one session, about three hours

1. Pick your three write-up cases and your two presentation cases. For each, list what is missing: pre-treatment records, radiographs, photographs, models or scans, progress notes, debond records, PAR scoring. **2.** Send every request the same afternoon — patient recalls, radiograph retrieval, trainer sign-off. **3.** Ask the Programme Management Office when the three written-up cases are due. If that date falls before 5 November, it outranks everything below. **4.** Sit Question Bank Blocks **H** and **AD** cold. 55 questions, ethics and governance — the likeliest thin spot and the cheapest marks in the paper.

The spine is the curriculum, not the book. Volume II's thirty sections and Volume III's thirty blocks run one-to-one — §1 is Block A, §30 is Block AD — so the weeks below advance through them in order. The Volume I column then names the units whose *main* treatment sits in that week's sections. **Volume I has two separate series and they are not the same thing**: the twenty essay **topics** of Part II and the sixteen **chapters** of Part III. Chapter 1 is Evidence-Based Orthodontics; Topic 1 is the biological basis of tooth movement. Every chapter and every topic appears exactly once below.

Week	Dates	Curriculum	Read in Volume I	Alongside
1	14–20 Sep	§1–§5 · Blocks A–E	Chapters 2–6 · Topic 1	Viva V1–V20 aloud
2	21–27 Sep	§6–§10 · Blocks F–J	Chapters 1, 7 · Topic 2 · Part V.1	V21–V48. First diagnostic drill
3	28 Sep–4 Oct	§11–§15 · Blocks K–O	Chapters 8, 11 · Topics 3–5, 7	V49–V80. Write up case 1
4	5–11 Oct	§16–§20 · Blocks P–T	Chapters 9, 10, 12–14 · Topics 6, 8, 13–15, 20	V81–V111. Write up case 2
5	12–18 Oct	§21–§25 · Blocks U–Y	Topics 11, 12, 16, 17, 19	V112–V151. Write up case 3
6	19–25 Oct	§26–§30 · Blocks Z–AD	Chapters 15, 16 · Topics 9, 10, 18 · Part V.2–V.4	V152–V207
7	26 Oct–1 Nov	Re-sit every block scored under 70%	Only what the mark sheet sends you back to	Mock oral. Present both cases aloud
Final	2–4 Nov	Two or three blocks a session at examination pace	Crash notebook · number cards	Nothing new

Week 5 lists no chapters because §21–§25 are Chapter 13 territory, read in week 4; that week is topic-led. Part V has no chapter or topic of its own — it is new in this edition.

The daily loop — about two hours

One **Volume I** unit (45 min) → the **Volume II** section its teal strap names (20 min) → that **Volume III** block, timed at 60 seconds a question (30 min) → read the explanation for *every* question, including the ones you got right (25 min).

Log the score on the mark sheet. Anything under 70 per cent goes on the week 7 list. Close each session with one viva section, aloud, to the end, before reading the model.

THREE RULES THAT DECIDE THIS

1. **Speak from week one, not week seven.** The oral tests whether you can produce a structured spoken answer under pressure — a separate skill from knowing the material. Question blocks are measurable and satisfying, which is exactly why they expand to fill the time and crowd out the one thing you cannot cram.
2. **Protect the case deadline.** You cannot revise your way out of a missing pre-treatment photograph. A patient recall is a three-to-four week loop you do not control.
3. **Do not re-read Volume I cover to cover.** Use it as a textbook — when a Volume III explanation does not land, follow the strap back to that one unit.

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How to use this volume

This is the practice volume. It contains 1016 single best answer questions arranged in thirty blocks, each mapped to one section of the Candidate's Revision Handbook, together with answers, explanations and references; and 207 viva questions with model answers.

The three-pass method

Pass	What to do
First	Work one block at a time, under timed conditions, at 60 seconds per question. Mark the whole block before reading any answer. Read the explanation for *every* question in the block, including the ones you got right.
Second	Two weeks later, repeat any block you scored under 70% on. If you score under 70% again, stop and re-read the corresponding Handbook section before trying a third time.
Third	In the final fortnight, work the answer-key grid backwards: cover the key, sit two or three blocks in one session at examination pace, and mark them together. This trains stamina, which is the part of a two-hour paper nobody practises.

Reading the answers

Each answer gives the correct option, the reasoning, and a reference. **The reasoning is more examinable than the fact**: a question you answered correctly for the wrong reason will be answered wrongly when the stem is rephrased. References use the abbreviations set out at the head of Part 2, and every question additionally carries a Handbook section number, so a block you score poorly on tells you exactly what to re-read.

The viva bank

Read the question, close the book, and answer it **out loud** to the end before reading the model. Answering in your head is not practice. The model answers show the shape of a good spoken answer — a direct opening sentence, the reasoning, the number where a number exists, and a close that says what you would actually do — and the italic note beneath each says what the question is really testing.

> **A NOTE ON THE ANSWER KEY**
>
> The correct option has been distributed evenly across A to E, so approximately one fifth of the answers sit on each letter. Do not read anything into a run of the same letter, and do not use letter frequency as a guessing strategy — in this bank it will not help you.

Finding your way around

This is **Volume III** of three, and it is built to be used *with* the other two on your desk, not instead of them. Every block of questions names the **Volume II** section it tests and the part of **Volume I**, *Notes In Orthodontics*, that the material came from, so a block you score badly on tells you exactly what to re-read — and in which book.

If you know what you want, start here

What you want to do	Where to go
Test one curriculum topic	Part 1, the block named in the index overleaf
Test one *Notes In Orthodontics* essay topic or chapter	The concordance overleaf
Mark a block quickly	Part 2, the answer-key grid at the end
Understand why you got one wrong	Part 2, the explanation for that question
Find which Handbook section to re-read	The last column of the mark sheet, or the strap at the head of the block
Practise speaking	Part 3, the viva bank
Rehearse the set-piece answers	Revision Handbook Part 5 first, then Part 3 here
Track what you have done	The mark sheet at the end of Part 2, and the viva tracker at the end of Part 3

How each block is laid out

Element	What it is for
Block head	Each block starts on a fresh page and its name runs along the top of every page, so you can flip to it
Rust strap	Where this material sits in *Notes In Orthodontics*; which Revision Handbook section to re-read; how many questions and how long to allow
Questions	Numbered continuously 1–1016 across the whole bank, so a question number is unambiguous
Answers, Part 2	In the same block order. **The answer number is a link back to the question** in the electronic version. Each answer gives the option, the reasoning and a reference, and each block closes with the Handbook section to re-read
Answer-key grid	Ten answers to a row, for marking a block in under a minute
Mark sheet	One row per block, three attempts, with the Handbook section in the last column

THE ONE HABIT THAT MATTERS

Mark a whole block before reading any answer, then read the explanation for *every* question in it — including the ones you got right. A question answered correctly for the wrong reason will be answered wrongly when the stem is rephrased, and rephrasing is what examiners do.

Concordance with *Notes In Orthodontics*

If you have been revising topic by topic from the Companion, use these two tables to find the questions that test what you have just read. Question ranges are generated, so they stay correct.

The twenty essay topics

	Essay topic	Questions	Viva
1	The biological basis of tooth movement	Block D, Q96–140	V21–V30
2	Cephalometry: uses and limitations	Block J, Q286–325	V59–V68
3	Factors in formulating a treatment plan	Blocks K and L, Q326–350, Q351–390	V69–V80
4	Factors affecting the choice of extraction	Block M, Q391–430	V81–V90
5	Anchorage and its clinical significance	Block N, Q431–481	V91–V102
6	The three orders and the evolution of fixed appliances	Block Q, Q567–606	V112–V121
7	The ideal archwire; steel versus nickel titanium	Block O, Q482–525	V103–V111
8	Functional appliances	Block P, Q526–566	V112–V121
9	Benefits and complications of treatment	Block AC, Q962–991	V49–V58, V174–V183
10	Medical conditions relevant to orthodontics	Block AC, Q962–991	V174–V183
11	Overbite and methods of overbite reduction	Block Y, Q842–871	V162–V173
12	Reducing overjet in a growing child	Block V, Q747–786	V152–V161
13	Management of a missing maxillary central incisor	Block T, Q677–721	V140–V151
14	Management of an impacted maxillary canine	Block T, Q677–721	V140–V151
15	Management of an impacted lower second molar	Block T, Q677–721	V140–V151
16	Methods of crossbite correction	Block X, Q817–841	V162–V173
17	Correcting a Class III incisor relationship	Block W, Q787–816	V152–V161
18	Malocclusion and temporomandibular disorder	Block Z, Q872–901	V174–V183
19	Mild facial asymmetry in a growing child	Block X, Q817–841	V162–V173
20	Late lower incisor crowding	Block S, Q642–676	V130–V139

The sixteen chapters

	Chapter	Questions	Viva
1	Evidence-Based Orthodontics	Block G, Q196–225	V39–V48
2	Occlusion and Malocclusion	Blocks E and F, Q141–170, Q171–195	V31–V48
3	Prenatal Development of the Craniofacial Region	Block B, Q26–55	V1–V10
4	Postnatal Growth of the Craniofacial Region	Blocks A and B, Q1–25, Q26–55	V1–V10
5	Development of the Dentition	Block C, Q56–95	V11–V20

6	Orthodontic Tooth Movement	Blocks D and N, Q96–140, Q431–481	V21–V30, V91–V102
7	The Orthodontic Patient: Examination and Diagnosis	Blocks I and J, Q256–285, Q286–325	V59–V68
8	The Orthodontic Patient: Treatment Planning	Blocks K–M, Q326–350–Q391–430	V69–V90
9	Functional Jaw Orthopaedics	Blocks P and X, Q526–566, Q817–841	V112–V121, V162–V173
10	Contemporary Fixed Appliances	Blocks O, Q, R, Q482–525, Q567–606–Q607–641	V103–V129
11	Skeletal Anchorage	Block N, Q431–481	V91–V102
12	Management of the Developing Dentition	Block T, Q677–721	V140–V151
13	Management of the Permanent Dentition	Blocks R–Y, Q607–641–Q842–871	V122–V173
14	Clear Aligners	Block Q, Q567–606	V112–V121
15	Orthodontics and Orthognathic Surgery	Block AA, Q902–931	V184–V193
16	Cleft Lip and Palate, and Syndromes	Block AB, Q932–961	V184–V193

THE TWO BLOCKS THE FIRST EDITION LEFT UNCOVERED

Block **AD** (Q992–1016, orthodontic practice, infection control and governance) had no equivalent unit in the first edition of *Notes In Orthodontics*, and much of Block **H** (Q226–255, ethics, consent and communication) was only touched on there. Both are explicit curriculum items — items 63, and 9 and 10 — and both are cheap marks in the written paper. The revised Companion now covers them in **Part V**: V.1 ethics and consent, V.2 radiation protection, V.3 infection control, V.4 governance and audit. Sit these two blocks early rather than last, and read Part V alongside them.

Part 1 — 1016 single best answer questions

How to use this bank. Work through one block at a time under timed conditions — allow **60 seconds per question**, which is the real pace of a two-hour paper covering sixty-five curriculum items. Answers, explanations and references for every question are in Part 2, together with a complete answer-key grid for rapid marking. Resist the temptation to look up each answer as you go: mark a whole block, then read the explanation for *every* question in it, including the ones you got right — the reasoning is more examinable than the fact.

Each block maps to one section of the Revision Handbook, so a block you score poorly on tells you exactly which section to re-read.

The blocks

Block	Questions	Hbk	Subject, and where it sits in *Notes In Orthodontics*
A	1–25	§1	Concepts of growth and development
B	26–55	§2	Chapter 4 (Postnatal Growth) Craniofacial growth, prenatal and postnatal
C	56–95	§3	Chapters 3 and 4 (Prenatal Development; Postnatal Growth) Development of the dentition
D	96–140	§4	Chapter 5 (Development of the Dentition) The biology of tooth movement, eruption and root resorption
E	141–170	§5	Topic 1 (The biological basis of tooth movement); Chapter 6 Occlusion, malocclusion and aetiology
F	171–195	§6	Chapter 2 (Occlusion and Malocclusion) Treatment need, indices and epidemiology
G	196–225	§7	Chapter 2, pp. 106–112 (IOTN and PAR) Evidence-based orthodontics
H	226–255	§8	Chapter 1 (Evidence-Based Orthodontics); Appendix D Ethics, consent, communication and the psychosocial patient
I	256–285	§9	Part V, unit V.1 (Ethics, consent and capacity); Topic 9 in part; Chapter 7 Clinical examination and records
J	286–325	§10	Chapter 7 (Examination and Diagnosis) Imaging and cephalometrics
K	326–350	§11	Topic 2 (Cephalometry); Chapter 7, pp. 411–482 Study models, space analysis and the digital set-up
L	351–390	§12	Chapter 7 (study models); Chapter 8, Box 8.6 Diagnosis, the problem list and treatment planning
M	391–430	§13	Topic 3 (Formulating a treatment plan); Chapter 8 Extraction decisions and space management
N	431–481	§14	Topic 4 (Choice of extraction); Chapter 8, pp. 505–530 Biomechanics, force control and anchorage
			Topics 5 and 6 (Anchorage; the three orders); Chapters 6 and 11

O	482–525	§15	Orthodontic materials, wires, bonding and instruments Topic 7 (The ideal archwire); Chapter 10, pp. 645–665
P	526–566	§16	Removable and functional appliances Topic 8 (Functional appliances); Chapter 9
Q	567–606	§17	Fixed appliances — labial, lingual and aligners Topic 6 (Evolution of fixed appliances); Chapters 10 and 14
R	607–641	§18	Treatment mechanics: Stage I, II and III Chapters 10 and 13 (mechanics; definitive treatment)
S	642–676	§19	Retention, relapse and stability Topic 20 (Late lower incisor crowding); Chapter 13, pp. 975–992
T	677–721	§20	Interceptive orthodontics and the developing dentition Topics 13, 14 and 15 (missing central; impacted canine; impacted 7); Chapter 12
U	722–746	§21	Class I malocclusion and bimaxillary protrusion Chapter 13 (Class I and the permanent dentition)
V	747–786	§22	Class II malocclusion Topic 12 (Reducing overjet in a growing child); Chapters 9, 12 and 13
W	787–816	§23	Class III malocclusion Topic 17 (Correcting a Class III incisor relationship); Chapters 8, 9 and 13
X	817–841	§24	Transverse problems and facial asymmetry Topics 16 and 19 (Crossbite correction; facial asymmetry); Chapters 9, 12, 13 and 15
Y	842–871	§25	Vertical problems: deep bite and open bite Topic 11 (Overbite and overbite reduction); Chapter 13, pp. 928–945
Z	872–901	§26	TMD, condylar trauma, the airway and myofunctional therapy Topic 18 (Malocclusion and TMD); Chapter 2, pp. 102–105
AA	902–931	§27	Adult orthodontics and orthognathic surgery Chapters 13 and 15 (adult treatment; orthognathic surgery)
AB	932–961	§28	Cleft lip and palate and craniofacial syndromes Chapter 16 (Cleft Lip and Palate, and Syndromes)
AC	962–991	§29	Risks of orthodontic treatment and the medical history Topics 9 and 10 (Benefits and complications; Medical conditions); Chapters 2 and 7
AD	992–1016	§30	Orthodontic practice, infection control and governance Part V, units V.2–V.4 (radiation protection; infection control; governance and audit)

Block A — Concepts of growth and development (Q1–25)

Notes In Orthodontics: Chapter 4 (Postnatal Growth)
Supplements: Revision Handbook **§1** — read it again if you score under 70%
Practise this: 25 questions · allow 25 minutes · answers begin on the page named in the contents

1 Which single term best describes an increase in size produced by hyperplasia, hypertrophy and secretion of extracellular matrix?

A Growth
B Development
C Maturation
D Differentiation
E Remodelling

2 A 10-year-old girl attends for assessment. Which single measure is least useful in deciding when to begin functional appliance therapy?

A Cervical vertebral maturation stage
B Hand-wrist radiograph
C Chronological age
D Standing height velocity
E Tanner stage of sexual maturation

3 At approximately what age does peak height velocity occur in girls?

A 10 years
B 8 years
C 14 years
D 16 years
E 12 years

4 A girl reports that menarche occurred 8 months ago. What does this tell you about her growth?

A The adolescent spurt has not yet begun
B Peak height velocity has already passed
C Peak height velocity is imminent
D Growth has entirely ceased
E Growth status cannot be inferred

5 A lateral cephalogram shows cervical vertebral maturation stage CS3. What is the correct interpretation?

A Peak height velocity is 2 years away
B Peak height velocity is 1 year away
C Peak height velocity passed 1 to 2 years ago
D Peak height velocity will occur during this year
E Growth is complete

6 Which sequence correctly describes the order in which craniofacial dimensions complete their growth?

A Heights, then depths, then widths
B Depths, then widths, then heights
C Widths, then depths, then heights
D All three complete simultaneously
E Widths, then heights, then depths

7 Which statement about growth of the chin is correct?

A There is no distinct anteroposterior growth spurt of the chin
B There is a marked anteroposterior growth spurt of the chin at puberty
C The chin completes growth before the cranial base
D Chin growth is entirely determined by the nasal septum
E Chin prominence decreases throughout adolescence

8 Which theory of growth control provides the intellectual basis for functional appliance therapy?

A Genetic theory of Brodie
B Sutural theory of Sicher
C Nasal septum theory of Scott
D Cybernetic theory of Petrovic
E Functional matrix theory of Moss

9 Which single statement about the condylar cartilage is correct?

A It is a primary growth centre comparable to an epiphyseal plate
B It ceases growing at 12 years in both sexes
C It is entirely genetically determined and cannot be modified
D It is a secondary, adaptive cartilage
E It grows by intramembranous ossification

10 What is the correct distinction between a site and a centre of growth?

A A site is genetically controlled; a centre is not
B A site is a location where growth occurs; a centre is a location of independent, genetically controlled growth
C A centre is a location where growth occurs; a site is where it is controlled
D The terms are interchangeable
E A site refers to cartilage; a centre refers to bone

11 Which of the following is the best chairside proxy for the timing of the adolescent growth spurt in a boy?

A Chronological age
B Dental age from a panoramic radiograph
C Number of erupted permanent teeth
D Shoe size
E Standing height measured at successive visits

12 Puberty is initiated by which single event?

A Increased adrenal androgen secretion
B Pulsatile release of gonadotrophin-releasing hormone from the hypothalamus
C Closure of the spheno-occipital synchondrosis
D A rise in circulating growth hormone alone

E Ossification of the sesamoid of the thumb

13 Which statement best describes the cephalocaudal gradient of growth?

A Growth proceeds from the feet upwards
B Growth is uniform across all body segments
C The cranium and the mandible grow at identical rates
D Structures nearer the head mature earlier than those further from it
E Limb growth precedes cranial growth

14 At what mean age does cervical vertebral maturation stage CS4 occur, and what does it indicate?

A 13.4 years; peak passed 1 to 2 years ago
B 12.0 years; peak this year
C 10.5 years; peak within 1 year
D 9.7 years; peak 2 years away
E 15.0 years; growth complete

15 Which single caveat should always accompany a statement about maturational staging?

A It is only valid in females
B It requires a CBCT
C None of the underlying principles is universally accepted
D It is more accurate than serial height measurement
E It predicts the amount of remaining growth precisely

16 Which is the correct definition of the pattern of growth?

A The absolute increase in size over time
B The timing of the adolescent spurt
C The variability of growth between individuals
D The proportional relationships between parts at a point in time, and their orderly change over time
E The rate of growth per unit time

17 A juvenile growth spurt is typically observed at which age?

A 3 to 4 years
B 6 to 8 years
C 10 to 11 years
D 12 to 14 years
E It does not occur

18 Which statement about mandibular growth in Class III patients is correct?

A It ceases earlier than in Class I patients
B It follows the neural growth curve
C It is unaffected by sex
D It is complete by the time of menarche in both sexes
E It may continue into the twenties, particularly in males

19 Which of the following growth curves best describes the growth of the cranial vault?

A Neural curve
B General somatic curve
C Lymphoid curve
D Genital curve

E Linear throughout life

20 Which single statement about adult craniofacial change is correct?

A Growth ceases entirely at 18 years
B Only the soft tissues change after 18 years
C Slow rotational growth continues throughout adult life in the individual's original pattern
D Change occurs only in patients treated orthodontically
E The mandible resorbs uniformly after 25 years

21 Dental age is most usefully assessed by which single method?

A Number of erupted teeth alone
B Stage of tooth formation on a radiograph
C Chronological age plus 6 months
D Hand-wrist radiograph
E Tanner staging

22 Which statement about the spheno-occipital synchondrosis is correct?

A It is a suture and grows by apposition
B It is the principal site of mandibular growth
C It fuses at birth
D It grows endochondrally and fuses at approximately 13 to 15 years
E It fuses later in females than in males

23 Which is the most defensible position to state in an oral examination about the modifiability of mandibular growth?

A It can be modified within genetically determined limits, and the effect is modest and largely transient
B It can be modified without limit if treatment starts early enough
C It cannot be modified at all
D It can be modified only by surgery
E It can be modified only in females

24 Which single feature most reliably indicates that the adolescent growth spurt has been completed?

A A CS3 cervical vertebral maturation stage
B Eruption of the second permanent molars
C Cessation of increase in standing height on serial measurement
D Menarche in a girl
E Attaining the age of 16 years

25 Which of the following best describes the relationship between statural growth and mandibular growth?

A They are entirely independent
B Mandibular growth precedes statural growth by about 2 years
C Mandibular growth ceases at menarche
D Both cease at exactly the same time
E Statural growth slightly precedes mandibular growth, and mandibular growth continues after statural growth appears complete

Block B — Craniofacial growth, prenatal and postnatal (Q26–55)

Notes In Orthodontics: Chapters 3 and 4 (Prenatal Development; Postnatal Growth)
Supplements: Revision Handbook **§2** — read it again if you score under 70%
Practise this: 30 questions · allow 30 minutes · answers begin on the page named in the contents

26 At which week of intrauterine development do the five facial processes appear around the stomodeum?

A Week 4
B Week 2
C Week 6
D Week 8
E Week 10

27 The upper lip is formed by fusion of which structures?

A Maxillary processes with the lateral nasal processes
B Frontonasal process with the mandibular processes
C Maxillary processes with the medial nasal processes
D The two palatine shelves
E Medial nasal processes with the lateral nasal processes

28 By which week is the human face complete?

A Week 4
B Week 6
C Week 12
D Week 10
E Week 8

29 Isolated cleft palate results from which single developmental failure?

A Failure of the maxillary process to fuse with the medial nasal process
B Failure of the palatine shelves to elevate or fuse, between weeks 6 and 10
C Failure of neural crest migration at week 3
D Failure of the mandibular processes to fuse in the midline
E Premature fusion of the incisive suture

30 A newborn has an isolated cleft of the palate. What is the approximate probability that this is part of a recognised syndrome?

A 10 per cent
B 30 per cent
C 70 per cent
D 50 per cent
E 90 per cent

31 Which cell population forms practically all of the mesenchyme of the face?

A Mesodermal cells
B Endodermal cells
C Neural crest cells
D Trophoblastic cells
E Primordial germ cells

32 What is the reported incidence of cleft lip with or without cleft palate?

A 1 in 700
B 1 in 250
C 1 in 2000
D 1 in 5000
E 1 in 10000

33 Which sex ratio applies to isolated cleft palate?

A Male 2 to 1
B Female 2 to 1
C Equal
D Male 4 to 1
E Female 4 to 1

34 By what mechanism does the cranial vault principally grow after birth?

A Endochondral ossification at synchondroses
B Displacement by the maxillary tuberosity
C Cartilaginous growth at the nasal septum
D Sutural apposition driven by the expanding brain, plus surface remodelling
E Interstitial growth within bone

35 The maxilla grows principally by which combination of mechanisms?

A Endochondral ossification only
B Displacement forwards and downwards by the surrounding sutures, plus surface remodelling
C Condylar growth plus surface remodelling
D Sutural growth only, with no remodelling
E Interstitial expansion

36 Which surface of the maxilla characteristically undergoes resorption during growth?

A The posterior surface of the tuberosity
B The orbital floor
C The palatal surface
D The zygomatic process
E The anterior surface

37 Bjork's implant studies demonstrated which single phenomenon?

A That the mandible does not grow after 14 years
B That the mandible rotates internally during growth, largely masked by surface remodelling
C That cephalometric growth prediction is accurate
D That the maxilla grows only by sutural apposition
E That the condyle is a primary growth centre

38 Which equation correctly relates the three mandibular rotations?

A Total equals matrix minus intramatrix
B Intramatrix equals total plus matrix
C Matrix equals total plus intramatrix
D Total equals matrix plus intramatrix
E The three are independent

39 A patient shows a forward-inclined condyle, a curved mandibular canal, a thick symphysis and a shallow antegonial notch. Which growth pattern do these signs suggest?

A Forward mandibular rotation
B Backward mandibular rotation
C No rotation
D Transverse maxillary deficiency
E Vertical maxillary excess

40 Backward mandibular rotation is characteristically associated with which occlusal feature?

A Deep overbite
B Class III molar relationship
C Anterior open bite
D Bilateral crossbite
E Reduced overjet

41 Which single feature of a forward-rotating mandible explains late lower incisor crowding?

A Continued condylar growth
B Mesial drift of the first molars alone
C Increased intercanine width
D Progressive uprighting of the lower labial segment
E Loss of the leeway space

42 Which is the principal site of postnatal mandibular growth?

A The symphysis
B The condyle, with extensive surface remodelling
C The coronoid process
D The alveolar process alone
E The mandibular canal

43 Which single anatomical structure carries the infraorbital nerve and forms part of the maxilla?

A The zygomatic arch
B The palatine shelf
C The pterygoid plates
D The nasal septum
E The roof of the maxillary sinus, which is the orbital floor

44 Which statement about the cranial base angle is correct?

A An obtuse cranial base angle tends to position the glenoid fossa more posteriorly, contributing to an apparent Class II relationship
B It has no influence on the apparent jaw relationship
C It is fixed from birth
D It is measured between porion and orbitale
E It is unaffected by the spheno-occipital synchondrosis

45 In cleft lip with or without cleft palate, what approximate proportion of cases are unilateral?

A 20 per cent
B 50 per cent
C 80 per cent
D 65 per cent
E 95 per cent

46 Which single environmental exposure has the strongest evidence for reducing the risk of orofacial clefting?

A Vitamin D supplementation
B Folic acid supplementation
C Iron supplementation
D Avoidance of caffeine
E Zinc supplementation

47 Which anticonvulsant drugs are associated with an increased risk of orofacial clefting?

A Carbamazepine and gabapentin
B Topiramate only
C Levetiracetam and lamotrigine alone
D Phenytoin and sodium valproate
E None are associated

48 Which statement about the palatine shelves is correct?

A They appear vertically either side of the tongue and later elevate
B They appear horizontally and then descend
C They form from the frontonasal process
D They fuse before the lip
E They fuse at week 4

49 Which of the following best describes the growth of the nasomaxillary complex relative to the anterior cranial base?

A Upwards and backwards
B Purely downwards
C Downwards and forwards
D Purely forwards
E It does not move relative to the cranial base

50 A soft tissue change commonly observed in adult life is which of the following?

A Increasing incisor show at rest
B Increasing upper lip vermilion display
C Thickening of the vermilion border
D Reduction in nasal projection
E Decreasing incisor show at rest, with lengthening of the nose and upper lip

51 Craniosynostosis presents early in life principally because of which fact?

A The cranial vault follows the neural growth curve and is approximately 90 per cent complete by age 5
B The mandible grows fastest in infancy
C The cranial base fuses at birth
D The sutures are the only sites of growth in the body
E Brain growth ceases at 2 years

52 Which single statement about the nasal septum theory of Scott is correct?

A It proposed that sutures are the primary determinants of facial growth
B It proposed that soft tissue function determines bone growth
C It proposed that cartilage is the primary determinant and that sutures respond passively
D It proposed that growth is entirely genetically predetermined
E It proposed a cybernetic feedback model

53 The functional matrix theory would predict which of the following?

A That removing a functional matrix has no effect on the associated bone
B That sutures are growth centres
C That growth is entirely determined at conception
D That the condyle drives all mandibular growth
E That bone growth follows the growth of the soft tissue capsules and functional spaces it serves

54 Which statement about the sex ratio and syndromic proportion of clefting is the best examination answer?

A The two cleft types share the same sex ratio because they are the same condition
B The opposite sex ratios indicate different developmental failures occurring at different times
C Sex ratio is not a useful discriminator
D Both types are equally likely to be syndromic
E Isolated cleft palate is more common than cleft lip with or without palate

55 A 3-year-old is noted to have severe midface hypoplasia, exophthalmos and syndactyly of the hands and feet. Which single diagnosis is most likely?

A Crouzon syndrome
B Pierre Robin sequence
C Treacher Collins syndrome
D Apert syndrome
E Cleidocranial dysplasia

Block C — Development of the dentition (Q56–95)

Notes In Orthodontics: Chapter 5 (Development of the Dentition)
Supplements: Revision Handbook **§3** — read it again if you score under 70%
Practise this: 40 questions · allow 40 minutes · answers begin on the page named in the contents

56 What is the approximate incisor liability in the maxillary arch?

A 3 mm
B 5 mm
C 7 mm
D 10 mm
E 12 mm

57 What is the approximate leeway space per quadrant in the mandibular arch?

A 2.5 mm
B 1.5 mm
C 0.5 mm
D 4.0 mm
E 5.0 mm

58 Which statement about primate spaces is correct?

A They lie distal to the maxillary canine and mesial to the mandibular canine
B They are pathological
C They are present only in the permanent dentition
D They develop at 6 years of age
E They lie mesial to the maxillary canine and distal to the mandibular canine

59 A 4-year-old has no spacing anywhere in the primary dentition. What is the most appropriate interpretation?

A This is normal and reassuring
B This indicates macrodontia of the primary teeth
C This indicates hypodontia
D This predicts crowding in the permanent dentition
E This requires immediate intervention

60 A tooth typically erupts when what proportion of its root is formed?

A One quarter
B Two thirds to three quarters
C One half
D The root is fully formed
E The root has begun to resorb

61 In the mandibular arch, which tooth is most commonly left crowded or displaced because of the eruption sequence?

A Canine
B First premolar
C Lateral incisor
D First molar
E Second premolar

62 In the maxillary arch, which tooth is most commonly left crowded or displaced because of the eruption sequence?

A Central incisor
B Canine
C Lateral incisor
D First premolar
E Second molar

63 A distal step terminal plane in the primary dentition most commonly develops into which molar relationship?

A Class I
B It self-corrects to Class I in most cases
C Class III
D Class II
E It cannot be predicted

64 Transient crowding of the lower permanent incisors shortly after their eruption is best managed by which single approach?

A Reassurance and review
B Extraction of the primary canines
C Immediate lower arch expansion
D Extraction of a lower incisor
E Fitting a lower lingual arch

65 What is the approximate prevalence of hypodontia, excluding third molars?

A 1 per cent
B 3 per cent
C 6.4 per cent
D 12 per cent
E 20 per cent

66 Which tooth is most commonly congenitally absent, excluding third molars?

A Maxillary lateral incisor
B Mandibular central incisor
C Maxillary second premolar
D Mandibular second premolar
E Maxillary canine

67 What is the approximate prevalence of supernumerary teeth in the permanent dentition?

A 0.1 per cent
B 2 per cent
C 10 per cent
D 15 per cent
E 25 per cent

68 Which type of supernumerary tooth most typically prevents eruption of a permanent incisor?

A Conical
B Mesiodens of conical form
C Supplemental
D Odontome

E Tuberculate

69 An 8-year-old has an unerupted maxillary central incisor with a tuberculate supernumerary above it. What is the most appropriate first step after confirming the diagnosis?

A Remove the supernumerary and create or maintain space, then observe for eruption
B Extract the permanent incisor
C Expose and bond the incisor immediately
D Await spontaneous eruption for 3 years
E Fit a removable partial denture

70 An ectopic maxillary first permanent molar is resorbing the distal aspect of the second primary molar in a 7-year-old. Which statement is most accurate?

A It will never resolve spontaneously
B The permanent molar must be extracted
C It may reverse spontaneously up to approximately 8 years of age
D The primary molar must be extracted immediately
E A CBCT is mandatory

71 What is the approximate prevalence of tooth transposition?

A 0.03 per cent
B 0.3 per cent
C 3 per cent
D 8 per cent
E 15 per cent

72 Which transposition is the most common?

A Maxillary central and lateral incisor
B Maxillary canine and lateral incisor
C Mandibular canine and lateral incisor
D Maxillary canine and first premolar
E Mandibular first and second premolar

73 Molar incisor hypomineralisation has an approximate prevalence of which value?

A 15 per cent
B 5 per cent
C 1 per cent
D 30 per cent
E 50 per cent

74 A first permanent molar of poor prognosis is to be extracted to allow the second permanent molar to erupt into its place. What is the ideal radiographic timing?

A Before the second molar crown is complete
B After the second molar has fully erupted
C When the bifurcation of the second permanent molar is just calcifying, at about 8 to 10 years of dental age
D At 6 years of age
E When the third molar has erupted

75 Spontaneous space closure after first permanent molar extraction is generally better in which arch?

A The maxilla
B It depends only on age
C Both are equal
D Neither closes spontaneously
E The mandible

76 Which extraction is most likely to be indicated to prevent over-eruption of an opposing tooth?

A A compensating extraction
B A balancing extraction
C A serial extraction
D A therapeutic extraction of a third molar
E An enforced extraction

77 A balancing extraction is principally performed to prevent which single problem?

A Over-eruption
B Root resorption
C Centre line shift
D Ankylosis
E Impaction of the second molar

78 An infra-occluded primary molar with a permanent successor present and eruption obstructed should be managed how?

A Left indefinitely
B Ankylosed deliberately
C Restored with a stainless steel crown
D Built up with composite
E Extracted

79 What is the approximate prevalence of infra-occlusion of primary molars?

A Under 1 per cent
B Up to 9 per cent
C 20 per cent
D 35 per cent
E 50 per cent

80 Finding a peg-shaped maxillary lateral incisor should prompt which single action?

A Immediate restoration
B A CBCT of the whole dentition
C Extraction
D A search for other associated dental anomalies, including palatally displaced canines and hypodontia
E Referral for genetic testing

81 Which best describes gemination?

A Fusion of two separate tooth germs
B Union of two teeth by cementum only
C Incomplete division of a single tooth germ
D Excess cementum deposition

E An extra cusp on a molar

82 What is the approximate prevalence of a double tooth in the permanent dentition?

A 0.5 per cent
B 0.05 per cent
C 5 per cent
D 10 per cent
E 20 per cent

83 Which of the following describes the increase in intercanine width that normally occurs during the mixed dentition?

A Increase occurs only in the maxilla
B Approximately 10 mm in each arch
C No increase occurs
D Approximately 1 mm in each arch
E Approximately 3 mm in the mandible and 4.5 mm in the maxilla

84 Which single statement about leeway space is correct?

A It is a deficit that must be met
B It is present only if the primary dentition is spaced
C It is larger in the maxilla
D It is an excess that can be used, and it is larger in the mandible
E It is created by extraction of the primary canines

85 Preservation of the leeway space to relieve mild lower arch crowding is best achieved with which appliance?

A A quadhelix
B A lower lingual arch
C A twin block
D A Nance button
E A transpalatal arch

86 Oligodontia is conventionally defined as the absence of how many teeth, excluding third molars?

A Two or more
B Four or more
C All permanent teeth
D Ten or more
E Six or more

87 Ectodermal dysplasia characteristically produces which combination?

A Multiple supernumerary teeth and delayed eruption
B Hypodontia or anodontia with conical teeth, sparse hair and hypohidrosis
C Absent clavicles and delayed fontanelle closure
D Downward-sloping palpebral fissures and malar hypoplasia
E Macroglossia and midface hypoplasia

88 Cleidocranial dysplasia is caused by a mutation in which gene?

A FGFR2
B TCOF1
C PTH1R

D RUNX2
E MSX1

89 Primary failure of eruption is associated with a loss-of-function mutation in which gene?

A PTH1R
B RUNX2
C FGFR2
D TCOF1
E PAX9

90 Which statement about the dental follicle is correct?

A It is not required for eruption
B It becomes the periodontal ligament only
C It coordinates the osteoclastic resorption that forms the eruption pathway
D It is removed at the bell stage
E It determines crown morphology

91 At approximately what age do the first permanent molars erupt?

A 4 years
B 10 years
C 8 years
D 6 years
E 12 years

92 The primary dentition is normally complete by which age?

A 18 months
B 3 years
C 2 years
D 4 years
E 5 years

93 What proportion of children show a flush terminal plane in the primary dentition?

A Under 20 per cent
B Approximately 30 per cent
C It is never seen
D Over 90 per cent
E Approximately 60 to 75 per cent

94 A megadont maxillary central incisor in an otherwise Class I dentition creates which principal problem?

A A tooth-size discrepancy detectable on Bolton analysis
B Anchorage loss
C Root resorption
D Ankylosis
E Delayed eruption of the canine

95 Which single number pair should be quoted for the Bolton analysis?

A Anterior 85 per cent and overall 95 per cent
B Anterior 91.3 per cent and overall 77.2 per cent
C Anterior 77.2 per cent and overall 91.3 per cent
D Anterior 70 per cent and overall 80 per cent

E Anterior 65 per cent and overall 88 per cent

Block D — The biology of tooth movement, eruption and root resorption (Q96–140)

Notes In Orthodontics: Topic 1 (The biological basis of tooth movement); Chapter 6
Supplements: Revision Handbook **§4** — read it again if you score under 70%
Practise this: 45 questions · allow 45 minutes · answers begin on the page named in the contents

96 Which single observation is the cleanest evidence that orthodontic tooth movement is mediated by the periodontal ligament rather than by bone alone?

A Teeth move faster in younger patients
B Neither an ankylosed tooth nor an osseointegrated implant moves under orthodontic force
C Root resorption occurs during treatment
D Movement is faster in the maxilla
E Hyalinisation occurs at low force levels

97 Which two workers independently founded the pressure-tension theory?

A Reitan and Storey
B Baumrind and Frantz
C Bjork and Skieller
D Sandstedt and Oppenheim
E Ericson and Kurol

98 What is the approximate average width of the periodontal ligament?

A 0.2 mm
B 0.02 mm
C 0.5 mm
D 1.0 mm
E 2.0 mm

99 Osteocytes constitute approximately what proportion of bone cells?

A 10 to 15 per cent
B 30 to 40 per cent
C 85 to 90 per cent
D 50 to 60 per cent
E Under 5 per cent

100 Which molecule secreted by osteocytes inhibits osteoblast differentiation and canonical Wnt signalling?

A RANKL
B Osteoprotegerin
C Interleukin 1 beta
D Cathepsin K
E Sclerostin

101 Which statement about RANKL and osteoprotegerin during orthodontic tooth movement is correct?

A RANKL is raised on the compression side and OPG on the tension side
B RANKL is raised on the tension side and OPG on the compression side
C Both are raised equally on both sides
D Neither is involved in orthodontic tooth movement
E OPG stimulates osteoclast differentiation directly

102 At approximately what force level does hyalinisation occur?

A It occurs only above 500 g
B It occurs only above 150 g
C It occurs even at 30 g
D It never occurs with light forces
E It occurs only with bodily movement

103 What is the approximate duration of the lag phase of orthodontic tooth movement?

A 2 days
B 7 days
C 6 months
D 3 months
E 20 to 30 days

104 Undermining resorption differs from frontal resorption in which respect?

A It proceeds directly from the periodontal ligament surface
B It proceeds from adjacent marrow spaces because osteoclasts cannot resorb beneath a hyalinised zone
C It occurs only on the tension side
D It occurs only in adults
E It is faster than frontal resorption

105 What is the approximate normal rate of orthodontic tooth movement?

A 0.1 mm per month
B 1 mm per week
C 3 mm per month
D 1 mm per month
E 5 mm per month

106 Which force range is currently considered optimal for bodily tooth movement?

A 10 to 20 cN
B 150 to 200 cN
C 50 to 100 cN
D 250 to 400 cN
E Over 500 cN

107 Which single statement best represents the examinable position on optimal force?

A No universal optimal force exists, because of large inter-individual variation and differing loaded root surface areas
B An optimal force of 150 g has been established for all movements
C Heavier forces always move teeth faster
D Force magnitude is irrelevant
E Optimal force is determined by bracket slot size

108 Which force is appropriate for intrusion of a four-incisor segment?

A Approximately 500 g in total
B Approximately 150 g in total
C Approximately 250 g in total
D Approximately 400 g in total
E Approximately 60 g in total

109 Storey and Smith in 1952 proposed which force range as optimal?

A 30 to 60 g
B 400 to 500 g
C 250 to 300 g
D 150 to 200 g
E Under 25 g

110 Ren and colleagues modelled the fastest rate of canine retraction at approximately which force?

A 72 cN
B 272 cN
C 150 cN
D 500 cN
E 1000 cN

111 What is the approximate threshold of force duration below which orthodontic tooth movement is not clinically useful?

A 1 hour per day
B 24 hours per day
C 12 hours per day
D 18 hours per day
E 4 to 8 hours per day

112 RANKL expression during orthodontic tooth movement peaks at approximately which time?

A 6 hours
B 3 days
C 14 days
D 30 days
E 60 days

113 Which experimental finding most directly demonstrates that RANKL is essential for orthodontic tooth movement?

A Hyalinisation occurs at 30 g
B Movement is faster in the maxilla
C OPG rises on the tension side
D Deletion of RANKL in the periodontium of transgenic mice reduces osteoclasts and tooth movement
E Photobiomodulation accelerates discrete phases

114 Osteoprotegerin gene transfer to the periodontium produces which effect?

A Reduced tooth movement
B Increased tooth movement
C No change in tooth movement

D Increased root resorption
E Accelerated eruption

115 Which is the correct sequence of Henneman's four-step mechanotransduction model?

A Cell recruitment, matrix strain, tissue remodelling, fluid flow
B Tissue remodelling, cell differentiation, matrix strain, fluid flow
C Matrix strain and fluid flow, strain transferred to cells, cell recruitment and differentiation, tissue remodelling
D Fluid flow, tissue remodelling, matrix strain, cell recruitment
E Strain transferred to cells, matrix strain, tissue remodelling, fluid flow

116 What is the current status of the bone-bending hypothesis?

A It is the accepted mechanism of tooth movement
B It explains the lag phase
C It has been proved by implant studies
D There is little evidence that orthodontic forces bend bone to any clinically meaningful degree
E It replaced the pressure-tension theory

117 Which statement about piezoelectricity in orthodontic tooth movement is correct?

A It is the principal mechanism of tooth movement
B The currents generated are too small and too short-lived to be clinically significant
C It explains hyalinisation
D It is the basis of photobiomodulation
E It has been demonstrated to double the rate of movement

118 What is the approximate radiographic incidence of orthodontically induced inflammatory root resorption?

A Under 10 per cent
B Approximately 30 per cent
C 100 per cent
D Up to 95 per cent
E Up to 73 per cent

119 What proportion of teeth show severe root resorption, defined as loss of at least one third of root length?

A 1 to 5 per cent
B Under 0.1 per cent
C 10 to 15 per cent
D 25 per cent
E Over 40 per cent

120 Which single risk factor for orthodontically induced root resorption has substantive evidence?

A Treatment duration
B Class II elastics
C Heavy force
D Asthma
E Abnormal root morphology

121 Significant root resorption is identified on a mid-treatment radiograph. What is the recommended management?

A Debond immediately
B Pause force application for 3 to 6 months
C Increase the force to complete treatment faster
D Root treat the affected teeth
E Continue unchanged and review at debond

122 Which teeth are most susceptible to orthodontically induced root resorption?

A Mandibular first molars
B Maxillary second molars
C Mandibular premolars
D Maxillary incisors
E Mandibular canines

123 Which statement about the long-term significance of severe root resorption is correct?

A It does not necessarily cause greater mobility or discolouration in the long term
B It reliably causes increased mobility and tooth loss
C It always requires extraction of the affected tooth
D It reverses completely after debond
E It causes loss of vitality in most cases

124 At approximately what stage of root formation does a tooth normally erupt?

A One quarter formed
B One half formed
C Two thirds to three quarters formed
D Fully formed
E After apexification

125 Which structure is essential for formation of the eruption pathway?

A The reduced enamel epithelium alone
B Hertwig's epithelial root sheath
C The periodontal ligament
D The alveolar crestal fibres
E The dental follicle

126 A 14-year-old has a lower second molar that has failed to erupt and does not respond to orthodontic traction, becoming ankylosed. Which diagnosis should be considered?

A Primary failure of eruption
B Mechanical obstruction by the third molar
C Ectopic eruption
D Ankylosis of the primary predecessor
E Dentine dysplasia

127 Alveolar bone exists in a site only under which condition?

A Where the alveolar nerve runs
B Where a graft has been placed
C Where a tooth has erupted
D Where the periosteum is intact
E In the anterior maxilla only

128 Which single statement about corticotomy, micro-osteoperforation and piezocision is best supported?

A They reliably halve total treatment time
B They are contraindicated in adults
C They have no biological effect
D They act by increasing OPG expression
E They can increase the local rate through the regional acceleratory phenomenon, but the effect is short-lived and evidence for shorter overall treatment is sparse

129 What did El-Angbawi and colleagues conclude about photobiomodulation in 2023?

A It halves overall treatment time
B It may accelerate discrete phases but does not influence overall treatment time
C It increases root resorption
D It is more effective than corticotomy
E It works only in adults

130 On the tension side of a tooth undergoing orthodontic movement, which combination of changes occurs?

A RANKL up, OPG down, osteoclast recruitment
B Hyalinisation and undermining resorption
C Both RANKL and OPG down
D OPG up, RANKL down, osteoblast activity and bone deposition
E Cathepsin K release and Howship's lacunae

131 Osteoclasts resorb bone by which principal mechanism?

A Secretion of alkaline phosphatase
B Phagocytosis of intact hydroxyapatite crystals
C Release of cathepsins, matrix metalloproteinases and hydrogen ions, forming Howship's lacunae
D Secretion of osteoprotegerin
E Deposition of osteoid

132 The basic multicellular unit describes which concept?

A The coupled apposition and resorption unit of bone remodelling
B The smallest measurable unit of orthodontic force
C The unit of periodontal ligament width
D The bracket-wire interface
E A single osteocyte and its canaliculi

133 Which is the approximate composition of bone by proportion?

A Matrix 60 per cent, mineral 30 per cent, cells 10 per cent
B Matrix 20 per cent, mineral 70 per cent, cells 10 per cent
C Matrix 10 per cent, mineral 80 per cent, cells 10 per cent
D Matrix 45 per cent, mineral 45 per cent, cells 10 per cent
E Matrix 30 per cent, mineral 60 per cent, cells 10 per cent

134 Which cells give rise to osteoclasts?

A Mesenchymal stem cells
B Epithelial rests of Malassez
C Periodontal ligament fibroblasts
D Haematopoietic stem cells, circulating as monocytes

E Cementoblasts

135 Which of the following is a constituent of the periodontal ligament that is specific to it?

A Type I collagen
B Oxytalan fibres
C Fibroblasts
D Blood vessels
E Nerve endings

136 Which movement produces the greatest amount of hyalinisation for a given force?

A Bodily movement
B Root torque
C Extrusion
D Rotation
E Tipping

137 A patient asks how likely root resorption is. Which is the best answer to give at consent?

A It essentially never happens
B Some root shortening occurs in most people treated with fixed appliances, usually of no consequence, but in about 1 to 5 per cent of teeth it exceeds a third of the root
C It happens in about half of patients and always causes tooth loss
D It occurs only in adults
E It can be prevented entirely by light forces

138 During the initial phase of tooth movement, what is principally happening?

A Frontal resorption
B Osteoblast recruitment
C Undermining resorption
D Displacement of the tooth within the periodontal ligament space, with fluid expressed
E Hyalinisation clearance

139 Which is the correct force range for tipping a single tooth with a removable appliance?

A 25 to 40 g
B 10 to 20 g
C 100 to 150 g
D 200 to 250 g
E Over 300 g

140 Substance P and calcitonin gene-related peptide are released early in tooth movement from which source?

A Osteoclasts
B Osteocytes
C Distorted nerve endings in the periodontal ligament
D Cementoblasts
E Epithelial rests of Malassez

Block E — Occlusion, malocclusion and aetiology (Q141–170)

Notes In Orthodontics: Chapter 2 (Occlusion and Malocclusion)
Supplements: Revision Handbook **§5** — read it again if you score under 70%
Practise this: 30 questions · allow 30 minutes · answers begin on the page named in the contents

141 Which definition of malocclusion is the most defensible in an examination?

A Any deviation from ideal occlusion
B An occlusion requiring orthodontic treatment
C Any Class II or Class III relationship
D A variation from ideal occlusion of sufficient degree to be, or to have the potential to be, of aesthetic or functional significance to the individual
E An occlusion with crowding of more than 4 mm

142 In Angle's Class I molar relationship, the mesiobuccal cusp of the upper first permanent molar occludes with which structure?

A The mesial marginal ridge of the lower first molar
B The buccal groove of the lower first permanent molar
C The distal marginal ridge of the lower second premolar
D The central fossa of the lower first molar
E The buccal groove of the lower second molar

143 Which is a principal limitation of Angle's classification?

A It cannot be applied to adults
B It is not reproducible
C It requires a lateral cephalogram
D It cannot describe Class III cases
E It assumes the first permanent molar is in a stable, correct anteroposterior position, and ignores the vertical and transverse planes

144 The British Standards Institute classification of incisor relationship is based on which relationship?

A The relationship of the lower incisor edges to the cingulum plateau of the upper central incisors
B The molar relationship
C The relationship of the canines
D The overjet in millimetres
E The skeletal pattern

145 In a Class II division 2 incisor relationship, which single feature is characteristic?

A The overjet is always increased
B The lower incisors are proclined
C The upper central incisors are retroclined
D The overbite is reduced
E The molars are always Class II

146 Andrews derived his six keys of occlusion from what sample?

A 100 treated orthodontic cases
B 120 non-orthodontic normal occlusions
C 500 randomly selected patients
D A cadaveric collection
E A single family pedigree

147 Which is not one of Andrews' six keys of occlusion?

A Correct molar relationship
B Correct crown angulation
C Correct crown inclination
D Canine-guided excursion
E Flat or minimal curve of Spee

148 What proportion of the population develops an acceptable occlusion naturally?

A 30 to 40 per cent
B Under 10 per cent
C 60 per cent
D 80 per cent
E Over 90 per cent

149 What proportion of individuals have retruded contact position coincident with intercuspal position?

A Approximately 50 per cent
B Approximately 30 per cent
C Approximately 10 per cent
D Approximately 75 per cent
E Over 90 per cent

150 What proportion of malocclusion has a specific, identifiable cause?

A Nearly all
B About 20 per cent
C About 50 per cent
D About 75 per cent
E Under 5 per cent

151 Equilibrium theory explains which observation?

A That teeth sit where resting soft tissue forces balance, so light sustained forces move teeth while heavy brief forces do not
B That heavy chewing forces determine tooth position
C That the mandible grows in response to function
D That expansion is stable
E That extraction narrows the airway

152 A digit-sucking habit characteristically produces which combination?

A Retroclined uppers, proclined lowers and a deep overbite
B Bilateral scissors bite and increased overbite
C Proclined uppers, retroclined lowers, an asymmetric anterior open bite and a posterior crossbite
D Class III incisors with a reverse overjet
E A symmetrical anterior open bite with no transverse effect

153 Which single statement about the airway and facial form is the most defensible?

A Chronic nasal obstruction causes long face syndrome
B Mouth breathing has been shown to be entirely harmless
C The airway has no relationship to facial form
D Adenoidectomy reliably normalises facial growth
E Chronic nasal obstruction is associated with, but not proved to cause, increased face height and a narrow maxilla

154 The prevalence of Class II malocclusion in United States adolescents is approximately which range?

A 2 to 5 per cent
B 15 to 22 per cent
C 8 to 10 per cent
D 30 to 40 per cent
E Over 50 per cent

155 Worldwide Class III prevalence is approximately which value, with marked regional variation?

A 1.2 per cent
B 22 per cent
C 15 per cent
D 6.8 per cent
E 30 per cent

156 Which is the normal range for overbite?

A 0 to 1 mm
B 5 to 7 mm
C 2 to 4 mm, or one third to one half of the lower incisor crown
D Complete to the palatal mucosa
E There is no normal range

157 Which single trait is the strongest evidence of a genetic contribution to malocclusion?

A Familial recurrence of a Class III skeletal pattern
B Crowding of the lower incisors
C Anterior open bite
D A midline diastema
E A unilateral crossbite

158 The line of occlusion, as defined by Angle, passes through which structures?

A The buccal cusps of the upper teeth and the central fossae of the lower
B The interproximal contact points
C The incisal edges of both arches only
D The gingival margins of all teeth
E The central fossae and cingula of the upper arch, and the buccal cusps and incisal edges of the lower

159 Centric relation is best defined as which position?

A Maximum intercuspation of the teeth
B The position adopted during swallowing
C The most retruded position of the mandible
D The condyle in its most superior and anterior position against the articular eminence, independent of tooth contact
E The rest position of the mandible

160 Which is the ideal overjet range?

A 0 to 1 mm
B 2 to 4 mm
C 5 to 6 mm
D Reverse by 1 mm
E It has no defined range

161 A Class II division 1 malocclusion is perpetuated by a lower lip trap through which mechanism?

A The lip retroclines the upper incisors
B The lip causes a posterior crossbite
C The lip extrudes the lower molars
D The lip prevents mandibular growth
E The lip rests behind the upper incisors, proclining them and retroclining the lowers

162 What proportion of children with an overjet of 5 mm or more still have it as adults?

A 13 per cent of children rises to 23 per cent of adults
B 23 per cent of children falls to 13 per cent of adults
C Both figures are 23 per cent
D Both figures are 5 per cent
E Over half persist unchanged

163 Which of the following is the correct definition of ideal occlusion?

A The occlusion most people have
B Any Class I occlusion
C An occlusion with no crowding
D A theoretical concept describing the anatomically perfect relationship of the dental arches, rarely found
E An occlusion after orthodontic treatment

164 Which condition is most likely to produce a Class II malocclusion of acquired rather than developmental origin?

A A childhood condylar fracture with growth disturbance
B A digit habit
C A midline supernumerary
D Early loss of a primary molar
E Hypodontia of the lower second premolars

165 Which single statement about Class II division 2 incisors is correct?

A The lateral incisors are usually retroclined with the centrals
B The overbite is usually reduced
C The lateral incisors are frequently proclined and mesiolabially rotated
D The interincisal angle is usually decreased

E The lower face height is usually increased

166 Which is the correct approximate prevalence of Class II division 2 in Caucasian populations?

A 1 per cent
B 30 per cent
C 20 per cent
D 5 to 10 per cent
E 45 per cent

167 A patient has a 3 mm discrepancy between retruded contact position and intercuspal position with an anterior slide. Which single conclusion follows?

A This is entirely normal and requires no comment
B This exceeds the normal 0 to 2 mm range and should be recorded, since it may indicate a pseudo-Class III or a displacement
C It indicates temporomandibular joint disease
D It is an indication for occlusal adjustment
E It contraindicates orthodontic treatment

168 Which of the following is described as a seventh key of occlusion, additional to Andrews' original six?

A Canine guidance
B A Class I canine relationship
C Absence of a curve of Wilson
D Coincident centre lines
E Correct crown size, shape and tooth-size harmony

169 Neural crest cells are relevant to the aetiology of malocclusion principally because of which fact?

A They form practically all the mesenchyme of the face
B They form the enamel organ
C They determine tooth eruption timing
D They form the periodontal ligament exclusively
E They control the growth spurt

170 The equilibrium theory would predict that which of the following is unlikely to be stable?

A A tooth moved into the neutral zone
B Correction of a rotation with a bonded retainer
C Lower incisors proclined 4 mm beyond their original position
D Closure of a midline diastema with permanent retention
E Levelling of a curve of Spee in a growing child

Block F — Treatment need, indices and epidemiology (Q171–195)

Notes In Orthodontics: Chapter 2, pp. 106–112 (IOTN and PAR)
Supplements: Revision Handbook **§6** — read it again if you score under 70%
Practise this: 25 questions · allow 25 minutes · answers begin on the page named in the contents

171 The Index of Orthodontic Treatment Need Dental Health Component is applied in which way?

A All qualifying traits are summed
B Only the single worst trait is recorded
C The mean of all traits is taken
D Traits are weighted and averaged
E Only overjet is considered

172 What does the acronym MOCDO stand for?

A Molar, overjet, crossbite, displacement, overbite
B Missing teeth, occlusion, crowding, deviation, overbite
C Midline, occlusion, crowding, displacement, overjet
D Missing teeth, overjet, crossbite, displacement of contact points, overbite
E Molar, occlusion, canine, displacement, overjet

173 An overjet of 7 mm falls into which IOTN Dental Health Component grade?

A 4a
B 3a
C 5a
D 2a
E 4b

174 A crossbite with a mandibular displacement of 3 mm falls into which grade?

A 3c
B 5c
C 4c
D 2c
E 4d

175 Impeded eruption of a tooth, other than by third molar crowding, falls into which grade?

A 3i
B 4i
C 4t
D 2i
E 5i

176 The IOTN Aesthetic Component requires the clinician to select the photograph closest in which respect?

A Aesthetic impairment
B Morphological similarity to the patient
C Overjet magnitude
D Crowding severity
E Molar relationship

177 Which Aesthetic Component grades correspond to a definite need for treatment?

A 1 to 4
B 5 to 7
C 8 to 10
D Only 10
E Any grade above 3

178 Which statement about the Peer Assessment Rating is correct?

A It measures need for treatment before treatment
B It replaced IOTN
C It is hierarchical and records only the worst trait
D It is scored from photographs
E It is an accumulative score with no maximum cut-off, used to assess the outcome of treatment

179 What PAR point reduction is required for a case to be classed as greatly improved?

A At least 10 points
B At least 22 points
C At least 30 points
D At least 50 points
E Any reduction

180 A PAR score reduction of 25 per cent indicates which outcome?

A Greatly improved
B Improved
C Cannot be interpreted
D Worse or no different
E Requires retreatment by definition

181 The PAR index is best used for which purpose?

A Judging an individual clinician on a single case
B Determining eligibility for treatment
C Auditing an operator or a service across many cases
D Predicting relapse
E Assessing facial aesthetics

182 Which index is used to assess the quality of orthodontic finishing across eight criteria?

A ABO Objective Grading System
B PAR
C ICON
D IOTN
E Little's Irregularity Index

183 Which index detected a significant difference in occlusal outcome between aligners and fixed appliances?

A PAR
B IOTN Dental Health Component
C Little's Irregularity Index
D ICON
E ABO Objective Grading System

184 Little's Irregularity Index measures which quantity?

A The overjet
B The intercanine width
C The depth of the curve of Spee
D The sum of the linear displacements of the anatomical contact points of the mandibular incisors
E The Bolton discrepancy

185 Which value of lower incisor irregularity has been proposed as an acceptable level of relapse?

A Under 1 mm
B Under 3.5 mm
C Under 6 mm
D Under 10 mm
E Any value is acceptable

186 The ICON index combines which elements?

A Need and outcome only
B Overjet and overbite only
C Aesthetics and function
D Skeletal and dental factors
E Need, complexity and outcome

187 Which IOTN grade corresponds to a reverse overjet greater than 3.5 mm with no reported masticatory or speech difficulty?

A 3b
B 4b
C 5b
D 2b
E 5m

188 A patient has 3 mm of contact point displacement in the lower labial segment and no other traits. Which DHC grade applies?

A 2d
B 5d
C 4d
D 3d
E 1

189 Which single statement about the relationship between treatment need and treatment demand is correct?

A Demand is a separate question from need, and correlates only moderately with it
B They are the same thing
C Demand always exceeds need
D Need always exceeds demand
E Demand is measured by the Dental Health Component

190 Defects of cleft lip and palate fall into which IOTN grade?

A 3p
B 4p
C 5p

D 2p
E Not covered by IOTN

191 An increased and complete overbite causing gingival or palatal trauma falls into which grade?

A 3f
B 2f
C 5f
D 4f
E 4a

192 Which is the correct use of the Aesthetic Component in relation to the Dental Health Component?

A The two are summed to give a total score
B They are recorded independently, and either may indicate need
C The AC overrides the DHC
D The DHC is used only if the AC is above 8
E The AC applies only to adults

193 Which epidemiological figure best describes the prevalence of dental trauma by age 12?

A Under 2 per cent
B 5 per cent
C Over 50 per cent
D 35 per cent
E Up to 18 per cent

194 Traumatic loss of a maxillary central incisor affects approximately what proportion of children?

A 3 per cent
B 0.3 per cent
C 8 per cent
D 15 per cent
E 25 per cent

195 Which single justification for the existence of indices of orthodontic treatment need is the most defensible?

A They determine which treatment technique to use
B They predict treatment outcome
C They allow the allocation of a scarce, largely publicly funded resource according to defined criteria
D They measure patient satisfaction
E They replace clinical judgement

Block G — Evidence-based orthodontics (Q196–225)

Notes In Orthodontics: Chapter 1 (Evidence-Based Orthodontics); Appendix D
Supplements: Revision Handbook **§7** — read it again if you score under 70%
Practise this: 30 questions · allow 30 minutes · answers begin on the page named in the contents

196 Evidence-based practice is best defined as the integration of which three elements?

A Randomised trials, meta-analyses and guidelines
B Best available evidence, clinical expertise and patient values
C Textbooks, expert opinion and audit
D Systematic reviews, cohort studies and consensus
E Research, teaching and practice

197 Which sits highest in the conventional hierarchy of evidence?

A A large randomised controlled trial
B A case series
C A prospective cohort study
D A systematic review with meta-analysis
E Expert opinion

198 What does the acronym PICO denote?

A Population, intervention, comparison, outcome
B Patient, indication, cost, outcome
C Prevalence, incidence, comparison, odds
D Protocol, intervention, control, observation
E Population, incidence, cohort, odds

199 Randomisation in a clinical trial principally achieves which single aim?

A It blinds the operator
B It increases the sample size
C It makes groups comparable in known and unknown confounders
D It removes the need for a control group
E It guarantees statistical significance

200 Allocation concealment differs from blinding in which respect?

A Allocation concealment occurs after randomisation
B Allocation concealment applies only to crossover trials
C They are the same procedure
D Blinding applies only to statisticians
E Allocation concealment prevents the recruiter knowing the next allocation; blinding conceals group identity after allocation

201 In an orthodontic trial, blinding of which party is usually most feasible and most important?

A The outcome assessor
B The operator
C The patient
D The statistician alone
E The funder

202 Intention-to-treat analysis means which of the following?

A Analysing only those who completed treatment
B Analysing per protocol
C Analysing participants in the group to which they were randomised, whatever they actually received
D Excluding dropouts
E Analysing only compliant participants

203 A type I error is best described as which of the following?

A Failing to detect a real effect
B Recruiting too few participants
C Using the wrong statistical test
D Measuring the wrong outcome
E Concluding there is an effect when there is none

204 Statistical power is conventionally set at which level?

A 50 per cent
B 80 to 90 per cent
C 65 per cent
D 95 per cent
E 99 per cent

205 An I squared value of 78 per cent in a meta-analysis indicates which of the following?

A Low heterogeneity
B Moderate heterogeneity
C A statistically significant result
D High heterogeneity
E Publication bias

206 A relative risk of 0.58 for new incisor trauma after early overjet reduction corresponds to what reduction in risk?

A 58 per cent
B 29 per cent
C 42 per cent
D 142 per cent
E It cannot be expressed as a percentage

207 A funnel plot is used to assess which problem?

A Publication bias
B Heterogeneity
C Confounding
D Regression to the mean
E Loss to follow-up

208 A 95 per cent confidence interval for a relative risk that runs from 0.32 to 1.14 should be interpreted how?

A The result is statistically significant
B Heterogeneity is high
C The intervention is harmful
D The sample size was adequate
E The result is compatible with no effect, because the interval crosses 1

209 Which trial overturned the belief that early Class II treatment improves the final result?

A Mandall 2016
B Scott 2008
C Benson 2021
D O'Brien 2009
E Penning 2017

210 Which studies established that self-ligating brackets are not faster than conventional brackets?

A Tulloch and Keeling
B Scott and Fleming
C Ericson and Kurol
D Baumrind and Frantz
E Kerr and Kochel

211 Which body of work established that orthodontic treatment does not cause temporomandibular disorder?

A The Cochrane reviews of 2021
B The Royal London space analysis studies
C The Florida trial
D The Manchester twin block trial
E The American Association of Orthodontists funded studies of 1992

212 The 2021 Cochrane review on interceptive extraction of the primary canine concluded what?

A Strong evidence that it prevents impaction
B Very weak evidence that it increases eruption, and no evidence that it reduces the need for surgery
C That double extraction is superior
D That it should be performed in all cases
E That it increases the need for surgery

213 Penning and colleagues in 2017 studied customised appliances and found what?

A A significant reduction in treatment time
B Fewer appointments
C Superior occlusal outcomes
D Comparable outcomes with no reduction in treatment time, more breakages and more planning time
E Reduced root resorption

214 Which conclusion about the effect of extraction on the airway is supported by the evidence?

A There is no evidence that extraction-based treatment reduces the oropharyngeal airway or increases sleep apnoea
B Extraction reliably narrows the oropharyngeal airway
C Extraction increases apnoea-hypopnoea index by 30 per cent
D Only premolar extraction affects the airway
E The evidence is strongly positive for harm

215 Baumrind's 1984 work is cited in relation to which conclusion?

A Optimal force levels
B Anchorage loss during space closure
C The unreliability of cephalometric growth prediction
D Root resorption incidence
E Self-ligation

216 Number needed to treat is calculated as which of the following?

A 1 divided by the relative risk
B The relative risk multiplied by 100
C The reciprocal of the odds ratio
D 1 divided by the absolute risk reduction
E The absolute risk multiplied by the sample size

217 Which single statement about the orthodontic evidence base is the most defensible in an oral examination?

A It is as robust as that of cardiology
B It is thin, with few large randomised trials, and several long-held beliefs have not survived it
C It is entirely absent
D It consists only of case reports
E It has been settled by consensus statements

218 Richardson's work of 2002 relates to which conclusion?

A Aligners are inferior on ABO-OGS
B Self-ligation is not faster
C Extraction does not damage the profile
D Growth prediction is unreliable
E Third molars are not a primary cause of late lower incisor crowding

219 Luther's 2007 conclusion concerned which belief?

A That an ideal functional occlusion improves stability and prevents TMD
B That extraction narrows the airway
C That early treatment shortens overall treatment
D That RME improves nasal breathing
E That headgear is more effective than a functional appliance

220 Bowman and Johnston in 2000 concluded what about extraction and the profile?

A Extraction reliably flattens the profile
B Extraction always improves the profile
C Lip retraction must be considerable before a profile is regarded as unattractive
D The profile is unaffected by incisor position
E Only non-extraction treatment preserves the profile

221 Which study design is most susceptible to selection bias?

A Randomised controlled trial
B Retrospective case-control study
C Systematic review of randomised trials
D Meta-analysis
E Split-mouth randomised trial

222 A split-mouth design is particularly appropriate for which type of orthodontic question?

A Comparing surgical with non-surgical Class III correction
B Comparing early with late Class II treatment
C Comparing two extraction patterns
D Comparing two bonding adhesives
E Comparing retention regimens

223 The GRADE approach is used to assess which of the following?

A The certainty of the evidence for an outcome
B The size of a treatment effect
C The cost of an intervention
D The prevalence of a condition
E The statistical power of a study

224 Which single factor most commonly downgrades the certainty of evidence in orthodontic Cochrane reviews?

A Publication bias
B Indirectness of population
C Risk of bias and imprecision arising from small, unblinded trials
D Excessive sample sizes
E Absence of a comparator

225 An orthodontic claim is supported only by a series of 15 consecutively treated cases with no control group. Where does this sit in the hierarchy of evidence?

A Above a cohort study
B It has no place in the hierarchy
C At the level of a randomised trial
D At the level of a systematic review
E Below a cohort study and above expert opinion

Block H — Ethics, consent, communication and the psychosocial patient (Q226–255)

Notes In Orthodontics: Part V, unit V.1 (Ethics, consent and capacity); Topic 9 in part; Chapter 7
Supplements: Revision Handbook **§8** — read it again if you score under 70%
Practise this: 30 questions · allow 30 minutes · answers begin on the page named in the contents

226 Which three elements are required for consent to be valid?

A Capacity, sufficient information and voluntariness
B A signed form, a witness and a chaperone
C Capacity, cost disclosure and a cooling-off period
D Written information, a second opinion and a signature
E Age over 16, a parent present and a written estimate

227 The Montgomery standard requires disclosure of which risks?

A Only risks occurring in more than 1 per cent of cases
B Only risks the clinician considers material
C Risks to which a reasonable patient in the patient's position would attach significance
D Only risks specified by the regulator
E All conceivable risks without exception

228 Which alternative must always be discussed as part of orthodontic consent?

A Treatment by a specialist only
B Surgical treatment in every case
C Treatment abroad
D Treatment with aligners specifically
E The alternative of no treatment, and what will happen if nothing is done

229 Which single statement about a signed consent form is correct?

A It constitutes consent
B It is evidence that a consent process took place, but is not itself consent
C It removes liability for complications
D It is required only for surgical procedures
E It is valid indefinitely

230 Gillick competence describes which situation?

A Emergency treatment without consent
B A parent's right to override a child's refusal
C An adult lacking capacity
D A child under 16 with sufficient understanding to consent to treatment
E Consent by a sibling

231 Capacity is best described as which of the following?

A A fixed attribute of an individual
B Determined solely by age
C Decision-specific and time-specific, assessed by a four-part functional test
D Determined by the treating clinician's judgement of the patient's intelligence
E Present only in adults

232 What is the approximate prevalence of body dysmorphic disorder among orthodontic patients?

A 5 to 13 per cent
B 1 to 3 per cent
C Under 1 per cent
D 20 to 25 per cent
E Over 30 per cent

233 Which single feature is the strongest red flag for body dysmorphic disorder in an orthodontic consultation?

A A request for aligners rather than fixed appliances
B A history of two previous courses of successful treatment
C A request for a second opinion
D Anxiety about the cost
E Preoccupation with a defect that is trivial or invisible to the clinician, with serial dissatisfaction with previous treatment

234 Which statement about motivation and cooperation is best supported?

A Cooperation can be reliably predicted at the first visit
B Cooperation correlates with IOTN grade
C Girls cooperate better than boys
D Internally motivated patients cooperate better than those brought by a parent or dentist
E Cooperation is unrelated to attendance history

235 Which appliance type is associated with the highest discontinuation rates in children and adolescents?

A Fixed labial appliances
B Removable appliances
C Bonded retainers
D Transpalatal arches
E Lingual appliances

236 Which benefit of orthodontic treatment has the best supporting evidence?

A Improved periodontal health
B Reduced caries incidence
C Improved speech
D Improved masticatory efficiency
E Psychosocial benefit and improved dental aesthetics

237 Seehra and colleagues in 2011 reported which finding?

A Orthodontic treatment improves long-term psychological health
B Increased overjet makes a child more susceptible to teasing and bullying
C Aligners are preferred by adolescents
D Extraction improves self-esteem
E Retention compliance is high

238 Kenealy and colleagues in 2007 concluded what about the long-term psychological effects of orthodontic treatment?

A There is strong evidence of lasting benefit
B Benefit is confined to females
C Treatment worsens psychological health

D There is little objective evidence that treatment improves long-term psychological health
E Benefit is proportional to PAR reduction

239 Javidi and colleagues in 2017 found that orthodontic treatment improves which domains of oral health related quality of life?

A The emotional and social well-being domains
B The functional limitation and physical pain domains
C The psychological disability domain only
D All domains equally
E No domains

240 A 13-year-old wants fixed appliances but her plaque control is poor and there is generalised marginal gingivitis. What is the most appropriate action?

A Place the appliance and reinforce hygiene at each visit
B Place the appliance but use glass ionomer for all bonds
C Defer active treatment, institute a documented oral hygiene programme with review, and re-assess
D Refer for periodontal surgery
E Offer aligners instead as they are removable

241 Which of the four principles of biomedical ethics is most directly engaged by the existence of indices of orthodontic treatment need?

A Autonomy
B Beneficence
C Non-maleficence
D Justice
E Veracity

242 Which single item must be discussed at consent and is most frequently omitted by candidates?

A The cost of treatment
B That retention is lifelong if the result is to be held, and that this is agreed before treatment begins
C The name of the bracket system
D The number of archwires that will be used
E The manufacturer of the adhesive

243 A child assents to treatment but the parent refuses. What is the correct position?

A Proceed on the child's assent alone
B Proceed if the IOTN grade is 4 or above
C Refer to a court immediately
D Proceed if the child is over 12
E Treatment should not proceed without consent from a person with parental responsibility

244 Which single statement about a child's assent is correct?

A It is clinically decisive, because a technically ideal plan the child will not wear is not a plan
B It is legally sufficient for elective treatment in all cases
C It is irrelevant if the parent consents
D It is required only for surgical procedures
E It replaces the need for parental consent after age 14

245 Which is the correct way to check understanding at the end of a consent discussion?

A Ask whether the patient has any questions
B Provide a leaflet and ask them to sign
C Ask the patient to tell you back, in their own words, what has been agreed
D Repeat the information a second time
E Ask the parent whether the child understood

246 Which single figure should be quoted at consent about demineralisation?

A It affects fewer than 1 in 100 patients
B It affects up to half of fixed appliance patients and can be visible within 6 months
C It affects only patients with poor diet
D It affects around 5 per cent
E It affects only the lower incisors

247 An adult self-refers requesting aligner treatment, having been dissatisfied with two previous courses of orthodontics elsewhere and presenting extensive photographic documentation of a discrepancy you cannot see. What is the most appropriate response?

A Begin aligner treatment as requested
B Offer surgery
C Refer directly to an orthodontist for immediate treatment
D Explore the patient's expectations carefully, consider body dysmorphic disorder, and do not commit to treatment at this visit
E Discharge the patient without explanation

248 Which item of information about cost should be given at the outset?

A The total fee and the cost of replacing lost or broken retainers
B Only the total treatment fee
C Only the deposit
D Nothing, until treatment is complete
E Only the cost of records

249 Which single statement about consent as a process is correct?

A Consent obtained at the assessment visit covers all subsequent changes of plan
B Consent may be delegated to a nurse
C A material change of plan during treatment requires the consent discussion to be revisited and recorded
D Consent is required only before the first appliance is fitted
E Consent expires after 12 months

250 Which of the following best describes the role of the patient's own concern in treatment planning?

A It is subordinate to the occlusal analysis
B It determines the appliance but not the aims
C It is relevant only in adults
D It should be recorded but not acted upon
E It is the outcome measure against which treatment will ultimately be judged

251 Which is the most appropriate response to a parent who has read online that orthodontic extractions cause facial collapse?

A Explain that comparable cases treated with and without extraction show little or no facial difference, and that lip retraction must be considerable before a profile is judged unattractive
B Agree and plan non-extraction treatment regardless
C Decline to discuss the point
D Refer the family elsewhere
E Provide a leaflet without discussion

252 A patient asks whether orthodontic treatment will cure their jaw clicking. What is the most appropriate reply?

A It usually does
B It will cure it if the occlusion is perfected
C There is no evidence that orthodontic treatment reduces temporomandibular disorder signs and symptoms, and none that it causes them
D It will worsen it
E Orthodontics should be avoided in anyone with a click

253 A 16-year-old asks that her parents not be told the details of her orthodontic treatment. What is the most appropriate approach?

A Refuse and inform the parents in full
B Continue treatment without records
C Discharge her from care
D Inform the parents only in writing
E Explore her reasons, respect confidentiality where she has capacity, and encourage but not compel parental involvement

254 Which single statement about eating disorders and orthodontic treatment is correct?

A They are an absolute contraindication for life
B Elective treatment should be deferred while the disorder is active, with liaison and preventive care in the interim
C They have no relevance to orthodontics
D They contraindicate only removable appliances
E They require antibiotic cover

255 Which statement about the elective nature of orthodontic treatment is correct?

A It lowers the standard of consent required
B It removes the need to discuss alternatives
C It has no bearing on consent
D It raises the standard, because the patient is healthy and the treatment is chosen rather than needed
E It applies only to adult patients

Block I — Clinical examination and records (Q256–285)

Notes In Orthodontics: Chapter 7 (Examination and Diagnosis)
Supplements: Revision Handbook **§9** — read it again if you score under 70%
Practise this: 30 questions · allow 30 minutes · answers begin on the page named in the contents

256 In which position should a patient be seated for extra-oral examination?

A Supine with the head supported
B Reclined at 45 degrees
C Upright, Frankfort plane horizontal, in natural head position, teeth in centric occlusion and lips at rest
D Standing with the head tilted forward
E Upright with the mouth open

257 Which single finding most changes the urgency of treating a crossbite?

A The presence of a mandibular displacement on closing
B The number of teeth involved
C The patient's age
D The IOTN aesthetic component grade
E The overjet

258 Competent lips are best defined as which of the following?

A Lips that meet with muscular effort
B Lips that cover the upper incisors completely
C Lips that meet only on swallowing
D Lips with an interlabial gap under 6 mm
E Lips that meet at rest with no muscular effort

259 What interlabial gap is regarded as the key soft tissue sign of vertical excess?

A Over 1 mm
B Over 12 mm
C Over 8 mm
D Over 3 to 4 mm
E Any measurable gap

260 What is the approximate normal maxillary-mandibular plane angle?

A 17 degrees
B 27 degrees
C 22 degrees
D 34 degrees
E 40 degrees

261 Lower anterior face height normally represents approximately what proportion of total anterior face height?

A 45 per cent
B 50 per cent
C 75 per cent
D 65 per cent
E 55 per cent

262 What proportion of the upper incisor crown is revealed by an average smile?

A 25 to 50 per cent
B 75 to 100 per cent
C 50 to 75 per cent
D 100 per cent plus 4 mm of gingiva
E Under 25 per cent

263 The smile curve is parallel to the lower lip in approximately what proportion of people?

A 40 per cent
B 55 per cent
C 70 per cent
D 85 per cent
E Over 95 per cent

264 A nasolabial angle of 75 degrees in a patient with a bimaxillary protrusion suggests which of the following?

A Incisor retraction is likely to improve the profile
B Incisor retraction will worsen the profile
C Incisor proclination is indicated
D The nasolabial angle is irrelevant to planning
E Surgery is mandatory

265 From what age should the maxillary canine be palpated in the buccal sulcus at every examination?

A 6 years
B 12 years
C 9 to 10 years
D 14 years
E Only if the primary canine is mobile

266 What is the normal range of maximum mouth opening in an adult?

A 20 to 25 mm
B 25 to 35 mm
C At least 60 mm
D At least 40 mm
E There is no normal range

267 Which single record is most useful for permitting formal space analysis?

A Extra-oral photographs
B Study casts or an intra-oral scan
C A lateral cephalogram
D A panoramic radiograph
E A periapical of the incisors

268 Which is the correct indication for a lateral cephalogram?

A Every orthodontic patient
B To predict growth
C Only in adults
D Only in Class III cases
E Where incisors will be moved significantly, growth will be modified, or surgery is planned

269 Which parallax rule identifies the position of an unerupted tooth?

A The object that moves with the tube is lingual or palatal
B The object that moves with the tube is buccal
C The object that moves against the tube is palatal
D Movement direction is unrelated to position
E Parallax cannot be used for canines

270 Which imaging modality is required to assess the temporomandibular disc?

A Panoramic radiograph
B Cone beam computed tomography
C Magnetic resonance imaging
D Lateral cephalogram
E Submentovertex view

271 Which single principle governs the taking of any orthodontic radiograph?

A As many views as possible for completeness
B Justification for the individual patient, optimisation and limitation
C Routine annual radiography
D Radiography only when requested by the patient
E Radiography at every visit

272 Which examination finding would most strongly suggest a pseudo-Class III?

A A reverse overjet of 5 mm with no displacement
B Proclined lower incisors
C A Class III molar relationship a full unit bilaterally
D An anterior displacement from retruded contact position to intercuspal position, with an edge-to-edge incisor relationship in centric relation
E An increased lower face height

273 The rule of thirds divides the face using which landmarks?

A Glabella, subnasale and menton, with trichion
B Nasion, gonion and menton
C Porion, orbitale and menton
D Sella, nasion and A point
E Trichion, nasion and pogonion

274 Which single element of the history sets the outcome against which treatment will be judged?

A The medical history
B The family history
C The presenting complaint and what the patient expects treatment to achieve
D The dental attendance history
E The social history

275 Which finding on examination of a Class III patient most influences prognosis for camouflage?

A The molar relationship
B The presence of third molars
C The overbite
D The centre line
E The degree of existing dental compensation, particularly lower incisor inclination

276 Which is the appropriate set of extra-oral photographs for an orthodontic record?

A Frontal at rest, frontal smiling, profile and three-quarter
B Frontal only
C Profile only
D Frontal smiling and occlusal
E Three-quarter only

277 Which single statement about vitality testing before orthodontic treatment is correct?

A It is required for every patient
B It is required only in adults
C It is required for any traumatised or discoloured tooth before force is applied, and the baseline recorded
D It is required only if the tooth is mobile
E It should be performed after appliance placement

278 What is the normal incisor show at rest in a young adult?

A 0 to 1 mm
B The upper incisors should not show at rest
C 6 to 8 mm
D 10 mm
E 2 to 4 mm

279 Which is the appropriate order of examination?

A Intra-oral, then extra-oral, then history
B History, then extra-oral, then intra-oral
C Radiographs, then history, then examination
D Extra-oral, then history, then radiographs
E Any order, provided all elements are covered

280 Which single feature should be recorded when describing an overbite?

A Its depth in millimetres only
B Its relationship to the curve of Spee
C Whether it is Class I or Class II
D Whether it is increased, average or reduced; complete or incomplete; and traumatic or not
E Its IOTN grade only

281 A CBCT is being considered for an impacted canine. Which single justification is adequate?

A The patient has requested one
B It provides a more attractive image for the record
C Conventional imaging has not established the relationship of the canine to the lateral incisor root, and the answer will change management
D It is included in the practice fee
E It is routine for all impactions

282 Which single statement about buccal corridors is correct?

A They are one component of smile aesthetics, and extraction does not reliably widen them
B They should always be eliminated
C They are widened by every extraction
D They determine the extraction decision
E They are measured on a lateral cephalogram

283 Which single feature of the lower lip most influences a Class II division 1 malocclusion?

A Its vermilion thickness
B Its relationship to the E-line only
C Its colour
D Its length relative to the upper lip
E Whether it rests behind the upper incisors, forming a lip trap

284 Which of the following should be recorded before any orthodontic appliance is placed in an adult?

A The patient's occupation
B The patient's preferred appointment times
C The brand of toothpaste used
D Pre-existing gingival recession, with probing of every tooth
E The number of previous dental visits

285 Which single statement about the path of closure is correct?

A It is relevant only in Class III cases
B A deviation from retruded contact position to intercuspal position is one of the most important findings that can be made, and must be recorded
C It cannot be assessed clinically
D It is assessed only on mounted casts
E It is relevant only in adults

Block J — Imaging and cephalometrics (Q286–325)

Notes In Orthodontics: Topic 2 (Cephalometry); Chapter 7, pp. 411–482
Supplements: Revision Handbook **§10** — read it again if you score under 70%
Practise this: 40 questions · allow 40 minutes · answers begin on the page named in the contents

286 Standardisation of a lateral cephalogram is achieved by which device?

A A bite jig
B A rectangular collimator
C A phantom head
D A grid
E A cephalostat

287 For a lateral cephalogram, the teeth should be in which position?

A Maximum intercuspation
B Retruded contact position
C The rest position
D Edge to edge
E Wide open

288 Which landmark is defined as the deepest concavity on the anterior maxilla between the anterior nasal spine and the alveolar crest?

A Nasion
B Pogonion
C B point
D A point
E Prosthion

289 What is the normal value for SNA?

A 81 degrees plus or minus 3
B 78 degrees plus or minus 3
C 75 degrees plus or minus 3
D 85 degrees plus or minus 3
E 90 degrees plus or minus 3

290 An ANB of 6 degrees indicates which skeletal pattern?

A Class I
B Class III
C Class II
D It cannot be determined from ANB
E Bimaxillary retrusion

291 The Wits appraisal is measured between which structures?

A Sella and nasion
B Gonion and menton
C Porion and orbitale
D Perpendiculars from A and B points to the functional occlusal plane
E Nasion and pogonion

292 Which normal value applies to the Wits appraisal in males?

A Plus 1 mm
B Minus 1 mm
C Minus 4 mm
D Plus 4 mm
E Zero in both sexes

293 The Eastman correction adjusts ANB for which factor?

A The vertical dimension
B The position of porion
C Magnification error
D The occlusal plane cant
E The anteroposterior position of nasion and the inclination of the sella-nasion plane

294 Which is the normal value for upper incisor inclination to the maxillary plane?

A 109 degrees plus or minus 6
B 102 degrees plus or minus 6
C 95 degrees plus or minus 6
D 118 degrees plus or minus 6
E 125 degrees plus or minus 6

295 Which is the normal value for the interincisal angle?

A 110 degrees plus or minus 10
B 120 degrees plus or minus 10
C 135 degrees plus or minus 10
D 150 degrees plus or minus 10
E 160 degrees plus or minus 10

296 Houston's 1983 statement about cephalometrics was which of the following?

A Cephalometrics is the most accurate diagnostic tool available
B Cephalometrics is definitely not an exact science
C Cephalometrics should replace clinical examination
D Cephalometrics predicts growth reliably
E Cephalometrics is unnecessary in orthodontics

297 Baumrind and Frantz in 1971 demonstrated which finding about landmark identification?

A All landmarks have equal reliability
B Error is random and cancels out
C Error is eliminated by digitisation
D Each landmark has a characteristic non-circular envelope of error in x and y
E Only soft tissue landmarks are unreliable

298 Why are A and B points criticised as cephalometric landmarks?

A They are convenient rather than anatomical, and both remodel as the incisors move
B They are difficult to see on a radiograph
C They are bilateral structures
D They lie outside the mid-sagittal plane
E They are affected by head position only

299 Which single problem affects the Frankfort horizontal plane as a reference?

A It is easy to locate but variable between individuals
B It cannot be identified on a lateral cephalogram
C Porion and orbitale are both difficult to locate, are bilateral and must be averaged, and the plane does not lie in the mid-sagittal plane
D It changes with tooth movement
E It requires a CBCT to identify

300 Why must the enlargement factor of a cephalostat be known?

A To calculate radiation dose
B To choose the exposure time
C To identify landmarks
D To superimpose serial films
E To permit comparison of linear values between machines

301 Which of the following statements about cephalometric superimposition is correct?

A No method has been proved to deliver completely accurate results, and all carry appreciable error
B One method has been proved completely accurate
C Superimposition should be performed on outlines
D Superimposition is required for all patients
E Superimposition eliminates landmark error

302 Superimposition on the mandible should use which structures?

A The lower border and the gonial angle
B The mandibular plane and menton
C The internal cortical outline of the symphysis, the mandibular canal and the tooth germ crypts
D The condylar outline
E The incisal edges

303 In what order should the three superimpositions be performed?

A Mandible, maxilla, cranial base
B The order is immaterial
C Maxilla, mandible, cranial base
D Cranial base, mandible, maxilla
E Cranial base, maxilla, mandible

304 Cephalometric growth prediction should be regarded in which way?

A As reliable in Class II cases
B As at best unreliable, and never a justification for a radiograph on its own
C As accurate to within 1 mm
D As mandatory before functional appliance therapy
E As superior to serial films

305 What is the constructive alternative to cephalometric growth prediction?

A A single CBCT
B A three-dimensional facial scan alone
C Hand-wrist radiography alone
D Serial films approximately one year apart
E Genetic testing

306 Which single limitation of CBCT is most relevant to temporomandibular assessment?

A It has a low radiation dose
B It cannot image cortical bone
C It cannot image the disc or the muscles
D It requires a cephalostat
E It is unavailable in most centres

307 What proportion of CBCT scans are reported to contain incidental findings?

A From about a quarter to nearly all
B Approximately 10 per cent
C Under 5 per cent
D Approximately 15 per cent
E None

308 Which is a valid justification for a CBCT in orthodontics?

A To obtain a more attractive record
B To assess oral hygiene
C To predict growth
D To measure the overjet
E To assess an impacted tooth where conventional imaging is inadequate and where the result will change management

309 Which statement about applying two-dimensional cephalometric norms to three-dimensional measurements is correct?

A They may be applied directly
B They apply only in adults
C They apply only to skeletal measurements
D They cannot be applied directly, because landmark definitions differ and no agreed normative database exists
E They apply only to angular measurements

310 Three-dimensional facial surface imaging by stereophotogrammetry has which principal advantage?

A It shows bone in detail
B It involves no ionising radiation and permits serial soft tissue superimposition
C It replaces the lateral cephalogram
D It images the temporomandibular disc
E It measures tooth movement directly

311 Which regions give good reproducibility for serial three-dimensional facial superimposition?

A The cheeks and the chin
B The ears
C The lips and the nasolabial folds
D The lower border of the mandible
E The forehead, mid-nose and infraorbital regions

312 Which is the correct positioning for a posteroanterior cephalogram?

A Mid-sagittal plane perpendicular to the beam and parallel to the film
B Mid-sagittal plane parallel to the beam and perpendicular to the film
C Frankfort plane vertical

D The head rotated 45 degrees
E The patient supine

313 Which is the principal difficulty in assessing asymmetry on a posteroanterior cephalogram?

A Radiation dose
B Patient cooperation
C Magnification
D Identifying the midsagittal plane
E Film size

314 Which of the following is the normal range for the Holdaway angle?

A 7 to 15 degrees
B 0 to 5 degrees
C 20 to 25 degrees
D 30 to 40 degrees
E 45 degrees plus

315 What is the normal position of the lower incisor relative to the A-pogonion line?

A Minus 4 mm
B Plus 4 mm
C 0 mm plus or minus 2
D Plus 8 mm
E It is not a recognised measurement

316 A radiograph is described as unjustified if which condition applies?

A The dose exceeds 1 mSv
B A previous radiograph exists
C The patient is under 16
D There is no specific clinical question that the radiograph will answer and that will change management
E It is taken in general practice

317 Which is the appropriate first-line radiograph for a survey of the developing dentition?

A Lateral cephalogram
B Panoramic radiograph
C CBCT
D Full-mouth periapicals
E Occlusal view alone

318 Which single cephalometric threshold most strongly favours surgery in a Class III patient?

A ANB of minus 1 degree
B SNA of 79 degrees
C MMPA of 30 degrees
D Overjet of 0 mm
E Wits appraisal of minus 11 mm or worse

319 Which cephalometric threshold favours surgery rather than camouflage in a Class II patient?

A Pogonion to nasion perpendicular of 18 mm or more
B Overjet greater than 6 mm

C MMPA less than 20 degrees
D ANB of 5 degrees
E SNB of 76 degrees

320 Which of the following best describes the correct role of cephalometric analysis?

A It replaces clinical examination
B It is the primary diagnostic tool
C It supplements a thorough clinical examination and does not attempt to replace it
D It determines the appliance to be used
E It is required for all patients

321 Which measure would be most affected by an error in locating porion?

A SNA
B Frankfort-mandibular plane angle
C SNB
D Wits appraisal
E Interincisal angle

322 Which single measure is least affected by cranial base morphology?

A SNA
B SNB
C ANB
D Wits appraisal
E Sella-nasion to mandibular plane

323 Which statement about artificial intelligence assisted cephalometric landmarking is correct?

A It reduces error, and the main practical advantage of software is versatility rather than accuracy
B It has eliminated measurement error
C It is less reliable than manual tracing in all studies
D It removes the need for justification of the radiograph
E It is not yet available

324 Which single step best reduces measurement error in a cephalometric research study?

A Increasing the exposure
B Using a larger sample
C Hand-tracing then digitising, repeating measurements, calibrating examiners and knowing the magnification
D Using only angular measurements
E Using digital rather than film-based images

325 What is the approximate exposure time of a modern solid-state cephalostat using horizontal scanning?

A 0.1 seconds
B 1 second
C 60 seconds
D 30 seconds
E 10 seconds

Block K — Study models, space analysis and the digital set-up (Q326–350)

Notes In Orthodontics: Chapter 7 (study models); Chapter 8, Box 8.6
Supplements: Revision Handbook **§11** — read it again if you score under 70%
Practise this: 25 questions · allow 25 minutes · answers begin on the page named in the contents

326 Space analysis assesses how many areas of space requirement?

A Six
B Four
C Five
D Three
E Eight

327 Approximately how much arch length is consumed by levelling a curve of Spee?

A 0.5 mm per 1 mm of depth
B 3 mm per 1 mm of depth
C Approximately 1 mm of arch length per 1 mm of curve depth
D Levelling does not consume arch length
E It depends only on the wire used

328 How much usable space does arch expansion yield relative to the amount of expansion?

A The full amount
B Expansion yields no usable space
C Twice the amount
D One quarter
E Approximately half the amount of expansion

329 How much space is required for each millimetre of incisor retraction?

A 1 mm
B 2 mm
C 3 mm
D 4 mm
E 0.5 mm

330 Approximately how much space is consumed by 10 degrees of palatal root torque of the upper labial segment?

A 0.25 mm
B 5 mm
C 3 mm
D 1 mm
E None

331 What is the maximum interproximal reduction generally regarded as acceptable per contact in the buccal segments?

A 0.1 mm
B 0.25 mm
C Up to 1 mm

D 2 mm
E 3 mm

332 The relationship between the space requirements of the two arches should reflect which of the following?

A The molar relationship to which the case will be finished
B The patient's age
C The anchorage available
D The appliance chosen
E The overbite

333 Which named method of space analysis should be cited in an examination?

A The Bolton analysis
B The Little analysis
C The Moyers analysis
D The Tanaka-Johnston analysis
E The Royal London Hospital space analysis

334 What benefit of formal space analysis was reported by Al-Abdallah and colleagues?

A Shorter treatment times
B Fewer appointments
C Reduced root resorption
D A more disciplined and less varied approach, with more realistic aims and greater anticipation of anchorage requirements
E Better patient satisfaction scores

335 The Tanaka-Johnston prediction for the mandibular arch adds what value to half the sum of the mandibular incisor widths?

A 9.5 mm
B 10.5 mm
C 11.0 mm
D 12.0 mm
E 13.5 mm

336 Moyers probability tables predict the combined width of which teeth?

A The permanent incisors
B The third molars
C The permanent molars
D The primary molars
E The unerupted canine and premolars

337 When performing a mixed dentition space analysis, which item must be added to the space available?

A The Bolton discrepancy
B The leeway space
C The curve of Spee depth
D The overjet
E The intercanine width

338 Which single statement about the digital set-up is correct?

A It predicts the biological response to treatment
B It shows root position without additional imaging
C It replaces the space analysis
D It is a geometric prediction only, containing no biology, no anchorage and no soft tissue response
E It guarantees the predicted outcome

339 In a case with 9 mm of crowding and a thin mandibular symphysis, a digital set-up showing full alignment without extraction fails to account for which factor?

A The alveolar envelope and the biological limit of lower incisor advance
B Bracket prescription
C The archwire sequence
D The bonding adhesive
E The retention regimen

340 Which single phrase best captures the principal limitation of computer-aided planning?

A More is better
B Measure twice, cut once
C Garbage in, garbage out
D Form follows function
E Less is more

341 Which measurement is used to quantify a tooth-size discrepancy?

A Little's Irregularity Index
B The ICON index
C The PAR index
D The Bolton analysis
E The Wits appraisal

342 A Bolton overall ratio of 95 per cent indicates which of the following?

A Maxillary tooth-size excess
B Mandibular tooth-size excess
C No discrepancy
D A measurement error
E A Class III skeletal pattern

343 Which impression material is dimensionally unstable and must be cast promptly?

A Addition silicone
B Impression compound
C Polyether
D Condensation silicone
E Alginate

344 Which impression material offers the greatest dimensional stability for appliance construction?

A Addition silicone
B Alginate
C Impression plaster
D Zinc oxide eugenol
E Agar

345 Which is the correct method of measuring the depth of the curve of Spee on a study cast?

A From the incisal edge to the gingival margin
B From the mesial to the distal of the first molar
C From the deepest cusp tip to a plane joining the incisal edges and the distal cusp of the last molar
D From the occlusal plane to the mandibular plane
E It cannot be measured on a cast

346 Which of the following is not a source of space in a space analysis?

A Extraction
B Space opening for a prosthesis
C Molar distal movement
D Interproximal reduction
E Differential growth

347 Uprighting a distally angulated canine has which effect on the space analysis?

A It creates space
B It creates space only in the mandible
C It has no effect
D It consumes space
E It creates space only in extraction cases

348 Intra-oral scanning compared with conventional impressions offers which principal advantage?

A Better patient tolerance, immediate transmissibility and digital storage, with comparable full-arch accuracy
B Superior accuracy for subgingival margins
C Lower equipment cost
D The ability to record the airway
E Elimination of the need for a laboratory

349 Which single use of the digital set-up is most defensible in patient communication?

A Guaranteeing the final result
B Predicting treatment duration
C Demonstrating the space requirement and the consequence of extracting versus expanding
D Predicting relapse
E Predicting root resorption

350 Crowding of 6 mm in an arch is classified as which severity?

A Mild
B Not classifiable
C Severe
D Very severe
E Moderate

Block L — Diagnosis, the problem list and treatment planning (Q351–390)

Notes In Orthodontics: Topic 3 (Formulating a treatment plan); Chapter 8
Supplements: Revision Handbook **§12** — read it again if you score under 70%
Practise this: 40 questions · allow 40 minutes · answers begin on the page named in the contents

351 Which is the correct first stage of the treatment planning sequence?

A Diagnosis, comprising examination, records and the patient's own concerns
B Space analysis
C Anchorage planning
D Appliance selection
E Extraction decision

352 Where does orthodontic treatment planning begin, according to the modern position?

A With the mandibular incisor position
B With the molar relationship
C With the position of the dentition, particularly the maxillary incisors, within the face
D With the space analysis
E With the skeletal pattern on the cephalogram

353 Which structures define the limits of orthodontic treatment in each individual?

A The alveolar bone alone
B The periodontal ligament
C The cranial base
D The temporomandibular joints
E The soft tissues

354 The problem list should be ranked in which order?

A Alphabetically
B By order of importance to the patient
C By order in which they will be treated
D By anatomical region
E By IOTN grade

355 Which single feature makes a treatment objective measurable?

A That it is written down
B That it has a time limit
C That it is agreed with the patient
D That a colleague could assess whether it had been achieved
E That it refers to the occlusion

356 Growth modification is much more effective for which malocclusion?

A Class I
B Class III
C Class II
D Anterior open bite
E Bimaxillary protrusion

357 Which single factor determines whether growth modification is available as an option?

A Whether the patient is still growing
B The patient's motivation
C The severity of crowding
D The IOTN grade
E The appliance chosen

358 For every 1 mm of maxillary incisor retraction, the lips retract approximately how far?

A 0.3 mm
B 2.0 mm
C 1.0 mm
D 1.5 mm
E 0.7 mm

359 For every 1 mm of maxillary incisor retraction, the nasolabial angle increases by approximately how much?

A 0.5 degrees
B 5 degrees
C 3 degrees
D 1.6 degrees
E It decreases

360 The envelope of discrepancy describes which concept?

A The range of cephalometric normality
B The range of movement achievable by orthodontics alone, with growth modification, and by surgery, in each direction
C The limits of the alveolar bone
D The variation in landmark identification
E The range of acceptable overjet

361 Which single statement about the envelope of discrepancy is correct?

A It is symmetrical in all directions
B It applies only to Class II cases
C Growth modification extends it equally in all patients
D Soft tissue limits are always wider than hard tissue limits
E Maxillary incisors can be retracted further than they can be advanced

362 A compromised tooth sits in a less convenient position than a healthy one. Which should be extracted?

A The healthy tooth, for symmetry
B The compromised tooth, even though its position is less convenient
C Neither; treat non-extraction
D Both
E The decision depends on the IOTN grade

363 Which stage of the planning sequence follows directly from the space analysis?

A Appliance selection
B Phasing
C Retention
D Anchorage planning
E Diagnosis

364 Which is the correct phasing of a fixed appliance treatment?

A Alignment and levelling, overjet and overbite correction, space closure, detailing and finishing
B Space closure, alignment, levelling, detailing
C Detailing, alignment, space closure, levelling
D Overbite correction, detailing, alignment, space closure
E Levelling, detailing, alignment, space closure

365 At which point in the plan should retention be decided?

A At debond
B At the mid-treatment review
C Before active treatment begins, and agreed with the patient
D When the retainer is fitted
E When relapse occurs

366 A non-growing patient has a moderate skeletal Class II discrepancy and asks for a non-surgical solution. Which route is being described?

A Growth modification
B Interceptive treatment
C Orthognathic surgery
D Camouflage
E Adjunctive treatment

367 Which of the following is not one of Proffit's cephalometric thresholds favouring surgery in Class II?

A Overjet greater than 10 mm
B MMPA greater than 40 degrees
C Mandibular body length less than 70 mm
D Anterior face height greater than 125 mm
E Pogonion to nasion perpendicular of 18 mm or more

368 Which single element of the plan is most frequently omitted by candidates in a treatment planning question?

A The extraction pattern
B The phasing
C The anchorage plan
D The appliance
E The facial aims

369 Which single statement about compliance in treatment planning is correct?

A It is a planning factor, because a technically ideal plan the patient will not wear is not a plan
B It should be assessed at the mid-treatment review
C It is irrelevant with fixed appliances
D It can be reliably predicted from the IOTN grade
E It applies only to functional appliances

370 Which single element must be included when presenting treatment options?

A Only the option you recommend
B Only the cheapest option
C All reasonable options, including no treatment and its consequences

D Only options available in your own practice
E Only surgical options if the discrepancy is severe

371 Which single sentence best summarises a case for an examiner?

A A statement of the extraction pattern
B A statement giving age, growth status, the three planes of space, incisor and molar class, the key numbers and the soft tissue driver
C A statement of the IOTN grade
D A statement of the patient's chief complaint alone
E A statement of the appliance to be used

372 Which is the correct interpretation of a heavily dentally compensated Class III case?

A It will be easier to camouflage
B Compensation always indicates a growing patient
C Compensation is irrelevant to planning
D There is less room left for camouflage, so surgery becomes more likely
E Compensation improves stability

373 Which patient factor most often changes the plan between a 12-year-old and a 32-year-old with the same malocclusion?

A The absence of growth, which removes growth modification and changes overbite mechanics
B The appliance available
C The IOTN grade
D The extraction pattern alone
E The retention regimen alone

374 The Class III thresholds for surgery include which value for lower incisor inclination?

A 100 degrees or more
B 93 degrees
C 80 degrees or less
D 110 degrees
E Lower incisor inclination is not used

375 Discriminant analysis of the Class III surgical thresholds predicts the need for surgery with approximately what accuracy?

A 50 per cent
B 65 per cent
C 75 per cent
D 100 per cent
E Over 90 per cent

376 Which statement about the interaction between space analysis and anchorage is correct?

A The space analysis is completed first, and the anchorage requirement follows from it
B Anchorage is decided first and constrains the space analysis
C The two are unrelated
D Anchorage depends only on the appliance
E Anchorage is decided at the working wire stage

377 A patient presents with an unfavourable Class III pattern in the mixed dentition. What is often the correct answer?

A Immediate facemask therapy in all cases
B Immediate camouflage with fixed appliances
C Monitor into the late teens, avoiding mandibular extractions if surgery is likely
D Immediate surgical referral
E Extraction of the lower first premolars

378 Which single question should be answered before choosing an extraction pattern?

A Which brackets will be used
B What the IOTN aesthetic component grade is
C What the patient's occupation is
D Which retainer will be fitted
E What the space requirement is in each arch, and what molar relationship the case will finish to

379 Which single term describes limited tooth movement undertaken to serve a larger periodontal or restorative goal?

A Comprehensive treatment
B Adjunctive treatment
C Interceptive treatment
D Camouflage
E Growth modification

380 Which statement best describes the relationship between facial and occlusal aims?

A Occlusal aims always take precedence
B They are independent
C Facial aims are relevant only in surgical cases
D They must be reconciled against the skeletal pattern, and a Class I occlusion must not be bought at the cost of the profile
E Facial aims apply only to adults

381 Which single statement about treatment planning in the presence of a tooth of poor prognosis is correct?

A The tooth should be retained and the plan built around it
B The tooth should be root treated and retained
C The tooth should be removed in preference to a healthy tooth, even in a less convenient position
D The tooth should be extracted only after appliance removal
E The tooth should be extracted only if it is symptomatic

382 The database, problem list, ranked problem list, possible solutions and plan sequence is attributed to which author?

A Proffit
B Andrews
C Angle
D Begg
E Tweed

383 Which is the correct approach when facial aims and occlusal aims conflict in a non-growing patient with a severe skeletal discrepancy?

A Prioritise the occlusion and accept the profile
B Extract four premolars and reassess
C Begin camouflage and reassess at 12 months
D Refer only if the patient complains about the profile
E Recognise that the case is likely to require surgery, and make that decision before treatment begins

384 A 14-year-old has 4 mm of lower crowding, 7 mm of upper crowding and a half-unit Class II molar relationship. Which extraction pattern best reflects the space requirement?

A Four first premolars
B Lower first premolars only
C Upper second premolars and lower first premolars
D Upper first premolars and lower second premolars
E Four second premolars

385 Which factor should be assessed before any decision about arch expansion in the lower arch?

A The patient's age
B That the mandibular arch is the template and lower intercanine expansion relapses
C The bracket prescription
D The overbite
E The IOTN grade

386 Which is the correct definition of camouflage in orthodontics?

A Treatment that disguises a poor result
B Treatment that avoids retention
C Treatment using aesthetic appliances
D Treatment without extraction
E Acceptance of the skeletal pattern with positioning of the teeth to disguise it

387 Which single element should close a treatment planning answer in an oral examination?

A The bracket prescription
B What would change your mind, and where you would refer
C The cost
D The appointment interval
E The archwire sequence

388 Which is the correct order of the last three stages of the planning sequence?

A Retention, phasing, appliance
B Retention, appliance, phasing
C Phasing, retention, appliance
D Appliance, phasing, retention
E Appliance, retention, phasing

389 In a diagnostic case, which single failure is most commonly seen?

A An unranked list of findings with no argument
B Excessive detail on the cephalometric analysis
C Failure to name the appliance
D Failure to quote the IOTN grade

E Excessive discussion of retention

390 A space analysis shows an upper requirement of 12 mm and a lower requirement of 5 mm. What does this imply for the finish?

A Class I molars
B Class III molars
C Class II molars
D The analysis is in error
E The molar relationship is unaffected

Block M — Extraction decisions and space management (Q391–430)

Notes In Orthodontics: Topic 4 (Choice of extraction); Chapter 8, pp. 505–530
Supplements: Revision Handbook **§13** — read it again if you score under 70%
Practise this: 40 questions · allow 40 minutes · answers begin on the page named in the contents

391 Which method of creating space is the most predictable?

A Expansion
B Distalisation
C Interproximal reduction
D Extraction
E Proclination

392 Crowding of 3 mm in an arch is classified as which severity?

A Moderate
B Mild
C Severe
D Very severe
E Not classifiable

393 Proffit's threshold above which extraction is almost always required is which value?

A 4 mm
B 6 mm
C 20 mm
D 15 mm
E 10 mm

394 In a Class II malocclusion, where should the mandibular extraction be sited relative to the maxillary one?

A Further back
B Further forward
C In the same position
D In the lower arch only
E Extraction should be avoided

395 In a Class III malocclusion, where should the mandibular extraction be sited relative to the maxillary one?

A Further back
B In the same position
C Further forward
D In the upper arch only
E Extraction should be avoided

396 Which single principle governs the choice of extraction site?

A Symmetry above all
B Put the space where the movement is needed
C Always extract first premolars
D Always extract in all four quadrants
E Extract the tooth with the longest root

397 Which extraction gives the best spontaneous improvement, particularly in the mixed dentition?

A First permanent molar
B Lower incisor
C Second premolar
D First premolar
E Second permanent molar

398 Second premolar extraction is preferred over first premolar extraction in which circumstance?

A Mild to moderate crowding sited posteriorly, or where anchorage should favour the anterior segment
B Severe anterior crowding
C A large overjet
D Maximum anchorage cases
E Severe proclination

399 Extraction of a second permanent molar requires which condition to be met?

A The third molar must be absent
B The first molar must be restored
C A third molar must be present, of good size, correctly angulated and at the right stage of development
D The patient must be over 18
E The overbite must be reduced

400 Lower incisor extraction is most appropriate in which situation?

A Severe upper arch crowding
B Bimaxillary protrusion
C Class II division 1 with a large overjet
D Anterior open bite
E Severe lower labial segment crowding, a lower incisor of poor prognosis, or mild Class III camouflage in a non-grower

401 NICE guidance on prophylactic removal of third molars to prevent late lower incisor crowding states what?

A It is not recommended
B It is recommended in all cases
C It is recommended only in females
D It is recommended after age 25
E It is recommended if the third molars are impacted

402 A balancing extraction is most relevant for which tooth type?

A Molars
B Second premolars
C Canines
D Second permanent molars
E Third molars

403 A compensating extraction is performed for which purpose?

A To prevent a centre line shift
B To allow eruption of the third molar
C To relieve crowding
D To create anchorage
E To preserve the buccal occlusion and prevent over-eruption of the opposing tooth

404 The evidence on soft tissue differences between extraction and non-extraction treatment shows what?

A Extraction reliably flattens the profile
B Differences are minimal, and both groups are perceived to benefit
C Non-extraction reliably worsens the profile
D The evidence is entirely absent
E Only extraction of first premolars affects the profile

405 Meyer, Christou and Cheng examined which claim about extraction?

A That extraction narrows the airway
B That extraction causes root resorption
C That extraction causes TMD
D That extraction produces larger buccal corridors and a less attractive smile
E That extraction shortens treatment

406 Pont's index of arch width is best regarded in which way?

A As an accurate predictor of ideal arch width
B As applicable only to Class I cases
C As not an acceptable predictor of arch width
D As the standard for expansion planning
E As superior to the space analysis

407 A canine requiring less than half a unit of distal movement is best managed how?

A Distalisation
B Extraction of the first premolar
C Extraction of the second premolar
D Interproximal reduction alone
E Expansion

408 Which single factor most often determines that a first permanent molar rather than a premolar should be extracted?

A Its position in the arch
B The presence of third molars
C The patient's age
D The overbite
E Its poor long-term prognosis, for example from caries or molar incisor hypomineralisation

409 Which four factors govern the choice of extraction?

A Age, sex, arch form and overbite
B IOTN grade, PAR score, overjet and overbite
C Anchorage, appliance, arch form and retention
D Condition of the teeth, site and degree of crowding, type of malocclusion, and age
E Cost, compliance, aesthetics and function

410 Single-arch extraction commits the clinician to which outcome?

A A Class I molar relationship
B Accepting a Class II or Class III molar relationship
C An anterior open bite
D A deepened overbite in all cases
E A crossbite

411 Which single statement about the timing of extraction in the mixed dentition is correct?

A Timing is irrelevant
B Extraction should follow eruption of the third molars
C Extraction should always be deferred to the permanent dentition
D Extraction in the mixed dentition always requires a space maintainer
E Well-timed extraction permits spontaneous alignment, whereas in the permanent dentition appliances become obligatory

412 Which of the following is the least appropriate reason to extract a healthy premolar?

A Severe crowding requiring space
B A patient request to shorten treatment time
C A large overjet requiring retraction
D Anchorage planning in a maximum anchorage case
E Correction of a centre line discrepancy

413 What total interproximal reduction is realistically available across a full arch?

A Around 2 mm
B Unlimited
C Around 15 mm
D A modest amount, in the order of 4 to 8 mm across all contacts
E None

414 Serial extraction is contraindicated in which circumstance?

A A Class II or Class III skeletal pattern, deep overbite, open bite or hypodontia
B Severe crowding with a Class I skeletal pattern
C A well-motivated patient
D Adequate space for the canines
E The presence of all permanent teeth

415 What is the classical sequence of serial extraction?

A First premolars, then primary canines, then first primary molars
B First permanent molars, then premolars, then canines
C Primary canines, then first primary molars, then first premolars as they erupt
D Primary molars, then permanent molars, then premolars
E Primary canines, then first premolars, then primary molars

416 Which is a recognised disadvantage of serial extraction?

A It always requires surgery
B It increases the risk of root resorption
C It cannot be performed before age 14
D A long period of unattractive spacing, a deepening overbite, retroclination of the lower incisors, and a fixed appliance usually still required
E It requires a lateral cephalogram at each visit

417 Which single statement about extraction and temporomandibular disorder is correct?

A Premolar extraction causes TMD
B There is no evidence that extraction causes temporomandibular disorder
C Extraction cures TMD
D Extraction should be avoided in any patient with a click
E Only four-premolar extraction is implicated

418 Which single statement about the airway and extraction is supported by the evidence?

A Extraction reduces the oropharyngeal airway
B The evidence is equivocal
C Extraction increases the risk of apnoea by 20 per cent
D Only maxillary extraction affects the airway
E There is no evidence that extraction-based treatment reduces the oropharyngeal airway or increases sleep apnoea

419 Extraction of a first permanent molar in the mandible requires more careful timing than in the maxilla because of which fact?

A The maxillary second molar drifts mesially readily at almost any age, whereas the mandibular one does not
B The mandibular third molar erupts earlier
C The mandibular first molar has a longer root
D Anchorage is greater in the maxilla
E The mandibular arch is the template

420 A patient has a heavily restored, hypomineralised lower first permanent molar and 5 mm of lower crowding. Which extraction is most appropriate?

A Lower first premolar
B Lower second premolar
C Lower first permanent molar
D Lower incisor
E Lower second permanent molar

421 Which single consideration should be confirmed radiographically before extracting a first permanent molar?

A The overbite
B The presence, position and development of all permanent teeth including third molars
C The bone density
D The width of the periodontal ligament
E The condylar morphology

422 Which of the following is the correct effect of proclining the lower incisors by 1 mm?

A It gains 1 mm of arch space
B It loses 2 mm
C It gains 4 mm
D It gains 2 mm of arch space
E It has no effect on space

423 Which single statement about extraction in the presence of a missing lateral incisor is correct?

A The decision is set by the existing occlusion, which determines whether the space is opened or closed

B The canine should always be extracted
C Extraction of premolars is always required
D The space must always be opened
E Implants should be placed immediately

424 A Class I case has 8 mm of crowding in both arches and an acute nasolabial angle. Which extraction pattern is most appropriate?

A Four second premolars
B Upper first premolars only
C Four first premolars
D Lower first premolars only
E No extraction, with expansion

425 A Class I case has 4 mm of crowding in both arches, an obtuse nasolabial angle and a flat profile. Which approach is most appropriate?

A Four first premolar extraction
B Upper first premolar extraction only
C Four second premolar extraction
D Lower incisor extraction
E A non-extraction approach with interproximal reduction and minor arch development

426 Which single statement about anchorage loss and extraction site is supported by the evidence?

A Anchorage loss is the same for first and second premolar extraction
B Anchorage loss is greater with first premolar extraction
C Anchorage loss is greater with second premolar extraction
D Anchorage loss occurs only in the maxilla
E Anchorage loss does not occur if a lingual arch is fitted

427 Which is the correct approach to a canine requiring more than one unit of distal movement?

A Distalise alone
B Extract alone
C Both distalise and extract
D Accept the position
E Expand the arch

428 Which extraction pattern would you expect in a Class I case with 6 mm of upper crowding, 6 mm of lower crowding and Class I molars?

A Upper first and lower second premolars
B Lower first premolars only
C Upper second and lower first premolars
D Upper first premolars only
E Corresponding premolars in both arches

429 Which single statement about extraction in the adult patient is correct?

A Extraction spaces close spontaneously as in children
B Spontaneous closure does not occur, so appliances are obligatory and anchorage must be planned from the outset
C Extraction is contraindicated in adults
D Only second premolars may be extracted in adults

E Adult extraction sites do not require root parallelism

430 Which single number should be quoted when explaining the cost of unsupported anchorage?

A One quarter of the space is lost
B One third of the space is lost
C Nine-tenths of the space is lost
D Around two-thirds of the space is lost
E No space is lost

Block N — Biomechanics, force control and anchorage (Q431–481)

Notes In Orthodontics: Topics 5 and 6 (Anchorage; the three orders); Chapters 6 and 11
Supplements: Revision Handbook **§14** — read it again if you score under 70%
Practise this: 51 questions · allow 51 minutes · answers begin on the page named in the contents

431 The centre of resistance of a single-rooted tooth lies approximately where?

A At the apex
B At the incisal edge
C One third to one half down the root from the alveolar crest
D At the cementoenamel junction
E At the bracket

432 Which statement about the centre of rotation is correct?

A It moves with the force system applied
B It is fixed by anatomy
C It always coincides with the centre of resistance
D It lies at the apex in all movements
E It is determined by bracket slot size

433 A couple is best defined as which of the following?

A A single force applied at the bracket
B The product of force and distance
C Two forces in the same direction
D A force applied through the centre of resistance
E Two equal, opposite, parallel, non-collinear forces producing rotation with no net force

434 Which moment-to-force ratio produces bodily movement of a tooth?

A 0 to 1
B Approximately 7 to 1
C Approximately 2 to 1
D Approximately 10 to 1
E Approximately 20 to 1

435 A moment-to-force ratio of approximately 7 to 1 produces which movement?

A Uncontrolled tipping
B Controlled tipping
C Translation
D Root torque
E Pure rotation

436 A single force applied at the bracket with no counterbalancing moment produces which movement?

A Bodily movement
B Pure rotation
C Root torque
D Pure intrusion
E Uncontrolled tipping, with crown and root moving in opposite directions

437 Periodontal attachment loss has which effect on the mechanics of tooth movement?

A The centre of resistance moves coronally, reducing tipping
B The centre of resistance moves apically, so the same force produces a larger moment and more tipping
C The centre of resistance is unchanged
D The centre of rotation becomes fixed
E Force levels must be increased

438 A one-couple system is best described as which of the following?

A Statically indeterminate
B Applicable only to removable appliances
C Impossible to construct clinically
D Statically determinate, with a calculable and measurable force system
E Always producing pure translation

439 Most clinical orthodontics is statically indeterminate. What does this mean?

A The force system cannot be calculated, only estimated
B The forces are zero
C The appliance is unstable
D The forces are always excessive
E Equilibrium does not apply

440 Which law of physics underlies the concept of orthodontic anchorage?

A Hooke's law
B Wolff's law
C Newton's third law
D Pascal's law
E Boyle's law

441 Maximum anchorage is best defined as which of the following?

A The anchorage unit may move mesially freely
B Both units move similar amounts
C Equal and opposite movement of two units
D No, or almost no, mesial movement of the anchorage unit
E Anchorage derived from the opposing arch

442 Reciprocal anchorage describes which situation?

A Anchorage from a miniscrew
B Equal and opposite movement of two units, which is acceptable and intended
C Anchorage from the opposing arch
D Anchorage from a headgear
E Anchorage in the vertical plane only

443 Which single factor most determines the anchorage value of a group of teeth?

A The number of teeth
B The patient's age
C The bracket prescription
D The archwire material
E The summed root surface area

444 Stationary anchorage is more efficient than simple anchorage because of which fact?

A The anchor unit is being moved bodily while the active unit tips, using the same absolute anchorage more efficiently
B The anchor unit is bonded to bone
C Fewer teeth are involved
D The wire is stiffer
E Elastics are avoided

445 Unless anchorage is supported, approximately what proportion of the space created is lost?

A One quarter
B One third
C Two-thirds
D Nine-tenths
E None

446 What did Diar-Bakirly and colleagues conclude about the transpalatal arch?

A It provides substantial sagittal anchorage
B It provides very little additional anchorage in the sagittal plane
C It reliably distalises molars
D It expands the maxilla by 4 mm
E It prevents root resorption

447 Rebellato and colleagues reported which finding about the lower lingual arch?

A It provides substantial anchorage
B It is contraindicated in the mixed dentition
C It reliably preserves leeway space and provides maximum anchorage
D It provides little anchorage support and can procline the lower incisors
E It prevents late lower incisor crowding

448 Melsen and Verna advised which precaution in maximum anchorage cases?

A Anchor teeth should not be subject to initial aligning forces
B Use the heaviest wire available from the start
C Bond second molars last
D Avoid transpalatal arches
E Use Class II elastics from the outset

449 Proffit's ratio of anterior retraction to posterior protraction for reciprocal space closure without reinforcement is approximately which value?

A 3 to 1
B 2 to 1
C 1 to 1
D 4 to 1
E 10 to 1

450 Which direction of headgear pull is indicated in a patient with increased vertical proportions?

A Cervical or low pull
B Straight or combination pull
C Headgear is contraindicated
D Any direction

E High pull or occipital

451 Which force and duration are appropriate for headgear used for anchorage support?

A 250 to 350 g per side for 10 to 12 hours per day
B 150 g for 8 hours per day
C 450 to 500 g per side for 14 hours per day
D 600 g per side for 24 hours per day
E 100 g per side for 6 hours per day

452 How many independent safety mechanisms are mandatory on a headgear?

A None
B One
C Two
D Three
E Four

453 Samuels described which two mechanisms of headgear injury?

A Direct trauma and thermal injury
B Traction and torsion
C Chemical burn and allergic reaction
D Compression and shearing
E Catapult injury and nocturnal disengagement

454 A patient reports that the headgear facebow came out during the night. What is the correct instruction?

A Continue wearing it and report at the next visit
B Stop wearing it immediately and attend for review
C Reduce the wear time by half
D Wear it only during the day
E Replace the neck strap and continue

455 A temporary anchorage device changes which element of the mechanical system?

A The force system
B The moment-to-force ratio
C Newton's third law
D The biological response to the force system, by placing the reaction in bone
E The centre of resistance

456 In direct skeletal anchorage, which requirement applies?

A The screw must be ligated to the anchor teeth
B A rectangular steel wire must be used
C The line of force must pass through the screw
D The screw must be loaded immediately
E The screw must be in the mandible

457 In indirect skeletal anchorage, the anchor teeth should be ligated to the screw using which wire?

A A large rectangular stainless steel wire
B A light elastomeric chain
C A superelastic nickel titanium wire
D A 0.012 inch round wire

E Floss ligature

458 What is the recommended insertion torque range for an orthodontic miniscrew?

A 1 to 3 Ncm
B Over 40 Ncm
C 15 to 20 Ncm
D 25 to 30 Ncm
E 5 to 10 Ncm

459 Miniscrew failure rates rise appreciably below which dimensions?

A 6 mm length and 1.0 mm diameter
B 12 mm length and 2.0 mm diameter
C 10 mm length and 1.6 mm diameter
D 8 mm length and 1.2 mm diameter
E Dimensions do not affect failure

460 Which is the single commonest cause of miniscrew failure?

A Excessive force
B Root contact
C Patient age
D Screw material
E Immediate loading

461 Which miniscrew site has the lowest reported failure rate?

A Mandibular buccal shelf
B Maxillary buccal interradicular
C Zygomatic buttress
D Maxillary between the first and second molars
E Median palate

462 Which miniscrew site has the highest reported failure rate?

A Median palate
B Mandibular buccal shelf
C Maxillary buccal interradicular
D Zygomatic buttress
E Paramedian palate

463 What is the approximate pooled failure rate for orthodontic miniscrews across all sites?

A 2 per cent
B 6 per cent
C 25 per cent
D 12 per cent
E 40 per cent

464 A palatal implant requires a minimum bone thickness of which value?

A 4 mm
B 2 mm
C 1 mm
D 8 mm
E 10 mm

465 After what interval should an orthodontic miniplate be loaded?

A Immediately
B At 3 months
C At about 3 weeks
D At 6 months
E After 12 months

466 A miniscrew is mobile at four weeks. What is the correct management?

A Tighten it and continue
B Splint it to the adjacent tooth
C Reduce the force and continue
D Remove it, identify the cause, and either re-site it or change the anchorage plan
E Prescribe antibiotics and continue

467 Which single factor most governs the success of a temporary anchorage device?

A The material of the screw
B Cortical bone thickness at the site
C The patient's age
D The archwire in use
E The bracket prescription

468 Which force is appropriate for space closure using sliding mechanics?

A 50 g
B Over 600 g
C 300 g
D 500 g
E 150 g

469 Which effect does bonding the second molars have on anchorage and levelling?

A It brings the second molars into the system, aiding bite opening and increasing the anchorage unit
B It reduces anchorage
C It has no effect
D It increases root resorption
E It prevents sliding mechanics

470 A laceback is best described as which of the following?

A An elastomeric chain from molar to canine
B A bend placed distal to the molar tube
C A passive 0.010 inch wire ligature from the molar to the canine
D A stainless steel wire tied to a miniscrew
E A stop placed mesial to the molar

471 Without lacebacks, the lower incisor advances by approximately how much during levelling?

A 0.2 mm
B 1.4 mm
C 3 mm
D 5 mm
E It does not advance

472 A bendback serves which purpose?

A To increase anchorage in the vertical plane
B To open the bite
C To derotate the molar
D To minimise labial tipping of the incisors as the wire straightens
E To retain the arch form

473 Class II elastics produce which combination of side effects?

A Lower incisor proclination, upper incisor extrusion and an increased occlusal plane cant
B Lower incisor retroclination and upper incisor intrusion
C Molar intrusion in both arches
D Reduction in lower face height
E Maxillary expansion

474 Class III elastics are particularly hazardous in which patient?

A A low-angle patient
B A patient with a deep overbite
C A high-angle patient with an increased lower face height
D A growing patient
E A patient with a Class III skeletal base

475 Which single statement about anchorage in the maxillary arch is supported by the evidence?

A The maxillary arch resists anchorage loss better than the mandibular arch
B Anchorage is unnecessary in the maxilla
C Anchorage loss occurs only in the mandible
D Anchorage loss is independent of arch
E The maxillary arch is particularly susceptible to anchorage loss

476 The equilibrium equations governing an orthodontic force system state which of the following?

A The sum of forces equals zero and the sum of moments equals zero
B The sum of forces equals the sum of moments
C Forces must exceed moments
D Moments must exceed forces
E Neither forces nor moments need balance

477 Which anchorage classification allows the anchorage unit to move mesially freely?

A Maximum, or Group A
B Moderate, or Group B
C Minimum, or Group C
D Reciprocal
E Absolute

478 Cortical anchorage refers to which concept?

A Anchorage from the cortical plate of the palate
B Anchorage in the vertical plane
C Anchorage from a miniplate
D Anchorage from the opposing arch
E Increased resistance to movement when roots lie against cortical bone

479 Which single statement about a Nance button is correct?

A It provides absolute anchorage
B It provides palatal anchorage but carries a risk of ulceration beneath the acrylic button
C It is used only in the mandible
D It expands the maxilla
E It cannot be combined with a fixed appliance

480 Which is the correct force for intrusion of a single tooth?

A 150 g
B 35 to 60 g
C 100 g
D 10 to 20 g
E 250 g

481 Which single statement about placing a miniscrew as a general dentist with an interest in orthodontics is most defensible?

A All sites are equally appropriate
B Miniscrews should never be placed outside a specialist practice
C Placement in the maxillary buccal interradicular region with adequate space and imaging is within scope with appropriate training, whereas the mandibular buccal shelf is not
D Miniscrews may be placed without imaging
E Miniscrews are contraindicated in general practice

Block O — Orthodontic materials, wires, bonding and instruments

(Q482–525)

Notes In Orthodontics: Topic 7 (The ideal archwire); Chapter 10, pp. 645–665
Supplements: Revision Handbook **§15** — read it again if you score under 70%
Practise this: 44 questions · allow 44 minutes · answers begin on the page named in the contents

482 Which property is defined as the gradient of the elastic portion of the stress-strain curve?

A Stiffness
B Strength
C Range
D Formability
E Resilience

483 Springback is calculated as which ratio?

A Stiffness divided by range
B Resilience divided by strength
C Modulus of elasticity divided by yield strength
D Range multiplied by stiffness
E Yield strength divided by modulus of elasticity

484 Which identity relates the three principal wire properties?

A Strength equals stiffness plus range
B Stiffness equals strength multiplied by range
C Range equals strength multiplied by stiffness
D Strength equals stiffness multiplied by range
E The three are independent

485 Which combination describes the ideal orthodontic archwire?

A High strength, high stiffness, low range, low formability
B High strength, low stiffness, high range, high formability
C Low strength, low stiffness, high range, low formability
D High strength, high stiffness, high range, low formability
E Low strength, high stiffness, low range, high formability

486 Which is the composition of austenitic stainless steel used in orthodontics?

A Iron 50 per cent, chromium 30 per cent, nickel 20 per cent
B Titanium and molybdenum
C Iron 90 per cent, chromium 5 per cent, nickel 5 per cent
D Titanium 55 per cent, nickel 45 per cent
E Iron 70 per cent, chromium 17 to 20 per cent, nickel 8 to 12 per cent

487 The modulus of elasticity of nickel titanium relative to stainless steel is approximately which value?

A 0.05
B 0.26
C 0.5
D 1.0

E 2.0

488 The stiffness of beta titanium relative to stainless steel is approximately which value?

A One tenth
B Equal
C One half
D One third
E Twice

489 Which property of nickel titanium prevents it from being used for a bonded retainer or a finishing wire with bends?

A Its inability to be bent, welded or soldered
B Its high stiffness
C Its low range
D Its high cost
E Its high friction

490 Superelasticity in a nickel titanium archwire refers to which phenomenon?

A A high modulus of elasticity
B Shape memory on heating
C A load-deflection plateau with hysteresis, arising from a stress-induced phase transformation
D A low yield strength
E High formability

491 Which archwire comparison was supported by Cochrane evidence?

A Copper nickel titanium is superior to conventional nickel titanium
B Beta titanium aligns faster than nickel titanium
C Stainless steel aligns faster than nickel titanium
D Coaxial superelastic nickel titanium produces greater movement over 12 weeks than single-strand nickel titanium
E No archwire has ever been compared

492 Papageorgiou and colleagues in 2014 reported which finding about copper nickel titanium?

A It is significantly faster than conventional nickel titanium
B It is slightly slower to reach a working wire and more painful
C It causes less root resorption
D It is stiffer than stainless steel
E It cannot be used in the mandible

493 Which is the standard etchant concentration and time for enamel bonding?

A 10 per cent phosphoric acid for 60 seconds
B Hydrofluoric acid for 20 seconds
C 50 per cent phosphoric acid for 5 seconds
D 17 per cent EDTA for 2 minutes
E 37 per cent phosphoric acid for 15 to 30 seconds

494 Self-etching primers compared with conventional etch and bond show which characteristic?

A A shallower etch pattern, faster application, and comparable clinical bond failure rates
B Significantly higher bond failure rates
C Superior bond strength in all studies
D Incompatibility with light curing
E A requirement for a separate etchant

495 Ceramic brackets bond to composite by which principal mechanism?

A Mechanical retention alone
B Ionic exchange
C Silane coupling, providing a chemical bond
D Micromechanical retention in the enamel only
E Adhesion by van der Waals forces

496 Which is a recognised disadvantage of ceramic brackets?

A Poor aesthetics
B Brittleness, higher friction, abrasion of opposing enamel, and risk of enamel fracture at debond
C Inability to accept a rectangular wire
D Incompatibility with elastomeric modules
E Excessive flexibility

497 Which cement is preferred for orthodontic bands because of its fluoride release?

A Zinc phosphate
B Zinc polycarboxylate
C Composite resin
D Glass ionomer
E Calcium hydroxide

498 White spot lesions become visible within approximately what period of appliance placement?

A 6 months
B 2 weeks
C 18 months
D 3 years
E Only after debond

499 Which teeth are the commonest sites for white spot lesions during fixed appliance treatment?

A Lower incisors and lower premolars
B Upper central incisors and upper canines
C First molars, upper lateral incisors and lower canines
D Second molars only
E Lower molars only

500 Benson and colleagues in 2019 recommended which preventive regimen?

A Fluoride varnish annually
B No specific regimen
C Chlorhexidine mouthwash twice daily indefinitely
D Silver diamine fluoride at debond

E Daily high-fluoride toothpaste plus professional high-concentration fluoride every 6 to 8 weeks

501 Which ligation method is preferred in a patient with poor plaque control?

A Stainless steel ligatures
B Elastomeric modules
C Elastomeric chain
D Self-ligating clips only
E Any method is equivalent

502 Which instrument and technique is recommended for adhesive clean-up at debond?

A A diamond bur at high speed
B An ultrasonic scaler
C A tungsten carbide bur at low speed, followed by polishing
D A steel scaler alone
E Air abrasion

503 Which is the typical archwire used for initial alignment?

A 0.019 by 0.025 inch stainless steel
B 0.016 by 0.022 inch cobalt chromium
C 0.021 by 0.025 inch beta titanium
D 0.036 inch stainless steel
E 0.012 to 0.018 inch round nickel titanium or multistrand

504 Which is the standard MBT working wire in a 0.022 inch slot?

A 0.016 inch stainless steel
B 0.019 by 0.025 inch stainless steel
C 0.017 by 0.025 inch beta titanium
D 0.021 by 0.025 inch stainless steel
E 0.018 inch nickel titanium

505 A 0.016 by 0.022 inch wire deflects approximately how much more than a 0.019 by 0.025 inch wire in the same slot?

A 10 per cent
B 25 per cent
C 80 per cent
D 47 per cent
E They deflect equally

506 Popular superelastic nickel titanium archwires may expand the lower intercanine width by up to which amount?

A 1 mm
B 2.4 mm
C 5.9 mm
D 9 mm
E They do not expand it

507 Cobalt chromium wire such as Elgiloy has which distinctive property?

A It is supplied soft and formable, then heat-treated to raise its stiffness
B It cannot be bent
C It is superelastic

D It cannot be soldered
E It has a lower modulus than nickel titanium

508 Which single statement about aesthetic coated archwires is correct?

A They match the mechanical properties of uncoated wires
B They eliminate friction
C They are stiffer than stainless steel
D They cannot be used with ceramic brackets
E Their mechanical properties are inferior and the coating degrades in service

509 Nickel allergy affects approximately what proportion of females?

A 1 to 3 per cent
B 60 per cent
C 40 to 50 per cent
D 10 to 30 per cent
E Under 1 per cent

510 A patient with a documented nickel allergy requires fixed appliance treatment. What is the correct approach?

A Orthodontic treatment is contraindicated
B Avoid nickel titanium and use stainless steel, beta titanium or alternatives, and avoid extra-oral nickel components
C Use nickel titanium but prescribe antihistamines
D Use only ceramic brackets
E Treat with aligners only

511 The prevalence of latex allergy in healthcare workers is approximately which value?

A 1.2 per cent
B 4.3 per cent
C 35 per cent
D 20 per cent
E 9.7 per cent

512 Which is the correct activation regimen for a removable expansion plate?

A 1 mm daily
B 0.25 mm twice weekly
C 0.25 mm daily
D 2 mm weekly
E 0.5 mm monthly

513 Which is the correct activation regimen for a rapid maxillary expander?

A 0.25 mm twice weekly
B 2 mm per day for 1 week
C 1 turn per week for 3 months
D 1 to 4 turns per day, giving 0.25 to 1 mm per day, over 2 to 4 weeks
E 0.1 mm per day for 6 months

514 What force may be generated by a rapid maxillary expander?

A Up to 10 kg
B Up to 2 kg
C Up to 500 g

D Up to 50 kg
E Under 100 g

515 Which appliance was found by Cochrane review to be approximately 3 months faster than an expansion plate?

A The transpalatal arch
B The Nance button
C The quadhelix
D The lingual arch
E The rapid maxillary expander

516 The diode laser used in orthodontics operates at which wavelength range?

A 400 to 500 nm
B 2940 nm
C 1064 nm only
D 810 to 980 nm
E 10600 nm

517 Which is an appropriate orthodontic use of a diode laser?

A Cutting enamel to create space
B Exposure of a partially erupted tooth, gingival recontouring, frenectomy or operculectomy
C Removing composite adhesive
D Cutting an archwire
E Debonding ceramic brackets

518 Alginate impressions must be cast within approximately what period?

A Timing is unimportant
B Within 24 hours
C Within 3 days
D Within a week
E Immediately or within about 10 minutes, or stored in 100 per cent humidity

519 Which is the principal advantage of glass ionomer over composite for band cementation?

A Fluoride release
B Higher bond strength
C Better aesthetics
D Faster setting
E Lower cost only

520 Which orthodontic instrument should be used to prevent a cut archwire end from being inhaled or swallowed?

A A ligature cutter
B A Weingart plier
C A distal end cutter that retains the fragment
D A How plier
E A bird beak plier

521 Which is the correct storage and maintenance procedure for orthodontic pliers after sterilisation?

A Store wet in disinfectant
B Dry thoroughly and oil the hinges after sterilisation

C Store in the autoclave
D Wrap in gauze while damp
E No maintenance is required

522 Resilience is best defined as which quantity?

A The gradient of the elastic portion
B The permanent deformation tolerated
C The maximum load before fracture
D The area under the elastic portion of the stress-strain curve
E The distance the wire can be deflected

523 Formability is best defined as which quantity?

A The amount of permanent deformation the wire tolerates before fracture
B The stiffness of the wire
C The energy stored elastically
D The yield strength
E The modulus of elasticity

524 Which properties are not represented on the stress-strain curve but matter clinically?

A Stiffness and range
B Yield strength and resilience
C Biocompatibility, environmental stability, joinability, friction and cost
D Strength and springback
E Modulus and formability

525 Which single statement about intra-oral scanning accuracy is correct?

A It is inferior to alginate for full arches
B It cannot produce a digital set-up
C It is unaffected by moisture
D It is more accurate for edentulous spans than for dentate arches
E It is comparable to addition silicone for full arches, with deep undercuts and subgingival margins remaining harder

Block P — Removable and functional appliances (Q526–566)

Notes In Orthodontics: Topic 8 (Functional appliances); Chapter 9
Supplements: Revision Handbook **§16** — read it again if you score under 70%
Practise this: 41 questions · allow 41 minutes · answers begin on the page named in the contents

526 Removable appliance design should be considered under which four headings?

A Active components, retention, anchorage, baseplate
B Aims, retention, anchorage, baseplate
C Alignment, retraction, anchorage, bite plane
D Anchorage, reciprocation, adjustment, baseplate
E Aesthetics, retention, adjustment, bonding

527 Which movement can a removable appliance deliver?

A Bodily movement
B Root torque
C Uncontrolled tipping
D Rotation of a canine
E Intrusion of a molar

528 What wire gauge is conventionally used for an Adams clasp on a first permanent molar?

A 0.5 mm
B 0.6 mm
C 1.2 mm
D 0.9 mm
E 0.7 mm

529 Which wire gauge is used for a palatal finger spring?

A 0.7 mm
B 0.5 mm
C 0.9 mm
D 1.0 mm
E 0.3 mm

530 By approximately how much should a palatal finger spring be activated?

A One quarter of the wire diameter
B 5 mm
C Twice the wire diameter
D Approximately half the diameter of the spring wire
E Activation is not required

531 What force is appropriate for tipping a single tooth with a removable appliance?

A 10 g
B 100 g
C 25 to 40 g
D 200 g
E 400 g

532 The two fatal limitations of removable appliances are which of the following?

A Compliance and two-point contact
B Cost and aesthetics
C Anchorage and retention
D Speech and hygiene
E Breakage and adjustment time

533 A flat anterior bite plane in a removable appliance serves which purpose?

A Crossbite correction
B Retention of expansion
C Rotation of incisors
D Anchorage reinforcement only
E Overbite reduction

534 Posterior bite planes in a removable appliance are used principally for which purpose?

A Overbite reduction
B Retention
C Anchorage
D Disocclusion during anterior or posterior crossbite correction
E Molar distalisation

535 Functional appliances produce which predominant effect?

A Skeletal change of 8 to 10 mm
B Dentoalveolar change, with a small and largely transient skeletal effect
C Pure condylar growth with no dental change
D Maxillary expansion only
E Vertical change only

536 Which is the most sophisticated explanation of how functional appliances work?

A They stimulate condylar cartilage directly
B They reposition the glenoid fossa
C They restrain maxillary growth only
D They alter muscle fibre type permanently
E They unlock the occlusion and permit normal mandibular growth to express itself, alongside dentoalveolar change

537 At what angle to the occlusal plane are twin block bite blocks conventionally constructed?

A 30 degrees
B 70 degrees
C 45 degrees
D 90 degrees
E 110 degrees

538 Which is the principal advantage of the twin block over a one-piece activator?

A It produces greater skeletal change
B It is fixed to the teeth
C It requires no compliance
D Its two-piece design allows full-time wear and eating with the appliance in place
E It corrects the transverse dimension

539 Posterior open bites developing during twin block treatment should be managed how?

A Trim the bite blocks progressively to allow posterior eruption
B Abandon the appliance
C Add posterior acrylic
D Fit a fixed appliance immediately
E Increase the wear time

540 Which functional appliance does not depend on patient compliance?

A Twin block
B Andresen activator
C Herbst
D Bionator
E Frankel

541 Over what period does a Herbst appliance typically reduce the overjet?

A 2 to 3 weeks
B 3 years
C 18 to 24 months
D 6 to 8 months
E It does not reduce the overjet

542 The Pacha randomised trial reported which failure rates for twin block and Herbst?

A Twin block 17 per cent, Herbst 37 per cent
B Twin block 37 per cent, Herbst 17 per cent
C Both 25 per cent
D Twin block 5 per cent, Herbst 50 per cent
E Neither appliance failed

543 The Pacha trial reported which mean overjet reductions for twin block and Herbst?

A 3 mm and 4 mm respectively
B 7.1 mm and 5.8 mm respectively
C Both 6 mm
D Both 9 mm
E 5.8 mm and 7.1 mm respectively

544 Pancherz reported what proportion of overjet correction remained stable at 32 years?

A 86 per cent
B 64 per cent
C 46 per cent
D 95 per cent
E 100 per cent

545 Which retention regimen did Pancherz recommend after Herbst treatment?

A None
B 6 weeks of vacuum-formed retention
C At least 2 years with an activator
D A bonded retainer only
E Night-time headgear

546 Which is the optimal timing for functional appliance therapy?

A CS1
B CS3, the circumpubertal period
C CS2
D CS5
E Timing is immaterial

547 The Frankel appliance differs from the twin block in which respect?

A It is fixed
B It produces no dentoalveolar change
C It cannot be worn at night
D It is tissue-borne, using buccal shields and lip pads to hold the soft tissues away from the dentition
E It is made of metal only

548 Voudouris and colleagues in 2003 reported which finding using implanted electrodes?

A A reduction in lateral pterygoid activity on protrusion
B Increased lateral pterygoid activity on protrusion
C No measurable muscle activity
D Increased masseter activity only
E Increased temporalis activity only

549 Koretsi and colleagues in 2015 reported which effect of functional appliance therapy?

A A large skeletal effect persisting into adulthood
B No effect on the profile
C A positive effect on the soft tissue profile in Class II subjects
D Increased root resorption
E Improved periodontal health

550 Santana and colleagues in 2020 compared maximal with incremental mandibular advancement and found what?

A Incremental advancement was significantly superior
B Neither approach was effective
C Maximal advancement caused condylar resorption
D Incremental advancement doubled the skeletal effect
E Little clinical difference between the two

551 Before functional appliance therapy in a patient with a retroclined upper labial segment, which step is required?

A Procline the upper labial segment to convert it to a division 1 pattern
B Extract upper first premolars
C Fit a transpalatal arch
D Reduce the overbite with a bite plane only
E Begin with Class II elastics

552 Which appliance combination is used for vertical control in a high-angle Class II patient?

A Cervical pull headgear with a twin block
B A quadhelix with Class II elastics
C High-pull headgear combined with a functional appliance
D A lower lingual arch alone
E A Nance button with Class III elastics

553 Which is the correct force and duration for facemask protraction therapy?

A 150 g per side for 8 hours per day
B 50 g per side at night only
C 100 g per side for 24 hours per day
D 800 g per side for 6 hours per day
E Approximately 400 g per side for 12 to 14 hours per day

554 Approximately what proportion of the overjet correction achieved by facemask therapy is dental rather than skeletal?

A 10 per cent
B About half
C 90 per cent
D None
E 100 per cent

555 Mandall and colleagues in 2016 reported what proportion of facemask-treated patients avoided surgery?

A 36 per cent
B 100 per cent
C 85 per cent
D Over 60 per cent
E None

556 Proffit reports what recurrence rate of anterior crossbite 8 years after facemask therapy?

A Under 5 per cent
B 10 per cent
C 25 to 33 per cent
D 60 per cent
E 90 per cent

557 The chin cup is best described in which way?

A A device whose short-term gain is not maintained after growth, with no agreement that it inhibits mandibular growth
B An effective inhibitor of mandibular growth
C The treatment of choice for Class III in all children
D A fixed functional appliance
E An anchorage device

558 Which single statement about myofunctional therapy is best supported?

A Muscle exercises reliably correct malocclusion
B Myofunctional appliances have strong randomised evidence
C Tongue exercises close a skeletal open bite
D Lip exercises correct a Class II skeletal pattern
E Interrupting a digit habit before the permanent incisors are established allows spontaneous correction of an anterior open bite

559 Which is the correct first-line approach to a digit-sucking habit in a motivated 8-year-old?

A A palatal crib immediately
B A twin block
C Extraction of the upper incisors
D Positive reinforcement and reminder therapy

E No intervention at any age

560 A habit-breaking appliance should be explained to a child in which way?

A As a punishment for the habit
B As a reminder rather than a punishment
C As a temporary brace to straighten the teeth
D As a retainer
E As a device that will be fitted permanently

561 Which of the following removable appliance components provides retention on an incisor?

A Adams clasp
B T-spring
C Roberts retractor
D Z-spring
E Southend clasp

562 A Roberts retractor is constructed in which gauge of wire?

A 0.7 mm without tubing
B 0.5 mm in tubing
C 0.9 mm in tubing
D 1.0 mm without tubing
E 0.3 mm in tubing

563 A midline expansion screw in a removable appliance delivers approximately what expansion per quarter turn?

A 0.1 mm
B 1.0 mm
C 0.5 mm
D 0.25 mm
E 2.0 mm

564 Which single statement about the Andresen activator is correct?

A It is a tooth-borne, passive, one-piece appliance worn principally at night
B It is a fixed appliance
C It is tissue-borne
D It permits eating in situ
E It corrects transverse discrepancies reliably

565 Which appliance is attached to a fixed appliance and used mid-treatment for Class II correction?

A Twin block
B Frankel
C Forsus or similar semi-rigid fixed functional
D Andresen activator
E Bionator

566 Which single statement about proprietary prefabricated myofunctional appliance systems is correct?

A They have strong randomised evidence of skeletal effect
B They correct skeletal open bite reliably
C They are superior to the twin block

D There is no evidence for the stronger claims made for them
E They eliminate the need for fixed appliances

Block Q — Fixed appliances — labial, lingual and aligners (Q567–606)

Notes In Orthodontics: Topic 6 (Evolution of fixed appliances); Chapters 10 and 14
Supplements: Revision Handbook **§17** — read it again if you score under 70%
Practise this: 40 questions · allow 40 minutes · answers begin on the page named in the contents

567 First order movement refers to which of the following?

A Tip
B In and out
C Torque
D Rotation
E Intrusion

568 Third order movement is built into which feature of a pre-adjusted bracket?

A The slot-to-base distance
B The mesiodistal slot angulation
C The bracket base curvature
D The tie wings
E The buccolingual slot inclination

569 Angle's edgewise appliance of 1928 differed from the ribbon arch in which respect?

A The slot was turned horizontal, giving greater control of tooth movement
B The slot was made vertical
C The wire was made round
D Brackets were bonded rather than banded
E Ligation was elastomeric

570 Standard edgewise treatment traditionally left results undertreated by approximately what proportion?

A Under 5 per cent
B 30 to 40 per cent
C 10 to 20 per cent
D 50 per cent
E It did not undertreat

571 The pre-adjusted edgewise appliance moved the source of error from where to where?

A From the bracket to the wire
B From the wire to the bracket position
C From the patient to the operator
D From the archwire to the ligature
E From the slot to the base

572 Which is the correct chronological order of Angle's appliances?

A Edgewise, ribbon arch, pin and tube, E-arch
B Pin and tube, edgewise, E-arch, ribbon arch
C Ribbon arch, E-arch, edgewise, pin and tube
D E-arch, pin and tube, ribbon arch, edgewise
E E-arch, ribbon arch, edgewise, pin and tube

573 The Begg appliance is best characterised as which of the following?

A An anchorage philosophy based on differential force, with deliberate tipping then uprighting
B A torque-driven appliance
C A lingual appliance
D An aligner system
E A functional appliance

574 The Tip-Edge appliance operates by which principle?

A Full three-dimensional control from the outset
B Continuous heavy force
C Limited tipping, with full prescription expressed in rectangular wire at finishing
D Tissue-borne mechanics
E Self-ligation

575 The MBT prescription differs from the Andrews prescription principally in which respects?

A Increased anterior tip and less incisor torque
B Absence of second order values
C Larger slot size
D Vertical slot orientation
E Reduced anterior tip, more palatal root torque on upper incisors, and three canine torque options

576 Which statement about 0.018 inch and 0.022 inch slot sizes is correct?

A There is no evidence that either is superior
B The 0.022 slot is proven superior
C The 0.018 slot is proven superior
D The 0.022 slot cannot accept a rectangular wire
E The 0.018 slot cannot express torque

577 What is the appropriate finishing wire in an 0.018 inch slot?

A 0.016 inch round steel
B 0.019 by 0.025 inch
C 0.017 by 0.025 inch
D 0.021 by 0.025 inch
E 0.014 inch nickel titanium

578 The MBT principle for varying wire stiffness at the finishing stage is which of the following?

A Change the dimension, not the material
B Use only beta titanium
C Use the same wire throughout
D Use only nickel titanium
E Change the material, not the dimension

579 Scott, Fleming and colleagues studied self-ligating brackets and found what?

A Faster tooth movement and shorter treatment
B No difference in rate of movement, treatment time, number of visits or occlusal outcome
C Superior occlusal outcomes
D Significant expansion of the arches
E Reduced root resorption

580 The distinction between active and passive self-ligating brackets is which of the following?

A Active brackets are bonded and passive brackets are banded
B Passive brackets have a larger slot
C Active brackets are ceramic
D Active clips store energy against the wire; passive slides do not
E Active brackets require elastomeric ligation

581 Penning and colleagues studied customised appliances and reported which finding on treatment time?

A A reduction of 4 months
B A reduction of 8 months
C No reduction in treatment time
D A doubling of treatment time
E Treatment time was not measured

582 Which is the principal biomechanical difference between lingual and labial fixed appliances?

A The lingual bracket lies palatal to the centre of resistance and closer to it vertically, altering intrusive and torquing effects
B The lingual bracket is further from the centre of resistance
C Lingual appliances cannot deliver torque
D Lingual appliances have a larger interbracket span anteriorly
E Lingual appliances use only round wires

583 The principal obstacle to a stock lingual appliance is which of the following?

A The cost of the brackets
B Bond strength
C Patient acceptance
D Archwire availability
E The very variable lingual anatomy of the teeth

584 Speech disturbance following placement of a lingual appliance typically lasts how long?

A Permanently
B 12 months
C 6 months
D 1 to 3 weeks
E It does not occur

585 Which single statement about the evidence for lingual appliances is correct?

A They produce superior outcomes to labial appliances
B Comparative trials are few and small, so superiority of outcome should not be claimed
C They are proven faster
D They cause less root resorption
E They eliminate the need for retention

586 Clear aligners work by which mechanical principle?

A A constant force from a rigid shell
B A shape-memory alloy effect
C A magnetic force
D Continuous heavy force
E A viscoelastic plastic shell acting as a spring that relaxes, with force decaying through the wear period

587 Which tooth movement do clear aligners perform least predictably?

A Anterior intrusion
B Root torque, extrusion and rotation of round-sectioned teeth
C Tipping
D Arch expansion by tipping
E Mild crowding relief

588 Bonded attachments in aligner therapy serve which purpose?

A To improve aesthetics
B To reduce friction
C To increase retention of the aligner only
D To give the plastic a feature to push against, which is essential for extrusion and rotation
E To provide anchorage in bone

589 Invisalign achieves the predicted outcome with approximately what accuracy?

A Around 41 per cent
B Around 65 per cent
C Around 85 per cent
D Around 95 per cent
E 100 per cent

590 What proportion of aligner cases complete treatment without a refinement?

A 41 per cent
B 17 per cent
C 6 per cent
D 65 per cent
E 90 per cent

591 What proportion of aligner cases convert to fixed appliances to finish?

A Under 2 per cent
B 50 per cent
C 35 per cent
D Over 17 per cent
E None

592 Which index shows aligners to be significantly worse than fixed appliances?

A PAR
B ABO Objective Grading System
C IOTN Dental Health Component
D ICON
E Little's Irregularity Index

593 Which outcomes favour aligners over fixed appliances?

A Occlusal outcome and root angulation
B Reduced relapse
C Severe crowding correction
D Torque expression
E Less pain in the first few days, better periodontal index scores, less eating disturbance and far less visibility

594 What daily wear time is required for aligner therapy?

A 20 to 22 hours
B 14 to 16 hours
C 8 to 10 hours
D 24 hours
E Wear time is immaterial

595 Which is the unsolved biomechanical problem in aligner therapy?

A Bond failure
B Wire selection
C The influence of adjacent teeth, since the aligner acts on every tooth simultaneously and reaction forces are accepted rather than designed
D Ligation
E Bracket positioning

596 Cobourne's assessment of aligner performance at three years after treatment reported what?

A Less post-treatment change than fixed appliances
B Greater post-treatment change than fixed appliances
C No difference
D Complete stability
E Improvement over time

597 Which patient is the most appropriate candidate for aligner therapy in general practice?

A An adolescent with 10 mm of crowding and a skeletal Class II pattern
B A patient requiring molar uprighting
C A patient requiring extraction space closure
D An adult with mild to moderate crowding, a Class I skeletal pattern and good motivation
E A patient requiring significant torque correction

598 Which single statement should be made at consent for aligner treatment?

A That fewer than one in ten cases finishes without a refinement and around one in six converts to fixed appliances
B That the predicted outcome is guaranteed
C That treatment will be faster than with fixed appliances
D That no attachments will be needed
E That retention will not be required

599 Which single element of the pre-adjusted appliance cannot compensate for a bracket placed incorrectly?

A The archwire sequence
B The ligation method
C The prescription

D The slot size
E The elastic wear

600 Which single statement about Roth's contribution to prescription design is correct?

A He introduced the vertical slot
B He introduced lingual brackets
C He invented the edgewise appliance
D He introduced self-ligation
E He sought one appliance rather than an inventory, with over-corrected values built in

601 Andrews derived the pre-adjusted prescription in which year?

A 1979
B 1956
C 1972
D 1928
E 1988

602 Which single feature distinguishes true customised lingual appliances from stock lingual systems?

A Slot size
B Bonding technique
C CAD-CAM brackets and robotically-bent wires that compensate for individual lingual anatomy
D Wire material
E Ligation method

603 Which is a recognised drawback of lingual appliances?

A Poor aesthetics
B Incompatibility with elastics
C Inability to correct rotations
D Excessive visibility
E Difficult access and hygiene, longer chairside time, tongue discomfort and a steep learning curve

604 Which single statement about clear aligners and severe crowding is correct?

A Aligners perform as well as fixed appliances
B Fixed appliances are better for severe crowding and complex cases
C Aligners are superior for severe crowding
D Severity is irrelevant to appliance choice
E Aligners should be used for all extraction cases

605 The claim that self-ligating brackets produce arch expansion is best described how?

A Well supported by randomised evidence
B Supported only in adults
C Supported only in the mandible
D Unsupported
E Proven by Cochrane review

606 Which statement about ligation method and rate of tooth movement is correct?

A Elastomeric ligation is significantly faster
B Self-ligation is significantly faster
C No difference in rate of movement has been demonstrated
D Steel ligatures are significantly faster
E Rate depends only on the ligation method

Block R — Treatment mechanics: Stage I, II and III (Q607–641)

Notes In Orthodontics: Chapters 10 and 13 (mechanics; definitive treatment)
Supplements: Revision Handbook **§18** — read it again if you score under 70%
Practise this: 35 questions · allow 35 minutes · answers begin on the page named in the contents

607 At what stage of treatment is anchorage principally spent or saved?

A During levelling and aligning, before any active mechanics begin
B During space closure
C During finishing
D During retention
E At debond

608 When should second molars be bonded?

A At the finishing stage
B Never
C Only in extraction cases
D Only in adults
E As early as possible, because they are needed for bite opening and anchorage

609 How long should a rectangular working wire remain passive before active force is applied?

A Immediately active
B Six months
C Three months
D Approximately one month
E Passive periods are unnecessary

610 At what interval should tiebacks be reactivated during space closure?

A Weekly
B Every 4 to 6 weeks
C Every 3 months
D Every 6 months
E Once only

611 Which is the principal advantage of loop mechanics over sliding mechanics?

A Simplicity
B Better patient comfort
C Lower cost
D Shorter chairside time
E Absence of friction

612 Coil springs compared with elastomeric modules for space closure show which characteristic?

A Slower and less consistent movement
B More consistent and slightly faster translation
C Identical performance
D Greater anchorage loss
E Greater root resorption

613 Every 1 mm of lower incisor advance yields how much space?

A 0.5 mm
B 1 mm
C 4 mm
D 2 mm
E None

614 The conventional limit for lower incisor proclination is which value?

A 100 degrees to the mandibular plane
B 93 degrees to the mandibular plane
C 85 degrees to the mandibular plane
D 115 degrees to the mandibular plane
E There is no limit

615 Which single question should be asked before planning overjet reduction?

A Which bracket prescription is in use
B What the patient's age is
C Whether lower arch anchorage is available
D Which retainer will be used
E What the IOTN grade is

616 The commonest single reason a case does not finish well is which of the following?

A Poor archwire selection
B Excessive force
C Inadequate elastic wear
D Bracket positioning
E Poor bonding technique

617 At which stage should brackets be reassessed and repositioned?

A At bond-up
B At the working wire stage, before finishing wires
C At debond
D At the first review after debond
E They should never be repositioned

618 Root parallelism at extraction sites should be confirmed by which means?

A Clinical inspection
B Intra-oral photographs
C A lateral cephalogram
D Study casts
E A panoramic radiograph at the finishing stage

619 For how long should a case be settled in light round wires with vertical elastics before retention?

A Approximately 6 weeks
B 1 week
C 3 months
D 6 months
E Settling is unnecessary

620 Which arch form should be retained at the end of treatment?

A The treated arch form
B A standard ovoid form
C The patient's original arch form
D A standard tapered form
E Whichever is most aesthetic

621 The distribution of arch forms in a Caucasian orthodontic practice is approximately which of the following?

A Ovoid 50 per cent, tapered 42 per cent, square 8 per cent
B Tapered 50 per cent, ovoid 42 per cent, square 8 per cent
C Square 50 per cent, ovoid 30 per cent, tapered 20 per cent
D Equal thirds
E Tapered 80 per cent, ovoid 15 per cent, square 5 per cent

622 Buccal uprighting of the posterior segments relapses in approximately what proportion of cases?

A 10 per cent
B 30 per cent
C 95 per cent
D 70 per cent
E It does not relapse

623 The connector area rule for anterior aesthetics is which of the following?

A 50, 40, 30 from the midline distally
B 30, 40, 50 from the midline distally
C Equal connectors throughout
D 60, 40, 20
E 20, 40, 60

624 Incisal embrasures should show which pattern?

A Equal size throughout
B Decreasing size from the midline distally
C Increasing size from the midline distally
D Absence of embrasures
E Largest between the centrals

625 The gingival zenith of the lateral incisor should sit where relative to the centrals and canines?

A On the same line
B Position is immaterial
C 2 mm above the line
D 3 mm below the line
E 0.5 to 1 mm below the line joining the central and canine zeniths

626 In which five planes should the occlusion be checked at finishing?

A Horizontal, vertical, transverse, dynamic and cephalometric or aesthetic
B Sagittal, vertical, transverse, rotational and axial
C Anterior, posterior, left, right and midline
D Static, dynamic, functional, aesthetic and periodontal
E Occlusal, incisal, buccal, lingual and apical

627 Which is the correct first-line management of an anterior crossbite in Stage I, where there is space and a positive overbite will result?

A Extraction of the lower incisor
B Class III elastics
C Proclination of the upper incisor with an opening coil or a removable appliance with a Z-spring and posterior bite planes
D Rapid maxillary expansion
E Referral for surgery

628 What force should be used for traction on an exposed impacted canine?

A 10 g
B 400 g
C 100 g
D 200 g
E 30 to 50 g

629 Which single element must precede exposure of an impacted tooth?

A Retention planning
B Creation of space
C Debonding of the opposing arch
D A CBCT in all cases
E Extraction of the contralateral tooth

630 Which single check should be made before every space closure activation?

A The patient's oral hygiene score alone
B The bracket prescription
C The archwire material
D Whether the anchorage is holding, measured objectively
E The elastic size

631 Class II elastics are contraindicated in which circumstance?

A A deep overbite
B A Class II molar relationship
C Lower incisors already proclined beyond the alveolar envelope
D A growing patient
E Mild crowding

632 Which is the correct sequence of the three treatment stages?

A Alignment and levelling, buccal segment correction and space closure, finishing and detailing
B Space closure, alignment, finishing
C Finishing, alignment, space closure
D Levelling, finishing, space closure
E Space closure, finishing, alignment

633 Which single feature is most often left imperfect at the end of treatment and should be checked deliberately?

A Molar relationship
B Overbite
C Overjet
D Centre line

E Vertical position and torque of the canines

634 Interproximal reduction at the finishing stage is used for which purposes?

A Anchorage control
B Molar distalisation
C Overbite reduction
D Tooth-size discrepancy, black triangles and marginal ridge discrepancies
E Root parallelism

635 Which single statement about detailing bends is correct?

A They should be avoided entirely in a pre-adjusted appliance
B They comprise first order in and out, second order tip, third order torque and artistic positioning bends
C They are placed only in nickel titanium
D They replace bracket repositioning
E They are placed before space closure

636 Which is the correct approach to a residual space at an extraction site during settling?

A Leave it to close spontaneously
B Debond and refer
C Open it to allow settling
D Place a temporary pontic
E Keep every space closed through settling

637 Which single statement about the lower arch during Stage I is correct?

A The lower arch should be expanded to relieve crowding
B The lower arch is the template, and its original form should be preserved
C The lower arch should be aligned last
D Lower incisor inclination is freely modifiable
E The lower arch requires no anchorage planning

638 Which auxiliary is used to level a deep curve of Spee by true intrusion in a non-growing patient?

A A reverse curve archwire
B Class II elastics
C A quadhelix
D A utility or bypass intrusion arch, or segmental mechanics
E A transpalatal arch

639 Which single statement about anchorage measurement is correct?

A Anchorage loss should be measured objectively, on casts or by superimposed photographs
B Anchorage loss can be judged by clinical impression
C Anchorage loss cannot be measured
D Anchorage loss is measured only on a cephalogram
E Anchorage loss is irrelevant once the working wire is placed

640 What proportion of cases did the MBT authors treat without extraction?

A 20 to 30 per cent
B 40 to 50 per cent
C 70 to 80 per cent
D Over 95 per cent

E None

641 Which is the correct role of a laceback in a maximum anchorage case?

A To advance the incisors
B To expand the arch
C To open the bite
D To distalise the canine without tipping it, preventing incisor advance during levelling
E To derotate the molar

Block S — Retention, relapse and stability (Q642–676)

Notes In Orthodontics: Topic 20 (Late lower incisor crowding); Chapter 13, pp. 975–992
Supplements: Revision Handbook **§19** — read it again if you score under 70%
Practise this: 35 questions · allow 35 minutes · answers begin on the page named in the contents

642 The elastic recoil of gingival and supracrestal fibres continues for approximately how long after tooth movement?

A 1 week
B 6 to 12 months
C 3 years
D 10 years
E Indefinitely

643 Which movement relapses most readily because of gingival fibre recoil?

A Extrusion
B Tipping
C Intrusion
D Bodily movement
E Rotation

644 Which are Proffit's three causes of relapse?

A Elastic recoil of gingival fibres, soft tissue pressures and continued growth
B Poor bracket position, inadequate torque and early debond
C Anchorage loss, root resorption and demineralisation
D Extraction, expansion and proclination
E Poor compliance, breakage and missed appointments

645 Beyond approximately what degree of lower incisor advance is permanent retention very likely to be required?

A 0.5 mm
B 5 mm
C 2 mm
D 8 mm
E Advance never requires permanent retention

646 Houston's rule for overbite stability requires which condition?

A A Class I molar relationship
B Correction of the interincisal angle and establishment of an occlusal stop
C A bonded retainer
D Extraction of premolars
E Levelling by intrusion only

647 Which retainer was found more effective than a Hawley in the lower labial segment?

A A bonded retainer
B A Begg wrap-around retainer
C A positioner
D A vacuum-formed retainer
E A spring retainer

648 Forde and Martin reported which comparison between fixed and removable retainers?

A Fixed retainers are marginally more effective in the lower arch for the first 12 months, especially for rotations, and equivalent thereafter
B Fixed retainers are superior indefinitely
C Removable retainers are superior throughout
D There is no difference at any time point
E Fixed retainers cause more relapse

649 By 4 to 5 years after treatment, what proportion of patients are still wearing their retainers?

A Nearly all
B A majority
C A minority
D Approximately half
E All patients

650 Which single complication is associated with long-term bonded retainers?

A Root resorption
B Condylar resorption
C Loss of vitality
D Ankylosis
E Bond failure, wire fracture, unwanted torque or wire syndrome, calculus and caries

651 Which is the most honest sentence to give a patient about the duration of retention?

A If you want the teeth to stay where they are, you wear a retainer for as long as you want that; there is no endpoint
B Retention is required for two years
C Retention is required until growth ceases
D Retention is optional after debond
E Retention is required only for the lower arch

652 When should the retention plan be agreed?

A At debond
B At the mid-treatment review
C Before active treatment begins
D When relapse occurs
E At the first review after debond

653 Which type of crowding is described as tertiary?

A Tooth size versus jaw size
B Crowding due to tongue, lip and cheek pressures
C Crowding due to supernumeraries
D Crowding due to early loss of primary teeth
E Crowding developing from adolescence onwards

654 Which is the major factor driving late lower incisor crowding?

A Third molars
B Mandibular growth rotation uprighting the lower labial segment
C The anterior component of occlusal force
D Reduced interproximal wear
E Periodontal attachment loss

655 The anterior component of occlusal force as a cause of late lower crowding is best described how?

A Proven
B Relevant only in males
C The principal cause
D Doubted, because the forces are too brief
E Relevant only after extraction

656 Sinclair and Little reported which finding in untreated subjects?

A Arch dimensions increase with age
B Crowding resolves spontaneously
C Mandibular arch width and length decrease with age, with increasing anterior crowding
D Third molars prevent crowding
E No change occurs after 18 years

657 A 27-year-old presents with 2 mm of lower incisor irregularity 10 years after orthodontic treatment. What is the correct interpretation?

A Normal age-related change rather than relapse, occurring in treated and untreated people alike
B Treatment failure requiring retreatment at no charge
C Evidence of an unstable extraction pattern
D Evidence of inadequate torque expression
E Evidence of third molar pressure

658 Circumferential supracrestal fiberotomy is performed at which point?

A After debond
B Only in adults
C Six months after debond
D At the time of bonding
E Before appliance removal

659 Which is the correct sequence when closing a midline diastema associated with a prominent fraenum?

A Frenectomy first, then close the space
B Neither is required
C Frenectomy alone will close the space
D Close the space first, then perform a frenectomy if needed
E Extract the fraenum-bearing tooth

660 Which retainer permits posterior settling of the occlusion?

A A vacuum-formed retainer
B A Hawley or Begg wrap-around retainer
C A bonded retainer alone
D A positioner
E None permit settling

661 Which retention appliance simultaneously settles the occlusion and retains?

A A Hawley retainer
B A vacuum-formed retainer
C A bonded retainer
D A spring retainer

E A positioner

662 Which single appliance is used to correct minor relapse of lower incisor alignment?

A A quadhelix
B A spring retainer or an aligner used as an active retainer
C A twin block
D A transpalatal arch
E A facemask

663 An anterior crossbite corrected with a positive overbite typically requires which retention?

A Permanent bonded retention
B A functional appliance
C Six months of full-time wear
D Usually no retention, because the correction is self-retaining
E A fixed retainer for 12 months

664 Which is the correct retention strategy after lower intercanine expansion?

A Permanent retention, since lower intercanine expansion reliably relapses
B No retention is required
C Six weeks of retention
D Retention only until growth ceases
E A Hawley retainer for 3 months

665 Which is the standard vacuum-formed retainer material thickness?

A 0.25 mm
B 2 mm
C 0.75 to 1 mm
D 3 mm
E 4 mm

666 A common retention regimen after fixed appliance treatment is which of the following?

A Full-time for 5 years, then stop
B Full-time indefinitely
C Nights for 6 weeks, then stop
D Full-time for approximately 3 months, then nights indefinitely
E Alternate nights for 12 months, then stop

667 Retention cannot be abandoned until which condition is met?

A The patient reaches 18 years
B Growth is essentially completed
C The retainer is lost
D Two years have elapsed
E The occlusion has settled

668 Which single item related to retention is most often omitted from the consent discussion and most often disputed later?

A The type of retainer
B The impression material
C The colour of the retainer
D The laboratory used
E The cost of replacing lost or broken retainers

669 Which of the following is the correct place for a bonded retainer in the lower arch?

A From 6 to 6
B From 3 to 3, or 2 to 2
C On the incisors only, spot-bonded to two teeth
D On the canines only
E Bonded to the molars

670 Which single statement about teeth moved outside the zone of soft tissue equilibrium is correct?

A They are stable once the periodontal ligament reorganises
B They stabilise after 12 months
C They return towards the neutral zone and need permanent retention
D They require no retention if the occlusion is ideal
E They relapse only in adults

671 Which statement about growth and retention is correct?

A Growth ceases at 18 and retention can then stop
B Very slow growth continues throughout adult life in the original pattern, so retention cannot be abandoned on the basis of age alone
C Growth affects only the maxilla after 18
D Growth-related change is confined to Class III patients
E Growth has no relevance to retention

672 Wire syndrome associated with a bonded retainer refers to which problem?

A Fracture of the wire
B Gingival recession
C Caries beneath the composite
D Unwanted torque or positional change developing in one or more bonded teeth from an unnoticed partial bond failure or wire distortion
E Loss of vitality

673 Which is the correct advice about retainer wear in a patient who reports the retainer feels tight after a week without wear?

A Attend for review rather than forcing it, since tightness indicates movement has occurred
B Force it into place and continue
C Discard the retainer
D Wear it full time for a month without review
E Reduce wear to alternate nights

674 Which is the correct sequence of the three timescales of relapse?

A Growth over months, fibres over years, soft tissues over decades
B Soft tissues over weeks, growth over months, fibres over years
C Fibres over months, soft tissues continuously, growth over a lifetime
D All three act over the same period
E Only fibre recoil is time-limited

675 Which single measure best prevents relapse of a closed midline diastema?

A A vacuum-formed retainer worn at night
B Interproximal reduction
C A Hawley retainer for 6 months
D A frenectomy alone

E A permanent bonded retainer

676 Which is the correct interpretation of retention as it appears in a treatment plan?

A A planned and agreed element of the plan, decided before treatment begins
B A single sentence in the conclusion
C A decision made at debond
D A decision for the patient alone
E An optional extra

Block T — Interceptive orthodontics and the developing dentition (Q677–721)

Notes In Orthodontics: Topics 13, 14 and 15 (missing central; impacted canine; impacted 7); Chapter 12
Supplements: Revision Handbook **§20** — read it again if you score under 70%
Practise this: 45 questions · allow 45 minutes · answers begin on the page named in the contents

677 Which interceptive intervention has the strongest supporting evidence?

A Arch expansion for crowding
B Two-phase Class II treatment
C Correction of a posterior crossbite with a displacement at ages 7 to 11
D Interceptive extraction of the primary canine
E Serial extraction

678 Localised space loss of what magnitude can generally be regained?

A Up to 1 mm
B Any amount
C Up to 6 mm
D Up to 10 mm
E 3 mm or less

679 Which single question should be asked at every interceptive decision?

A Which appliance is available
B Does waiting cost growth I cannot get back
C What the patient prefers
D What the IOTN grade is
E Whether the parents can afford treatment

680 A review interval for an interceptive decision should be specified how?

A As keep an eye on it
B As at the discretion of the general dentist
C As whenever the patient returns
D As a defined date, with a defined endpoint
E No review is required

681 The prevalence of maxillary canine impaction is approximately which value?

A Under 1 per cent
B 8 per cent
C 2 to 3 per cent
D 15 per cent
E 25 per cent

682 What proportion of impacted maxillary canines are palatally positioned?

A Up to 85 per cent
B 45 per cent
C 20 per cent
D 95 per cent
E All of them

683 Bilateral occurrence of maxillary canine impaction is reported in what proportion of affected patients?

A Under 5 per cent
B Always bilateral
C 60 per cent
D 80 per cent
E 17 to 45 per cent

684 The eruptive path of the maxillary canine is approximately how long?

A 5 mm
B 10 mm
C 40 mm
D Over 20 mm
E It has no defined length

685 The real threat posed by a palatally displaced canine is which of the following?

A Loss of the primary canine
B Resorption of the lateral and central incisor roots
C Ankylosis of the first molar
D Loss of arch length only
E Development of a dentigerous cyst in all cases

686 The RCS guidance interval for interceptive extraction of the primary canine is which age range?

A 6 to 8 years
B After 16 years
C 14 to 16 years
D Any age
E 10 to 13 years

687 Which angulation of the canine to the midsagittal plane is regarded as favourable for interception?

A Under 10 degrees
B 20 to 30 degrees
C 40 to 50 degrees
D Over 55 degrees
E Angulation is irrelevant

688 Ericson and Kurol found what proportion of canines normalise when the overlap of the lateral incisor root is less than half?

A 54 per cent
B 64 per cent
C 78 per cent
D 91 per cent
E 100 per cent

689 Naoumova's randomised work identified which two most influential predictors of successful eruption?

A Alpha angle and sector position
B Age and sex
C Overjet and overbite

D Arch length and intercanine width
E Skeletal pattern and vertical proportion

690 The 2017 Cochrane review by Parkin compared open and closed exposure and found what?

A Closed exposure was significantly superior
B Open exposure was significantly superior
C No difference in success or other outcomes, on low to very low quality evidence
D Open exposure caused more ankylosis
E Closed exposure was faster

691 Which surgical technique is indicated for a buccally displaced canine lying below the mucogingival junction?

A Closed exposure with a gold chain
B Extraction
C Open exposure with a dressing
D An apically repositioned flap
E Autotransplantation

692 The reported success rate of canine autotransplantation is approximately which value?

A 55 per cent
B 88 per cent
C 70 per cent
D 97 per cent
E 100 per cent

693 Which prognostic feature most worsens the outlook for aligning an impacted canine?

A A crown position below the apical third of the incisor roots
B Presence of the primary canine
C A palatal position
D An age of 12 years
E A crown position above the apical third of the incisor roots

694 The prevalence of mandibular second molar impaction is approximately which value?

A Just over 2 per cent
B 0.2 per cent
C 8 per cent
D 15 per cent
E 25 per cent

695 Which radiographic sign warns of impending mandibular second molar impaction?

A A short root on the first molar
B A widened periodontal ligament space
C A third molar follicle sitting above the occlusal table of the developing second molar
D An enlarged mandibular canal
E A deep antegonial notch

696 Which is the correct escalating ladder for an impacted mandibular second molar?

A Extract the second molar, then observe
B Observe, disimpact, remove the third molar, surgically upright, then extract the second molar

C Surgically upright first, then observe
D Extract the third molar, then extract the second molar
E Apply orthodontic traction immediately

697 Extraction of the impacted mandibular second molar is described in which way?

A The treatment of choice
B Contraindicated in all cases
C The fastest option with the best results
D The least successful outcome, because it commits the patient to unpredictable third molar eruption
E The first-line intervention

698 Where should a separator or open coil be placed to disimpact a mesially tilted mandibular second molar?

A Below the contact point
B Above the contact point
C At the gingival margin of the first molar
D On the distal of the third molar
E On the occlusal surface

699 After avulsion of a maxillary central incisor in a 9-year-old, what is the correct immediate approach?

A Leave the socket to heal and plan a prosthesis
B Extract the contralateral incisor for symmetry
C Always attempt re-implantation, even with a poor long-term prognosis
D Close the space orthodontically at once
E Place an immediate implant

700 Implant placement to replace a missing maxillary central incisor in an adolescent should be deferred until when?

A Age 16 in girls and 18 in boys as a rule
B Age 14 in both sexes
C Immediately after space creation
D Immediately after debond
E Growth is complete, confirmed by serial records

701 Which single argument favours letting a maxillary central incisor space close and reopening it later?

A It preserves alveolar bone
B It shortens treatment
C It avoids the need for orthodontics
D It avoids retention
E It is cheaper

702 After an uncomplicated crown fracture, how long should orthodontic force be delayed?

A No delay
B 6 months
C 3 months
D 12 months
E 24 months

703 After re-implantation following avulsion, how long should orthodontic force be delayed?

A 3 months
B 6 months
C No delay is needed
D 24 months
E 12 months

704 After a complicated crown fracture, orthodontic force should be delayed until which point?

A 3 months have elapsed
B A hard tissue barrier is seen
C The tooth is root treated
D 6 weeks have elapsed
E The tooth is asymptomatic

705 A root-fractured tooth healing by connective tissue rather than by dentine and cementum should be treated in which way?

A As a normal tooth
B With heavy forces to overcome the fracture
C With no orthodontic force ever
D As a short-rooted tooth, with light forces
E By extraction

706 During orthodontic movement of a previously traumatised tooth, radiographs should be repeated at what interval after starting movement?

A 2 weeks
B 18 months
C 6 to 9 months
D At debond only
E Radiographs are unnecessary

707 Which are the three complications of moving a previously traumatised tooth?

A Loss of vitality, pulpal calcification and root resorption where resorption has already occurred
B Ankylosis, caries and periodontal disease
C Fracture, discolouration and mobility
D Impaction, transposition and ectopia
E Gingival recession, hyperplasia and abscess

708 If root resorption appears on a mid-treatment radiograph in a previously traumatised tooth, force should be paused for how long?

A 1 month
B Force need not be paused
C 9 months
D 12 months
E 3 months

709 Serial extraction is indicated in which circumstance?

A Class II division 2 with a deep overbite
B Hypodontia of the lower second premolars
C Anterior open bite with a digit habit
D Severe crowding with a Class I skeletal pattern and no vertical problem

E Mild crowding of 3 mm

710 Which appliance is used to preserve the leeway space to relieve mild lower arch crowding?

A A quadhelix
B A lower lingual arch
C A twin block
D A transpalatal arch
E A Nance button

711 An anterior crossbite with a displacement in an 8-year-old, with space available and a positive overbite anticipated, should be treated in which way?

A Observation until the permanent dentition
B Extraction of the lower incisor
C Rapid maxillary expansion
D Facemask therapy
E Proclination of the upper incisor, for example with a removable appliance and posterior bite planes

712 Which single sign indicates that a posterior crossbite requires treatment on diagnosis?

A Involvement of three or more teeth
B A mandibular displacement on closing
C Bilateral involvement
D Presence in the primary dentition
E Association with a digit habit

713 An anterior open bite caused by a digit habit that ceases before the permanent incisors are established will usually do what?

A Persist unchanged
B Require a twin block
C Require surgery
D Close spontaneously
E Worsen with growth

714 A skeletal anterior open bite, if treated early at all, should be treated at approximately which age?

A 7 to 9 years
B 4 to 5 years
C 11 to 12 years
D 14 to 16 years
E Only after growth is complete

715 Which is the correct management of a tuberculate supernumerary preventing eruption of a maxillary central incisor?

A Extract the permanent incisor
B Expose and bond the incisor at the same visit in all cases
C Remove the supernumerary, create or maintain space, and observe for eruption for about 12 months
D Observe for 3 years
E Extract the primary incisor only

716 An ectopic maxillary first permanent molar in a 7-year-old resorbing the second primary molar may be managed initially by which approach?

A Immediate extraction of the second primary molar
B Rapid maxillary expansion
C Immediate extraction of the first permanent molar
D Observation, since spontaneous disimpaction may occur up to about 8 years
E Headgear

717 Which interceptive intervention has weak evidence and should be presented cautiously?

A Correction of a crossbite with a displacement
B Arch expansion undertaken purely to relieve crowding
C Leeway space maintenance
D Removal of a supernumerary preventing eruption
E Habit interception before the permanent incisors erupt

718 Two-phase Class II treatment is best described in which way?

A Superior to single-phase treatment
B Contraindicated in all cases
C Superior in all high-angle cases
D Mandatory when the overjet exceeds 6 mm
E Not superior to a single phase started in the late mixed dentition, except for trauma risk or significant teasing

719 Which is the correct approach to transient crowding of the lower permanent incisors in an 8-year-old?

A Reassure and review
B Extract the primary canines
C Fit a lower lingual arch immediately
D Begin lower arch expansion
E Extract a lower incisor

720 A firm, unresorbed primary maxillary canine in a 14-year-old should be regarded in which way?

A As normal and reassuring
B As an indication for immediate extraction without imaging
C As a red flag for an ectopic permanent canine
D As an indication for a CBCT in all cases
E As a contraindication to orthodontic treatment

721 Which is the correct first step in the five-step interceptive triage?

A Choose the appliance
B Ask whether there is a problem at all, or whether this is normal development
C Take a lateral cephalogram
D Refer to a specialist
E Extract the primary canines

Block U — Class I malocclusion and bimaxillary protrusion (Q722–746)

Notes In Orthodontics: Chapter 13 (Class I and the permanent dentition)
Supplements: Revision Handbook **§21** — read it again if you score under 70%
Practise this: 25 questions · allow 25 minutes · answers begin on the page named in the contents

722 In a Class I malocclusion, which statement is correct?

A The skeletal base is the principal problem
B Surgery is frequently required
C Growth modification is always indicated
D The skeletal base is not the problem, so the problem is space, transverse or vertical
E Extraction is always required

723 A midline diastema in a 9-year-old with unerupted canines is best managed how?

A Reassurance and review, since it usually closes as the canines erupt
B Immediate closure with a fixed appliance
C Frenectomy
D Extraction of the primary canines
E Bonded retainer placement

724 Which investigation should precede attributing a persistent midline diastema to a prominent fraenum?

A A lateral cephalogram
B A CBCT
C An occlusal or periapical radiograph to exclude a midline supernumerary
D A study cast analysis
E A blanching test alone

725 Which single feature most reliably indicates that retraction will improve the profile in bimaxillary protrusion?

A An obtuse nasolabial angle
B An increased lower face height
C A Class II molar relationship
D A deep overbite
E An acute nasolabial angle with lip procumbency and strain on closure

726 Bimaxillary protrusion is significantly more common in which populations?

A African and East Asian
B Northern European
C South American only
D Middle Eastern only
E It shows no population variation

727 Which is the usual treatment plan for a significant bimaxillary protrusion?

A Non-extraction alignment
B Four second premolar extractions with minimum anchorage
C Four first premolar extractions with maximum anchorage and bodily retraction of the labial segments
D Expansion

E Distalisation with headgear alone

728 The evidence associates gingival recession most strongly with which movement?

A Proclination in a low-angle patient
B Expansion of the maxilla
C Intrusion of the lower incisors
D Extrusion of the molars
E Retroclination in a high-angle Class III patient with a narrow symphysis and a thin biotype

729 Which assessment should precede planning significant incisor retraction in a bimaxillary protrusion?

A Bracket prescription selection
B Assessment of gingival biotype and symphyseal width, considering CBCT of the symphysis
C Selection of the archwire sequence
D Selection of the retention regimen
E Assessment of the third molars

730 Which surgical option is available for a severe skeletal bimaxillary protrusion?

A Le Fort I advancement
B Genioplasty alone
C Bilateral sagittal split advancement
D Anterior segmental osteotomies, or a two-jaw procedure with setback
E Surgically assisted rapid palatal expansion

731 Which single element should drive the decision in a bimaxillary protrusion?

A The incisor angle in isolation
B The molar relationship
C The lips at rest and on smiling, the nasolabial angle, and the periodontal support
D The overbite
E The arch form

732 Generalised spacing in a Class I adult with small teeth is best managed how?

A Closure and permanent retention, or redistribution and restorative build-up
B Extraction of premolars
C Expansion
D Distalisation
E No treatment in any case

733 Which single feature makes a Class I crowded case suitable for non-extraction management?

A Crowding of 9 mm
B A thin mandibular symphysis
C A deep overbite
D A Class II molar relationship
E Crowding of 3 to 4 mm with an obtuse nasolabial angle and a flat profile

734 In a Class I case with 8 mm of crowding in both arches, which extraction pattern maintains the buccal segment relationship?

A Upper first and lower second premolars
B Upper first premolars only

C Upper second and lower first premolars
D Corresponding premolars in both arches
E Second permanent molars

735 Which single statement about the periodontal management of a bimaxillary protrusion case is correct?

A Periodontal referral should follow debond
B A periodontist should be involved early, with light forces, bodily movement and consideration of grafting
C Periodontal assessment is unnecessary in young patients
D Recession cannot be prevented
E Retraction should always be avoided

736 Which movement should be used to retract the labial segments in a bimaxillary protrusion?

A Uncontrolled tipping
B Intrusion alone
C Extrusion
D Pure rotation
E Bodily movement, with a high moment-to-force ratio

737 Which is the correct sequence when a persistent midline diastema requires both closure and a frenectomy?

A Frenectomy, then closure
B Closure, then frenectomy before appliance removal
C Frenectomy after debond
D Both simultaneously at bond-up
E Neither is required

738 Which single feature would make canine substitution for a missing lateral incisor inadvisable?

A A Class II molar relationship
B A well-aligned lower arch
C A spaced upper arch
D A high smile line with a dark, large and prominent canine
E A young patient

739 Canine substitution for a missing lateral incisor requires the case to finish in which molar relationship?

A Class II
B Class I
C Class III
D Any relationship
E A cusp-to-cusp relationship

740 Which additional procedures are usually required when substituting a canine for a lateral incisor?

A None
B Root canal treatment of the canine
C Recontouring and often composite build-up, gingival recontouring of the canine zenith, and palatal root torque on the canine

D Extraction of the first premolar
E Surgical repositioning

741 Which retention regimen applies while awaiting a definitive restoration in a space-opening case?

A Night-time wear only
B Bonded retention alone
C No retention
D Full-time retention until the definitive restoration is placed
E Six weeks of retention

742 The soft tissue response of 0.7 mm of lip retraction per 1 mm of incisor retraction is most relevant in which case type?

A Class III camouflage
B Bimaxillary protrusion
C Anterior open bite
D Deep overbite
E Transverse discrepancy

743 Which single statement about Class I malocclusion and the lower labial segment is correct?

A The lower labial segment can be freely repositioned
B The lower labial segment should always be retroclined
C The lower labial segment should always be proclined
D The lower labial segment position is irrelevant in Class I
E The lower labial segment is the fixed point around which the plan is built

744 A Class I case presents with a displaced upper lateral incisor in crossbite with a displacement on closing. What is the priority?

A Correct the crossbite, since a displacement causes attrition and recession
B Align the arches first and correct the crossbite last
C Refer for surgical assessment
D Extract the lateral incisor
E Observe until growth is complete

745 Which anchorage arrangement is required in a maximum anchorage bimaxillary protrusion case?

A Reciprocal anchorage
B Minimum anchorage
C Skeletal anchorage, headgear or a stabilising arch, with lacebacks during levelling
D A lower lingual arch alone
E A transpalatal arch alone

746 Which single statement about treating bimaxillary protrusion in an adult with a thin symphysis is correct?

A Retraction should proceed as planned regardless
B The symphyseal morphology may limit the retraction achievable safely, and this changes the plan before treatment begins
C A CBCT is never justified
D Recession is unavoidable
E Surgery is always required

Block V — Class II malocclusion (Q747–786)

Notes In Orthodontics: Topic 12 (Reducing overjet in a growing child); Chapters 9, 12 and 13
Supplements: Revision Handbook **§22** — read it again if you score under 70%
Practise this: 40 questions · allow 40 minutes · answers begin on the page named in the contents

747 Class II malocclusion most commonly arises from which skeletal component?

A Maxillary excess
B Transverse maxillary deficiency
C Vertical maxillary excess
D Mandibular retrognathia
E Cranial base shortening

748 Which single variable should choose the appliance in Class II treatment?

A The vertical dimension
B The anteroposterior discrepancy
C The overjet magnitude
D The patient's age alone
E The molar relationship

749 The randomised trials of early Class II treatment showed which single benefit?

A Improved final skeletal outcome
B Improved final occlusal outcome
C Reduction in incisor trauma
D Shorter overall treatment
E Fewer total appointments

750 Which caveat should accompany the trauma-reduction argument for early Class II treatment?

A Trauma is rare in children
B Headgear was less effective than a functional appliance
C Trauma affects only girls
D The relative risk is above 1
E Much incisor trauma occurs before age 9, so even early treatment may come too late

751 O'Brien's 2009 trial reported which finding about the late-treated group?

A They had a slightly better final occlusal result
B They had a worse final occlusal result
C They had more attendances
D They had a longer overall treatment
E They required surgery more often

752 Which patients should be treated early for Class II?

A All patients with an overjet above 6 mm
B All patients in the mixed dentition
C Children at high risk of trauma, with a very large overjet, vertical excess and incompetent lips, or where there is significant teasing
D All patients with a Class II skeletal base
E No patients under any circumstances

753 Lateral open bites developing during twin block treatment should be regarded in which way?

A As a complication requiring abandonment
B As a sign of condylar resorption
C As evidence of poor compliance
D As an indication for extraction
E As expected, because overjet reduction outpaces overbite reduction

754 In a hyperdivergent Class II patient, which mechanics should be avoided?

A High-pull headgear
B Anything extrusive, including cervical headgear, Class II elastics and buccal segment extrusion
C Molar intrusion
D Skeletal anchorage
E Extraction

755 In a hypodivergent Class II patient with a deep overbite, which manoeuvre is appropriate in a grower?

A True incisor intrusion with miniscrews
B Proclination of the upper incisors only
C Molar intrusion with TADs
D Levelling by relative intrusion, allowing buccal segment extrusion
E Class III elastics

756 The mechanical difficulty in Class II division 2 is which of the following?

A An excessive overjet
B A transverse discrepancy
C An increased interincisal angle with a deep, complete overbite
D An anterior open bite
E Severe crowding

757 Which sequence is correct in Class II division 2 treatment?

A Procline the upper labial segment to convert to a division 1 pattern, reduce the overbite, then retract if space has been created
B Retract, then procline, then reduce the overbite
C Reduce the overbite, then extract, then procline
D Extract first, then reduce the overbite
E Retract first, then torque

758 Why are many Class II division 2 cases treated non-extraction in the upper arch?

A Because the overbite prevents extraction
B Because the molars are Class I
C Because anchorage is always maximum
D Because the canines are always crowded
E Because there is often little or no overjet, so extraction space in the upper arch has nowhere to go

759 Which bracket prescription feature is required for the upper central incisors in Class II division 2?

A Increased labial root torque
B Reduced in and out

C Increased mesial tip
D Extra palatal root torque
E Zero torque

760 Which retainer is classically used to hold a corrected deep overbite?

A A vacuum-formed retainer
B A Hawley retainer with an anterior bite plane, worn long-term
C A bonded retainer alone
D A positioner
E A Begg wrap-around retainer

761 Which is Proffit's threshold overjet above which surgery rather than camouflage is indicated in Class II?

A 6 mm
B 8 mm
C 20 mm
D 14 mm
E 10 mm

762 Which mandibular body length favours surgery in Class II?

A Less than 80 mm
B Less than 70 mm
C Less than 90 mm
D More than 70 mm
E Mandibular body length is not used

763 Which anterior face height favours surgery in Class II?

A Greater than 105 mm
B Greater than 115 mm
C Greater than 145 mm
D Greater than 125 mm
E Anterior face height is not used

764 Which surgical procedure is usually performed for mandibular advancement?

A Bilateral sagittal split osteotomy
B Le Fort I osteotomy
C Vertical subsigmoid osteotomy
D Genioplasty alone
E Surgically assisted rapid palatal expansion

765 Which is added to a mandibular advancement where there is vertical maxillary excess and a gummy smile?

A A genioplasty alone
B Surgically assisted rapid palatal expansion
C A Le Fort I impaction
D A vertical subsigmoid osteotomy
E Distraction osteogenesis

766 Pre-surgical orthodontics in a Class II case has which effect on the overjet?

A It reduces it
B It reverses it
C It has no effect
D It increases it, because decompensation removes the dental compensation
E It corrects the molar relationship

767 Which is the correct approach to a Class II patient who develops temporomandibular symptoms during treatment?

A Complete treatment as planned
B Pause active mechanics, counsel on natural history, and manage conservatively and reversibly
C Perform an occlusal adjustment
D Refer for surgery
E Debond immediately and discharge

768 What proportion of Caucasians have a Class II malocclusion, in the range usually quoted for adolescents?

A 5 to 8 per cent
B Over 60 per cent
C 30 to 40 per cent
D 45 to 55 per cent
E 15 to 27 per cent

769 Which is the correct force and duration for headgear used for distalisation or an orthopaedic effect?

A 450 to 500 g per side for 14 hours per day
B 250 g per side for 10 hours per day
C 150 g per side for 8 hours per day
D 700 g per side for 6 hours per day
E 100 g per side for 24 hours per day

770 Molar distalisation corrects which type of Class II relationship?

A A skeletal Class II
B Any Class II
C A dentoalveolar Class II only
D A Class II with a vertical excess
E A Class II with a transverse deficiency

771 What distal force is appropriate for molar distalisation?

A 50 g
B 150 to 250 g
C 400 g
D 600 g
E Over 800 g

772 Molar derotation alone can gain approximately how much space per side?

A 0.5 mm
B 8 mm
C 5 mm
D 2 to 3 mm

E None

773 Which single statement about transitioning from a functional appliance to a fixed appliance is correct?

A Overcorrect, then transition carefully, since the correction relapses if the posterior open bite is not settled
B The functional appliance can be abandoned as soon as the overjet is corrected
C Fixed appliances should be placed at the same time as the functional appliance
D The posterior open bite always resolves spontaneously without attention
E Retention is unnecessary after functional appliance therapy

774 Which cephalometric measurement is used with pogonion in Proffit's Class II surgical thresholds?

A The Frankfort plane
B The sella-nasion plane
C The nasion perpendicular
D The occlusal plane
E The mandibular plane

775 Which single feature of a Class II division 1 malocclusion perpetuates it?

A A deep overbite
B Crowding of the lower incisors
C A Class II molar relationship
D An increased lower face height
E A lower lip trap resting behind the upper incisors

776 Which is the correct extraction pattern for a Class II case requiring maximum upper retraction and with mild lower crowding?

A Upper first premolars and lower second premolars
B Four first premolars
C Upper second premolars and lower first premolars
D Lower first premolars only
E Upper second molars

777 Which single statement about the Class II division 2 lateral incisors is correct?

A They are retroclined with the centrals
B They are usually absent
C They are frequently proclined and mesiolabially rotated
D They are usually in crossbite
E They require no correction

778 In a Class II patient with an increased lower face height, which extraction strategy may assist?

A Extraction of lower incisors
B Extraction to allow the mandible to rotate backwards
C Non-extraction with expansion
D Extraction of second molars only
E Extraction to allow the mandible to rotate forwards

779 Which single risk should be quoted at consent for bilateral sagittal split osteotomy?

A Loss of vitality of the incisors
B Altered sensation in the distribution of the inferior alveolar nerve, commonly transient and occasionally permanent
C Ankylosis of the temporomandibular joint
D Loss of the mandibular canine
E Failure of the maxillary graft

780 Which is the correct timing for functional appliance therapy in relation to skeletal maturation?

A CS1
B After growth is complete
C CS5
D CS2 to CS3, with the greatest response in the circumpubertal period
E Timing is immaterial

781 Which is the correct approach when a Class II patient will not comply with a removable functional appliance?

A Abandon treatment
B Increase the prescribed wear time
C Consider a fixed functional appliance such as a Herbst, which does not depend on wear-time compliance
D Extract four premolars immediately
E Refer for surgery

782 Which single statement about mandibular advancement surgery and stability is correct?

A It is very stable in the low- and average-angle patient
B It is the least stable of the common movements
C It is unstable in all patients
D Its stability depends only on the fixation used
E It is more stable than maxillary impaction

783 A 9-year-old boy has an overjet of 12 mm, incompetent lips and vertical excess. Which single consideration most justifies early treatment?

A Improved final occlusal result
B Improved skeletal outcome
C Shorter overall treatment
D Fewer appointments
E Reduction in the risk of incisor trauma

784 Which single statement about promising TMD improvement from Class II correction is correct?

A It should be promised, since correction reliably improves symptoms
B It should be promised only in adults
C It should be promised only in division 2 cases
D It should not be promised, since there is no evidence that orthodontics reduces TMD signs or symptoms
E It should be promised only where a click is present

785 Which is the correct approach to documenting joint status in a Class II patient before treatment?

A No documentation is required
B Screen and document range of opening, deviation, joint sounds and tenderness before treatment begins
C Document only if the patient reports pain
D Refer all patients for MRI
E Document at debond only

786 Which single feature distinguishes a Class II division 2 face?

A An increased lower face height
B A prognathic mandible
C An anterior open bite
D A transverse maxillary deficiency
E A reduced lower face height with a forward mandibular growth rotation

Block W — Class III malocclusion (Q787–816)

Notes In Orthodontics: Topic 17 (Correcting a Class III incisor relationship); Chapters 8, 9 and 13
Supplements: Revision Handbook **§23** — read it again if you score under 70%
Practise this: 30 questions · allow 30 minutes · answers begin on the page named in the contents

787 Which single fact makes Class III harder to manage than Class II?

A Class III is more common
B Residual mandibular growth helps in Class II and harms in Class III
C Class III patients are less compliant
D Class III cannot be camouflaged
E Class III has no skeletal component

788 Before diagnosing a Class III malocclusion, which manoeuvre is essential?

A Taking a lateral cephalogram
B Taking a panoramic radiograph
C Measuring the overjet in intercuspal position
D Manipulating the mandible into centric relation
E Assessing the third molars

789 The characteristic dental compensation in Class III is which of the following?

A Upper incisors proclined, lower incisors retroclined
B Upper incisors retroclined, lower incisors proclined
C Both retroclined
D Both proclined
E No compensation occurs

790 Which finding in the mixed dentition favours early intervention in Class III?

A A reduced overbite
B Retroclined lower incisors
C An average or increased overbite with an anterior displacement on closing
D An increased lower face height
E Class III molars with no displacement

791 Which finding in the mixed dentition indicates monitoring rather than early intervention in Class III?

A An anterior displacement on closing
B An increased overbite
C A mild Class III skeletal base
D Retroclined lower incisors with proclined uppers and no displacement
E A reduced lower face height

792 Facemask protraction therapy should ideally be started before which age?

A 6 years
B 10 years
C 13 years
D 16 years
E Age is immaterial

793 Which is the correct daily wear time for a facemask?

A 6 to 8 hours
B 4 hours
C 18 hours
D 24 hours
E 12 to 14 hours

794 To what overjet should a Class III correction be overcorrected?

A Plus 3 to 4 mm
B Plus 1 mm
C 0 mm
D Plus 8 mm
E Overcorrection is unnecessary

795 Mandall reported what proportion of facemask-treated patients avoided surgery compared with untreated controls?

A Both groups 50 per cent
B 36 per cent against 60 per cent
C Over 60 per cent against 36 per cent
D 90 per cent against 70 per cent
E No difference

796 Proffit reports what proportion of facemask cases show recurrence of the anterior crossbite at 8 years?

A Under 5 per cent
B 25 to 33 per cent
C 10 per cent
D 60 per cent
E 90 per cent

797 Bone-anchored maxillary protraction using Class III elastics to miniplates is suited to which age range?

A 6 to 8 years
B 8 to 10 years
C 16 to 18 years
D 11 to 14 years
E Adults only

798 A parent asks whether a chin cup will stop their child's lower jaw growing forwards. Which reply is correct?

A That there is no agreement that it inhibits mandibular growth, and that it largely redirects the mandible downwards and backwards with no lasting gain
B That it reliably inhibits mandibular growth
C That it will shorten the mandible by 5 mm
D That it works only after the growth spurt
E That it is the treatment of choice for all Class III children

799 Which Wits appraisal value most strongly favours surgery in Class III?

A Minus 2 mm
B Minus 5 mm
C Minus 11 mm or worse

D Minus 8 mm
E Minus 1 mm

800 Which ANB value forms part of the Class III surgical threshold set?

A Minus 1 degree
B ANB is not used
C Plus 2 degrees
D Minus 2 degrees
E Minus 4 degrees or worse

801 Which gonial angle value forms part of the Class III surgical threshold set?

A 120 degrees or less
B 130 degrees or more
C 110 degrees
D 140 degrees
E Gonial angle is not used

802 Which lower anterior face height proportion forms part of the Class III surgical threshold set?

A 50 per cent or less
B 55 per cent
C 66 per cent or more
D 60 per cent
E It is not used

803 A Class III camouflage plan typically uses which extraction pattern?

A Upper first premolars and lower second premolars
B Upper first premolars only
C Four first premolars
D Four second premolars
E Upper second premolars and lower first premolars

804 Why should mandibular extractions be avoided in a Class III case where surgery is likely?

A They increase root resorption
B They consume the decompensation space the surgeon will need
C They cause anchorage loss
D They increase the overbite
E They prolong treatment

805 Class III elastics should be avoided in which patient?

A A low-angle patient
B A patient with a deep overbite
C A growing patient
D A high-angle patient with an increased lower face height
E A patient with mild crowding

806 Pre-surgical orthodontics in a Class III case involves which movements?

A Proclining the uppers and retroclining the lowers
B Intruding the lowers only
C Retroclining the uppers and proclining the lowers, worsening the reverse overjet
D Expanding the mandible

E Levelling only

807 When should orthognathic surgery be undertaken in a Class III patient?

A After growth is complete, confirmed by serial cephalometric superimposition
B At 14 years in girls and 16 in boys
C As soon as the diagnosis is made
D After the third molars erupt
E At 18 years in both sexes

808 Discriminant analysis of the Class III surgical threshold variables predicts the need for surgery with what accuracy?

A 60 per cent
B 75 per cent
C It cannot predict
D 100 per cent
E Over 90 per cent

809 Which population shows a markedly higher prevalence of Class III malocclusion?

A Northern European
B Indigenous Australian
C Sub-Saharan African
D East Asian
E South Asian

810 Which single feature is characteristic of the Class III component in East Asian populations?

A Mandibular prognathism alone
B Maxillary retrusion is the more common component
C An anterior open bite in all cases
D A transverse deficiency in all cases
E A reduced lower face height

811 An anterior crossbite in a 7-year-old with space available and a positive overbite anticipated corrects in approximately what period using an incline plane?

A It does not correct
B 3 months
C 6 months
D 12 months
E 3 to 4 weeks

812 Which single family history feature is most relevant in Class III?

A A history of hypodontia
B A family history of Class III skeletal pattern
C A family history of caries
D A history of orthodontic treatment in a sibling
E A family history of periodontal disease

813 Which is the correct approach to parents requesting immediate treatment of a Class III crossbite in a 7-year-old?

A Treat all cases immediately with a facemask
B Refer for immediate surgical assessment
C Refuse all treatment until 16 years

D Distinguish an anterior crossbite with a displacement, which is treated on diagnosis, from a skeletal Class III, where early treatment buys a 60 versus 36 per cent chance of avoiding surgery but most skeletal effect is lost
E Extract the lower incisors

814 Which single statement about rapid maxillary expansion combined with facemask therapy is correct?

A It may be used to disarticulate the maxilla and improve protraction
B It is contraindicated
C It replaces the facemask
D It prevents relapse entirely
E It is used only in adults

815 Which single statement about lower incisor extraction in Class III camouflage is correct?

A It is the first-line approach in all cases
B It is contraindicated in all cases
C It is occasionally used in a mild non-growing case, at the cost of a tooth-size discrepancy and a deepened overbite
D It creates 8 mm of space
E It improves the profile reliably

816 A Class III patient with a reverse overjet of 4 mm, ANB of minus 5 degrees, Wits of minus 8 mm, lower incisors at 82 degrees and a chief complaint about the profile is best managed how?

A Camouflage with lower incisor extraction
B Facemask therapy
C Observation only
D Combined orthodontic and orthognathic treatment, with camouflage offered honestly and its limitations stated
E Rapid maxillary expansion alone

Block X — Transverse problems and facial asymmetry (Q817–841)

Notes In Orthodontics: Topics 16 and 19 (Crossbite correction; facial asymmetry); Chapters 9, 12, 13 and 15
Supplements: Revision Handbook **§24** — read it again if you score under 70%
Practise this: 25 questions · allow 25 minutes · answers begin on the page named in the contents

817 A unilateral posterior crossbite with a mandibular displacement most commonly represents which underlying problem?

A A true unilateral maxillary deficiency
B A bilateral maxillary constriction that the patient closes out of
C A mandibular asymmetry
D A dental crossbite of one tooth
E A vertical discrepancy

818 Which single finding determines the urgency of treating a posterior crossbite?

A The number of teeth involved
B The molar relationship
C The patient's age
D The overbite
E The presence of a mandibular displacement

819 Which appliance did Cochrane evidence favour for crossbite correction at ages 7 to 11?

A A quadhelix
B A removable expansion plate
C A rapid maxillary expander
D A transpalatal arch
E A twin block

820 Rapid maxillary expansion achieves approximately what proportion of the screw expansion at the suture?

A 10 per cent
B 25 per cent
C 50 per cent
D 85 per cent
E 100 per cent

821 What proportion of the skeletal change achieved by rapid maxillary expansion relapses at 4 to 8 months?

A Approximately 2 per cent
B Approximately 10 per cent
C Approximately 30 per cent
D Approximately 60 per cent
E None

822 What proportion of initial dental expansion is maintained with slow maxillary expansion compared with rapid expansion?

A Both 70 per cent
B 85 per cent with slow and 95 per cent with rapid
C Both 90 per cent

D 95 per cent with slow and 85 per cent with rapid
E Neither is maintained

823 Beyond approximately how much required expansion in a non-growing patient is surgically assisted expansion indicated?

A 5 mm
B 2 mm
C 10 mm
D 15 mm
E Surgery is never required

824 Which single statement about the timing of maxillary expansion is correct?

A Sutural fusion correlates closely with chronological age
B Expansion is impossible after 12 years
C Sutural fusion occurs by the late teens, but the correlation between sutural maturation and chronological age is poor
D Expansion is impossible before 7 years
E The suture never fuses

825 Proffit's position on rapid versus slow expansion is which of the following?

A Rapid expansion produces a much greater skeletal component
B Only rapid expansion opens the suture
C Slow expansion produces no skeletal change
D Rapid expansion is contraindicated
E At 4 months, rapid and slow expansion have produced about the same skeletal to dental split

826 Slow expansion at a rate below which value matches the maximum speed of bone formation?

A 2 mm per week
B 0.5 mm per week
C 4 mm per week
D 1 mm per day
E There is no such rate

827 Rapid maxillary expansion is recommended as a treatment for which condition?

A Nasal obstruction
B Sleep-disordered breathing
C Neither nasal obstruction nor sleep-disordered breathing on current evidence
D Temporomandibular disorder
E Class II malocclusion

828 In which order should crossbites be corrected?

A Anterior before posterior
B Neither should be corrected before alignment
C Simultaneously in all cases
D Whichever is more severe first
E Posterior before anterior

829 What makes an anterior crossbite correction self-retaining?

A A Class I molar relationship
B A positive overbite resulting from the correction
C A bonded retainer
D An increased overjet
E Extraction of the lower incisor

830 What proportion of the population has clinically significant asymmetric jaw development?

A 25 per cent
B 3 per cent
C 10 per cent
D Under 0.5 per cent
E Over 40 per cent

831 Which is the commonest single cause of significant mandibular asymmetry?

A Hemifacial microsomia
B Condylar hyperplasia
C Condylar fracture in childhood
D Juvenile idiopathic arthritis
E Cleft lip and palate

832 Which term describes a functional shift producing an apparent asymmetry?

A Laterocclusion
B Laterognathy
C Hemimandibular elongation
D Hemimandibular hyperplasia
E Plagiocephaly

833 Hemimandibular hyperplasia in the Obwegeser and Makek classification is characterised by which feature?

A A horizontal vector with the chin deviated to the unaffected side
B An anterior open bite only
C A purely transverse discrepancy
D Bilateral symmetrical enlargement
E A vertical vector with a true occlusal cant and a unilateral open bite

834 Hemimandibular elongation is characterised by which feature?

A A true occlusal cant
B Bilateral involvement
C A unilateral open bite
D Chin deviation to the unaffected side with a level occlusal plane
E Absence of any occlusal effect

835 Which investigation for condylar hyperplasia is sensitive but non-specific?

A Panoramic radiography
B Bone scintigraphy
C Magnetic resonance imaging
D Posteroanterior cephalometry
E Cone beam computed tomography

836 Which findings suggest a progressive asymmetry requiring referral?

A A stable 2 mm chin deviation since childhood
B A Class II molar relationship on one side only
C A unilateral crossbite with a displacement
D A midline diastema
E Accelerating chin deviation, a new occlusal cant, or new temporomandibular symptoms in the teens

837 How many histological subtypes of condylar hyperplasia did Slootweg and Muller describe?

A Two
B Four
C Three
D Five
E Six

838 Which is the appropriate management of a mild asymmetry in a growing child?

A Immediate surgery
B Chin cup therapy
C Immediate camouflage with fixed appliances
D Watch, with a defined review interval and records that can be superimposed
E Extraction of premolars

839 Which single step should be taken early where an asymmetry is associated with a functional shift?

A Elimination of the shift, since it can drive asymmetric growth
B Surgical assessment
C Chin cup therapy
D Observation until growth is complete
E Extraction of the deviated side

840 Which is the correct clinical test to distinguish a true occlusal cant from a functional shift?

A Palpation of the condyles
B Measuring the overjet
C Asking the patient to bite on a spatula, and assessing the path of closure from retruded contact position
D Assessing the third molars
E Measuring maximum opening

841 Which imaging modality is indicated for a significant transverse asymmetry?

A Lateral cephalogram
B Bitewings
C Periapical radiographs
D Posteroanterior cephalogram
E Occlusal view

Block Y — Vertical problems: deep bite and open bite (Q842–871)

Notes In Orthodontics: Topic 11 (Overbite and overbite reduction); Chapter 13, pp. 928–945
Supplements: Revision Handbook **§25** — read it again if you score under 70%
Practise this: 30 questions · allow 30 minutes · answers begin on the page named in the contents

842 Which are the three mechanisms of overbite reduction?

A Extraction, expansion and distalisation
B Intrusion of incisors, extrusion of buccal segments, and proclination of the labial segments
C Torque, tip and rotation
D Anchorage, retention and settling
E Levelling, aligning and finishing

843 Buccal segment extrusion for overbite reduction is stable in which patient?

A A non-growing adult
B No patient
C A high-angle adult
D Any patient
E A growing patient, because condylar growth compensates

844 True incisor intrusion is required in which situation?

A A high-angle patient or an adult in whom face height must not increase
B A growing patient with a low mandibular plane angle
C A patient with an anterior open bite
D A patient with a transverse discrepancy
E A patient with a Class III skeletal base

845 Which force is appropriate for segmental intrusion of the four incisors?

A 20 g total
B 150 g total
C 60 g total, at 15 g per tooth
D 300 g total
E 500 g total

846 Ng and colleagues reported what magnitude of incisor intrusion achievable with segmental mechanics in non-growing patients?

A 0.5 mm maxillary and 0.6 mm mandibular
B 1.5 mm maxillary and 1.9 mm mandibular
C 3 mm maxillary and 4 mm mandibular
D 5 mm in both arches
E Intrusion is not achievable

847 Miniscrews achieve how much more intrusion than segmental mechanics?

A 0.1 to 0.2 mm
B 4 to 5 mm
C 2 to 3 mm
D 0.67 to 0.95 mm
E No additional intrusion

848 Proclination of the incisors changes the overbite by approximately how much per degree?

A 0.1 to 0.2 mm
B 0.01 mm
C 0.5 mm
D 1 mm
E 2 mm

849 What single condition must be met for a corrected overbite to remain stable?

A A Class I molar relationship
B A bonded retainer
C Correction of the interincisal angle and establishment of an occlusal stop
D Extraction of four premolars
E A flat curve of Spee only

850 The Dahl concept re-establishes posterior occlusal contacts over approximately what period?

A 2 weeks
B Contacts do not re-establish
C 18 months
D 3 years
E 6 months

851 Dake and Sinclair compared segmental with continuous reverse curve mechanics and reported what?

A Segmental mechanics gave greater true intrusion, with a Ricketts utility arch producing more lower incisor proclination
B Continuous mechanics gave greater intrusion
C Both were equivalent
D Neither produced intrusion
E Reverse curve mechanics caused root resorption

852 Burstone's intrusion arch is best described as which type of system?

A A two-couple system
B A continuous archwire system
C A one-couple bypass system
D A tissue-borne system
E A self-ligating system

853 Which is the key soft tissue sign of a skeletal anterior open bite?

A A deep labiomental fold
B A prominent chin
C A thin upper lip
D An acute nasolabial angle
E An interlabial gap greater than 3 mm

854 One millimetre of molar intrusion produces approximately how much anterior open bite reduction?

A 0.5 mm
B 3 mm
C 1 mm
D 6 mm

E No change

855 One millimetre of molar extrusion reduces incisor overlap by approximately how much?

A 0.2 mm
B 6 mm
C 4 mm
D 1.5 to 2.5 mm
E No change

856 What proportion of anterior open bite corrections relapse, whether surgical or non-surgical?

A Under 5 per cent
B Approximately 50 per cent
C Approximately 20 per cent
D Approximately 80 per cent
E None

857 A tongue thrust observed in a patient with an anterior open bite is most commonly which of the following?

A Adaptive to the open bite rather than causative
B The primary cause of the open bite
C An endogenous tongue thrust
D A sign of macroglossia
E A neurological disorder

858 Which single feature distinguishes a habit-related from a skeletal anterior open bite?

A Symmetry and increased lower face height in the habit-related type
B They cannot be distinguished clinically
C The habit-related type is always larger
D Only the skeletal type has a posterior crossbite
E Asymmetry, often localised to the digit, in the habit-related type, with a symmetrical open bite and increased face height in the skeletal type

859 Which iatrogenic cause of anterior open bite should be recognised in advance?

A Extraction of premolars
B Bonded retention
C Interproximal reduction
D Extrusive mechanics in a high-angle patient, including Class II elastics and untrimmed functional appliance blocks
E Rapid maxillary expansion

860 Approximately how much dentoalveolar height separates a deep bite face from an open bite face?

A 1 mm
B 5 mm
C 12 mm
D 20 mm
E There is no measurable difference

861 A short face pattern is associated with which condylar growth direction?

A Posterior condylar growth
B Lateral condylar growth
C Vertical condylar growth only
D No condylar growth
E Forward condylar growth

862 Which instruction is used to record the lip at rest for vertical planning?

A Ask the patient to smile broadly
B Ask the patient to say Emma or Mississippi
C Ask the patient to open wide
D Ask the patient to clench
E Ask the patient to swallow

863 Which retainer is classically used to retain a corrected deep overbite?

A A vacuum-formed retainer
B A positioner
C A bonded retainer alone
D A Hawley retainer with an anterior bite plane, worn into the early twenties
E A quadhelix

864 Which single decision should be made first in planning treatment of an open bite?

A Which occlusal plane the case will be treated to
B The bracket prescription
C The retention regimen
D The extraction pattern
E The archwire sequence

865 In a growing patient with a deep overbite and a low mandibular plane angle, which appliance for overbite reduction is appropriate?

A A high-pull headgear
B A miniscrew-supported intrusion arch
C A flat anterior bite plane, allowing posterior eruption
D Posterior bite blocks
E A quadhelix

866 In a non-growing patient, buccal segment extrusion for overbite reduction produces which effect?

A Stable overbite reduction
B Forward mandibular rotation
C Reduced face height
D Downward and backward mandibular rotation, worsening the Class II and increasing face height
E No change in the mandibular position

867 Which is the appropriate approach to an anterior open bite in a 6-year-old with an active digit habit?

A Immediate surgical assessment
B Interrupt the habit before the permanent incisors are established, expecting spontaneous closure
C Fit a twin block

D Extract the primary molars
E Place a fixed appliance

868 Which single mechanical option is most efficient for correcting a skeletal anterior open bite non-surgically?

A Class II elastics
B Rapid maxillary expansion
C Buccal segment extrusion
D Anterior box elastics alone
E Molar intrusion with temporary anchorage devices

869 Which surgical procedure corrects a skeletal anterior open bite of maxillary vertical excess origin?

A Posterior maxillary impaction with mandibular autorotation
B Bilateral sagittal split advancement
C Genioplasty
D Surgically assisted rapid palatal expansion
E Vertical subsigmoid osteotomy

870 Which is the appropriate advice to a patient with a skeletal anterior open bite who asks about muscle exercises?

A That exercises will close the bite
B That exercises are the first-line treatment
C That there is no evidence that muscle exercises alone will close a skeletal open bite
D That exercises will prevent relapse
E That exercises replace surgery

871 Which statement about extrusion in a long-face patient is correct?

A Extrusion is the treatment of choice
B Extrusion is contraindicated where the face is already long
C Extrusion has no vertical effect
D Extrusion reduces face height
E Extrusion is indicated only with TADs

Block Z — TMD, condylar trauma, the airway and myofunctional therapy (Q872–901)

Notes In Orthodontics: Topic 18 (Malocclusion and TMD); Chapter 2, pp. 102–105
Supplements: Revision Handbook **§26** — read it again if you score under 70%
Practise this: 30 questions · allow 30 minutes · answers begin on the page named in the contents

872 Which single position on orthodontics and temporomandibular disorder should be stated in an examination?

A Orthodontics causes TMD
B Orthodontics cures TMD
C Orthodontics prevents TMD if canine guidance is achieved
D Orthodontics neither causes nor prevents TMD, and only weak statistical associations exist
E The relationship is unstudied

873 Which body of work reported that orthodontic treatment does not cause temporomandibular disorder?

A The American Association of Orthodontists funded studies of 1992
B The Florida trial
C The Manchester twin block trial
D The 2021 Cochrane review on canine extraction
E The Royal London space analysis studies

874 What was the outcome of Brimm versus Malloy in 1987?

A The case was dismissed
B An award of 85000 dollars
C An award of 850000 dollars following temporomandibular disorder after four-premolar extraction treatment
D A criminal conviction
E A finding that orthodontics prevents TMD

875 Luecke and Johnstone in 1992 reported which finding?

A Incisor retraction displaces the condyles distally
B Occlusal adjustment repositions the condyles
C Extraction causes condylar resorption
D Class II elastics displace the condyles
E Conventional incisor retraction does not produce distal displacement of the condyles

876 Approximately what proportion of the population has at least one sign of temporomandibular disorder?

A 75 per cent
B 33 per cent
C 50 per cent
D 10 per cent
E 95 per cent

877 At what stage of life do temporomandibular symptoms most commonly begin?

A Early childhood
B Middle age
C Late adolescence to early adulthood
D Old age
E Symptoms have no typical onset

878 Which single investigation is required to image the temporomandibular disc?

A Panoramic radiography
B Submentovertex view
C Cone beam computed tomography
D Arthrography of the joint space
E Magnetic resonance imaging

879 Which single measure of mouth opening is regarded as the lower limit of normal?

A 25 mm
B 40 mm
C 30 mm
D 50 mm
E 60 mm

880 Which management approach for temporomandibular disorder is best supported?

A Occlusal adjustment
B Immediate surgery
C Immediate orthodontic treatment
D Reassurance, explanation of the natural history, jaw rest, analgesia, physiotherapy and a stabilisation splint
E Full-mouth rehabilitation

881 Which single systemic condition can destroy the condyle in a child, producing a progressive Class II with an open bite?

A Rheumatic fever
B Type 1 diabetes
C Juvenile idiopathic arthritis
D Coeliac disease
E Asthma

882 Which is the correct management of a child with a condylar fracture?

A Conservative functional management with early mobilisation, soft diet, analgesia and physiotherapy
B Prolonged intermaxillary fixation
C Immediate open reduction in all cases
D Condylectomy
E No treatment at all

883 A childhood condylar fracture is the commonest cause of which orthodontic problem?

A Anterior open bite
B Hypodontia
C Class III malocclusion
D Transverse maxillary deficiency
E Significant mandibular asymmetry

884 Which clinical sign is characteristic of a healed unilateral condylar injury with growth disturbance?

A Deviation to the contralateral side on opening
B An increased overjet only
C Bilateral posterior open bite
D Deviation to the affected side on opening, with reduced ramus height and an occlusal cant
E Symmetrical reduction in opening

885 The classic adenoid facies is associated with which combination?

A A broad maxilla and reduced face height
B A narrow maxilla, increased face height, backward mandibular rotation and an anterior open bite
C A prognathic mandible with a deep bite
D Bilateral scissors bite
E A Class II division 2 pattern

886 What is the correct position to state about nasal obstruction and facial form?

A Nasal obstruction causes long face syndrome
B Adenoidectomy normalises facial growth
C There is no association
D Nasal obstruction reliably causes Class III
E The association is real but weak, and causation is not established

887 Adenoid tissue peaks at approximately which age?

A 1 year
B 3 to 5 years
C 8 to 10 years
D 13 to 15 years
E It does not involute

888 Which is the approximate prevalence of obstructive sleep apnoea in children?

A 0.1 per cent
B 15 per cent
C 8 per cent
D 2 per cent
E 25 per cent

889 Which single step is required before treating suspected obstructive sleep apnoea orthodontically?

A A sleep study and appropriate medical referral
B A lateral cephalogram
C A CBCT
D A panoramic radiograph
E Rapid maxillary expansion

890 Koretsi and colleagues in 2018 reported that mandibular advancement appliances are effective in which group?

A Severe obstructive sleep apnoea
B Children under 5 years
C Mild to moderate obstructive sleep apnoea
D Patients with central sleep apnoea

E Patients with a Class III skeletal pattern

891 Which is the most effective surgical treatment for severe adult obstructive sleep apnoea?

A Uvulopalatopharyngoplasty
B Rapid maxillary expansion
C Adenotonsillectomy
D Maxillomandibular advancement
E Genioplasty

892 Which symptoms in a child should prompt referral to ENT or paediatrics?

A A midline diastema
B Snoring, witnessed apnoeas, restless sleep, daytime somnolence or behavioural problems
C A deep overbite
D An anterior crossbite
E Delayed eruption of the canines

893 Which single statement about occlusal adjustment for temporomandibular disorder is correct?

A It is the treatment of choice
B It is recommended by Cochrane review
C It should precede orthodontic treatment
D It reliably eliminates clicking
E It has little beneficial value and is irreversible

894 Manfredini and colleagues in 2017 concluded what about dental occlusion and temporomandibular disorder?

A There is a lack of evidence implicating dental occlusion in the pathophysiology of TMD
B Occlusion is the principal cause
C Occlusion causes TMD only in adults
D Occlusal interferences reliably predict TMD
E Occlusion prevents TMD

895 Proffit's strongest argument against an occlusal cause of temporomandibular disorder is which of the following?

A That TMD is untreatable
B That the occlusion cannot be measured
C That TMD is no more prevalent now than before modern orthodontics, and symptoms disappear long before any occlusal change could explain it
D That clicking is normal
E That MRI cannot image the disc

896 Which single element should be documented before orthodontic treatment in every patient, in relation to the temporomandibular joints?

A A CBCT
B Range of opening, deviation, joint sounds and muscle tenderness
C An MRI
D A stabilisation splint prescription
E An occlusal adjustment record

897 Which is the correct management if temporomandibular symptoms appear during orthodontic treatment?

A Complete the treatment as planned
B Perform an occlusal adjustment
C Debond immediately and discharge
D Suspend active mechanics and manage conservatively and reversibly
E Refer for surgery

898 A parent asks whether tongue and lip exercises will correct their child's malocclusion. Which reply is best supported?

A That there is no good evidence that muscle exercises alone correct a malocclusion, although interrupting a digit habit early does help
B That exercises reliably correct malocclusion
C That exercises will close a skeletal open bite
D That exercises replace appliance treatment
E That proprietary myofunctional systems have strong randomised evidence

899 A lip trap in a Class II division 1 malocclusion is best described in which way?

A An irrelevant finding
B A cause of temporomandibular disorder
C A soft tissue factor that perpetuates the malocclusion and is eliminated by a functional appliance
D A cause of anterior open bite
E A contraindication to treatment

900 Which single statement about temporomandibular disorder classification is correct?

A All cases are internal derangements
B Only degenerative disease is treatable
C All cases involve the disc
D Classification is not clinically useful
E The main categories are myofascial pain, internal derangement, degenerative joint disease and systemic inflammatory arthropathies

901 Which is the correct reply to a patient asking whether orthodontic treatment will cure their jaw clicking?

A There is no evidence that orthodontic treatment reduces temporomandibular signs and symptoms, and none that it causes them
B It usually will
C It will if canine guidance is established
D It will worsen the clicking
E Treatment is contraindicated with a click

Block AA — Adult orthodontics and orthognathic surgery (Q902–931)

Notes In Orthodontics: Chapters 13 and 15 (adult treatment; orthognathic surgery)
Supplements: Revision Handbook **§27** — read it again if you score under 70%
Practise this: 30 questions · allow 30 minutes · answers begin on the page named in the contents

902 Which single factor most changes the mechanics of tooth movement in a periodontally reduced dentition?

A The bracket prescription
B Increased bone density
C Apical migration of the centre of resistance, producing a larger moment for the same force
D Reduced periodontal ligament width
E Increased root surface area

903 Which single record should be made in every adult before appliance placement?

A A CBCT
B An MRI of the joints
C A sleep study
D A hand-wrist radiograph
E Probing of every tooth with recording of pre-existing recession

904 Adjunctive orthodontic treatment is best defined as which of the following?

A Treatment that corrects the whole malocclusion
B Limited tooth movement serving a larger periodontal or restorative goal
C Treatment with removable appliances only
D Treatment without extraction
E Treatment in a growing patient

905 What force is appropriate for intrusion during molar uprighting?

A 5 g per tooth
B 100 g per tooth
C 50 g per tooth
D 10 to 15 g per tooth
E 200 g per tooth

906 Which patient is a poor candidate for molar uprighting?

A A brachycephalic patient with a deep bite
B A patient with a Class I molar relationship
C A dolichocephalic patient, because the mechanics are extrusive
D A patient with a shallow curve of Spee
E A patient with good periodontal support

907 Until what endpoint should an uprighted molar be retained?

A Until the lamina dura reappears
B Six weeks
C Three months
D One year
E Retention is unnecessary

908 Which single principle governs the creation of alveolar bone at a deficient implant site?

A Bone can be grown by loading alone
B Only grafting can create alveolar bone
C Bone forms spontaneously after extraction
D Alveolar bone regenerates in adults without intervention
E Erupting and extruding teeth bring alveolar bone with them, so a tooth may be erupted through the site and then moved back

909 When must the camouflage-or-surgery decision be made?

A At the mid-treatment review
B After a trial of camouflage
C After alignment is complete
D Before treatment begins
E At debond

910 Pre-surgical orthodontics typically lasts approximately how long?

A 3 months
B 12 to 18 months
C 3 years
D 6 weeks
E It is not required

911 Which single statement best describes pre-surgical decompensation?

A It aligns the arches into a Class I relationship before surgery
B It is optional in most cases
C It levels the curve of Spee only
D It is performed after surgery
E It deliberately removes the dental compensation, making the occlusion worse, so that the surgeon has the movement needed to correct the face

912 Which osteotomy is used to impact the maxilla and correct vertical maxillary excess?

A Bilateral sagittal split osteotomy
B Le Fort I osteotomy
C Vertical subsigmoid osteotomy
D Genioplasty
E Segmental mandibular osteotomy

913 Which nerve is principally at risk during a bilateral sagittal split osteotomy?

A The lingual nerve
B The facial nerve
C The infraorbital nerve
D The inferior alveolar nerve
E The hypoglossal nerve

914 Which procedure is used for mandibular setback other than a sagittal split?

A Vertical subsigmoid osteotomy
B Le Fort I osteotomy
C Genioplasty
D Anterior segmental maxillary osteotomy
E Surgically assisted rapid palatal expansion

915 Which statement about producing a chin prominence is correct?

A A chin can be produced by orthodontic tooth movement
B A chin is produced by functional appliance therapy
C A chin cannot be produced orthodontically; a genioplasty is required
D A chin is produced by extraction
E A chin develops with a growth spurt in all patients

916 Which movement is the most stable in the hierarchy of orthognathic stability?

A Maxilla down
B Maxilla wide
C Mandible back
D Maxilla superiorly, that is impaction
E Mandible back with maxilla forward

917 Which movement is the least stable in the hierarchy of orthognathic stability?

A Maxillary impaction
B Maxillary down-graft and surgical maxillary expansion
C Mandibular advancement
D Genioplasty
E Maxillary advancement

918 In the hierarchy of stability, which factor matters most?

A The fixation used
B The type of wafer
C The surgical technique
D The surgeon's experience
E The direction of movement

919 Which patient group is at particular risk of condylar resorption after orthognathic surgery?

A Young women with a high-angle pattern and pre-existing joint disease
B Older men with a low-angle pattern
C Patients with Class III malocclusion
D Patients with a Class I skeletal base
E Patients undergoing genioplasty alone

920 Distraction osteogenesis follows which sequence?

A Corticotomy, immediate distraction, consolidation
B Immediate distraction at 3 mm per day
C Corticotomy, a latency period of about 5 to 7 days, distraction at about 1 mm per day, then consolidation
D Osteotomy with immediate rigid fixation
E Corticotomy with no distraction phase

921 Which is the appropriate first step in the orthognathic pathway?

A Placing a fixed appliance
B Joint assessment by the orthodontist and maxillofacial surgeon together, with the patient's own concern recorded first
C Taking a CBCT
D Extracting premolars
E Fitting a surgical wafer

922 Which single psychological screen should be performed before orthognathic surgery?

A Screening for depression only
B No screening is required
C Screening for anxiety only
D Screening for body dysmorphic disorder, which affects up to 17 per cent of orthognathic patients
E Screening for eating disorders only

923 Which single feature distinguishes comprehensive from adjunctive orthodontic treatment in an adult?

A Comprehensive treatment corrects the whole malocclusion and requires the full diagnostic workup and a full appliance
B The use of fixed rather than removable appliances
C Comprehensive treatment is always surgical
D Adjunctive treatment always requires extraction
E Adjunctive treatment cannot be delivered in general practice

924 Which single statement about tooth movement rate in adults is correct?

A It is faster than in children
B It is identical to that in children
C It is slower, with more prolonged hyalinisation, and treatment takes longer
D It cannot be predicted
E It is faster in the maxilla only

925 Which is the correct sequence when planning adult orthodontics with a restorative endpoint?

A Begin orthodontics and consult the restorative colleague later
B Debond before restorative planning
C Complete all restorative work first, then move the teeth
D Place implants before orthodontics
E Plan the restorative endpoint with the restorative colleague first, then plan the tooth movement to reach it

926 Which single statement about bonding to porcelain restorations is correct?

A Surface preparation with hydrofluoric acid or air abrasion and a silane coupling agent is required
B Conventional enamel etching is sufficient
C Bonding to porcelain is impossible
D Bands must always be used
E Glass ionomer should be used

927 Which is the appropriate response when an adult patient with a severe skeletal discrepancy declines surgery?

A Proceed with the surgical plan regardless
B Discharge the patient
C Offer camouflage honestly, stating clearly what it can and cannot achieve, and record the discussion
D Begin decompensation in case they change their mind
E Refer to another surgeon without discussion

928 Which surgical procedure is used for transverse expansion in a non-growing patient requiring more than about 5 mm?

A Le Fort I osteotomy alone
B Vertical subsigmoid osteotomy
C Bilateral sagittal split osteotomy
D Genioplasty
E Surgically assisted rapid palatal expansion

929 Which is the correct approach to the lower centre line during pre-surgical orthodontics?

A It should be aligned to the upper centre line
B It should be held on the chin
C It should be ignored until after surgery
D It should be deliberately displaced
E It should be aligned to the facial midline of the maxilla

930 Which is the correct duration of post-surgical orthodontics?

A 1 to 2 weeks
B 3 years
C 18 months
D 4 to 6 months
E None is required

931 Which single statement about anterior segmental osteotomies is correct?

A They are used for mandibular advancement
B They are used only in cleft patients
C They are used for bimaxillary protrusion and for closing an anterior open bite, and are described by Wunderer and Kole
D They are used for transverse expansion
E They have been abandoned

Block AB — Cleft lip and palate and craniofacial syndromes (Q932–961)

Notes In Orthodontics: Chapter 16 (Cleft Lip and Palate, and Syndromes)
Supplements: Revision Handbook **§28** — read it again if you score under 70%
Practise this: 30 questions · allow 30 minutes · answers begin on the page named in the contents

932 At what age is lip repair conventionally performed?

A 3 to 6 months
B At birth
C 9 to 12 months
D 18 months
E 3 years

933 What does the rule of tens refer to in cleft lip repair?

A 10 hours of fasting
B 10 months of age and 10 kg in weight
C 10 days of age
D A 10 mm cleft width
E 10 weeks of age, 10 pounds in weight and haemoglobin of 10 g per dL

934 At what age is alveolar bone grafting performed?

A 3 to 6 months
B 6 to 12 months
C 14 to 16 years
D 8 to 10 years
E After growth is complete

935 Alveolar bone grafting should be performed at what stage of canine root development?

A Before root formation begins
B When about two thirds of the root is formed
C After the root is complete
D After the canine has erupted
E Root development is irrelevant

936 Which is the correct sequence of arch expansion and alveolar bone grafting?

A Graft first, then expand
B Expansion is contraindicated in clefts
C Both simultaneously in all cases
D Neither is required
E Expand first, then graft

937 Which bone is most commonly used for an alveolar bone graft?

A Cortical bone from the mandible
B Cancellous bone from the iliac crest
C Calvarial bone
D Rib
E Synthetic substitute only

938 Which of the following is not a purpose of alveolar bone grafting?

A Providing bone for the canine to erupt into
B Stabilising the maxillary segments
C Closing the oronasal fistula
D Correcting the skeletal Class III relationship
E Supporting the alar base

939 Which dental anomaly is commonest in a patient with a unilateral cleft lip and palate?

A A missing lateral incisor on the cleft side
B An additional premolar
C A missing first molar
D An impacted second molar
E Macrodontia of the central incisor

940 The central incisor adjacent to a cleft is characteristically which of the following?

A Absent
B Macrodont
C Rotated and tipped towards the cleft, and often hypoplastic
D In crossbite only
E Unaffected

941 Dental anomalies in clefting are more common in which group?

A Unilateral cases
B Submucous clefts
C Isolated cleft palate
D Bilateral cases
E They occur equally in all groups

942 Glue ear in cleft palate patients requires which single measure?

A No intervention
B Hearing monitoring, with grommets as required
C Immediate antibiotics
D Adenoidectomy
E Tonsillectomy

943 Velopharyngeal incompetence after palate repair may require which procedure?

A A second palate repair only
B Adenoidectomy
C Alveolar bone grafting
D Le Fort I osteotomy
E A pharyngoplasty

944 Which single feature distinguishes Apert from Crouzon syndrome?

A Syndactyly of the hands and feet
B Craniosynostosis
C Midface hypoplasia
D Exophthalmos
E Relative mandibular prognathism

945 Pierre Robin is correctly described as which of the following?

A A syndrome
B An association
C A sequence, in which micrognathia causes glossoptosis and a U-shaped cleft palate
D A dysplasia
E A malformation of unknown origin

946 Treacher Collins syndrome is caused by a mutation in which gene?

A FGFR2
B TCOF1
C RUNX2
D PTH1R
E MSX1

947 Which combination is characteristic of Treacher Collins syndrome?

A Upward-sloping palpebral fissures and mandibular prognathism
B Syndactyly and craniosynostosis
C Absent clavicles and supernumerary teeth
D Downward-sloping palpebral fissures, malar and mandibular hypoplasia, ear anomalies and conductive deafness
E Macroglossia and midface hypoplasia

948 Cleidocranial dysplasia characteristically produces which dental features?

A Multiple supernumerary teeth and failure of eruption of permanent teeth
B Hypodontia and conical teeth
C Macrodontia and early eruption
D Anodontia
E Amelogenesis imperfecta

949 Which classification is used for the mandibular ramus in hemifacial microsomia?

A Obwegeser and Makek
B Veau
C Pruzansky
D Slootweg and Muller
E Angle

950 Which additional features distinguish Goldenhar syndrome from isolated hemifacial microsomia?

A Cleft palate and hearing loss
B Multiple supernumeraries
C Syndactyly and craniosynostosis
D Absent clavicles
E Epibulbar dermoid and vertebral anomalies

951 Which is the correct distinction between a Turner tooth and Turner syndrome?

A A Turner tooth is a localised enamel hypoplasia caused by infection or trauma to the primary predecessor; Turner syndrome is 45,X
B They are the same condition
C A Turner tooth occurs only in Turner syndrome
D Turner syndrome causes Turner teeth in all cases
E Both are caused by RUNX2 mutation

952 Which craniofacial features are characteristic of Down syndrome?

A Maxillary excess and mandibular retrognathia
B Craniosynostosis and syndactyly
C Midface hypoplasia, relative mandibular prognathism, macroglossia, hypodontia and delayed eruption
D Absent clavicles
E Downward-sloping palpebral fissures with malar hypoplasia

953 The 22q11 deletion syndrome characteristically includes which feature?

A Absent clavicles
B Epibulbar dermoid
C Syndactyly
D Multiple supernumeraries
E Cleft palate or velopharyngeal incompetence with cardiac anomalies and learning difficulties

954 The Veau classification of clefts uses how many classes?

A Two
B Four
C Three
D Five
E Six

955 The LAHSHAL notation records which structures?

A The nasal structures only
B Lip and palate only
C The dental arches only
D Lip, alveolus, hard palate and soft palate on both sides
E The velopharyngeal mechanism

956 Which single organisational change measurably improved cleft outcomes?

A Introduction of presurgical orthopaedics
B Earlier lip repair
C Centralisation of cleft care into multidisciplinary teams with national audit
D Use of resorbable sutures
E Introduction of CBCT

957 Orthognathic surgery in a cleft patient is characteristically required for which problem?

A Maxillary hypoplasia
B Mandibular prognathism
C Transverse mandibular excess
D Vertical maxillary excess
E An anterior open bite alone

958 Retention in the cleft patient is characteristically which of the following?

A Not required
B Required only in bilateral cases
C Six weeks
D Twelve months
E Prolonged, and usually permanent

959 Presurgical orthopaedics, including naso-alveolar moulding, is best described in which way?

A Standard of care in all centres with strong evidence
B Performed after palate repair
C Contraindicated
D Used in some centres, with contested evidence
E Performed at 8 to 10 years

960 Which is the approximate risk of an orofacial cleft where one parent and one sibling are affected?

A Unchanged from the population risk
B Substantially increased above the population risk, rising with the number of affected relatives
C Reduced
D 100 per cent
E Only relevant in isolated cleft palate

961 Which maternal exposure is associated with an increased risk of orofacial clefting?

A Folic acid
B Calcium supplementation
C Vitamin C
D Iron supplementation
E Maternal smoking

Block AC — Risks of orthodontic treatment and the medical history

(Q962–991)

Notes In Orthodontics: Topics 9 and 10 (Benefits and complications; Medical conditions); Chapters 2 and 7
Supplements: Revision Handbook **§29** — read it again if you score under 70%
Practise this: 30 questions · allow 30 minutes · answers begin on the page named in the contents

962 Which benefit of orthodontic treatment has the strongest evidence?

A Improved periodontal health
B Dental aesthetics and trauma prevention
C Reduced caries
D Improved speech
E Improved masticatory efficiency

963 The evidence that orthodontic treatment improves periodontal health is best described how?

A Strong
B Conclusive for adults only
C Moderate
D Weak or absent
E Conclusive for children only

964 The evidence that orthodontic treatment reduces caries is best described how?

A Weak
B Strong
C Moderate
D Conclusive
E Absent for all age groups

965 Which four risks should be named when discussing complications of orthodontic treatment?

A Pain, breakage, cost and time
B Ankylosis, impaction, transposition and infection
C Demineralisation, root resorption, recession and relapse
D Caries, periodontal disease, halitosis and staining
E Allergy, ulceration, bleeding and swelling

966 Orthodontic pain peaks at approximately what interval after appliance activation?

A 6 hours
B 10 days
C 5 days
D 48 hours
E It does not peak

967 Which analgesic is generally preferred during orthodontic treatment?

A Ibuprofen
B Paracetamol
C Aspirin
D Diclofenac
E Codeine

968 Severe root resorption affects approximately what proportion of treated teeth?

A Under 0.5 per cent
B Over 30 per cent
C 10 per cent
D 20 per cent
E 1 to 5 per cent

969 Which single risk factor should be identified before treatment because it increases the likelihood of severe root resorption?

A Previous trauma or previously identified resorption
B A Class II molar relationship
C A deep overbite
D Poor oral hygiene
E A high mandibular plane angle

970 Gingival hyperplasia during fixed appliance treatment is best managed how?

A Immediate excision
B Debonding immediately
C Oral hygiene measures, with excision reserved for persistence after debond
D Antibiotics
E Chlorhexidine indefinitely

971 NICE guidance on antibiotic prophylaxis for dental procedures states which of the following?

A It should be given to all patients with a heart murmur
B It should not routinely be provided solely to prevent infective endocarditis
C It should be given for band placement only
D It should be given for all extractions
E It should be given to all children

972 Mortality from infective endocarditis is approximately which value?

A 2 per cent
B 8 per cent
C 45 per cent
D 21 per cent
E 70 per cent

973 Which appliance type should be avoided in a patient with epilepsy?

A Removable appliances
B Fixed labial appliances
C Bonded retainers
D Transpalatal arches
E Aligners

974 A patient taking phenytoin, or one who has undergone organ transplantation, is best managed with which appliance feature?

A Bands on all molars
B A palatal arch
C Bonded tubes rather than bands
D A removable appliance
E Aligners exclusively

975 Which appliance feature should be avoided in a patient with asthma?

A Bonded tubes
B A bonded retainer
C A fixed labial appliance
D A transpalatal arch
E A palate-covering removable appliance

976 Bisphosphonate therapy has which effect on orthodontic tooth movement?

A It slows or inhibits tooth movement and delays eruption
B It accelerates tooth movement
C It has no effect
D It increases the risk of relapse only
E It increases the rate of eruption

977 Which single change should be made in a patient with sickle cell disease?

A Avoid all orthodontic treatment
B Use heavier forces to shorten treatment
C Use light forces and avoid extractions where possible
D Use only removable appliances
E Prescribe antibiotic prophylaxis

978 How long should elective orthodontic treatment be deferred after organ transplantation?

A 2 weeks
B 3 months
C Indefinitely
D 2 years
E At least 6 months

979 A patient with a documented nickel allergy should avoid which material?

A Stainless steel
B Nickel titanium
C Beta titanium
D Cobalt chromium after heat treatment
E Elastomeric modules

980 Which single statement about orthodontic treatment during pregnancy is correct?

A It is absolutely contraindicated
B Radiographs are contraindicated in all circumstances
C It should proceed unchanged
D Elective treatment should be deferred, with urgent care provided ideally in the second trimester
E Fixed appliances must be removed

981 Which single measure most reduces the risk of demineralisation during fixed appliance treatment?

A Chlorhexidine mouthwash
B Fluoride varnish at debond only
C Selecting patients on the basis of demonstrated plaque control, and not starting until it is adequate
D Using ceramic brackets
E Shortening the archwire sequence

982 Which fluoride regimen is recommended during fixed appliance treatment in a patient at risk?

A High-fluoride toothpaste at 2800 or 5000 parts per million daily, with professional fluoride every 6 to 8 weeks
B 1450 parts per million toothpaste alone
C Fluoride mouthwash weekly
D Fluoride varnish annually
E Silver diamine fluoride monthly

983 Which single statement about recession and incisor movement is supported by the evidence?

A Proclination reliably causes recession
B Recession is prevented by bonded retention
C Recession is unrelated to tooth movement
D Recession occurs only in adults
E Recession relates more to retroclination in high-angle Class III cases with a narrow symphysis and thin biotype

984 Which is the correct management of white spot lesions identified at debond?

A Immediate restoration of all affected teeth
B Immediate bleaching
C Immediate microabrasion
D Explain and record, institute intensive prevention, and review over several months before considering restorative intervention
E No action

985 Klaus and colleagues reported high discontinuation rates with which appliance type?

A Fixed labial appliances
B Removable appliances
C Bonded retainers
D Lingual appliances
E Aligners

986 Which single statement about headgear safety should be recorded in the notes?

A That the patient was shown how to fit the appliance
B That the appliance was CE marked
C That the force was measured
D That the patient's parent was present
E That the patient was warned in writing about injury, given two safety mechanisms, and instructed to stop and attend if the bow disengages

987 Which single statement about enamel loss at debond is correct?

A No enamel is removed
B Debonding and clean-up removes a small but real thickness of enamel
C All enamel damage is reversible
D Only ceramic brackets remove enamel
E Enamel loss occurs only with self-etching primers

988 Which single statement about the risk of failing to achieve the treatment aim is correct?

A It need not be discussed
B It arises only in surgical cases
C It arises only with removable appliances
D It is a real risk that must be consented for, together with what would be done if it occurred
E It is covered by the practice indemnity

989 A patient reports a possible allergy to latex. Which single change is required?

A Latex-free elastics, gloves and rubber dam
B No change
C Avoidance of nickel titanium
D Antibiotic prophylaxis
E Avoidance of fixed appliances

990 Which is the correct statement about orthodontic treatment in a patient with an active eating disorder?

A Treatment should proceed normally
B Treatment is absolutely contraindicated for life
C Elective treatment should be deferred while the disorder is active, with liaison, prevention and management of erosion
D Only removable appliances may be used
E Antibiotic prophylaxis is required

991 Which single statement about the psychosocial benefit of orthodontic treatment is most defensible?

A It reliably improves long-term psychological health
B It benefits only adults
C It has no measurable benefit
D It improves the emotional and social domains of oral health related quality of life, but there is little objective evidence of long-term psychological benefit
E It benefits only patients with severe malocclusion

Block AD — Orthodontic practice, infection control and governance

(Q992–1016)

Notes In Orthodontics: Part V, units V.2–V.4 (radiation protection; infection control; governance and audit)
Supplements: Revision Handbook **§30** — read it again if you score under 70%
Practise this: 25 questions · allow 25 minutes · answers begin on the page named in the contents

992 Which is the correct sequence for instrument decontamination?

A Sterilise, clean, inspect, store
B Clean, inspect, sterilise, store in the sterile pouch, and track by batch to the patient
C Inspect, store, clean, sterilise
D Clean, store, sterilise, inspect
E Sterilise, store, clean, track

993 At what temperature range does a vacuum or non-vacuum autoclave sterilise instruments?

A 100 to 110 degrees Celsius
B 121 degrees Celsius only
C 160 degrees Celsius
D 150 degrees Celsius
E 134 to 137 degrees Celsius

994 Which single statement about single-use items is correct?

A They must never be reprocessed
B They may be reprocessed if visibly clean
C They may be reprocessed once
D They may be reprocessed if autoclaved
E Reprocessing is permitted for orthodontic bands only

995 Bands removed from stock during band sizing should be managed how?

A Returned to stock after wiping
B Returned to stock after autoclaving
C Not returned to stock; a band-sizing kit sterilised between patients should be used
D Discarded only if fitted
E Returned to stock if not fitted

996 Impressions and appliances returning from the laboratory should be managed how?

A Fitted directly
B Rinsed, disinfected and labelled before dispatch and on receipt, with the process documented
C Autoclaved
D Discarded and remade
E No action is required

997 Which single procedure in orthodontics generates significant aerosol?

A Archwire placement
B Elastic module placement
C Impression taking
D Debonding and adhesive clean-up

E Study cast trimming in the surgery

998 Cut archwire ends should be regarded as which category?

A Sharps
B Clinical waste
C Domestic waste
D Recyclable metal
E Amalgam waste

999 The three principles governing every radiographic exposure are which of the following?

A Referral, consent and record keeping
B Indication, exposure and reporting
C Justification, optimisation and limitation
D Frequency, dose and duration
E Screening, diagnosis and audit

1000 Which single statement about radiation dose is correct?

A There is a safe threshold dose below which no risk exists
B Only CBCT carries a measurable risk
C Doses below 1 mSv carry no risk
D Children may receive higher doses safely
E Stochastic risk means there is no threshold dose, only a justified exposure

1001 Which named role justifies a radiographic exposure?

A The practitioner
B The referrer
C The operator
D The legal person
E The dental nurse

1002 Which single record must accompany every radiograph taken?

A The exposure time only
B The manufacturer of the equipment
C A written justification and a report on the film
D The patient's consent form
E The batch number of the receptor

1003 Which single element of the consent discussion must be recorded?

A The time the appointment started
B The next appointment date
C The bracket prescription
D The archwire size
E What was discussed, what was warned about, what was given in writing, who was present, and that the patient agreed

1004 Which items should be recorded at every routine orthodontic adjustment visit?

A The patient's mood only
B What was done, wire sizes and sequence, elastics prescribed, oral hygiene status, breakages, instructions given and the next interval
C The name of the nurse only
D The room used

E The fee charged only

1005 Which outcome measure is most appropriate for auditing an orthodontic service across many cases?

A Patient satisfaction alone
B Treatment duration alone
C IOTN Dental Health Component
D PAR or ICON scores
E Number of appointments

1006 Which of the following should form part of an orthodontic practice audit?

A Staff attendance
B The number of new patients only
C Failure and breakage rates, decalcification rates, treatment duration, discontinuation rates and radiograph rejects
D Stock levels
E Appointment book utilisation only

1007 Which case type should a Diploma holder normally refer rather than treat?

A A severe skeletal discrepancy where surgery is likely
B A Class I case with 5 mm of crowding
C An anterior crossbite with a displacement in a child
D A simple posterior crossbite
E A mild Class II division 1 in a growing child

1008 Which single statement demonstrates judgement most effectively in an oral examination?

A Describing the most sophisticated available technique
B Avoiding any mention of limits
C Quoting the largest number of references
D Describing the appliance in the greatest detail
E Stating what you would treat yourself and the point at which you would refer

1009 Which single element must be recorded at the assessment visit?

A The appointment interval
B The bracket prescription to be used
C The laboratory to be used
D The complaint in the patient's own words
E The archwire sequence

1010 Which single item should be recorded when a plan changes materially during treatment?

A Nothing, provided the outcome is achieved
B The reason for the change and evidence that the consent discussion was revisited
C The date only
D The new appliance only
E The patient's verbal agreement without detail

1011 Which single measure protects against Legionella in dental unit water lines?

A Annual autoclaving
B No measure is required
C Boiling the water supply
D Using bottled water only for impressions

E Flushing, water treatment and monitoring in accordance with local guidance, with a risk assessment

1012 Which waste streams must be segregated in an orthodontic practice?

A Clinical and domestic only
B Clinical, sharps, amalgam and domestic
C Sharps only
D Amalgam only
E No segregation is required

1013 Intra-oral scanner wands should be managed between patients how?

A Wiped only
B Immersed in disinfectant
C Autoclaved whole
D A disposable or sterilisable tip sleeve should be used and the wand wiped between patients
E No measure is required

1014 Which single record should be made at debond?

A Final records, the retention issued, the instructions given and the review arranged
B The bracket prescription used
C The number of visits only
D The total fee
E The archwire sequence used

1015 Which single statement about a radiograph quality assurance programme is correct?

A It is optional
B It applies only to CBCT
C It should include audit of rejected films, staff training and dose monitoring
D It applies only in hospital settings
E It is required only every five years

1016 Which is the correct approach where a Diploma holder identifies a case beyond their competence mid-treatment?

A Continue and hope for the best
B Refer without informing the patient
C Debond and discharge without explanation
D Stop, explain to the patient, record the discussion and refer
E Complete the treatment and refer at debond

Part 2 — Answers, explanations and references

Each answer gives the correct option, the reasoning, and a source reference so that you can read further. **Read the explanation for every question in a block, not only the ones you answered incorrectly.** A complete answer-key grid for rapid marking follows at the end of this Part.

References are given in the abbreviated form used throughout the Revision Handbook: *Companion* = Cobourne & DiBiase, *Handbook of Orthodontics* 3e; *Proffit* = *Contemporary Orthodontics* 6e; *Mosby* = *Mosby's Orthodontic Review*; *Cobourne cases* = *Clinical Cases in Orthodontics*; *MBT* = McLaughlin, Bennett & Trevisi; *B&M* = Bennett & McLaughlin; *Nanda* = *Biomechanics and Esthetic Strategies*; *Hbk* = this series' Revision Handbook, by section.

Block A — Concepts of growth and development (Q1–25 · Handbook §1)

1 A Growth is the anatomical increase in size; development is the physiological and behavioural progression towards maturity. Examiners use the distinction to test precision of language. (Companion Ch 4; Proffit Ch 2; Hbk 1.1)

2 C Chronological age correlates poorly with maturational status. Every clinically useful index of readiness is a maturational one. (Mosby ch. on growth; Hbk 1.2)

3 E Peak height velocity occurs at approximately 12 years in girls and 14 years in boys, with wide individual variation. (Companion Ch 4; Hbk 1.3)

4 B Menarche follows peak height velocity by approximately one year, so a girl who has begun menstruating has largely passed the window for orthopaedic change. (Companion Ch 4; Hbk 1.3)

5 D CS3 (mean age 12.0 years) indicates the peak occurs within the current year. This is the window for growth modification. (Baccetti, Franchi and McNamara; Hbk 1.3)

6 C Widths mature before depths, which mature before heights. This is why transverse problems are treated earliest. (Mosby ch. on growth; Proffit Ch 4; Hbk 1.5)

7 A A counter-intuitive but frequently examined point: there is no anteroposterior growth spurt of the chin. Apparent increases in chin projection are largely the result of vertical growth and rotation. (Mosby ch. on growth; Hbk 1.5)

8 E Moss proposed that bone grows in response to the soft tissue capsules and functional spaces it serves. Petrovic refined this into a servosystem model, but Moss is the theory usually named as the basis of functional appliances. (Companion Ch 4; Hbk 1.4)

9 D The condyle is a secondary, adaptive cartilage responding to its functional environment. This is precisely why functional appliances can influence it at all, and why the influence is modest. (Companion Ch 4; Hbk 1.4)

10 B The distinction underlies the whole nasal septum versus sutures debate. Sutures are sites; whether they are centres was the substance of Sicher's claim. (Companion Ch 4; Hbk 1.1)

11 E Serial height measurement gives height velocity, which is the direct measure. A single height is far less useful than a series. (Mosby ch. on growth; Hbk 1.2)

12 B Puberty begins in the brain, with pulsatile GnRH release. The peripheral changes follow. (Proffit Ch 4; Hbk 1.3)

13 D The cephalocaudal gradient describes the earlier maturation of structures nearer the head, which is why the cranium is near adult size long before the mandible. (Proffit Ch 2; Hbk 1.1)

14 A CS4 has a mean age of 13.4 years and indicates that peak height velocity occurred 1 to 2 years previously. Growth modification is losing its power. (Baccetti, Franchi and McNamara; Hbk 1.3)

15 C Maturational indices tell you where in the curve the patient is, not how much growth remains, and none of the underlying principles is universally accepted. (Companion Ch 4; Hbk 1.3)

16 D Pattern refers to proportional relationships and their orderly change. Pattern, variability and timing are the three properties of growth. (Proffit Ch 2; Hbk 1.1)

17 B A juvenile spurt occurs at about 6 to 8 years, followed by a prepubertal minimum and then the adolescent spurt. (Companion Ch 4; Hbk 1.3)

18 E Residual mandibular growth persists later in Class III patients and later in males, which is why surgery is deferred and confirmed by serial superimposition. (Companion Ch 13; Hbk 1.3)

19 A The cranial vault follows the neural growth curve, being approximately 90 per cent complete by about age 5. This is why craniosynostosis presents early. (Proffit Ch 2; Hbk 2.2)

20 C Behrents demonstrated continued slow rotational growth throughout adult life, and the cumulative effect is described as surprisingly large. (Behrents 1985; Hbk 2.4)

21 B Stage of tooth formation, using Demirjian's or Nolla's stages, is more reliable than eruption alone, which is influenced by local factors. (Mosby ch. on growth; Hbk 1.2)

22 D The synchondroses of the cranial base grow endochondrally. The spheno-occipital synchondrosis fuses at about 13 to 15 years, earlier in girls. (Companion Ch 4; Hbk 2.2)

23 A This is the modern synthesis position, following Petrovic and supported by the randomised trial evidence on functional appliances. (Companion Ch 9; Hbk 1.4)

24 C Cessation of increase in serial standing height is the direct evidence. All the other options are indirect or unreliable. (Mosby ch. on growth; Hbk 1.2)

25 E Statural growth slightly precedes mandibular growth, and the mandible continues to grow after height appears to have stopped. This matters when timing surgery. (Companion Ch 4; Hbk 1.3)

Scored under 70% on this block? Re-read Revision Handbook §1, then sit it again in a fortnight.

Block B — Craniofacial growth, prenatal and postnatal (Q26–55 · Handbook §2)

26 A Five processes appear at week 4: the single frontonasal process and the paired maxillary and mandibular processes. (Companion Ch 3; Hbk 2.1)

27 C The upper lip is formed by fusion of the maxillary processes with the medial nasal processes, by week 8. Failure of this fusion produces cleft lip. (Companion Ch 3; Hbk 2.1)

28 E The face is complete by week 8; the palate is complete by week 10. (Companion Ch 3; Hbk 2.1)

29 B Isolated cleft palate arises from failure of shelf elevation or fusion between weeks 6 and 10, a different failure at a different time from cleft lip. (Companion Ch 3 and Ch 16; Hbk 2.1)

30 D Approximately 50 per cent of isolated cleft palates are syndromic, against approximately 30 per cent of cleft lip with or without cleft palate. (Companion Ch 16; Hbk 2.1)

31 C Neural crest cells form practically all of the facial mesenchyme, which is why disorders of neural crest migration produce craniofacial syndromes. (Proffit Ch 5; Hbk 2.1)

32 A Cleft lip with or without cleft palate occurs in approximately 1 in 700 births; isolated cleft palate in approximately 1 in 2000. (Companion Ch 16; Hbk 2.1 and 28.1)

33 E Isolated cleft palate shows a female preponderance of approximately 4 to 1, the opposite of cleft lip with or without cleft palate, which is male 2 to 1. (Companion Ch 16; Hbk 2.1)

34 D The cranial vault grows by sutural apposition in response to the expanding brain, with surface remodelling. (Companion Ch 4; Hbk 2.2)

35 B The maxilla is displaced downwards and forwards by growth at its sutures, with surface remodelling including resorption anteriorly and apposition at the tuberosity. (Companion Ch 4; Hbk 2.2)

36 E Resorption occurs on the anterior surface of the maxilla while apposition occurs posteriorly at the tuberosity, which is how arch length increases posteriorly. (Companion Ch 4; Hbk 2.2)

37 B Bjork's metallic implant studies revealed internal rotation of the jaws, largely masked by remodelling of the lower border. (Companion Ch 4; Hbk 2.3)

38 D Total (true) rotation equals matrix (apparent) rotation plus intramatrix rotation, which is angular remodelling of the lower border. (Companion Ch 4; Hbk 2.3)

39 A These are Bjork's structural signs of forward (anterior) mandibular rotation, which is associated with a deep overbite and reduced lower face height. (Companion Ch 4; Hbk 2.3)

40 C Backward rotation increases anterior face height relative to posterior face height and is associated with anterior open bite. (Companion Ch 4; Hbk 2.3)

41 D Forward mandibular rotation drives progressive uprighting of the lower labial segment, which is the principal factor in tertiary crowding. (Behrents; Solow; Hbk 2.3 and 19.4)

42 B The condyle grows endochondrally, and the mandible additionally undergoes extensive surface remodelling with posterior remodelling of the ramus. (Companion Ch 4; Hbk 2.2)

43 E The roof of the maxillary sinus is the orbital floor, which carries the infraorbital nerve. (Standard anatomy; Hbk 2.2)

44 A An obtuse cranial base angle positions the glenoid fossa, and therefore the mandible, more posteriorly, contributing to a Class II relationship. (Companion Ch 4; Hbk 2.2)

45 C Approximately 80 per cent are unilateral, and the left side is more commonly affected than the right. (Companion Ch 16; Hbk 2.1 and 28.1)

46 B Folic acid supplementation reduces the risk of orofacial clefting. Smoking, alcohol, anticonvulsants and retinoids increase it. (Companion Ch 16; Hbk 28.1)

47 D Phenytoin and sodium valproate are the anticonvulsants classically associated with an increased risk of clefting. (Companion Ch 16; Hbk 28.1)

48 A The palatine shelves appear vertically either side of the tongue at week 6 and elevate above the tongue as the mandible grows forwards and the head extends. (Companion Ch 3; Hbk 2.1)

49 C The nasomaxillary complex is displaced downwards and forwards relative to the anterior cranial base. (Companion Ch 4; Hbk 2.2)

50 E With age the upper lip lengthens, the vermilion thins, the nose lengthens and incisor show at rest decreases. This is relevant to planning incisor display in adults. (Behrents; Hbk 2.4)

51 A Because cranial vault growth follows the neural curve and is largely complete by age 5, premature sutural fusion produces obvious deformity early. (Proffit Ch 2; Hbk 2.2)

52 C Scott proposed cartilage, and particularly the nasal septum, as the primary determinant, with sutures responding passively to displacement. (Companion Ch 4; Hbk 1.4)

53 E Moss proposed that bone grows in response to its functional matrix. This underpins both functional appliance therapy and the airway hypothesis of facial form. (Companion Ch 4; Hbk 1.4)

54 B The opposite sex ratios and different syndromic proportions are the clearest evidence that these are distinct developmental failures at different times. (Companion Ch 16; Hbk 2.1)

55 D Syndactyly distinguishes Apert from Crouzon; both involve FGFR2 mutations and craniosynostosis with midface hypoplasia. (Companion Ch 16; Hbk 28.4)

Scored under 70% on this block? Re-read Revision Handbook §2, then sit it again in a fortnight.

Block C — Development of the dentition (Q56–95 · Handbook §3)

56 C Incisor liability is approximately 7 mm in the maxilla and 5 mm in the mandible. Note that the larger liability is maxillary. (Companion Ch 5; Hbk 3.1)

57 A Leeway space is approximately 2.5 mm per quadrant in the mandible and 1.5 mm in the maxilla. Note that the larger leeway is mandibular. (Companion Ch 5; Hbk 3.1)

58 E Primate spaces lie mesial to the maxillary canine and distal to the mandibular canine, and are present from eruption of the primary dentition. (Companion Ch 5; Hbk 3.1)

59 D Developmental spacing in the primary dentition is normal and necessary. Its absence predicts crowding of the permanent dentition. (Companion Ch 5; Hbk 3.1)

60 B Eruption occurs when approximately two thirds to three quarters of the root is formed, and continues until occlusal contact. (Companion Ch 5; Hbk 3.2)

61 E In the mandible the sequence is canine, then first premolar, then second premolar, so the second premolar is the tooth left short of space. (Companion Ch 5; Hbk 3.2)

62 B In the maxilla the sequence is first premolar, second premolar, then canine, so the canine is the tooth left short of space. The last tooth to erupt in each arch is the one that ends up crowded. (Companion Ch 5; Hbk 3.2)

63 D A distal step becomes a Class II molar relationship and never self-corrects. A mesial step becomes Class I; a flush terminal plane may go either way. (Companion Ch 5; Hbk 3.2)

64 A Space for the mandibular incisors is negative for approximately two years after eruption. Transient early crowding is normal and self-correcting. (Proffit Ch 3; Hbk 3.1)

65 C Hypodontia excluding third molars affects approximately 6.4 per cent. Third molar agenesis occurs in up to 30 per cent. (Companion Ch 5 and Ch 12; Hbk 3.3)

66 D The mandibular second premolar is the most commonly absent, followed by the maxillary lateral incisor and the maxillary second premolar. (Companion Ch 5; Hbk 3.3)

67 B Supernumerary teeth affect approximately 2 per cent of the permanent dentition, with figures of up to 4 per cent quoted overall. About 90 per cent are maxillary. (Companion Ch 12; Hbk 3.3)

68 E Tuberculate supernumeraries typically prevent eruption; conical supernumeraries more often cause displacement or a diastema. (Companion Ch 12; Hbk 3.3)

69 A Removal of the obstruction with space creation or maintenance, then observation for spontaneous eruption, is the standard sequence. Expose and bond if there is no eruption within about 12 months. (Companion Ch 12; Hbk 20.2)

70 C Spontaneous disimpaction may occur up to approximately 8 years. Beyond that, a separator, brass wire or distalising spring is used. (Companion Ch 12; Hbk 3.3 and 20.2)

71 B Transposition affects approximately 0.3 per cent. True transposition involves crowns and roots; pseudo-transposition involves crowns only. (Proffit Ch 12; Hbk 3.3)

72 D Maxillary canine to first premolar transposition is the most frequently reported. (Companion Ch 12; Hbk 3.3)

73 A MIH affects approximately 15 per cent of children and frequently changes the extraction calculus, since a compromised first permanent molar may become the planned extraction. (Companion Ch 12; Hbk 3.3)

74 C Calcification of the second permanent molar bifurcation, at approximately 8 to 10 years of dental age, is the classical timing. (Companion Ch 12; Hbk 3.4)

75 E Spontaneous closure is better in the mandible in the sense that timing is critical there; the maxillary second molar drifts mesially readily at almost any age, whereas the mandibular one does not, which is why timing matters most in the lower arch. (Companion Ch 12; Hbk 3.4)

76 A A compensating extraction is the removal of the opposing tooth in the same quadrant, considered to preserve the buccal occlusion and prevent over-eruption. (Companion Ch 12; Hbk 13.5)

77 C A balancing extraction is the removal of the contralateral tooth in the same arch to prevent a centre line shift, and is most relevant for canines. (Companion Ch 12; Hbk 13.5)

78 E Where the successor is present and eruption is obstructed, or infra-occlusion is severe, extraction is indicated. Otherwise monitor. (Companion Ch 12; Hbk 3.3)

79 B Infra-occlusion affects up to approximately 9 per cent of primary molars. (Companion Ch 12; Hbk 3.3)

80 D Dental anomalies co-occur far more often than chance, reflecting a shared genetic field effect. One anomaly is a reason to look for the others. (Companion Ch 5; Cobourne cases; Hbk 3.3)

81 C Gemination is incomplete division of a single tooth germ; fusion is union of two separate germs. Concrescence is union by cementum only. (Companion Ch 5; Hbk 3.3)

82 A Double teeth affect approximately 0.5 per cent of the permanent dentition and rather more of the primary dentition. (Companion Ch 5; Hbk 3.3)

83 E Intercanine width increases by approximately 3 mm in the mandible and 4.5 mm in the maxilla, which is one of the ways the incisor liability is met. (Companion Ch 5; Hbk 3.1)

84 D Leeway space is the excess of the primary canine and molar widths over their successors. It is larger in the mandible at approximately 2.5 mm per quadrant. (Companion Ch 5; Hbk 3.1)

85 B A lower lingual arch preserves the leeway space by preventing mesial drift of the first permanent molars. It has good evidence as an interceptive measure. (Companion Ch 12; Hbk 20.2)

86 E Oligodontia is conventionally the absence of six or more teeth excluding third molars, and is frequently syndromic. (Companion Ch 5; Hbk 3.3)

87 B Ectodermal dysplasia produces hypodontia or anodontia, conical teeth, sparse hair and reduced sweating. (Companion Ch 16; Hbk 28.4)

88 D Cleidocranial dysplasia results from RUNX2 mutation, producing absent or hypoplastic clavicles, delayed fontanelle closure, multiple supernumeraries and failure of eruption. (Companion Ch 16; Hbk 28.4)

89 A Primary failure of eruption is caused by a loss-of-function mutation in PTH1R. Affected teeth do not respond to orthodontic traction, and attempting it causes ankylosis. (Decker 2008; Hbk 4.8)

90 C The follicle is essential: removing it prevents eruption, whereas removing the tooth and leaving the follicle still allows a pathway to form. (Companion Ch 5; Hbk 4.8)

91 D The first permanent molars erupt at approximately 6 years, marking the start of the mixed dentition. (Companion Ch 5; Hbk 3.2)

92 B The primary dentition erupts between about 6 months and 2.5 years and is complete by approximately 3 years. (Companion Ch 5; Hbk 3.2)

93 E A flush terminal plane is seen in approximately 60 to 75 per cent of children and may correct to Class I using leeway space or remain end to end. (Companion Ch 5; Hbk 3.2)

94 A Megadontia produces a tooth-size discrepancy, which is quantified by the Bolton analysis and usually managed by interproximal reduction or restorative addition. (Companion Ch 5; Hbk 11.1)

95 C The Bolton anterior ratio is 77.2 per cent and the overall ratio 91.3 per cent. A ratio above the norm indicates mandibular excess. (Companion Ch 7; Hbk 11.1)

Scored under 70% on this block? Re-read Revision Handbook §3, then sit it again in a fortnight.

Block D — The biology of tooth movement, eruption and root resorption (Q96–140 · Handbook §4)

96 B An intact periodontal ligament is an absolute prerequisite. Ankylosed teeth and implants have no ligament and do not move, which localises the response to the ligament. (Companion Ch 6; Hbk 4.1)

97 D Carl Sandstedt and Albin Oppenheim independently produced the light-microscopy studies that founded the pressure-tension theory at the turn of the twentieth century. (Companion Ch 6; Hbk 4.1)

98 A The periodontal ligament is approximately 0.2 mm wide on average, and is thinnest around the middle third of the root. (Companion Ch 6; Hbk 4.2)

99 C Osteocytes make up 85 to 90 per cent of bone cells and are the principal mechanosensors. (Companion Ch 6; Hbk 4.2)

100 E Sclerostin acts as a brake against excess bone formation. It is a refinement most candidates miss. (Companion Ch 6; Hbk 4.2)

101 A RANKL rises under compression and drives osteoclastic resorption; OPG, a soluble decoy receptor, rises under tension and opposes it. Reversing this is the single commonest slip in this topic. (Companion Ch 6; Hbk 4.4)

102 C Reitan showed in 1967 that hyalinisation occurs even at 30 g, and that it occurs more with tipping than with bodily movement. (Reitan 1967; Hbk 4.3)

103 E The lag phase lasts 20 to 30 days while hyalinised tissue is removed by undermining resorption. (Companion Ch 6; Hbk 4.3)

104 B Osteoclasts cannot act beneath the cell-free hyalinised zone, so resorption proceeds from the adjacent marrow spaces. It is slower than frontal resorption. (Companion Ch 6; Hbk 4.3)

105 D Approximately 1 mm per month, faster in the maxilla and in younger patients. (Companion Ch 6; Hbk 4.3)

106 C Theodorou and colleagues concluded that 50 to 100 cN is optimal for bodily movement, with similar rates achieved across 50 to 250 cN. (Theodorou 2019; Hbk 4.6)

107 A The examinable position is that no universal optimum exists. What matters more than absolute force is the differential stress generated. (Ren 2003, 2004; Hbk 4.6)

108 E Intrusion requires approximately 10 to 20 g per tooth, giving about 60 g total for four incisors, because force is dissipated through the small apical surface area. (Companion Ch 13; Nanda; Hbk 4.6)

109 D Storey and Smith's historical range of 150 to 200 g is the figure to attribute to them. (Storey and Smith 1952; Hbk 4.6)

110 B Ren's modelling gave a maximum at 272 cN, while also concluding that no universal ideal force can be defined. (Ren 2004; Hbk 4.6)

111 E Proffit gives a force duration threshold of approximately 4 to 8 hours per day, and notes that continuous force is more effective than intermittent force. (Proffit Ch 8; Hbk 4.6)

112 B RANKL peaks at 3 days and returns to baseline between days 6 and 58, implying that other pathways sustain the post-lag phase. (Danz 2021; Hbk 4.4)

113 D Yang and colleagues showed in 2018 that RANKL deletion in transgenic mice reduces osteoclast numbers and tooth movement. (Yang 2018; Hbk 4.4)

114 A Kanzaki and colleagues showed that OPG gene transfer reduces tooth movement, consistent with OPG acting as a decoy receptor for RANKL. (Kanzaki 2004; Hbk 4.4)

115 C Henneman's model runs from extracellular matrix strain and fluid flow, through strain transfer to cells, to cell recruitment and differentiation, and finally tissue remodelling. (Henneman 2008; Hbk 4.5)

116 D Baumrind proposed bone bending in 1969, but there is little evidence that orthodontic forces bend bone to a clinically meaningful degree. (Baumrind 1969; Hbk 4.5)

117 B Piezoelectric currents decay too rapidly and are too small to be clinically significant. Streaming potentials are of unclear significance. (Companion Ch 6; Hbk 4.5)

118 E Radiographic incidence is quoted as up to 73 per cent and histological incidence as up to 90 per cent. (Companion Ch 6; Hbk 4.7)

119 A Severe resorption affects 1 to 5 per cent of treated teeth, and is greatest in the anterior maxilla. (Sameshima and Sinclair 2004; Hbk 4.7)

120 C Heavy force is the only factor with substantive supporting evidence. The others are suggested but not established. (Weltman 2010; Hbk 4.7)

121 B Levander and colleagues recommended pausing force application for 3 to 6 months where significant resorption is identified. (Levander 1994; Hbk 4.7)

122 D The order of susceptibility is maxillary incisors, then canines, then mandibular incisors, then first permanent molars. (Companion Ch 6; Hbk 4.7)

123 A Severe resorption does not necessarily cause greater mobility or discolouration long term, although it should still be minimised and consented for. (Cobourne cases; Hbk 4.7)

124 C Eruption occurs at approximately two thirds to three quarters root formation, and continues to occlusal contact and slowly thereafter. (Companion Ch 5; Hbk 4.8)

125 E The dental follicle coordinates the osteoclastic resorption that forms the eruption pathway. Removing the follicle prevents eruption. (Companion Ch 5; Hbk 4.8)

126 A Primary failure of eruption, due to PTH1R mutation, does not respond to traction and attempts at traction cause ankylosis. Distinguish it from mechanical obstruction, which does respond. (Decker 2008; Hbk 4.8)

127 C Proffit's argument is that alveolar bone exists only where a tooth erupted. Erupting and extruding teeth bring alveolar bone with them. (Proffit Ch 13; Hbk 4.8)

128 E The regional acceleratory phenomenon is real but transient, and evidence that these techniques shorten overall treatment is sparse. (Companion Ch 6; Hbk 4.9)

129 B Photobiomodulation may accelerate discrete phases of movement but does not influence overall treatment time. (El-Angbawi 2023; Hbk 4.9)

130 D On the tension side OPG rises and RANKL falls, with osteoblast activity, TGF-beta, VEGF and FGF2 supporting deposition and angiogenesis. (Companion Ch 6; Hbk 4.3)

131 C Osteoclasts acidify the resorption lacuna and release cathepsin K and MMPs, forming Howship's lacunae. (Companion Ch 6; Hbk 4.2)

132 A The BMU is the remodelling unit in which apposition and resorption are tightly coupled. (Companion Ch 6; Hbk 4.2)

133 E Extracellular matrix is approximately 30 per cent, mineral approximately 60 per cent and cells approximately 10 per cent. (Companion Ch 6; Hbk 4.2)

134 D Osteoclasts are multinucleate cells derived from haematopoietic stem cells that circulate as monocytes. (Companion Ch 6; Hbk 4.2)

135 B Elastic oxytalan fibres are a characteristic constituent of the periodontal ligament, along with epithelial rests of Malassez and pluripotent stem cells. (Companion Ch 6; Hbk 4.2)

136 E Tipping concentrates stress at the alveolar crest and apex, producing more hyalinisation than bodily movement for the same force. (Reitan 1967; Hbk 4.3)

137 B This is the honest, quantified answer, with the risk-reduction measures of light forces, short treatment and a mid-treatment radiograph. (Companion Ch 2; Hbk 4.7)

138 D The initial phase lasts about 2 days and consists of displacement within the ligament space, before true bone remodelling begins. (Companion Ch 6; Hbk 4.3)

139 A Tipping a single tooth requires approximately 25 to 40 g, sometimes quoted as 35 to 60 g. Removable appliance springs are activated accordingly. (Companion Ch 9; Hbk 16.1)

140 C Matrix strain and altered fluid flow distort nerve endings, releasing substance P and CGRP as part of the early neurogenic inflammatory response. (Companion Ch 6; Hbk 4.4)

Scored under 70% on this block? Re-read Revision Handbook §4, then sit it again in a fortnight.

Block E — Occlusion, malocclusion and aetiology (Q141–170 · Handbook §5)

141 D Malocclusion is defined by significance to the individual, which is why demand and need are separate questions. (Companion Ch 2; Hbk 5.1)

142 B Angle's Class I is defined by the mesiobuccal cusp of the upper first molar occluding in the buccal groove of the lower first molar. (Companion Ch 2; Hbk 5.2)

143 E Angle's classification assumes a correctly positioned first molar and ignores the vertical and transverse dimensions, the skeletal pattern, and cases where the molar is missing or displaced. (Companion Ch 2; Hbk 5.2)

144 A The BSI classification is incisor-based, defined by the position of the lower incisor edges relative to the cingulum plateau of the upper centrals. (Companion Ch 2; Hbk 5.2)

145 C In Class II division 2 the upper central incisors are retroclined; the overjet is usually minimal but may be increased. (Companion Ch 2; Hbk 5.2)

146 B Andrews studied 120 non-orthodontic normal occlusions to derive the six keys and, subsequently, the pre-adjusted prescription. (Andrews 1972; Hbk 5.2)

147 D The six keys are molar relationship, crown angulation, crown inclination, no rotations, tight contacts and a flat curve of Spee. Canine guidance is not among them. (Andrews 1972; Hbk 5.2)

148 A An acceptable occlusion develops naturally in only 30 to 40 per cent, which is one reason the specialty exists. (Cobourne cases; Hbk 5.4)

149 C Posselt showed that only about 10 per cent have RCP coincident with ICP, and that a discrepancy of 0 to 2 mm is normal. (Posselt 1952; Hbk 5.1)

150 E Only 5 per cent or fewer of malocclusions have a specific identifiable cause such as a syndrome, cleft, trauma or pathology. (Proffit Ch 5; Hbk 5.3)

151 A Equilibrium theory states that light sustained forces such as resting lip and tongue posture move teeth, whereas heavy brief forces such as chewing do not. (Proffit Ch 5; Hbk 5.3)

152 C The classic digit-sucking pattern is proclined uppers, retroclined lowers, an asymmetric anterior open bite and a posterior crossbite from a lowered tongue position and cheek pressure. (Companion Ch 12; Hbk 5.3)

153 E The association is real but weak, and causation is not established. Stating the association without asserting the cause is the examinable answer. (Companion Ch 7; Hbk 26.5)

154 B Class II affects approximately 15 to 22 per cent of United States adolescents, and Class III approximately 5 to 6 per cent. (Mosby ch. on classification; Hbk 5.4)

155 D Worldwide Class III prevalence is approximately 6.8 per cent, with much higher figures in East Asian populations. (Mosby ch. on classification; Hbk 5.4)

156 C Normal overbite is 2 to 4 mm, or one third to one half of the lower incisor crown height. (Companion Ch 7; Hbk 5.4)

157 A Skeletal pattern, particularly Class III, and tooth size and number show the highest heritability. Occlusal detail is more environmentally determined. (Proffit Ch 5; Hbk 5.3)

158 E Angle's line of occlusion runs through the central fossae and cingula of the upper arch and the buccal cusps and incisal edges of the lower. (Companion Ch 2; Hbk 5.1)

159 D Centric relation is a condylar position independent of tooth contact. Intercuspal position is defined by the teeth. (Companion Ch 2; Hbk 5.1)

160 B Ideal overjet is 2 to 4 mm, corresponding to the aim of Class I incisors in a stable soft tissue environment. (Cobourne cases; Hbk 5.4)

161 E A lower lip trap resting behind the upper incisors proclines them and retroclines the lowers, perpetuating the malocclusion. (Companion Ch 12; Hbk 22.1)

162 B Overjet of 5 mm or more falls from 23 per cent of children to 13 per cent of adults, because the mandible catches up. This is a key argument against early treatment. (Proffit Ch 13; Hbk 22.1)

163 D Ideal occlusion is a theoretical construct. Normal occlusion is an occlusion within an accepted range of variation, functioning without pathology. (Companion Ch 2; Hbk 5.1)

164 A Condylar fracture in childhood is the commonest acquired cause of mandibular growth disturbance, and can produce a Class II relationship with asymmetry. (Companion Ch 15; Hbk 26.4)

165 C In Class II division 2, the laterals are frequently proclined and mesiolabially rotated while the centrals are retroclined, and the interincisal angle is increased. (Companion Ch 2; Hbk 22.4)

166 D Class II division 2 affects approximately 5 to 10 per cent of Caucasian populations. (Companion Ch 2; Hbk 5.4)

167 B A discrepancy of 0 to 2 mm is normal. A larger anterior slide should be recorded, and raises the possibility of a pseudo-Class III or a functional displacement. (Posselt 1952; Hbk 5.1 and 23.1)

168 E A seventh key covering crown size, shape and tooth-size harmony is often added to Andrews' original six. (Andrews 1972; Hbk 5.2)

169 A Neural crest cells form practically all the facial mesenchyme, so disorders of their migration produce craniofacial anomalies. (Proffit Ch 5; Hbk 5.3)

170 C Teeth moved outside the zone of soft tissue equilibrium return towards it. Lower incisor advance of more than about 2 mm very likely needs permanent retention. (Proffit Ch 17; Hbk 19.1)

Scored under 70% on this block? Re-read Revision Handbook §5, then sit it again in a fortnight.

Block F — Treatment need, indices and epidemiology (Q171–195 · Handbook §6)

171 B The DHC is hierarchical: only the single worst trait is recorded, following the MOCDO order. (Companion Ch 2; Hbk 6.1)

172 D MOCDO is the hierarchy: Missing teeth, Overjet, Crossbite, Displacement of contact points, Overbite. Work down the list and record the first trait met. (Companion Ch 2; Hbk 6.1)

173 A Overjet greater than 6 mm and up to 9 mm is grade 4a. Over 9 mm is 5a; over 3.5 mm up to 6 mm with incompetent lips is 3a. (Companion Ch 2; Hbk 6.1)

174 C A crossbite with a displacement of more than 2 mm is grade 4c. A displacement of 1 to 2 mm is 3c. (Companion Ch 2; Hbk 6.1)

175 E Impeded eruption is grade 5i, in the very great need category. (Companion Ch 2; Hbk 6.1)

176 A The AC is a rank order of attractiveness. The photograph selected should match the degree of aesthetic impairment, not the morphology. (Companion Ch 2; Hbk 6.1)

177 C Grades 1 to 4 indicate no need, 5 to 7 borderline need, and 8 to 10 need for treatment. (Companion Ch 2; Hbk 6.1)

178 E PAR is accumulative, scored from study casts before and after, and has no maximum. IOTN measures need; PAR measures outcome. (Companion Ch 2; Hbk 6.2)

179 B A reduction of at least 22 points, or greater than 70 per cent, is classed as greatly improved. (Companion Ch 2; Hbk 6.2)

180 D A reduction of less than 30 per cent is classed as worse or no different. Greater than 30 per cent is improved. (Companion Ch 2; Hbk 6.2)

181 C PAR is a coarse instrument best used to audit an operator or service over many cases rather than to judge a single case. (Companion Ch 2; Hbk 6.2)

182 A The ABO-OGS assesses alignment, marginal ridges, buccolingual inclination, occlusal relationship, occlusal contacts, overjet, interproximal contacts and root angulation. (Companion Ch 14; Hbk 6.3)

183 E PAR shows no difference; ABO-OGS shows aligners significantly worse. PAR is the coarser measure and the two do not correlate. (Companion Ch 14; Hbk 6.3 and 17.5)

184 D Little's Irregularity Index is the sum of contact point displacements of the mandibular incisors, used principally in relapse research. (Little 1975; Hbk 6.3)

185 B Little proposed under 3.5 mm of lower incisor irregularity as an acceptable level of relapse. (Little 1975; Hbk 19.4)

186 E ICON, the Index of Complexity, Outcome and Need, combines all three in a single instrument. (Companion Ch 2; Hbk 6.3)

187 B Reverse overjet greater than 3.5 mm without reported difficulty is 4b; with reported difficulties it becomes 5m. (Companion Ch 2; Hbk 6.1)

188 D Displacement of contact points greater than 2 mm and up to 4 mm is grade 3d. More than 4 mm is 4d. (Companion Ch 2; Hbk 6.1)

189 A Need and demand are distinct. Indices measure need; demand is driven principally by appearance and by social factors. (Proffit Ch 1; Hbk 6.1)

190 C Cleft lip and palate defects are grade 5p, in the very great need category. (Companion Ch 2; Hbk 6.1)

191 D An increased and complete overbite with gingival or palatal trauma is grade 4f. Complete to the gingiva without trauma is 3f. (Companion Ch 2; Hbk 6.1)

192 B The two components are recorded independently. Either may indicate need for treatment. (Companion Ch 2; Hbk 6.1)

193 E Up to approximately 18 per cent of children have experienced dental trauma by age 12, and increased overjet is a significant risk factor. (Companion Ch 2; Hbk 6.3 and 29.2)

194 A Jarvinen reported that traumatic loss of a maxillary central affects approximately 3 per cent of children, typically boys with an increased overjet in the mixed dentition. (Jarvinen 1978; Hbk 20.6)

195 C Indices exist principally to allow rational, defensible allocation of a limited resource, which is the justice principle applied to orthodontics. (Companion Ch 2; Hbk 6.1 and 8.3)

Scored under 70% on this block? Re-read Revision Handbook §6, then sit it again in a fortnight.

Block G — Evidence-based orthodontics (Q196–225 · Handbook §7)

196 B Evidence-based practice integrates the best available evidence with clinical expertise and the patient's own values and preferences. (Companion Ch 1; Hbk 7.1)

197 D Filtered secondary research sits above unfiltered primary research; clinical practice guidelines derived from high-quality meta-analysis sit at the top. (Companion Ch 1; Hbk 7.1)

198 A PICO structures an answerable clinical question: Population, Intervention, Comparison, Outcome. (Companion Ch 1; Hbk 7.2)

199 C Randomisation balances both known and unknown confounders between groups, which is what no observational design can achieve. (Companion Ch 1; Hbk 7.2)

200 E Allocation concealment protects the randomisation sequence at the point of recruitment. Blinding conceals group membership thereafter. (Companion Ch 1; Hbk 7.2)

201 A Operator and patient blinding is rarely possible in orthodontics. Blinding the outcome assessor is both feasible and the most important protection against detection bias. (Companion Ch 1; Hbk 7.2)

202 C Intention-to-treat preserves the benefit of randomisation and gives the more conservative and realistic estimate of effect. (Companion Ch 1; Hbk 7.2)

203 E A type I error is a false positive, conventionally controlled at an alpha of 0.05. A type II error is a false negative. (Companion Ch 1; Hbk 7.2)

204 B Power is conventionally 80 to 90 per cent, and equals 1 minus beta, the probability of a type II error. (Companion Ch 1; Hbk 7.2)

205 D Conventional thresholds are 25 per cent low, 50 per cent moderate and 75 per cent high heterogeneity. (Companion Ch 1; Hbk 7.2)

206 C A relative risk of 0.58 corresponds to a 42 per cent relative reduction in risk, although the confidence intervals in these trials are wide. (Cobourne 2022; Hbk 7.3 and 22.2)

207 A Asymmetry of a funnel plot suggests publication bias, in which positive results are preferentially published. (Companion Ch 1; Hbk 7.2)

208 E An interval for a ratio measure that crosses 1 is compatible with no effect. For a difference measure the equivalent value is 0. (Companion Ch 1; Hbk 7.2)

209 D O'Brien's Manchester twin block trial reported no long-term difference, with the late-treated group finishing slightly better and the early group having more attendances. (O'Brien 2009; Hbk 7.3)

210 B Scott, Fleming and colleagues found no difference in rate of movement, treatment time, number of visits or occlusal outcome. (Scott 2008; Fleming 2010; Hbk 7.3 and 17.3)

211 E Behrents and White reported the AAO-funded studies in 1992, showing no significant association and that TMD cannot be prevented by any intervention. (Behrents and White 1992; Hbk 7.3 and 26.2)

212 B Benson and colleagues found only very weak evidence for improved eruption and none for reducing surgical need, and no benefit from double extraction. (Benson 2021; Hbk 7.3 and 20.4)

213 D The randomised trial found comparable outcomes and no time saving, with more breakages, more complaints and more planning time. Treatment time was dictated by case difficulty and by the treating orthodontist. (Penning 2017; Hbk 7.3 and 17.3)

214 A Larsen and Orabi found no evidence of harm, and the claim has been described in the literature as scaremongering and marketing. (Larsen 2015; Orabi 2021; Hbk 7.3 and 13.6)

215 C Cephalometric growth prediction was shown to be at best unreliable and no better than chance. (Baumrind 1984; Hbk 7.3 and 10.4)

216 D NNT is the reciprocal of the absolute risk reduction, and is the measure patients understand most readily. (Companion Ch 1; Hbk 7.2)

217 B Acknowledging the weakness of the evidence base, and naming the studies that overturned tradition, is the fastest way to demonstrate scholarship. (Companion Ch 1; Hbk 7.3)

218 E Richardson's work, supported by NICE guidance, established that third molars are not a primary cause of late lower incisor crowding. (Richardson 2002; NICE; Hbk 7.3 and 19.4)

219 A Luther found no evidence that an ideal functional occlusion improves stability or prevents temporomandibular disorder. (Luther 2007; Hbk 7.3)

220 C Bowman and Johnston found that lip retraction must be considerable before the profile is judged unattractive, and comparable cases treated either way show little facial difference. (Bowman and Johnston 2000; Hbk 13.6)

221 B Retrospective case-control designs are particularly susceptible to selection bias in the choice of cases and controls. (Companion Ch 1; Hbk 7.1)

222 D A split-mouth design suits localised interventions with no carry-across effect, such as adhesive or bracket comparisons. It is inappropriate where the intervention affects the whole patient. (Companion Ch 1; Hbk 7.2)

223 **A** GRADE rates the certainty of evidence, downgrading for risk of bias, inconsistency, indirectness, imprecision and publication bias. (Companion Ch 1; Hbk 7.2)

224 **C** Orthodontic trials are typically small, with operator blinding impossible, so risk of bias and imprecision are the commonest reasons for downgrading. (Companion Ch 1; Hbk 7.2)

225 **E** A case series sits below cross-sectional and cohort designs and above expert opinion. It cannot establish causation. (Companion Ch 1; Hbk 7.1)

Scored under 70% on this block? Re-read Revision Handbook §7, then sit it again in a fortnight.

Block H — Ethics, consent, communication and the psychosocial patient (Q226–255 · Handbook §8)

226 **A** Valid consent requires capacity, sufficient information and voluntariness, and it is a process rather than a single event. (GDC Standards; Hbk 8.1)

227 **C** Montgomery replaced the professional standard with a patient-centred one: the material risks are those a reasonable patient in that position would attach significance to. (GDC Standards; Hbk 8.1)

228 **E** All reasonable alternatives must be discussed, including the option of no treatment and its consequences. (GDC Standards; Hbk 8.1)

229 **B** A signed form is evidence of consent, not consent itself. What matters is the discussion, and the record of it. (GDC Standards; Hbk 8.1 and 30.3)

230 **D** Gillick competence describes a child under 16 with sufficient understanding and intelligence to understand fully what is proposed. (GDC Standards; Hbk 8.2)

231 **C** Capacity is decision- and time-specific and is assessed by whether the person can understand, retain, weigh and communicate the decision. (Mental capacity legislation; Hbk 8.2)

232 **A** Body dysmorphic disorder affects 5 to 13 per cent of orthodontic patients, against 0.7 to 2.5 per cent of the general population and up to 17 per cent of those seeking orthognathic surgery. (Dons 2022; Hbk 8.3)

233 **E** Preoccupation with an imperceptible defect, serial dissatisfaction and refusal to accept reassurance are the classic features. (Dons 2022; Hbk 8.3)

234 **D** No reliable predictor of cooperation exists. Internal motivation is the best available proxy, along with attendance history and hygiene at assessment. (Mosby ch. on motivation; Hbk 8.3)

235 **B** Klaus and colleagues reported high discontinuation rates in children and adolescents, especially with removable appliances. (Klaus 2017; Hbk 8.3)

236 **E** Psychosocial benefit and dental aesthetics are the best-supported benefits. The functional claims all require qualifiers. (Proffit Ch 1; Hbk 29.1)

237 **B** Seehra found that increased overjet increases susceptibility to teasing and bullying, which is a legitimate treatment indication. (Seehra 2011; Hbk 8.3)

238 **D** Kenealy found little objective evidence of long-term psychological benefit. Do not overclaim this at consent. (Kenealy 2007; Hbk 8.3)

239 **A** Javidi reported improvement in the emotional and social well-being domains of OHRQoL. (Javidi 2017; Hbk 8.3)

240 **C** Demineralisation in elective treatment that the clinician chose to start is indefensible. Defer, document, review and re-assess. (Companion Ch 2; Hbk 8.3 and 29.2)

241 D Indices exist to allocate a scarce, largely publicly funded resource, which is the principle of justice applied to orthodontics. (Companion Ch 2; Hbk 8.3)

242 B Retention is agreed before active treatment begins, not mentioned at debond. It is one of the twelve traps that cost most marks. (Companion Ch 13; Hbk 8.1 and 19.3)

243 E Consent must be obtained from a person with parental responsibility for a child who is not Gillick competent. Elective treatment should not proceed without it. (GDC Standards; Hbk 8.2)

244 A Even where parental consent is legally sufficient, the child's assent determines whether the appliance is worn and therefore whether the plan will work. (Nanda; Hbk 8.2)

245 C Asking the patient to restate what has been agreed is the recognised method of checking understanding, and should be recorded. (GDC Standards; Hbk 8.1)

246 B White spot lesions affect up to half of fixed appliance patients and are visible within 6 months, commonest on first molars, upper laterals and lower canines. (Gorelick 1982; Hbk 8.1 and 29.2)

247 D Serial dissatisfaction with a defect invisible to the clinician is a red flag. Careful exploration of expectations, and declining to commit at that visit, is the safe response. (Dons 2022; Hbk 8.3)

248 A Cost is part of valid consent, and the cost of replacement retainers is the item most often omitted and most often disputed later. (GDC Standards; Hbk 8.1 and 19.3)

249 C A material change of plan, such as a decision to extract or to proceed to surgery, requires the consent discussion to be repeated and recorded. (GDC Standards; Hbk 8.1 and 30.3)

250 E The patient's own concern sets the outcome against which the treatment will be judged, which is why it is recorded first and ranked highly in the problem list. (Proffit Ch 6; Hbk 8.3 and 12.4)

251 A The comparative evidence should be explained plainly. Rushing, Paquette, Iared and Bowman and Johnston are the relevant sources. (Bowman and Johnston 2000; Hbk 13.6)

252 C The examinable position is that orthodontics is TMJ-neutral. Do not promise a cure, and do not let a click drive irreversible occlusal change. (Macfarlane 2009; Luther 2010; Hbk 26.2)

253 E A 16-year-old is presumed to have capacity. Confidentiality should be respected while encouraging, not compelling, parental involvement. (GDC Standards; Hbk 8.2)

254 B Active eating disorders are a reason to defer elective treatment, with liaison, prevention and management of erosion in the interim. (Companion Ch 7; Hbk 29.3)

255 D Elective treatment on a healthy dentition raises rather than lowers the consent standard, and makes foreseeable complications such as demineralisation harder to defend. (GDC Standards; Hbk 8.1)

Scored under 70% on this block? Re-read Revision Handbook §8, then sit it again in a fortnight.

Block I — Clinical examination and records (Q256–285 · Handbook §9)

256 C Natural head position with the Frankfort plane horizontal, teeth in occlusion and lips at rest is the reproducible standard. (Companion Ch 7; Hbk 9.2)

257 A A crossbite with a displacement is treated on diagnosis; one without a displacement is treated when convenient. (Companion Ch 12; Hbk 24.1)

258 E Competent lips meet at rest without muscular effort. Potentially competent lips would meet but for proclined incisors; incompetent lips do not meet at rest. (Companion Ch 7; Hbk 9.2)

259 D An interlabial gap greater than 3 to 4 mm is the key soft tissue sign of vertical excess and of a skeletal open bite tendency. (Nanda; Hbk 9.2 and 25.2)

260 B The maxillary-mandibular plane angle is approximately 27 degrees plus or minus 4, as is the Frankfort-mandibular plane angle. (Companion Ch 7; Hbk 10.2)

261 E Lower anterior face height is approximately 55 per cent plus or minus 2 of total anterior face height. (Companion Ch 7; Hbk 10.2)

262 B An average smile reveals 75 to 100 per cent of the upper incisor crown plus up to 2 mm of gingiva. (Nanda; Hbk 9.2 and 18.4)

263 D Approximately 85 per cent have a smile curve parallel to the lower lip, and approximately 90 per cent show a premolar as the terminal tooth in the smile. (Nanda; Hbk 18.4)

264 A An acute nasolabial angle means retraction is likely to help. An obtuse angle means retraction will harm the profile. (Mosby ch. on space analysis; Hbk 9.2 and 21.2)

265 C Palpation of the buccal sulcus from 9 to 10 years is the single most important screening manoeuvre for the palatally displaced canine. (Ericson and Kurol; Hbk 9.3 and 20.4)

266 D Maximum opening of at least 40 mm is regarded as normal. Deviation or deflection on opening should be recorded. (Companion Ch 2; Hbk 26.3)

267 B Study casts or a scan permit measurement, space analysis and a medico-legal record of the starting point. (Companion Ch 7; Hbk 9.4)

268 E A lateral cephalogram is not justified routinely, and specifically not for minor tooth movement, nor on the basis of growth prediction alone. (Companion Ch 7; Hbk 9.4 and 10.4)

269 A The SLOB rule states that the object which moves in the Same direction as the tube shift is Lingual, and the one moving in the Opposite direction is Buccal. (Companion Ch 7; Hbk 9.4)

270 C MRI is required for the disc and soft tissues. CBCT images bone but cannot image the disc or muscle. (Companion Ch 2; Hbk 10.5 and 26.3)

271 B Justification, optimisation and limitation govern all exposures. There is no threshold dose, so every exposure must be individually justified. (Companion Ch 7; Hbk 9.4 and 30.2)

272 D A pseudo-Class III manipulates back to an edge-to-edge relationship in centric relation. The prognosis is completely different from a true skeletal Class III. (Mosby ch. on Class III; Hbk 23.1)

273 A The rule of thirds uses trichion, glabella, subnasale and menton. The lower third divides into one third upper lip and two thirds lower lip and chin. (Companion Ch 7; Hbk 9.2)

274 C The patient's own concern is recorded first because it becomes the outcome measure. It also ranks highly in the problem list. (Proffit Ch 6; Hbk 9.1 and 12.4)

275 E A heavily compensated case has less room left for camouflage. Lower incisor inclination is the key measure. (Companion Ch 13; Hbk 23.1)

276 A Four extra-oral views are standard: frontal at rest, frontal smiling, profile and three-quarter, together with five intra-oral views. (Companion Ch 7; Hbk 9.4 and Part 4)

277 C Any traumatised or discoloured tooth requires vitality testing and a periapical radiograph before force is applied, with the baseline recorded. (Companion Ch 8; Hbk 9.4 and 20.7)

278 E Incisor show at rest is approximately 2 to 4 mm in a young adult, decreasing with age as the upper lip lengthens. (Nanda; Hbk 9.2)

279 B A fixed order is required for reproducibility: complaint and history, then extra-oral, then intra-oral, then a provisional problem list, then a decision about records. (Companion Ch 7; Hbk 9.1)

280 D A complete description records the degree, whether it is complete to tooth or tissue or incomplete, and whether there is trauma. (Companion Ch 7; Hbk 9.3)

281 C Justification requires a specific question the scan will answer and evidence that the answer will change management, with conventional imaging having failed to answer it. (Companion Ch 7; Hbk 10.5)

282 A Meyer, Christou and Cheng found that extraction does not produce larger buccal corridors or a less attractive smile. (Meyer 2014; Hbk 9.2 and 13.6)

283 E A lower lip trap resting behind the upper incisors proclines them and retroclines the lowers, perpetuating the malocclusion. (Companion Ch 12; Hbk 9.2 and 22.1)

284 D Probing every adult and recording pre-existing recession is essential, both clinically and medico-legally. (Nanda; Hbk 27.1)

285 B The path of closure changes urgency, prognosis and sometimes the diagnosis itself, as in pseudo-Class III and functional crossbite. (Companion Ch 7; Hbk 9.3)

Scored under 70% on this block? Re-read Revision Handbook §9, then sit it again in a fortnight.

Block J — Imaging and cephalometrics (Q286–325 · Handbook §10)

286 E The cephalostat holds the mid-sagittal plane at a fixed distance from source and film, keeping magnification constant and making films comparable. (Companion Ch 7; Hbk 10.1)

287 B The teeth are placed in retruded contact position with the lips at rest, in natural head position or with the Frankfort plane horizontal. (Companion Ch 7; Hbk 10.1)

288 D A point is the deepest concavity on the anterior maxilla. It represents the anterior limit of the apical base, but no specific anatomical structure exists there. (Companion Ch 7; Hbk 10.2)

289 A SNA is 81 degrees plus or minus 3, SNB is 78 degrees plus or minus 3, and ANB is 3 degrees plus or minus 2. (Companion Ch 7; Hbk 10.2)

290 C ANB of 2 to 4 degrees is Class I, greater than 4 degrees is Class II and less than 2 degrees is Class III, subject to the Eastman correction. (Companion Ch 7; Hbk 10.2)

291 D The Wits appraisal drops perpendiculars from A and B points to the functional occlusal plane, sidestepping cranial base error altogether. (Companion Ch 7; Hbk 10.2)

292 B The Wits appraisal is approximately minus 1 mm in males and 0 mm in females, both plus or minus 2 mm. (Companion Ch 7; Hbk 10.2)

293 E The Eastman correction adjusts ANB by 0.5 degrees for every degree that the sella-nasion to maxillary plane angle differs from 8 degrees, and for SNA differing from 81 degrees. (Cobourne cases; Hbk 10.2)

294 A Upper incisor to maxillary plane is 109 degrees plus or minus 6; lower incisor to mandibular plane is 93 degrees plus or minus 6. (Companion Ch 7; Hbk 10.2)

295 C The interincisal angle is 135 degrees plus or minus 10, and its correction is essential to overbite stability. (Companion Ch 7; Hbk 10.2)

296 B Houston's aphorism is the one to quote when discussing the limitations of cephalometric analysis. (Houston 1983; Hbk 10.3)

297 D Each landmark has its own non-circular envelope of error, and the error is cumulative because the same landmarks are reused across an analysis. (Baumrind and Frantz 1971; Hbk 10.3)

298 A A and B points represent the anterior limit of the apical bases but correspond to no specific anatomical structure, and they remodel as the incisors move. (Companion Ch 7; Hbk 10.3)

299 C Frankfort horizontal is sensitive to head positioning and to landmark error in porion and orbitale, both of which are bilateral and hard to locate. (Companion Ch 7; Hbk 10.3)

300 E Every machine has its own enlargement factor, which must be known before linear measurements are compared between machines. (Companion Ch 7; Hbk 10.3)

301 A Graf and colleagues concluded that no method has been proved completely accurate, and Baumrind and Ghafari found that all carry appreciable error with none clearly superior. (Graf 2022; Hbk 10.3)

302 C Bjork's stable mandibular structures are the internal cortical outline of the symphysis, the mandibular canal and the tooth germ crypts. (Companion Ch 7; Hbk 10.3)

303 E Cranial base first, showing total change of both jaws, then maxilla, then mandible. Always superimpose on structures, never on outlines. (Proffit Ch 6; Hbk 10.3)

304 B Ricketts' visualised treatment objective and Bjork's implant studies underpinned growth prediction, which has since been shown to be at best unreliable. (Baumrind 1984; Hbk 10.4)

305 D Serial films about a year apart give a more accurate assessment of the direction of change, and are particularly valuable in Class III where decisions are deliberately deferred. (Companion Ch 7; Hbk 10.4)

306 C CBCT images bone well but cannot image the disc, the retrodiscal tissues or muscle. That requires MRI. (Mosby ch. on imaging; Hbk 10.5 and 26.3)

307 A Reported rates of incidental findings on CBCT range from about a quarter to nearly all scans, depending on the definition and field of view. (Mosby ch. on imaging; Hbk 10.5)

308 E Impacted teeth where conventional imaging is inadequate, suspected resorption of adjacent incisor roots, clefts, complex asymmetry and orthognathic planning are the accepted indications. (Companion Ch 7; Hbk 10.5)

309 D Three-dimensional cephalometry removes projection error in principle, but landmark definitions differ from two-dimensional ones and there is no agreed normative database. (Mosby ch. on imaging; Hbk 10.5)

310 B Stereophotogrammetry uses no ionising radiation and is useful for soft tissue outcome assessment and serial monitoring of asymmetry. (Haner 2021; Hbk 10.5)

311 E Haner and colleagues reported good reproducibility using the forehead, mid-nose and infraorbital regions. (Haner 2021; Hbk 10.5 and 24.4)

312 B For a posteroanterior view the mid-sagittal plane is parallel to the beam and perpendicular to the film; the lateral view is the reverse. (Companion Ch 7; Hbk 10.1)

313 D Identifying a reliable midsagittal reference is the hard part of posteroanterior analysis of asymmetry. (Mosby ch. on asymmetry; Hbk 24.4)

314 A The Holdaway angle, describing the soft tissue profile, is normally 7 to 15 degrees. A value of 3.5 degrees or less favours surgery in Class III. (Companion Ch 13; Hbk 10.2 and 23.4)

315 C Lower incisor to A-pogonion of 0 mm plus or minus 2 is the classical stability reference. (Companion Ch 7; Hbk 10.2)

316 D Justification requires a specific question and a change in management. Curiosity is not a justification, and there is no threshold dose. (Companion Ch 7; Hbk 9.4 and 30.2)

317 B The panoramic radiograph is the default survey film in orthodontics. It is not suitable for fine detail, for which periapicals are used. (Companion Ch 7; Hbk 9.4)

318 E A Wits appraisal of minus 11 mm or worse is among the most important discriminators, with ANB of minus 4 degrees or worse and a reverse overjet of minus 5 mm or worse. (Kerr 1992; Hbk 23.4)

319 A Proffit's thresholds are overjet over 10 mm, pogonion to nasion perpendicular of 18 mm or more, mandibular body under 70 mm, or anterior face height over 125 mm. (Proffit 1992; Hbk 12.2)

320 C Cephalometric analysis supplements clinical examination. This is the correct closing stance in any answer on the subject. (Companion Ch 7; Hbk 10.3)

321 B Frankfort horizontal depends on porion and orbitale, so an error in porion directly affects the Frankfort-mandibular plane angle. (Companion Ch 7; Hbk 10.3)

322 D The Wits appraisal is measured to the functional occlusal plane and does not use the cranial base, which is why it is used when cranial base morphology is atypical. (Companion Ch 7; Hbk 10.2)

323 A Software reduces but does not eliminate error. Its main advantage is versatility: multiple analyses, superimposition and orthognathic prediction planning. (Companion Ch 7; Hbk 10.3)

324 C Error reduction requires good films, correct positioning, optimal viewing conditions, repeated measurement, examiner calibration and knowledge of magnification. (Companion Ch 7; Hbk 10.3)

325 E Modern solid-state cephalostats scan horizontally over an approximately 10-second exposure. (Companion Ch 7; Hbk 10.1)

Scored under 70% on this block? Re-read Revision Handbook §10, then sit it again in a fortnight.

Block K — Study models, space analysis and the digital set-up (Q326–350 · Handbook §11)

326 A The six areas are crowding or spacing, levelling the curve of Spee, arch width change, incisor anteroposterior change, angulation and inclination. (Companion Ch 8; Hbk 11.2)

327 C As a working rule, levelling consumes approximately 1 mm of arch length per 1 mm of curve depth. (Companion Ch 8; Hbk 11.2)

328 E Expansion creates space equal to only about half the amount of expansion, which is why expansion is a poor substitute for extraction in significant crowding. (O'Higgins and Lee 2000; Hbk 11.2)

329 B Each millimetre of incisor retraction requires 2 mm of space, since the movement occurs on both sides of the arch. (Companion Ch 8; Hbk 11.2)

330 D Approximately 1 mm of space is consumed per 10 degrees of torque, which is why torque requirements must be included in the space analysis. (Cobourne cases; Hbk 11.2)

331 C Up to approximately 1 mm per contact is acceptable in the buccal segments and approximately 0.5 mm anteriorly. (Companion Ch 8; Hbk 11.2)

332 A An equal requirement finishes Class I molars; a greater maxillary requirement finishes Class II molars. Getting this backwards makes the extraction pattern wrong before treatment starts. (Companion Ch 8; Hbk 11.2)

333 E Kirschen, O'Higgins and Lee described the Royal London Hospital space analysis, which assigns a millimetre value to every item. (Kirschen 2000; Hbk 11.2)

334 D Formal analysis produces a more disciplined and less varied approach, particularly valuable for trainees. (Al-Abdallah 2008; Hbk 11.2)

335 B Tanaka-Johnston adds 10.5 mm per quadrant for the mandibular arch and 11.0 mm for the maxillary arch. (Proffit Ch 6; Hbk 11.3)

336 E Moyers tables predict the combined mesiodistal width of the unerupted canine and premolars from the measured mandibular incisors, usually at the 75 per cent probability level. (Proffit Ch 6; Hbk 11.3)

337 B The leeway space must be added to the space available, since it becomes available as the primary molars and canines are shed. (Companion Ch 5; Hbk 11.3)

338 D A digital set-up shows what the teeth would look like if they went where you asked, not whether they will. It contains no biology and no anchorage. (Mosby ch. on digital planning; Hbk 11.4)

339 A A set-up cannot tell you that the lower incisors will be outside the alveolar envelope. The space analysis and the biological limit decide, not the software. (Mosby ch. on digital planning; Hbk 11.4)

340 C Errors enter at the scan, and in surgical planning at the bite registration. The output is only as good as the input. (Mosby ch. on digital planning; Hbk 11.4)

341 D Bolton's anterior ratio of 77.2 per cent and overall ratio of 91.3 per cent quantify tooth-size discrepancy. (Companion Ch 7; Hbk 11.1)

342 B A ratio above the norm of 91.3 per cent indicates relative mandibular excess; below it indicates relative maxillary excess. (Companion Ch 7; Hbk 11.1)

343 E Alginate is dimensionally unstable and should be cast within about 10 minutes or stored in 100 per cent humidity. (Companion Ch 10; Hbk 15.4)

344 A Addition silicone is dimensionally stable and accurate, and is used where accuracy matters, such as for bonded retainers and appliances. (Companion Ch 10; Hbk 15.4)

345 C The curve of Spee is measured from the deepest cusp tip to a plane joining the incisal edges and the distal cusp of the most posterior tooth. (Companion Ch 7; Hbk 11.1)

346 B Space opening for a prosthesis adds to the requirement, not the supply. The seven sources are tooth reduction or enlargement, extraction, prosthetic space opening as a requirement, molar distal movement, molar mesial movement, differential growth and residue. (Companion Ch 8; Hbk 11.2)

347 D Correction of angulation, or tip, consumes space, and is one of the six areas of space requirement. (Companion Ch 8; Hbk 11.2)

348 A Scanning is better tolerated, immediately transmissible and permits digital set-up and storage. Deep undercuts and subgingival margins remain harder than with an impression. (Mosby ch. on digital planning; Hbk 15.4)

349 C A set-up is useful for demonstrating geometry and the consequences of a choice, not for guaranteeing outcomes. (Mosby ch. on digital planning; Hbk 11.4)

350 E Mild crowding is 1 to 4 mm, moderate 5 to 8 mm and severe 9 mm or more. (Companion Ch 8; Hbk 13.1)

Scored under 70% on this block? Re-read Revision Handbook §11, then sit it again in a fortnight.

Block L — Diagnosis, the problem list and treatment planning (Q351–390 · Handbook §12)

351 A The nine-step sequence begins with diagnosis and ends with retention, and each stage constrains the next. (Companion Ch 8; Hbk 12.1)

352 C This is a deliberate shift from the classical approach of planning around mandibular incisor position and isolated static occlusal goals. (Companion Ch 8; Hbk 12.1)

353 E Ackerman and Proffit, and Holdaway, established that the soft tissues define the limits of orthodontic treatment. (Ackerman and Proffit 1997; Hbk 12.1)

354 B Proffit's discipline is to rank the problem list in order of importance to the patient. The chief complaint always ranks high. (Proffit Ch 6; Hbk 12.4)

355 D Rubber-stamp objectives such as achieve a good functional occlusion are not objectives, because no colleague could measure you against them. (Mosby ch. on treatment planning; Hbk 12.4)

356 C Growth modification relies on compliance and favourable growth and is much more effective for Class II than for Class III. (Companion Ch 8; Hbk 12.2)

357 A Growth status is the first discriminator. Its absence converts growth modification into camouflage or a surgical problem. (Companion Ch 8; Hbk 12.2)

358 E Konstantonis reported 0.7 mm of lip retraction per 1 mm of incisor retraction, with a 1.6 degree increase in nasolabial angle. (Konstantonis 2018; Hbk 12.2)

359 D The nasolabial angle increases by approximately 1.6 degrees per millimetre of incisor retraction. (Konstantonis 2018; Hbk 12.2)

360 B The envelope of discrepancy is asymmetric, extended modestly by growth modification in growers, and its soft tissue limits are frequently tighter than its hard tissue limits. (Proffit Ch 7; Hbk 12.3)

361 E The envelope is asymmetric: more movement is possible in some directions than others, and retraction of maxillary incisors exceeds advancement. (Proffit Ch 7; Hbk 12.3)

362 B Prognosis beats symmetry. A compromised tooth should always be removed in preference to a healthy one. (Companion Ch 8; Hbk 12.4 and 13.2)

363 D Anchorage arises directly from the space analysis and the extraction pattern, and cannot be decided before them. (Companion Ch 8; Hbk 12.1)

364 A The conventional phasing is alignment and levelling, then overjet and overbite correction, then space closure, then detailing and finishing. (Companion Ch 8; Hbk 12.1 and 18)

365 C Retention should be discussed prior to the commencement of active treatment, and the patient must agree to it. (Companion Ch 13; Hbk 12.1 and 19.3)

366 D Camouflage accepts the skeletal pattern and positions the teeth to disguise it. Its limits are the alveolar envelope, the soft tissue response and the existing dental compensation. (Companion Ch 8; Hbk 12.2)

367 B The four thresholds are overjet over 10 mm, pogonion to nasion perpendicular of 18 mm or more, mandibular body under 70 mm and anterior face height over 125 mm. (Proffit 1992; Hbk 12.2)

368 E Facial aims are routinely forgotten. A Class I incisor and canine relationship must not be achieved at the expense of the soft tissue profile. (Companion Ch 8; Hbk 12.1)

369 A Compliance is a planning factor, not an afterthought, and it influences the choice of appliance from the outset. (Companion Ch 8; Hbk 12.1)

370 C All reasonable alternatives must be presented, including no treatment. In a diagnostic case, offering three options and saying why two were rejected scores higher than asserting a single plan. (GDC Standards; Hbk 12.4 and Part 3)

371 B The one-sentence summary contains everything the examiner needs to place the case, and taking control of the frame makes everything afterwards easier. (Hbk Part 3)

372 D Existing compensation consumes the movement available for camouflage. It is often the deciding factor between camouflage and surgery. (Companion Ch 13; Hbk 12.2 and 23.4)

373 A Growth status changes the availability of growth modification and reverses the consequences of buccal segment extrusion for overbite reduction. (Companion Ch 13; Hbk 12.2 and 25.1)

374 C A lower incisor inclination of 80 degrees or less indicates that decompensation has already occurred and camouflage has little room left. (Kerr 1992; Hbk 23.4)

375 E Discriminant analysis on these cephalometric variables predicts the need for surgery with over 90 per cent accuracy. (Kochel 2011; Hbk 23.4)

376 A Anchorage requirement follows from the plan, not the other way round. This ordering is a frequent examination point. (Cobourne cases; Hbk 11.2 and 12.1)

377 C In an unfavourable Class III, monitoring is often correct because Class III worsens with growth, and mandibular extractions consume the decompensation space the surgeon will need. (Companion Ch 8; Hbk 23.2)

378 E The extraction pattern follows from the differential space requirement between the arches and the intended finishing molar relationship. (Companion Ch 8; Hbk 11.2 and 13.3)

379 B Adjunctive treatment serves a larger goal, such as uprighting a molar before a bridge, and is often appropriate in general practice. (Proffit Ch 18; Hbk 27.1)

380 D Reconciling facial and occlusal aims against the skeletal pattern is the third stage of the planning sequence and the one most often omitted. (Companion Ch 8; Hbk 12.1)

381 C A compromised tooth should always be removed in preference to a healthy one. This is one of the four factors governing extraction choice. (Companion Ch 8; Hbk 13.2)

382 A Proffit's problem-oriented approach runs from database through a ranked problem list and possible solutions to a single reconciled plan. (Proffit Ch 6; Hbk 12.4)

383 E The camouflage-or-surgery decision must be made before treatment begins, because pre-surgical orthodontics decompensates in the opposite direction from camouflage. (Proffit Ch 19; Hbk 12.2 and 27.2)

384 D The greater maxillary requirement finishes to a Class II molar relationship, which is achieved by extracting further back in the mandible. (Companion Ch 8; Hbk 13.3)

385 B The mandibular arch is the template. Lower intercanine width is among the least amenable dimensions to change, and expansion relapses. (Companion Ch 8; Hbk 12.1 and 19.1)

386 E Camouflage accepts the skeletal discrepancy and positions the teeth to disguise it, within the limits of the alveolar envelope and the soft tissues. (Companion Ch 8; Hbk 12.2)

387 B Saying what would change your mind and where you would refer demonstrates judgement and knowledge of your own limits. (Hbk Part 3 and Part 5)

388 D The sequence ends appliance, phasing, retention, having begun with diagnosis, problem list and aims. (Companion Ch 8; Hbk 12.1)

389 A The commonest failure is an unranked list of ten findings with no argument. Rank them, and say which two you would treat if you could treat only two. (Hbk Part 3)

390 C A greater maxillary space requirement means the maxillary buccal segments will move further forward relative to the mandibular ones, finishing to Class II molars. (Companion Ch 8; Hbk 11.2)

Scored under 70% on this block? Re-read Revision Handbook §12, then sit it again in a fortnight.

Block M — Extraction decisions and space management (Q391–430 · Handbook §13)

391 D Extraction yields the full mesiodistal width of the tooth and is the most predictable source of space, at the cost of an anchorage demand. (Companion Ch 8; Hbk 13.1)

392 B Mild crowding is 1 to 4 mm, moderate 5 to 8 mm and severe 9 mm or more. (Companion Ch 8; Hbk 13.1)

393 E Proffit gives fewer than 4 mm as rarely requiring extraction, 5 to 9 mm as a judgement, and 10 mm or more as almost always requiring extraction. (Proffit Ch 7; Hbk 13.6)

394 A Class II requires extraction further back in the mandible, typically maxillary first premolars with mandibular second premolars, so that the greater maxillary space allows retraction of the upper labial segment. (Companion Ch 8; Hbk 13.3)

395 C Class III requires extraction further forward in the mandible, typically mandibular first premolars with maxillary second premolars. (Companion Ch 8; Hbk 13.3)

396 B Space is most useful where the movement is needed, which is the rationale behind the differential extraction patterns by class. (Companion Ch 8; Hbk 13.3)

397 D First premolar extraction relieves anterior crowding most efficiently and gives the best spontaneous improvement, especially in the mixed dentition. (Companion Ch 8; Hbk 13.4)

398 A Second premolar extraction suits posteriorly sited crowding and cases where the anchorage balance should favour the anterior segment. It also eases Class II mechanics in the lower arch. (Companion Ch 8; Hbk 13.4)

399 C Second molar extraction requires a third molar present, of good size, angulated at 30 degrees or less to the second molar long axis, and at an appropriate developmental stage. (Companion Ch 8; Hbk 13.4)

400 E Lower incisor extraction is rarely indicated, and its costs are tooth-size discrepancy, residual spacing and a deepened overbite. (Companion Ch 8; Hbk 13.4)

401 A NICE advises against prophylactic removal of developing third molars to prevent late lower incisor crowding, and third molars are not a primary cause. (NICE; Richardson 2002; Hbk 13.4 and 19.4)

402 C Balancing extraction to prevent a centre line shift is most relevant for canines, and to a lesser extent incisors. It is rarely needed for molars. (Companion Ch 12; Hbk 13.5)

403 E Compensating extraction removes the opposing tooth in the same quadrant, and is considered principally for first permanent molars. (Companion Ch 12; Hbk 13.5)

404 B Rushing, Paquette and Iared found minimal soft tissue differences, with both groups perceived to benefit. (Rushing 1995; Iared 2017; Hbk 13.6)

405 D These studies found that extraction does not produce larger buccal corridors or a less attractive smile. (Meyer 2014; Christou 2019; Hbk 13.6)

406 C Proffit concluded that arch width indices such as Pont's are not acceptable predictors of arch width. (Proffit Ch 7; Hbk 13.6)

407 A The conventional rule is: less than half a unit distalise, more than half a unit extract, more than one unit both. (Companion Ch 8; Hbk 13.1)

408 E A first permanent molar is extracted only when its prognosis is poor. Prognosis beats symmetry and beats convenience of position. (Companion Ch 12; Hbk 13.4)

409 D These four factors are the standard structure for an extraction question, and prognosis is the first of them. (Cobourne cases; Hbk 13.2)

410 B Extracting in one arch only means accepting a non-Class I molar relationship, which must be a deliberate decision made in advance. (Cobourne cases; Hbk 13.3)

411 E Well-timed extraction in the mixed dentition permits spontaneous improvement. In the permanent dentition, appliances are required. (Companion Ch 8; Hbk 13.2)

412 B Extraction is not justified by a wish to shorten treatment. The four factors are prognosis, site and degree of crowding, malocclusion type and age. (Companion Ch 8; Hbk 13.2)

413 D Interproximal reduction offers a modest total, in the order of 0.5 mm per anterior contact and up to 1 mm per buccal contact. Enamel is finite. (Companion Ch 8; Hbk 13.1)

414 A Serial extraction requires severe crowding with a Class I pattern and no vertical problem. It is contraindicated in Class II or III, deep or open bite, hypodontia and mild crowding. (Proffit Ch 12; Hbk 20.3)

415 C The sequence is primary canines at about 8 to 9 years, first primary molars at about 9 to 10 years, then the first premolars as they erupt. (Proffit Ch 12; Hbk 20.3)

416 D The disadvantages are prolonged spacing, a deepening overbite, lower incisor retroclination, the usual need for a fixed appliance anyway, and irreversible commitment at a young age. (Proffit Ch 12; Hbk 20.3)

417 B The AAO-funded studies found no causal association, despite the 1987 Brimm versus Malloy award of 850000 dollars following premolar extraction treatment. (Behrents and White 1992; Hbk 13.6 and 26.1)

418 E Larsen and Orabi found no evidence for the claim, which has been described in the literature as scaremongering and marketing. (Larsen 2015; Orabi 2021; Hbk 13.6)

419 A Timing is critical in the mandible because the mandibular second molar does not drift mesially as readily as the maxillary one. (Companion Ch 12; Hbk 3.4 and 13.4)

420 C Prognosis beats convenience. The compromised molar should be removed in preference to a healthy premolar, with timing chosen against second molar development. (Companion Ch 12; Hbk 13.2 and 13.4)

421 B The presence and position of all permanent teeth, including third molars, must be confirmed before committing to first molar extraction. (Companion Ch 12; Hbk 3.4)

422 D Each millimetre of incisor advance yields 2 mm of arch space, but stability is lost beyond about 2 mm of advance. (Bennett and McLaughlin; Hbk 11.2 and 19.1)

423 A In hypodontia of the lateral incisors the existing molar relationship decides: Class II favours closure and canine substitution; Class I with ideal intercuspation favours opening. (Mosby ch. on hypodontia; Hbk 20.6 and Part 3 Drill 5)

424 **C** Corresponding premolar extraction in both arches maintains the Class I buccal segment, and first premolars give maximum anterior space. An acute nasolabial angle means retraction will help. (Companion Ch 8; Hbk 13.3 and 21.2)

425 **E** An obtuse nasolabial angle and a flat profile mean retraction will harm the face. Mild crowding is better managed without extraction. (Mosby ch. on space analysis; Hbk 13.2 and 21.2)

426 **A** Haque and colleagues found that anchorage loss is the same for first and second premolar extraction, and that two-thirds of the space is lost without support. (Haque 2017; Hbk 14.4)

427 **C** More than one unit of distal movement requires both extraction and distalisation. Less than half a unit is distalised; more than half a unit is extracted. (Companion Ch 8; Hbk 13.1)

428 **E** Equal space requirement in both arches maintains the Class I buccal segment relationship, so corresponding premolars are extracted. (Companion Ch 8; Hbk 13.3)

429 **B** In the permanent dentition, and particularly in adults, spontaneous improvement does not occur and appliance treatment with planned anchorage is obligatory. (Companion Ch 8; Hbk 13.2 and 27.1)

430 **D** Around two-thirds of the space created is lost unless anchorage is supported, which is the sharpest single fact on this topic. (Haque 2017; Hbk 14.4)

Scored under 70% on this block? Re-read Revision Handbook §13, then sit it again in a fortnight.

Block N — Biomechanics, force control and anchorage (Q431–481 · Handbook §14)

431 **C** For a single-rooted tooth the centre of resistance lies one third to one half down the root; for a multirooted tooth it lies 1 to 2 mm apical to the furcation. (Companion Ch 6; Hbk 14.1)

432 **A** The centre of resistance is fixed by the anatomy; the centre of rotation moves according to the moment-to-force ratio applied. (Companion Ch 6; Hbk 14.1)

433 **E** A couple produces pure rotation about the centre of resistance wherever it is applied, and has no net translational force. (Companion Ch 6; Hbk 14.1)

434 **D** A ratio of approximately 10 to 1 places the centre of rotation at infinity, giving translation. Zero gives uncontrolled tipping; 7 to 1 gives controlled tipping; 12 to 1 and above gives root torque. (Companion Ch 6; Hbk 14.2)

435 **B** Controlled tipping places the centre of rotation near the apex, so the crown moves while the root is held. (Companion Ch 6; Hbk 14.2)

436 **E** A moment-to-force ratio of zero gives uncontrolled tipping about a centre of rotation near the centre of resistance. This is what a removable appliance does. (Companion Ch 6; Hbk 14.2)

437 **B** Bone loss moves the centre of resistance apically, increasing the moment arm from the bracket. Lighter forces and higher moment-to-force ratios are required. (Companion Ch 6; Hbk 14.2)

438 **D** A cantilever or utility arch engaged in one bracket and tied as a point contact elsewhere is statically determinate. A continuous archwire engaged in several brackets is indeterminate. (Nanda; Hbk 14.3)

439 **A** Two-couple systems cannot be calculated, only estimated, which is why clinical experience matters so much in continuous archwire mechanics. (Nanda; Hbk 14.3)

440 **C** Every applied force produces an equal and opposite reaction. Anchorage cannot be avoided, only planned. (Companion Ch 6; Hbk 14.4)

441 D Maximum anchorage means the space is used entirely for anterior retraction. Minimum anchorage allows the posterior unit to use up to two-thirds of the space. (Companion Ch 6; Hbk 14.4)

442 B Reciprocal anchorage is used deliberately in midline diastema closure, expansion and crossbite correction with a fixed appliance. (Companion Ch 6; Hbk 14.4)

443 E Jepsen's measurements of root surface area are the basis of relative anchorage values. Root area matters more than tooth number. (Jepsen 1963; Hbk 14.4)

444 A In stationary anchorage the anchor unit resists bodily movement while the active unit tips, which is a more efficient use of the same anchorage. (Companion Ch 6; Hbk 14.4)

445 C Haque and colleagues found that around two-thirds of the space created is lost without anchorage support. (Haque 2017; Hbk 14.4)

446 B A transpalatal arch provides very little additional anchorage in the sagittal plane, although it is useful for molar derotation and transverse control. (Diar-Bakirly 2017; Hbk 14.4)

447 D A lingual arch gives little anchorage support and can procline the lower incisors, although it is effective for preserving leeway space. (Rebellato 1997; Hbk 14.4)

448 A In maximum anchorage cases, anchor teeth should be excluded from the initial aligning wire so that anchorage is not spent during levelling. (Melsen and Verna 1999; Hbk 14.4)

449 C Reciprocal closure without reinforcement is approximately 1 to 1, determined by periodontal ligament area rather than tooth number. A stabilising arch shifts it to about 2 to 1; headgear with elastics reaches 3 or 4 to 1. (Proffit Ch 15; Hbk 14.4)

450 E High-pull or occipital headgear is intrusive on the molar and is indicated in high-angle patients. Cervical pull is extrusive and suits reduced vertical proportions. (Companion Ch 6; Hbk 14.5)

451 A Anchorage support requires 250 to 350 g per side for 10 to 12 hours per day. Distalisation or an orthopaedic effect requires 450 to 500 g per side for 14 hours per day. (Companion Ch 6; Hbk 14.5)

452 C Two independent safety mechanisms are mandatory: a self-releasing snap-away neck strap and a locking facebow. (Companion Ch 6; Hbk 14.5)

453 E The catapult injury occurs when the bow disengages under tension and recoils into the face. Nocturnal disengagement occurs during sleep. (Samuels 1996; Hbk 14.5)

454 B The patient should stop wearing it and attend. Consequences of headgear injury include penetrating eye injury with loss of vision and intra-orbital infection. (Samuels 1996; Hbk 14.5)

455 D A TAD does not change the force system. It changes the biological response, by placing the reaction where nothing can move. (Mosby ch. on TADs; Hbk 14.6)

456 C In direct anchorage the active force acts on or through the screw, so the line of force must pass through it. (Mosby ch. on TADs; Hbk 14.6)

457 A Indirect anchorage requires a rigid connection, so a large rectangular stainless steel wire is used, stiff enough not to flex. (Mosby ch. on TADs; Hbk 14.6)

458 E Below 5 Ncm gives poor primary stability; above 10 Ncm risks microfracture and poor secondary stability. (Companion Ch 11; Hbk 14.6)

459 D Failure rates increase below 8 mm in length and 1.2 mm in diameter. (Companion Ch 11; Hbk 14.6)

460 B Root contact is the commonest cause of failure, followed by thin cortical bone, mobile mucosa and inflammation. (Companion Ch 11; Hbk 14.6)

461 E The median palate has a failure rate of approximately 1.3 per cent, because placement there is not restricted by the dentition. (Companion Ch 11; Hbk 14.6)

462 B The mandibular buccal shelf has a reported failure rate of approximately 65 per cent, against approximately 1.3 per cent in the median palate. (Companion Ch 11; Hbk 14.6)

463 D The pooled failure rate is approximately 12.3 per cent, corresponding to an overall survival of 80 to 90 per cent. (Companion Ch 11; Hbk 14.6)

464 A Nanda gives a minimum of 4 mm of bone thickness for a palatal implant, noting that the hard palate generally does not exceed 3 mm outside the paramedian region. (Nanda; Hbk 14.6)

465 C Nanda recommends loading a miniplate at approximately three weeks, not immediately, and notes a requirement for 2 to 3 mm of cortical bone depth. (Nanda; Hbk 14.6)

466 D A mobile screw will not become stable and perpetuates inflammation. Remove it, check for root contact, and re-site or change the plan. (Companion Ch 11; Hbk 14.6)

467 B Success is governed principally by cortical bone thickness, together with the avoidance of root contact and of mobile mucosa. (Proffit Ch 10; Hbk 14.6)

468 E Space closure is performed at approximately 150 g. Attempts to transfer traditional edgewise forces of 500 to 600 g produced the roller coaster effect, and 200 g is no better than 150 g. (McLaughlin, Bennett and Trevisi; Hbk 14.4 and 18.2)

469 A Second molars should be bonded as early as possible: they are needed in the system for bite opening and they add to the anchorage unit. (Bennett and McLaughlin; Hbk 18.1)

470 C A laceback is a passive wire ligature that distalises the canine without tipping it, and prevents incisor advance during levelling. (Bennett and McLaughlin; Hbk 18.1)

471 B Without lacebacks the lower incisor advances by approximately 1.4 mm; with them it retracts by approximately 1.0 mm. (McLaughlin, Bennett and Trevisi; Hbk 14.4 and 18.1)

472 D A bendback is placed in the wire distal to the molar tube to minimise labial tipping of the incisors during alignment. (Bennett and McLaughlin; Hbk 18.1)

473 A Class II elastics correct the molar relationship at the cost of lower incisor proclination, upper incisor extrusion and an increased occlusal plane cant. They are barred if the lower incisors are already forward. (Bennett and McLaughlin; Hbk 18.2)

474 C Class III elastics extrude the lower molars and rotate the mandible down and back, which is dangerous in a high-angle patient. (Bennett and McLaughlin; Hbk 18.2)

475 E Giannopoulou and colleagues found the maxillary arch particularly susceptible to anchorage loss. (Giannopoulou 2016; Hbk 14.4)

476 A Equilibrium requires that the sum of forces and the sum of moments both equal zero. Every applied force produces a reaction somewhere. (Nanda; Hbk 14.3)

477 C Minimum anchorage, Group C, allows the posterior segment to use up to two-thirds of the space. (Companion Ch 6; Hbk 14.4)

478 E Roots driven against cortical bone resist movement more, which increases anchorage but also increases the risk of root resorption. (Companion Ch 6; Hbk 14.4)

479 B The Nance button provides palatal anchorage, with a recognised risk of ulceration and tissue overgrowth beneath the acrylic. (Companion Ch 6; Hbk 15.5)

480 D Intrusion requires 10 to 20 g per tooth because force is concentrated at the small apical surface area. It is the lightest movement in orthodontics. (Proffit Ch 14; Hbk 14.4 and 25.1)

481 C Recognising which sites are reliable and which are not, and referring the whole case rather than the screw when the case depends on a difficult site, demonstrates the judgement examiners look for. (Companion Ch 11; Hbk 14.6)

Scored under 70% on this block? Re-read Revision Handbook §14, then sit it again in a fortnight.

Block O — Orthodontic materials, wires, bonding and instruments (Q482–525 · Handbook §15)

482 A Stiffness is Young's modulus, the gradient of the elastic portion. Low stiffness is desirable in an aligning wire. (Companion Ch 10; Hbk 15.1)

483 E Springback equals yield strength divided by modulus of elasticity, and is effectively the working range of the wire. (Companion Ch 10; Hbk 15.1)

484 D Proffit's identity is strength equals stiffness multiplied by range. You cannot raise one without paying elsewhere. (Proffit Ch 9; Hbk 15.1)

485 B The ideal wire has high strength, low stiffness, high range and high formability, plus biocompatibility, weldability, low friction and low cost. No single wire meets all of these. (Proffit Ch 9; Hbk 15.1)

486 E The 18-8 austenitic composition is iron approximately 70 per cent, chromium 17 to 20 per cent, nickel 8 to 12 per cent and carbon up to 0.08 per cent. (Companion Ch 10; Hbk 15.2)

487 B Nickel titanium has a modulus approximately 0.26 that of stainless steel, giving it much lower stiffness and a much larger working range. (Mosby ch. on materials; Hbk 15.2)

488 D Beta titanium, or TMA, has a stiffness of approximately one third that of stainless steel, with good formability and weldability but higher friction. (Companion Ch 10; Hbk 15.2)

489 A Nickel titanium cannot be permanently bent, welded or soldered, which limits it to alignment. (Companion Ch 10; Hbk 15.2)

490 C Superelasticity is a real, measurable phenomenon arising from a stress-induced austenite to martensite transformation, producing a plateau in the unloading curve. (Proffit Ch 9; Hbk 15.2)

491 D Wang and colleagues found greater movement with coaxial superelastic nickel titanium over 12 weeks; otherwise there is insufficient evidence that any archwire is superior. (Wang 2018; Hbk 15.2)

492 B Austenitic-active wires were slightly better than martensitic-stabilised ones, and copper nickel titanium sequences were slightly slower to a working wire and more painful. (Papageorgiou 2014; Hbk 15.2)

493 E Enamel is etched with 37 per cent phosphoric acid for 15 to 30 seconds, then rinsed and dried to a frosted appearance. (Mosby ch. on bonding; Hbk 15.3)

494 A Self-etching primers combine etch and prime, produce a shallower etch pattern, and give comparable clinical bond failure rates. (Companion Ch 10; Hbk 15.3)

495 C Silane coupling gives a chemical bond. The consequence is that ceramic brackets bond too well, and the enamel may pay at debond. (Mosby ch. on bonding; Hbk 15.3)

496 B Ceramic brackets are more brittle, have higher friction, abrade opposing enamel, and carry a risk of enamel fracture at debond. (Companion Ch 10; Hbk 15.3)

497 D Glass ionomer cement releases fluoride and is preferred for band cementation, particularly in patients at risk of demineralisation. (Companion Ch 10; Hbk 15.3)

498 A White spot lesions can be visible within 6 months and affect up to half of fixed appliance patients. (Gorelick 1982; Hbk 15.3 and 29.2)

499 C The commonest sites are first molars, upper lateral incisors and lower canines, and lesions are more prevalent in males. (Gorelick 1982; Hbk 15.3)

500 E Daily high-fluoride toothpaste, at 2800 or 5000 parts per million, plus professional high-concentration fluoride foam or varnish every 6 to 8 weeks, is the recommended regimen. (Benson 2019; Hbk 15.3)

501 A Steel ligatures rather than elastomeric modules reduce plaque retention, and are described as the cheapest intervention available in a poor brusher. (Mosby ch. on complications; Hbk 15.3)

502 C A tungsten carbide bur at low speed followed by polishing minimises enamel loss and iatrogenic damage. (Companion Ch 10; Hbk 15.3)

503 E Alignment uses light, flexible, large-range wires: 0.012 to 0.018 inch round nickel titanium or multistrand. (Companion Ch 10; Hbk 15.2 and 18.1)

504 B The 0.019 by 0.025 inch stainless steel wire in a 0.022 slot gives overbite control while remaining loose enough to slide. (McLaughlin, Bennett and Trevisi; Hbk 15.2 and 18.2)

505 D The MBT literature gives 47 per cent more deflection with 0.016 by 0.022 than with 0.019 by 0.025. (McLaughlin, Bennett and Trevisi; Hbk 15.2)

506 C Preformed superelastic nickel titanium archwires may expand the lower intercanine width by up to 5.9 mm if allowed to express fully, which is why arch form must be a deliberate decision. (McLaughlin, Bennett and Trevisi; Hbk 15.2 and 18.5)

507 A Cobalt chromium is supplied in a soft, formable state and is heat-treated after bending to raise its stiffness. (Companion Ch 10; Hbk 15.2)

508 E Coated and composite aesthetic wires have poorer mechanical properties and the coating degrades, so their role remains limited. (Companion Ch 10; Hbk 15.2)

509 D Nickel allergy affects approximately 10 to 30 per cent of females and 1 to 3 per cent of males. (Chakravarthi 2012; Hbk 15.6)

510 B Stainless steel is safe because the nickel is bound within the austenitic lattice. Avoid nickel titanium and extra-oral nickel-containing components. (Companion Ch 7; Hbk 15.6)

511 E Latex allergy affects approximately 4.3 per cent of the general population and 9.7 per cent of healthcare workers. (Wu 2016; Hbk 15.6)

512 B A removable expansion plate is activated 0.25 mm twice weekly, giving slow, largely dental expansion. (Companion Ch 12; Hbk 15.5 and 24.2)

513 D Rapid maxillary expansion uses 1 to 4 turns per day over 2 to 4 weeks, generating forces up to 10 kg. (Companion Ch 9; Hbk 15.5 and 24.2)

514 A Rapid maxillary expansion generates forces of up to approximately 10 kg, which is why it opens the midpalatal suture. (Companion Ch 9; Hbk 15.5 and 24.2)

515 C Ugolini and colleagues found that a quadhelix beats an expansion plate and is approximately 3 months faster, with high-certainty evidence at ages 7 to 11. (Ugolini 2021; Hbk 15.5 and 24.2)

516 D The orthodontic diode laser operates at 810 to 980 nm, absorbed by melanin and haemoglobin, cutting soft tissue while sparing enamel and metal. (Mosby ch. on lasers; Hbk 15.5)

517 B The diode laser is a soft tissue finishing instrument. The cause of any gingival disproportion should be diagnosed before it is used. (Mosby ch. on lasers; Hbk 15.5)

518 E Alginate is dimensionally unstable and should be cast promptly or stored in 100 per cent humidity. (Companion Ch 10; Hbk 15.4)

519 A Fluoride release is the principal advantage, and is relevant because bands are placed on teeth at risk of demineralisation. (Companion Ch 10; Hbk 15.3)

520 C A distal end cutter retains the fragment. Wire ends are sharps, and every wire end must be accounted for. (Companion Ch 10; Hbk 30.1)

521 B Orthodontic pliers are heat-tolerant but corrode. They must be dried thoroughly and the hinges oiled after sterilisation. (Local decontamination guidance; Hbk 30.1)

522 D Resilience is the energy stored, represented by the area under the elastic portion of the curve. (Companion Ch 10; Hbk 15.1)

523 A Formability describes how far a wire can be bent permanently without fracturing, and is why stainless steel remains indispensable for finishing. (Companion Ch 10; Hbk 15.1)

524 C Biocompatibility, environmental stability, joinability, friction, ductility and cost are all clinically important and are not on the curve. (Companion Ch 10; Hbk 15.1)

525 E Full-arch accuracy is comparable to addition silicone in most studies. Deep undercuts, subgingival margins and edentulous spans remain more difficult. (Mosby ch. on digital planning; Hbk 15.4)

Scored under 70% on this block? Re-read Revision Handbook §15, then sit it again in a fortnight.

Block P — Removable and functional appliances (Q526–566 · Handbook §16)

526 A The ARAB mnemonic covers Active components, Retention, Anchorage and Baseplate. (Companion Ch 9; Hbk 16.1)

527 C A removable appliance achieves only two-point contact on a tooth and therefore delivers uncontrolled tipping. Bodily movement, rotation and torque are beyond it. (Companion Ch 9; Hbk 16.1)

528 E An Adams clasp on a molar uses 0.7 mm wire; on a premolar 0.6 mm is used. (Companion Ch 9; Hbk 16.1)

529 B A palatal finger spring uses 0.5 mm wire, often with a guard. A buccal canine retractor uses 0.7 mm. (Companion Ch 9; Hbk 16.1)

530 D A palatal finger spring is activated by approximately half the diameter of the wire, delivering approximately 25 to 40 g. (Companion Ch 9; Hbk 16.1)

531 C Tipping a single tooth requires approximately 25 to 40 g. Excess force causes discomfort, appliance displacement and anchorage loss. (Companion Ch 9; Hbk 16.1)

532 A Proffit identifies compliance and two-point contact as the two fatal limitations of removable appliances. (Proffit Ch 10; Hbk 16.1)

533 E A flat anterior bite plane disoccludes the posterior teeth and allows them to erupt, reducing the overbite by relative intrusion in a growing patient. (Companion Ch 9; Hbk 16.1 and 25.1)

534 D Posterior bite planes disocclude the teeth to allow a crossbite to be corrected, particularly an anterior crossbite being corrected by proclination. (Companion Ch 9; Hbk 16.1)

535 B The dominant effect is dentoalveolar: upper incisors retrocline, lower incisors procline, and buccal segment eruption is guided. The skeletal effect is a few millimetres at best and essentially transient. (Companion Ch 9; Hbk 16.2)

536 E The occlusion is locked by intercuspation; the appliance unlocks it. This reads far better than saying the appliance grows the mandible. (Companion Ch 9; Hbk 16.2)

537 B Clark's twin block uses bite blocks inclined at approximately 70 degrees to the occlusal plane. (Companion Ch 9; Hbk 16.3)

538 D The two-piece design permits full-time wear and eating, which is why the twin block became the most widely used functional appliance. (Companion Ch 9; Hbk 16.3)

539 A Posterior open bites are expected and are managed by progressive trimming of the blocks to allow the posterior teeth to erupt. (Companion Ch 9; Hbk 16.3 and 22.3)

540 C The Herbst is a fixed functional appliance using bilateral telescopic rods and tubes, so it does not depend on wear-time compliance. (Companion Ch 9; Hbk 16.3)

541 D Pancherz reported overjet reduction in 6 to 8 months with the Herbst appliance. (Pancherz 1982; Hbk 16.4)

542 B Pacha reported 37 per cent failure with the twin block against 17 per cent with the Herbst, although the Herbst required more visits and produced more emergencies. (Pacha 2023; Hbk 16.4)

543 E The twin block achieved a smaller mean overjet reduction of 5.8 mm against 7.1 mm for the Herbst, and took 1.5 months longer. (Pacha 2023; Hbk 16.4)

544 A At 32 years, 64 per cent of molar correction and 86 per cent of overjet correction remained stable, with a recommendation of at least 2 years of activator retention. (Pancherz 2014; Hbk 16.4)

545 C Pancherz recommended at least 2 years of activator retention after Herbst treatment. (Pancherz 2014; Hbk 16.4)

546 B The greatest response occurs in the circumpubertal period, corresponding to cervical vertebral maturation stage CS3. (Mosby ch. on growth modification; Hbk 16.2 and 22.3)

547 D The Frankel is tissue-borne, using buccal shields and lip pads to allow transverse and sagittal dentoalveolar development. It is fragile. (Companion Ch 9; Hbk 16.3)

548 A Implanted electrodes showed a reduction in lateral pterygoid activity on protrusion, contradicting earlier surface electromyography. (Voudouris 2003; Hbk 16.2)

549 C Functional appliance therapy has a positive effect on the soft tissue profile in Class II subjects. (Koretsi 2015; Hbk 16.2)

550 E There is little clinical difference between maximal and incremental advancement. (Santana 2020; Hbk 16.4)

551 A Decompensation of a retroclined upper labial segment is required so that the mandible can be postured forwards. (Cobourne cases; Hbk 22.3)

552 C High-pull headgear combined with a functional appliance provides vertical control in the hyperdivergent Class II patient. (Companion Ch 9; Hbk 16.3 and 22.4)

553 E Facemask therapy uses approximately 400 g per side for 12 to 14 hours per day, ideally before age 10. (Companion Ch 9; Hbk 23.3)

554 B About half the overjet correction achieved with a facemask is dental. Overcorrection to a positive overjet of 3 to 4 mm is recommended. (Mosby ch. on Class III; Hbk 23.3)

555 D Over 60 per cent avoided surgery against 36 per cent without early treatment, although most of the initial skeletal effect is lost with growth. (Mandall 2016; Hbk 23.3)

556 C Proffit reports 25 to 33 per cent recurrence of the anterior crossbite at 8 years, mostly requiring surgery. (Proffit Ch 13; Hbk 23.3)

557 A The chin cup largely redirects the mandible downwards and backwards, and its short-term gain is not maintained after growth. (Nanda; Hbk 23.3)

558 E There is reasonable evidence that habit interruption before the permanent incisors are established permits spontaneous correction. There is no good evidence that exercises alone correct a malocclusion. (Companion Ch 12; Hbk 16.5)

559 D Positive reinforcement and reminder therapy come first. A habit-breaking appliance such as a palatal crib is used if that fails, and only in a motivated child. (Companion Ch 12; Hbk 16.5)

560 B The appliance is a reminder, not a punishment, and this framing improves acceptance and outcome. (Companion Ch 12; Hbk 16.5)

561 E The Southend clasp provides retention on incisors. Adams clasps are used on molars and premolars. (Companion Ch 9; Hbk 16.1)

562 B A Roberts retractor uses 0.5 mm wire supported in tubing, and retracts the upper labial segment by tipping. (Companion Ch 9; Hbk 16.1)

563 D A midline screw delivers approximately 0.25 mm per quarter turn, activated twice weekly for slow expansion. (Companion Ch 9; Hbk 16.1)

564 A The Andresen activator is the original tooth-borne, passive, one-piece functional appliance, bulky and worn at night, and wholly compliance-dependent. (Companion Ch 9; Hbk 16.3)

565 C Forsus and similar semi-rigid fixed functionals attach to a fixed appliance and are used mid-treatment. (Companion Ch 9; Hbk 16.3)

566 D There is no good evidence for the stronger claims made by proprietary myofunctional appliance systems. (Companion Ch 12; Hbk 16.5)

Scored under 70% on this block? Re-read Revision Handbook §16, then sit it again in a fortnight.

Block Q — Fixed appliances — labial, lingual and aligners (Q567–606 · Handbook §17)

567 B First order is in and out, built into the slot-to-base distance. Second order is tip, or angulation; third order is torque, or inclination. (Companion Ch 10; Hbk 17.1)

568 E Torque is expressed through the buccolingual inclination of the slot, and controls incisor root position and buccal root torque of molars. (Companion Ch 10; Hbk 17.1)

569 A Turning the slot horizontal gave greater control of tooth movement, but required all three orders to be bent into the wire tooth by tooth. (Angle 1928; Hbk 17.2)

570 C Bennett and McLaughlin record that standard edgewise results were traditionally left 10 to 20 per cent undertreated because of the wire bending required. (Bennett and McLaughlin; Hbk 17.2)

571 B Pre-adjustment built the three orders into the bracket, which moved the error from wire bending to bracket positioning. Bracket position is now where cases are lost. (Andrews 1979; Hbk 17.2)

572 D Angle worked through the E-arch, pin and tube, ribbon arch and finally the edgewise appliance of 1928. (Proffit Ch 10; Hbk 17.2)

573 **A** Begg inverted the ribbon arch slot and used differential force with three stages, deliberately tipping teeth and then uprighting them with auxiliary springs. (Begg 1956; Hbk 17.2)

574 **C** Kesling's Tip-Edge allows limited tipping and then expresses the full prescription in a rectangular finishing wire. (Kesling 1988; Hbk 17.2)

575 **E** MBT reduced anterior tip and increased palatal root torque on the upper incisors, with three canine torque options, because proven mechanics determined the bracket design. (McLaughlin, Bennett and Trevisi; Hbk 17.2)

576 **A** There is no evidence that either slot size is superior. The 0.018 slot expresses torque earlier in a smaller wire; the 0.022 slot allows more robust sliding mechanics. (Mosby ch. on appliances; Hbk 17.2)

577 **C** Finishing wires fill or nearly fill the slot: 0.017 by 0.025 inch in an 0.018 slot and 0.021 by 0.025 inch in an 0.022 slot. (Proffit Ch 16; Hbk 17.2 and 18.3)

578 **E** Stiffness should be varied by changing material rather than dimension, so that the slot remains filled. (Proffit Ch 16; Hbk 17.2)

579 **B** Multiple randomised trials found no difference in rate, treatment time, visits or occlusal outcome, with only a marginal pain benefit and unsupported expansion claims. (Scott 2008; Fleming 2010; Hbk 17.3)

580 **D** The distinction is a design difference, not a demonstrated clinical advantage. (Mosby ch. on appliances; Hbk 17.3)

581 **C** Outcomes were comparable with no time saving, and there were more breakages, more complaints and more planning time. Treatment time was dictated by case difficulty and by the treating orthodontist. (Penning 2017; Hbk 17.3)

582 **A** The changed bracket position alters the moment systems, and the shorter anterior interbracket span raises effective stiffness. (Companion Ch 10; Hbk 17.4)

583 **E** Lingual anatomy varies greatly between teeth and patients, which is why customised CAD-CAM lingual brackets and robotically-bent wires were developed. (Companion Ch 10; Hbk 17.4)

584 **D** Speech disturbance is usual but transient, typically resolving within 1 to 3 weeks. (Companion Ch 10; Hbk 17.4)

585 **B** The comparative evidence is limited. The indication is an adult whose sole objection is visibility and who accepts the trade-offs. (Companion Ch 10; Hbk 17.4)

586 **E** The plastic is viscoelastic: force decays through the wear period and decays faster when warm and moist. Plastic persuades; wire overpowers. (Companion Ch 14; Hbk 17.5)

587 **B** Root torque, extrusion and rotation of round teeth such as canines and premolars are the least predictable aligner movements. (Companion Ch 14; Hbk 17.5)

588 **D** Attachments are essential for movements that the smooth plastic shell cannot otherwise deliver, particularly extrusion and rotation. (Companion Ch 14; Hbk 17.5)

589 **A** Kravitz reported approximately 41 per cent accuracy against the predicted outcome, with rotation the least accurate movement. (Kravitz 2009; Hbk 17.5)

590 **C** Only about 6 per cent complete without a refinement, and over 17 per cent convert to fixed appliances to finish. (Companion Ch 14; Hbk 17.5)

591 **D** Over 17 per cent convert to fixed appliances. These are the two numbers to quote at consent. (Companion Ch 14; Hbk 17.5)

592 B PAR shows no difference; the ABO Objective Grading System shows aligners significantly worse, particularly for buccolingual inclination, occlusal contacts, overjet and root angulation. (Companion Ch 14; Hbk 17.5)

593 E Aligners perform better on pain in the first days, periodontal indices, eating disturbance and visibility. Fixed appliances are better on every occlusal measure. (Companion Ch 14; Hbk 17.5)

594 A Aligners require 20 to 22 hours per day. Compliance is the weakest element in the system. (Companion Ch 14; Hbk 17.5)

595 C The aligner is a single appliance acting on all teeth at once, so reaction forces are accepted rather than designed. This is the central unsolved problem. (Mosby ch. on aligners; Hbk 17.5)

596 B Cobourne's cases report greater post-treatment change at three years with aligners, alongside worse ABO grading. (Cobourne cases; Hbk 17.5)

597 D The appropriate candidate has mild to moderate crowding, a Class I skeletal pattern and good motivation, with the refinement and conversion rates explained at consent. (Companion Ch 14; Hbk 17.5)

598 A These are the two numbers that make the consent honest and defensible. (Companion Ch 14; Hbk 17.5)

599 C The prescription cannot fix a misplaced bracket. Bracket position, not prescription, decides the result. (Mosby ch. on appliances; Hbk 17.1 and 18.3)

600 E Roth's aim was a single prescription rather than an inventory of brackets, with overcorrected values built in. (McLaughlin, Bennett and Trevisi; Hbk 17.2)

601 A Andrews described the six keys in 1972 and the pre-adjusted prescription built into each bracket in 1979. (Andrews 1979; Hbk 17.2)

602 C Customised systems use CAD-CAM brackets and robotically-bent archwires to compensate for the variable lingual anatomy. (Companion Ch 10; Hbk 17.4)

603 E The drawbacks are speech disturbance, tongue discomfort and ulceration, access and hygiene difficulty, chairside time, cost and the learning curve. (Companion Ch 10; Hbk 17.4)

604 B Fixed appliances outperform aligners in severe crowding and complex cases, and in every occlusal outcome measure. (Companion Ch 14; Hbk 17.5)

605 D Expansion claims for self-ligating brackets are unsupported by the randomised evidence. (Scott 2008; Fleming 2010; Hbk 17.3)

606 C No difference in rate of movement has been demonstrated between ligation methods, once the trials are considered together. (Scott 2008; Fleming 2010; Hbk 17.3)

Scored under 70% on this block? Re-read Revision Handbook §17, then sit it again in a fortnight.

Block R — Treatment mechanics: Stage I, II and III (Q607–641 · Handbook §18)

607 A Anchorage is spent or saved during levelling and aligning. Lacebacks, bendbacks and the exclusion of anchor teeth from the initial wire all act at this stage. (McLaughlin, Bennett and Trevisi; Hbk 18.1)

608 E Second molars should be included as early as possible: the anchorage mechanics are the bite-opening mechanics. (Bennett and McLaughlin; Hbk 18.1)

609 D Approximately one month of passive rectangular wire allows the slots to line up. Level slot line-up is what makes sliding mechanics possible. (McLaughlin, Bennett and Trevisi; Hbk 18.2)

610 B Tiebacks are reactivated at 4 to 6 weeks, using a 0.010 inch wire ligature with a module stretched to about twice its resting length. (McLaughlin, Bennett and Trevisi; Hbk 18.2)

611 E Loop mechanics avoid friction but require wire bending and produce a force system that is harder to control. (Proffit Ch 15; Hbk 18.2)

612 B Comparative work shows coil springs give more consistent and slightly faster translation than modules. (Bennett and McLaughlin; Hbk 18.2)

613 D Each millimetre of lower incisor advance yields 2 mm of space, but stability is lost beyond about 2 mm of advance. (Bennett and McLaughlin; Hbk 18.2)

614 A The MBT literature quotes 100 degrees to the mandibular plane as the conventional limit for lower incisor proclination. (McLaughlin, Bennett and Trevisi; Hbk 18.2)

615 C Bennett's question is whether lower arch anchorage is available. If the lower incisors are already forward, Class II elastics are barred. (Bennett and McLaughlin; Hbk 18.2)

616 D Bracket positioning is the commonest reason a case does not finish. Every bracket should be reassessed at the working wire stage and repositioned before finishing wires. (Proffit Ch 16; Hbk 18.3)

617 B Reassess every bracket against marginal ridge levels and gingival contour at the working wire stage, and reposition before finishing wires. (Proffit Ch 16; Hbk 18.3)

618 E A panoramic radiograph at the finishing stage confirms root parallelism, particularly at extraction sites. (Proffit Ch 16; Hbk 18.3)

619 A Never go straight from rectangular wires to retainers. Settle in light round wires for approximately six weeks, keeping every space closed. (McLaughlin, Bennett and Trevisi; Hbk 18.3)

620 C The original arch form should be retained, not the treated one. Changing arch form results in relapse in up to 70 per cent of cases. (McLaughlin, Bennett and Trevisi; Hbk 18.5)

621 B Tapered 50 per cent, ovoid 42 per cent and square 8 per cent, with a recommended stock ratio of 45, 45 and 10. Japanese samples are almost the reverse. (McLaughlin, Bennett and Trevisi; Hbk 18.5)

622 D Buccal uprighting relapses in approximately 70 per cent of cases, which also means 30 per cent is stable, so the manoeuvre is unpredictable rather than worthless. (McLaughlin, Bennett and Trevisi; Hbk 18.5)

623 A Connector areas descend in size from central to central, then central to lateral, then lateral to canine, following the 50, 40, 30 rule. (Nanda; Hbk 18.4)

624 C Incisal embrasures increase in size from the midline distally, and this is a checkable finishing criterion. (Nanda; Hbk 18.4)

625 E The upper central and canine zeniths sit on a level line, with the lateral incisor zenith 0.5 to 1 mm below it. (Nanda; Hbk 18.4)

626 A The MBT finishing check covers horizontal, vertical, transverse, dynamic and cephalometric or aesthetic planes, finishing to ABO requirements deliberately rather than by luck. (McLaughlin, Bennett and Trevisi; Hbk 18.3)

627 C Proclination is the appropriate correction where there is room and a positive overbite will result. Posterior crossbites should be corrected before anterior ones. (Companion Ch 12; Hbk 18.1)

628 E Light traction of approximately 30 to 50 g along a controlled path, against adequate anchorage, is appropriate. Space must be created first. (Companion Ch 13; Hbk 18.1)

629 B Space must be created first, then exposure, then light traction against adequate anchorage. (Companion Ch 13; Hbk 18.1 and 20.4)

630 D Anchorage should be measured on the casts or with a superimposed photograph rather than by impression, and checked before every activation. (Bennett and McLaughlin; Hbk 18.2)

631 C Class II elastics procline the lower incisors, so they are barred if the lower incisors are already forward. (Bennett and McLaughlin; Hbk 18.2)

632 A Stage I is alignment and levelling, Stage II is buccal segment correction and space closure, Stage III is finishing and detailing. (Companion Ch 10; Hbk 18)

633 E Canines are frequently left slightly extruded or under-torqued. Check the gingival margin level against the central incisor and the cusp tip in the smile. (McLaughlin, Bennett and Trevisi; Hbk 18.3)

634 D Finishing interproximal reduction addresses tooth-size discrepancy, black triangles and marginal ridge discrepancies. (Proffit Ch 16; Hbk 18.3)

635 B Detailing bends cover all three orders plus artistic positioning, and are placed in a wire that fills or nearly fills the slot. (Proffit Ch 16; Hbk 18.3)

636 E Every space must be kept closed through the settling period, or it will reopen. (McLaughlin, Bennett and Trevisi; Hbk 18.3)

637 B The mandibular arch is the template, and intercanine width and lower incisor inclination are the least amenable to change. (Companion Ch 8; Hbk 18.5)

638 D True intrusion in a non-growing patient requires a one-couple intrusion arch or segmental mechanics, at approximately 60 g total. (Burstone 1977; Hbk 18.1 and 25.1)

639 A Measurement rather than impression is required, because anchorage loss is progressive and difficult to judge by eye. (Bennett and McLaughlin; Hbk 18.2)

640 C The MBT authors report 70 to 80 per cent of cases treated non-extraction, with extraction reserved for defined vertical and protrusion patterns. (McLaughlin, Bennett and Trevisi; Hbk 18.2)

641 D Lacebacks distalise the canine without tipping and are the single most effective way of preventing incisor advance during levelling. (Bennett and McLaughlin; Hbk 18.1)

Scored under 70% on this block? Re-read Revision Handbook §18, then sit it again in a fortnight.

Block S — Retention, relapse and stability (Q642–676 · Handbook §19)

642 B Periodontal ligament reorganisation takes about 3 to 4 months, but supracrestal collagen and elastic fibres take up to 6 to 12 months. (Proffit Ch 17; Hbk 19.1)

643 E Rotations relapse most readily, which is the rationale for circumferential supracrestal fiberotomy and for bonded retention. (Proffit Ch 17; Hbk 19.1)

644 A The three causes act on three different timescales: fibre recoil over months, soft tissue pressures continuously, and growth over a lifetime. (Proffit Ch 17; Hbk 19.1)

645 C If the lower incisors were advanced by more than about 2 mm, permanent retention is very likely to be needed. (Proffit Ch 17; Hbk 19.1)

646 B Overbite reduction relapses unless the interincisal angle is corrected and an occlusal stop established. (Houston 1989; Millett 2012; Hbk 19.1)

647 D Rowland and colleagues found vacuum-formed retainers more effective than Hawley retainers in the lower labial segment. (Rowland 2007; Hbk 19.2)

648 A Fixed retainers are marginally better for the first 12 months, particularly for rotations, and equivalent thereafter. (Forde 2017; Martin 2023; Hbk 19.2)

649 C Al-Moghrabi and Kramer found that by 4 to 5 years post-treatment the majority of patients have stopped wearing their retainers. (Al-Moghrabi 2018; Hbk 19.3)

650 E Bonded retainers require lifelong monitoring for bond failure, wire fracture, unwanted torque, calculus and caries. (Companion Ch 13; Hbk 19.2)

651 A There is no endpoint. This is the honest statement, and it must be made before active treatment begins, not at debond. (Companion Ch 13; Hbk 19.3)

652 C Retention should be discussed prior to the commencement of active treatment, and the patient must agree the plan. (Companion Ch 13; Hbk 19.3)

653 E Primary crowding is tooth size versus jaw size, secondary is environmental, and tertiary is the late crowding developing from adolescence onwards. (Companion Ch 5; Hbk 19.4)

654 B Continued mandibular growth rotation, uprighting the lower labial segment, is the major factor. Growth continues throughout adult life in the original pattern. (Behrents 1985; Solow 1980; Hbk 19.4)

655 D Southard proposed it, but the short duration of these forces makes it an unlikely cause. (Southard 1990; Hbk 19.4)

656 C Untreated subjects show decreasing mandibular arch width and length with increasing anterior crowding, which establishes late crowding as ageing rather than relapse. (Sinclair and Little 1983; Hbk 19.4)

657 A Late lower incisor crowding is age-related change, not relapse, and no extraction pattern or archwire sequence prevents it. The answer is long-term bonded retention. (Thilander 2009; Hbk 19.4)

658 E Both circumferential supracrestal fiberotomy and frenectomy are performed before appliance removal. The evidence for fiberotomy is weak but it is not harmful. (Cobourne cases; Hbk 19.1 and 21.1)

659 D A frenectomy performed before closure leaves scar tissue that resists closure. Close first, then operate if needed, and always retain permanently. (Cobourne cases; Hbk 21.1)

660 B A Hawley or Begg wrap-around retainer permits posterior settling, whereas a full-coverage vacuum-formed retainer holds the posterior teeth apart. (Companion Ch 13; Hbk 19.2)

661 E A positioner settles the occlusion and retains simultaneously, but is bulky and poorly tolerated. (Proffit Ch 17; Hbk 19.2)

662 B Active retainers, such as a spring retainer, a Barrer retainer or an aligner, can correct minor relapse of lower incisor alignment. (Companion Ch 13; Hbk 19.2)

663 D Wiedel and Bondemark found that an anterior crossbite corrected with a positive overbite is usually stable and requires no retention. (Wiedel and Bondemark 2015; Hbk 19.1 and 24.3)

664 A Lower intercanine width should not be expanded, and where it has been, permanent retention is required. (Proffit Ch 17; Hbk 19.1)

665 C Vacuum-formed retainers are made from 0.75 to 1 mm thermoplastic sheet. They wear out and must be replaced. (Companion Ch 13; Hbk 19.2)

666 D There is no universally accepted regimen. A common protocol is full-time for the first 3 months, or straight to nights with a vacuum-formed retainer, then nights indefinitely. (Companion Ch 13; Hbk 19.3)

667 **B** Retention cannot be abandoned until growth is essentially complete, and even then late crowding continues as an age-related change. (Proffit Ch 17; Hbk 19.1)

668 **E** The cost of replacement retainers should be stated at the outset, together with the consequences of loss. (GDC Standards; Hbk 19.3)

669 **A** A multistrand wire bonded 3 to 3, or 2 to 2, is the usual configuration in the lower arch. (Companion Ch 13; Hbk 19.2)

670 **C** Any tooth moved outside the neutral zone, such as by lower incisor proclination or lower arch expansion, needs permanent retention. (Proffit Ch 17; Hbk 19.1)

671 **B** Behrents demonstrated continued rotational growth throughout adult life, which is one of the three causes of relapse. (Behrents 1985; Hbk 19.1)

672 **D** Wire syndrome describes progressive unwanted tooth movement caused by an unnoticed change in a bonded retainer, and is the reason lifelong monitoring is required. (Companion Ch 13; Hbk 19.2)

673 **A** Tightness indicates that movement has occurred. Forcing a retainer can fracture it or move teeth unpredictably. Review is required. (Companion Ch 13; Hbk 19.3)

674 **C** Fibre recoil acts over months, soft tissue pressures act continuously, and growth acts over a lifetime. (Proffit Ch 17; Hbk 19.1)

675 **E** A closed diastema relapses reliably, so a bonded retainer is not optional. (Cobourne cases; Hbk 19.1 and 21.1)

676 **A** Retention should occupy a paragraph in the plan rather than a sentence in the conclusion, and must be agreed before treatment begins. (Companion Ch 8; Hbk 12.1 and 19.3)

Scored under 70% on this block? Re-read Revision Handbook §19, then sit it again in a fortnight.

Block T — Interceptive orthodontics and the developing dentition (Q677–721 · Handbook §20)

677 **C** Ugolini's Cochrane review provides high-certainty evidence that a quadhelix or expansion plate beats observation at ages 7 to 11 for crossbite with displacement. (Ugolini 2021; Hbk 20.2)

678 **E** Localised space loss of 3 mm or less is the space-regaining case. More than 4 mm generally means extraction. (Proffit Ch 12; Hbk 20.1)

679 **B** The interceptive question is always whether waiting costs growth that cannot be recovered. If not, waiting is usually correct. (Companion Ch 12; Hbk 20.1)

680 **D** Every observation period must have a defined endpoint. Indefinite observation is not a plan. (Proffit Ch 12; Hbk 20.1)

681 **C** The maxillary canine fails to erupt in approximately 2 to 3 per cent of Caucasian children, with up to 85 per cent of these palatal. (Ferguson 1990; Hbk 20.4)

682 **A** Up to 85 per cent of impacted maxillary canines are palatal, which is why palpation of the buccal sulcus is the key screening manoeuvre. (Ferguson 1990; Hbk 20.4)

683 **E** Peck reported bilateral occurrence in 17 to 45 per cent, with impaction up to three times more common in females and five times more common in Europeans than Asians. (Peck 1994; Hbk 20.4)

684 **D** The maxillary canine has the longest eruptive path in the dentition, over 20 mm, which is one reason it is so often displaced. (Companion Ch 12; Hbk 20.4)

685 B The principal threat is resorption of the adjacent incisor roots, which is why localisation matters and why CBCT is sometimes justified. (Companion Ch 12; Hbk 20.4)

686 E Interception is conventionally at 10 to 11 years, with RCS guidance giving 10 to 13 years, provided there is space and the canine position is favourable. (Ericson and Kurol; Hbk 20.4)

687 B An angulation of 20 to 30 degrees to the midsagittal plane, with the crown distal to the lateral incisor root midline, is favourable. Prognosis worsens beyond 55 degrees. (Ericson and Kurol; Hbk 20.4)

688 D Overlap of less than half the lateral incisor root gives 91 per cent normalisation; overlap of more than half gives 64 per cent after primary canine extraction. (Ericson and Kurol 1988; Hbk 20.4)

689 A Alpha angle and sector position are the most influential predictors of eruption after primary canine extraction. (Naoumova 2015, 2018; Hbk 20.4)

690 C Parkin found no difference in success or other outcomes. Cassina later reported that open exposure was associated with just over 2 months less time to alignment and lower odds of ankylosis. (Parkin 2017; Hbk 20.4)

691 D Kokich described the apically repositioned flap for buccal canines below the mucogingival junction, so that the tooth erupts through attached tissue. (Kokich 2004; Hbk 20.4)

692 B Patel reported success of approximately 88 per cent, with semi-rigid splinting for no more than 3 weeks followed by root treatment. (Patel 2009; Hbk 20.4)

693 E Prognosis worsens with height above the apical third of the incisor roots, with a horizontal position at or across the lateral incisor midline, and with angulation above about 55 degrees. (Companion Ch 13; Hbk 20.4)

694 A Mandibular second molar impaction affects just over 2 per cent, with evidence of a genetic basis at least in the mandible. (Shapira 2011; Hbk 20.5)

695 C A third molar follicle above the occlusal table of the developing second molar is the warning sign on the panoramic radiograph. (Companion Ch 12; Hbk 20.5)

696 B The ladder escalates from observation for 6 to 12 months, through disimpaction and third molar removal, to surgical uprighting, with second molar extraction as the least successful last resort. (Companion Ch 12; Hbk 20.5)

697 D Magnusson and Kjellberg found that extraction of the second molar itself gives the least successful outcome. (Magnusson and Kjellberg 2009; Hbk 20.5)

698 A A separator, brass wire or open coil is placed below the contact point in order to disimpact the tooth. (Companion Ch 12; Hbk 20.5)

699 C Re-implantation maintains the space and the alveolar bone and buys years, even when the long-term prognosis is poor. (Companion Ch 12; Hbk 20.6)

700 E Continued vertical alveolar growth leaves an implant crown in infra-occlusion, so implants must await confirmed completion of growth. (Companion Ch 12; Hbk 20.6)

701 A Space closure preserves alveolar bone height, at the cost of committing the patient to further orthodontics later. Space maintenance risks alveolar resorption. (Companion Ch 12; Hbk 20.6)

702 C Uncomplicated crown fracture, subluxation and minor lateral luxation each require a 3-month rest period before orthodontic force. (Companion Ch 8; Hbk 20.7)

703 E Re-implantation, intrusion, extrusion and moderate or severe lateral luxation all require a 12-month rest period. (Companion Ch 8; Hbk 20.7)

704 B A complicated crown fracture requires a delay until a hard tissue barrier is demonstrable. (Companion Ch 8; Hbk 20.7)

705 D Connective tissue healing means the coronal fragment behaves as a short-rooted tooth, so light forces are used. (Companion Ch 8; Hbk 20.7)

706 C Radiographs should be repeated 6 to 9 months after starting movement, with three-monthly vitality testing. (Companion Ch 8; Hbk 20.7)

707 A These three complications should be named specifically at consent for a case involving a previously traumatised tooth. (Companion Ch 8; Hbk 20.7)

708 E Pause for 3 months in a previously traumatised tooth, against 3 to 6 months for orthodontically induced resorption generally. (Companion Ch 8; Hbk 20.7)

709 D Serial extraction requires severe crowding with a Class I skeletal pattern and no vertical discrepancy. It is contraindicated in Class II or III, deep or open bite and hypodontia. (Proffit Ch 12; Hbk 20.3)

710 B A lower lingual arch preserves the leeway space by preventing mesial drift of the first permanent molars, and has good evidence as an interceptive measure. (Companion Ch 12; Hbk 20.2)

711 E An anterior crossbite with a displacement is treated on diagnosis, and correction is self-retaining if a positive overbite results. An incline plane corrects it in 3 to 4 weeks. (Companion Ch 12; Hbk 20.2 and 24.3)

712 B A displacement makes the case urgent because it causes attrition, is associated with asymmetric growth and with recession, and the case will not settle until corrected. (Ugolini 2021; Hbk 20.2 and 24.1)

713 D A habit-related open bite usually closes spontaneously if the habit stops before the permanent incisors are in occlusion. A skeletal open bite will not. (Companion Ch 12; Hbk 20.2 and 25.2)

714 A Where early treatment of a skeletal open bite is undertaken, it is done at 7 to 9 years, although the evidence is weak. (Mosby ch. on interceptive treatment; Hbk 20.2)

715 C Remove the obstruction, create and maintain space, then observe. Expose and bond if there is no eruption within about 12 months. (Companion Ch 12; Hbk 20.2)

716 D Spontaneous disimpaction may occur up to approximately 8 years. Beyond that a separator, brass wire or distalising spring is used. (Companion Ch 12; Hbk 20.2)

717 B Expansion yields only about half its width as usable space and lower arch expansion relapses, so expansion purely to relieve crowding has weak evidence. (Cobourne cases; Hbk 20.2)

718 E The randomised trials found no long-term advantage. The exceptions are high trauma risk and significant teasing. (O'Brien 2009; Hbk 20.2 and 22.2)

719 A Space for the mandibular incisors is negative for about two years after eruption. Transient early crowding is normal and self-correcting. (Proffit Ch 3; Hbk 20.1)

720 C A firm primary canine beyond 14 years is a red flag. Palpation from 9 to 10 years and radiographic localisation are indicated. (Companion Ch 12; Hbk 20.4)

721 B The triage begins by asking whether this is a problem at all, since transient lower incisor crowding, a midline diastema and the ugly duckling stage are all normal. (Proffit Ch 12; Hbk 20.1)

Scored under 70% on this block? Re-read Revision Handbook §20, then sit it again in a fortnight.

Block U — Class I malocclusion and bimaxillary protrusion (Q722–746 · Handbook §21)

722 D In Class I the base is not the problem, so the diagnosis must specify whether the problem is space, transverse or vertical. (Cobourne cases; Hbk 21.1)

723 A A midline diastema is normal in the mixed dentition, the so-called ugly duckling stage, and usually closes as the canines erupt. (Companion Ch 12; Hbk 21.1)

724 C A midline supernumerary must be excluded radiographically before the diastema is attributed to the fraenum. (Companion Ch 12; Hbk 21.1)

725 E An acute nasolabial angle with lip procumbency and strain on closure means retraction is likely to help. An obtuse angle means it will harm. (Mosby ch. on space analysis; Hbk 21.2)

726 A Bimaxillary protrusion is much commoner in patients of African and East Asian descent, and Caucasian cephalometric norms should not be applied uncritically. (Companion Ch 13; Hbk 21.2)

727 C Four first premolar extractions with maximum anchorage, using TADs, headgear or a stabilising arch, with bodily retraction, is the usual plan. (Companion Ch 13; Hbk 21.2)

728 E Joss-Vassalli, Aziz and Tepedino associate recession more with retroclination in high-angle cases with a narrow symphysis and thin biotype than with proclination. (Joss-Vassalli 2010; Hbk 21.2 and 29.2)

729 B Biotype and symphyseal width must be assessed before planning, since they determine whether the retraction can be delivered safely. (Nanda; Hbk 21.2)

730 D Anterior segmental osteotomies, described by Wunderer and Kole, or a two-jaw procedure with setback, are the surgical options. (Proffit Ch 19; Hbk 21.2 and 27.3)

731 C The decision is driven by the soft tissue and the periodontium, not by the cephalometric angle alone. (Nanda; Hbk 21.2)

732 A Generalised spacing from tooth-size discrepancy is managed by closure with permanent retention, or by redistribution and restoration, depending on the Bolton discrepancy. (Cobourne cases; Hbk 21.1)

733 E Mild crowding with an obtuse nasolabial angle and a flat profile is better managed without extraction, since retraction would harm the face. (Mosby ch. on space analysis; Hbk 21.1)

734 D Corresponding premolar extraction in both arches maintains the Class I buccal segment relationship. (Companion Ch 8; Hbk 21.1)

735 B Early periodontal involvement, light bodily forces, avoidance of over-retraction and consideration of soft tissue grafting are the recognised strategies. (Nanda; Hbk 21.2)

736 E Bodily movement is required, both for the aesthetic result and to protect the periodontium. Tipping concentrates stress at the crest and apex. (Nanda; Hbk 14.2 and 21.2)

737 B A frenectomy performed before closure leaves scar tissue that resists closure. Perform it before appliance removal, and retain permanently. (Cobourne cases; Hbk 21.1)

738 D Canine substitution is inadvisable where the canine is dark, large or prominent and the smile line is high. It is favoured in Class II molar cases where the lower arch needs no space. (Mosby ch. on hypodontia; Hbk 21.1 and Part 3 Drill 5)

739 A Closing the lateral incisor space by protracting the buccal segments finishes the case to a Class II molar relationship, which must be a deliberate choice. (Mosby ch. on hypodontia; Hbk 21.1)

740 C Recontouring, build-up, gingival recontouring and palatal root torque are all required so that the canine presents a flatter surface in a lateral incisor position. (Mosby ch. on hypodontia; Hbk 21.1)

741 D Retention is full-time until the definitive restoration is placed, and a bonded retainer thereafter. (Mosby ch. on hypodontia; Hbk 21.1)

742 B The Konstantonis figures are most directly applied in bimaxillary protrusion, where the whole plan is driven by the intended lip change. (Konstantonis 2018; Hbk 21.2)

743 E The lower labial segment position is the fixed point the plan is built around, because it is the least amenable to stable change. (Cobourne cases; Hbk 21.1)

744 A A crossbite with a displacement is treated on diagnosis, and correction is self-retaining if a positive overbite results. (Companion Ch 12; Hbk 24.3)

745 C Maximum anchorage requires reinforcement, and where the case needs 100 per cent of the space for the anterior teeth, only bone will do. (Nanda; Hbk 14.4 and 21.2)

746 B If the symphysis and biotype will not tolerate the retraction required, that changes the plan before a single bracket is chosen. (Nanda; Hbk 21.2)

Scored under 70% on this block? Re-read Revision Handbook §21, then sit it again in a fortnight.

Block V — Class II malocclusion (Q747–786 · Handbook §22)

747 D Class II is most often due to mandibular retrognathia, although the responsible jaw must be established in each case before planning. (Cobourne cases; Hbk 22.1)

748 A The appliance is chosen from the vertical dimension: cervical pull for the hypodivergent, high pull or occipital for the hyperdivergent. (Mosby ch. on Class II; Hbk 22.1)

749 C The only durable benefit was a reduction in incisor trauma, with a relative risk of 0.58, a 42 per cent reduction, though with wide confidence intervals. (Cobourne 2022; Hbk 22.2)

750 E Much trauma occurs before age 9, and headgear was marginally more effective than a functional appliance at reducing it. (Cobourne 2022; Kallunki 2021; Hbk 22.2)

751 A The late-treated group had a slightly better final occlusal result, and early-treated children had more attendances and longer treatment. (O'Brien 2009; Hbk 22.2)

752 C The pragmatic policy is to treat early only for trauma risk or significant teasing, and otherwise to treat in a single phase in the late mixed or early permanent dentition. (O'Brien 2009; Hbk 22.2)

753 E Overjet reduction outpaces overbite reduction, so lateral open bites are expected and are managed by trimming the blocks. (Cobourne cases; Hbk 22.3)

754 B In a high-angle patient the mandible rotates down and back at the slightest provocation, so extrusive mechanics must be avoided. (Companion Ch 13; Hbk 22.4)

755 D In a growing patient, buccal segment extrusion produces relative incisor intrusion and is stable because condylar growth compensates. (Companion Ch 13; Hbk 22.4 and 25.1)

756 C The increased interincisal angle with a deep complete overbite is the mechanical difficulty. Retroclined upper centrals must be torqued labially at the root, not merely tipped. (Companion Ch 13; Hbk 22.4)

757 A Proclination converts the case to a division 1 pattern, allowing overbite reduction and then retraction if space is available. (Companion Ch 13; Hbk 22.4)

758 **E** Where there is little overjet there is nowhere for upper extraction space to go, so many division 2 cases are treated non-extraction in the upper arch. (Companion Ch 13; Hbk 22.4)

759 **D** Extra palatal root torque is required on the upper centrals so that the roots are moved labially and the overbite can be held. (Companion Ch 13; Hbk 22.4)

760 **B** A Hawley retainer with an anterior bite plane, worn into the early twenties, is the classical answer for deep bite retention. (Mosby ch. on retention; Hbk 22.4 and 25.3)

761 **E** An overjet greater than 10 mm is one of Proffit's four cephalometric thresholds favouring surgery in Class II. (Proffit 1992; Hbk 22.5)

762 **B** A mandibular body length of less than 70 mm is one of Proffit's thresholds. (Proffit 1992; Hbk 22.5)

763 **D** An anterior face height greater than 125 mm is one of Proffit's four thresholds. (Proffit 1992; Hbk 22.5)

764 **A** The bilateral sagittal split osteotomy is the standard mandibular advancement, with a recognised risk to the inferior alveolar nerve. (Proffit Ch 19; Hbk 22.5 and 27.3)

765 **C** Le Fort I impaction corrects vertical maxillary excess and produces autorotation of the mandible. (Proffit Ch 19; Hbk 22.5 and 27.3)

766 **D** Decompensation deliberately makes the bite worse before surgery. Calling this aligning the arches is one of the costly slips. (Proffit Ch 19; Hbk 22.5 and 27.2)

767 **B** Kandasamy's guidance is to screen and document before treatment, counsel on natural history, suspend active mechanics if symptoms appear, and manage conservatively. (Kandasamy 2022; Hbk 22.6 and 26.2)

768 **E** Figures of 15 to 22 per cent for United States adolescents and up to 27 per cent in wider Caucasian samples are quoted. (Cobourne cases; Hbk 22.1)

769 **A** Distalisation or an orthopaedic effect requires 450 to 500 g per side for 14 hours per day; anchorage support requires 250 to 350 g for 10 to 12 hours. (Companion Ch 6; Hbk 22.3)

770 **C** Distalisation corrects a dentoalveolar Class II only, and always buys reciprocal mesial premolar movement unless a TAD is used. (Nanda; Hbk 22.3)

771 **B** Nanda gives a distal molar force of 150 to 250 g, with molar derotation alone gaining 2 to 3 mm per side. (Nanda; Hbk 22.3)

772 **D** Derotation of a mesially rotated maxillary first molar can gain 2 to 3 mm of arch length per side. (Nanda; Hbk 22.3)

773 **A** Overcorrection and careful transition are required, either directly into a fixed appliance or with a period of night-time functional appliance wear. (Cobourne cases; Hbk 22.3)

774 **C** Pogonion to the nasion perpendicular of 18 mm or more is one of Proffit's four thresholds. (Proffit 1992; Hbk 22.5)

775 **E** The lip trap proclines the uppers and retroclines the lowers, perpetuating the malocclusion, and its elimination is one of the soft tissue effects of a functional appliance. (Companion Ch 12; Hbk 22.1)

776 **A** Class II requires extraction further back in the mandible, giving a greater maxillary space requirement to retract the upper labial segment. (Companion Ch 8; Hbk 22.4)

777 **C** The laterals are frequently proclined and mesiolabially rotated, requiring first and second order correction. (Companion Ch 2; Hbk 22.4)

778 **E** In a high-angle case, extraction may allow the mandible to rotate forwards, reducing face height and improving the anteroposterior relationship. (Companion Ch 13; Hbk 22.4)

779 **B** Inferior alveolar nerve injury is the risk to quote and consent for, together with relapse, infection, unfavourable split and condylar resorption. (Proffit Ch 19; Hbk 22.5 and 27.5)

780 **D** The greatest response occurs in the circumpubertal period, around CS3, with preparation from late CS2. (Baccetti; Hbk 22.3)

781 **C** The Herbst is a fixed functional appliance with a 17 per cent failure rate against 37 per cent for the twin block, although it requires more visits. (Pacha 2023; Hbk 22.3)

782 **A** Mandibular advancement is very stable in the low- and average-angle patient. Maxillary down-grafting and surgical expansion are the least stable movements. (Proffit Ch 19; Hbk 27.4)

783 **E** Trauma reduction is the only durable benefit demonstrated, with a relative risk of 0.58, and is the accepted indication for early treatment. (Cobourne 2022; Hbk 22.2)

784 **D** There is no evidence that orthodontic intervention reduces TMD signs and symptoms, and none that it causes them. (Macfarlane 2009; Luther 2010; Hbk 22.6)

785 **B** Screening and documenting joint status before treatment is the recognised standard and protects both patient and clinician. (Kandasamy 2022; Hbk 22.6 and 26.3)

786 **E** Class II division 2 is characteristically associated with a reduced lower face height, a forward growth rotation, a high lower lip line and a deep overbite. (Companion Ch 2; Hbk 22.1)

Scored under 70% on this block? Re-read Revision Handbook §22, then sit it again in a fortnight.

Block W — Class III malocclusion (Q787–816 · Handbook §23)

787 **B** Unfavourable growth persists in Class III. Residual mandibular growth assists correction in Class II and worsens the discrepancy in Class III. (Cobourne cases; Hbk 23.1)

788 **D** A pseudo-Class III manipulates back to an edge-to-edge incisor relationship in centric relation, and the prognosis is completely different. (Mosby ch. on Class III; Hbk 23.1)

789 **A** Class III compensates by proclining the uppers and retroclining the lowers. The more compensated the case, the less room there is for camouflage. (Companion Ch 13; Hbk 23.1)

790 **C** Favourable features are a mild skeletal base, average or reduced lower face height, average or increased overbite, little compensation and an anterior displacement on closing. (Companion Ch 8; Hbk 23.2)

791 **D** Heavy compensation with no displacement, a moderate to severe skeletal base, increased lower face height and a reduced overbite are unfavourable features. (Companion Ch 8; Hbk 23.2)

792 **B** Facemask therapy is most effective before age 10. Bone-anchored maxillary protraction using Class III elastics to miniplates suits the 11 to 14 age range. (Mandall 2016; Hbk 23.3)

793 **E** Facemask therapy requires 12 to 14 hours per day, with an elastic force of approximately 400 g per side. (Companion Ch 9; Hbk 23.3)

794 **A** Overcorrection to a positive overjet of 3 to 4 mm is recommended, because relapse is expected as growth continues. (Mosby ch. on Class III; Hbk 23.3)

795 **C** Over 60 per cent avoided surgery against 36 per cent without early treatment, although most of the initial skeletal effect is lost with growth. (Mandall 2016; Hbk 23.3)

796 **B** Proffit reports 25 to 33 per cent recurrence at 8 years, mostly requiring surgery, which is the honest consent number. (Proffit Ch 13; Hbk 23.3)

797 D Miniplate-anchored protraction works through the adolescent spurt at about half the force, and suits the 11 to 14 age range; facemask suits 8 to 10 years. (Proffit Ch 13; Hbk 23.3)

798 A The chin cup's short-term gain is not maintained after growth, and it increases face height by rotating the mandible down and back. (Nanda; Hbk 23.3)

799 C A Wits appraisal of minus 11 mm or worse is among the most important discriminators favouring surgery. (Kerr 1992; Hbk 23.4)

800 E An ANB of minus 4 degrees or worse is one of the thresholds, alongside a Wits of minus 11 mm and a reverse overjet of minus 5 mm. (Kerr 1992; Hbk 23.4)

801 A A gonial angle of 120 degrees or less is one of the discriminant variables, with a lower anterior face height of 66 per cent or more. (Kerr 1992; Hbk 23.4)

802 C A lower anterior face height of 66 per cent or more of total anterior face height is one of the discriminant variables. (Kerr 1992; Hbk 23.4)

803 E Class III camouflage extracts further forward in the mandible, giving a greater mandibular space requirement to retract the lower labial segment. (Companion Ch 8; Hbk 23.4)

804 B Mandibular extractions consume the space required for pre-surgical decompensation, which retroclines the lowers further. (Companion Ch 8; Hbk 23.4)

805 D Class III elastics extrude the lower molars and rotate the mandible down and back, which is dangerous in a high-angle case. (Bennett and McLaughlin; Hbk 23.4)

806 C Decompensation reverses the natural compensation, retroclining the uppers and proclining the lowers, and makes the reverse overjet worse before surgery. (Proffit Ch 19; Hbk 23.4 and 27.2)

807 A Mandibular growth in Class III persists beyond adolescence and later in males, so surgery is deferred until serial superimposition confirms cessation. (Companion Ch 15; Hbk 23.4)

808 E Kerr, Stellzig-Eisenhauer and Kochel reported over 90 per cent accuracy for discriminant analysis of these variables. (Kochel 2011; Hbk 23.4)

809 D Class III prevalence is far higher in East Asian populations than the 5 to 6 per cent quoted for United States adolescents, with a worldwide figure of about 6.8 per cent. (Mosby ch. on classification; Hbk 23.1)

810 B Maxillary retrusion is the commoner component in East Asian Class III, which is relevant to the choice between protraction and setback. (Mosby ch. on Class III; Hbk 23.1)

811 E An anterior crossbite corrects in 3 to 4 weeks with an incline plane, and a face mask corrects a dental crossbite in 1 to 3 months. (Nanda; Hbk 23.3)

812 B Class III has the strongest hereditary component of the three classes, so a family history should be taken specifically. (Proffit Ch 5; Hbk 23.1)

813 D The two must be separated: the displacement is treated now; the skeletal problem requires a discussion of what early treatment can and cannot achieve. (Mandall 2016; Hbk 23.3)

814 A Rapid maxillary expansion may be used to disarticulate the maxilla and improve the response to protraction. (Companion Ch 9; Hbk 23.3)

815 C A single lower incisor extraction is occasionally used in mild Class III camouflage in a non-growing patient, with recognised costs. (Companion Ch 8; Hbk 13.4 and 23.4)

816 D The existing decompensation leaves little room for camouflage, and the complaint concerns the profile, which camouflage cannot change. (Kerr 1992; Hbk Part 3 Drill 3)

Scored under 70% on this block? Re-read Revision Handbook §23, then sit it again in a fortnight.

Block X — Transverse problems and facial asymmetry (Q817–841 · Handbook §24)

817 B A displacement indicates a bilateral constriction with a lateral shift into a unilateral crossbite in intercuspation. (Companion Ch 12; Hbk 24.1)

818 E A crossbite with a displacement is treated on diagnosis; one without a displacement is treated when convenient within the overall plan. (Ugolini 2021; Hbk 24.1)

819 A Ugolini found high-certainty evidence that a quadhelix or expansion plate beats observation, that the quadhelix beats a plate, and that it is about 3 months faster. (Ugolini 2021; Hbk 24.2)

820 C Rapid maxillary expansion achieves up to 4 mm at the suture, approximately 50 per cent of total screw expansion. (Liu 2015; Hbk 24.2)

821 B Approximately 10 per cent of the skeletal change relapses at 4 to 8 months post-retention. (Lo Giudice 2020; Hbk 24.2)

822 D Slow expansion retains approximately 95 per cent of the initial expansion against 85 per cent for rapid expansion, which produces slightly greater posterior skeletal expansion but more buccal tipping and more pain. (Bucci 2016; Rutili 2021; Hbk 24.2)

823 A Beyond approximately 5 mm of required expansion in a non-growing patient, surgically assisted rapid palatal expansion or bone-borne expansion is indicated. (Companion Ch 13; Hbk 24.2)

824 C The suture should be assessed rather than the birthday, since the correlation between sutural maturation and chronological age is poor. (Companion Ch 9; Hbk 24.2)

825 E Proffit's dissent is that rapid activation is not an effective way to minimise tooth movement, and that the net effect of rapid expansion is approximately equal skeletal and dental expansion. (Proffit Ch 13; Hbk 24.2)

826 A Proffit gives slow expansion at less than 2 mm per week as matching the maximum speed of bone formation. (Proffit Ch 13; Hbk 24.2)

827 C Rapid maxillary expansion is not recommended as a treatment for nasal breathing or obstructive sleep apnoea on current evidence. (Companion Ch 2; Hbk 24.2 and 26.5)

828 E Correct posterior before anterior. Correct the anterior first and you lose it when you expand. (Mosby ch. on crossbite; Hbk 24.3)

829 B A positive overbite makes the correction self-retaining, and such corrections are usually stable without retention. (Wiedel and Bondemark 2015; Hbk 24.3)

830 D Clinically significant asymmetric jaw development affects fewer than 0.5 per cent of the population, and the mandible is involved in 85 to 90 per cent of cases. (Proffit Ch 13; Hbk 24.4)

831 C A childhood condylar fracture is the commonest cause, not a syndrome. (Proffit Ch 13; Hbk 24.4 and 26.4)

832 A Laterocclusion describes a functional shift; laterognathy describes a true skeletal asymmetry. (Mosby ch. on asymmetry; Hbk 24.4)

833 E Hemimandibular hyperplasia, Type II, has a vertical vector with a true occlusal cant and often an undisturbed chin point and centreline. (Obwegeser and Makek 1986; Hbk 24.4)

834 D Hemimandibular elongation, Type I, has a horizontal vector: the chin deviates to the unaffected side with a level occlusal plane. (Obwegeser and Makek 1986; Hbk 24.4)

835 B Scintigraphy is sensitive but non-specific, with false positives. Serial three-dimensional facial superimposition is the non-invasive alternative. (Companion Ch 15; Hbk 24.4)

836 E Progression is the key feature. Accelerating deviation, a new cant or new joint symptoms in the teens should prompt referral. (Companion Ch 15; Hbk 24.4)

837 B Slootweg and Muller described four histological subtypes of condylar hyperplasia. (Slootweg and Muller 1986; Hbk 24.4)

838 D Mild asymmetry in a grower is a watch, not a treat, but the watching must be structured with defined reviews and superimposable records. (Cobourne cases; Hbk 24.4)

839 A A shift should be eliminated early, because it can drive asymmetric mandibular growth. (Cobourne cases; Hbk 24.4)

840 C Biting on a spatula reveals a true occlusal cant, and the path of closure reveals a functional displacement. (Companion Ch 15; Hbk 24.4)

841 D A posteroanterior cephalogram is indicated for significant transverse asymmetry, although identifying the midsagittal plane is the principal difficulty. (Mosby ch. on asymmetry; Hbk 24.4)

Scored under 70% on this block? Re-read Revision Handbook §24, then sit it again in a fortnight.

Block Y — Vertical problems: deep bite and open bite (Q842–871 · Handbook §25)

842 B The three mechanisms are incisor intrusion, buccal segment extrusion and labial segment proclination, and which is stable depends entirely on growth status. (Companion Ch 13; Hbk 25.1)

843 E In a grower, condylar growth compensates for the increased posterior vertical dimension. In a non-grower the mandible rotates down and back, worsening the Class II and relapsing. (Companion Ch 13; Hbk 25.1)

844 A True intrusion is required where face height must not increase, using a utility or bypass arch, segmental mechanics, or miniscrews. (Companion Ch 13; Hbk 25.1)

845 C Approximately 60 g total, at 15 g per tooth, is appropriate. Intrusion is the lightest movement in orthodontics. (Companion Ch 13; Hbk 25.1)

846 B Approximately 1.5 mm maxillary and 1.9 mm mandibular intrusion is achievable with segmental mechanics in non-growing patients. (Ng 2005; Hbk 25.1)

847 D Miniscrews give 0.67 to 0.95 mm more intrusion than segmental mechanics, with less molar extrusion, on low-quality evidence. (Sosly 2020; Shakti 2022; Hbk 25.1)

848 A Eberhart reported 0.1 to 0.2 mm of overbite change per degree of incisal angular change. (Eberhart 1990; Hbk 25.1)

849 C Houston's rule is that overbite reduction relapses unless the interincisal angle is corrected and an occlusal stop established. (Houston 1989; Hbk 25.1)

850 E Localised supra-occlusion re-establishes contacts over an average of 6 months, with adverse events rare, minor and transient. (Dahl 1975; Poyser 2005; Hbk 25.1)

851 A Segmental techniques give greater true intrusion; the Ricketts utility arch produces more lower incisor proclination. (Dake and Sinclair 1989; Weiland 1996; Hbk 25.1)

852 C Burstone's one-couple bypass intrusion arch is statically determinate, so the force system can be calculated and measured. (Burstone 1977; Hbk 14.3 and 25.1)

853 E An interlabial gap greater than 3 mm is the key soft tissue sign of a skeletal open bite. (Nanda; Hbk 25.2)

854 B One millimetre of molar intrusion gives approximately 3 mm of open bite reduction, which is why TADs transformed open bite management. (Nanda; Hbk 25.2)

855 D One millimetre of molar extrusion reduces overlap by 1.5 to 2.5 mm, which is why extrusion is contraindicated where the face is already long. (Nanda; Hbk 25.2)

856 C Open bites relapse in approximately 20 per cent of both surgical and non-surgical corrections, which is the number to state at consent. (Nanda; Hbk 25.2)

857 A Tongue posture is usually adaptive to the open bite rather than causative. An endogenous tongue thrust is rare, and such cases are unstable and need permanent retention. (Companion Ch 13; Hbk 25.2)

858 E A habit-related open bite is asymmetric and often localised, with an associated posterior crossbite, proclined uppers and retroclined lowers. A skeletal open bite is symmetrical with increased face height and a high mandibular plane angle. (Companion Ch 12; Hbk 25.2)

859 D Extrusive mechanics in a high-angle patient are a recognised iatrogenic cause, and recognising the high-angle face before choosing the mechanics prevents it. (Companion Ch 13; Hbk 25.2)

860 B Approximately 5 mm of dentoalveolar height separates a deep bite face from an open bite face, which is why small vertical changes matter. (Mosby ch. on vertical control; Hbk 25.3)

861 E Short face equals forward condylar growth and a deep bite. Long face equals posterior condylar growth, backward rotation and an open bite. (Mosby ch. on vertical control; Hbk 25.3)

862 B Rest position is recorded by asking the patient to say Emma or Mississippi; the posed smile is recorded by biting together and saying cheese. (Nanda; Hbk 25.3)

863 D A Hawley retainer with an anterior bite plane, worn into the early twenties, is the classical deep bite retention appliance. (Mosby ch. on retention; Hbk 25.3)

864 A Nanda's principle is to choose the treatment occlusal plane first, then design the force system to reach it. (Nanda; Hbk 25.2)

865 C A flat anterior bite plane disoccludes the posterior teeth and allows them to erupt, giving relative incisor intrusion, which is stable in a grower. (Companion Ch 9; Hbk 25.1)

866 D In a non-grower the mandible rotates down and back, so the manoeuvre worsens the Class II, increases face height and relapses. (Companion Ch 13; Hbk 25.1)

867 B A habit-related open bite usually closes spontaneously if the habit ceases before the permanent incisors are in occlusion. (Companion Ch 12; Hbk 25.2)

868 E Molar intrusion with TADs is the most efficient lever available, giving approximately 3 mm of open bite reduction per millimetre of intrusion. (Nanda; Hbk 25.2)

869 A Posterior maxillary impaction with mandibular autorotation is the standard correction, undertaken once growth is complete. (Proffit Ch 19; Hbk 25.2)

870 C There is no good evidence that muscle exercises alone correct a malocclusion. Do not blame the tongue reflexively or promise that exercises will help. (Companion Ch 12; Hbk 25.2)

871 B Extrusion increases face height and rotates the mandible down and back, so it is contraindicated in the long-face patient. (Nanda; Hbk 25.2)

Scored under 70% on this block? Re-read Revision Handbook §25, then sit it again in a fortnight.

Block Z — TMD, condylar trauma, the airway and myofunctional therapy (Q872–901 · Handbook §26)

872 D Weak statistical associations exist but there is no evidence of causation in either direction. Orthodontics is TMJ-neutral. (Behrents and White 1992; Hbk 26.2)

873 A Behrents and White reported the AAO-funded studies in 1992, showing no significant association and that TMD cannot be prevented by any intervention. (Behrents and White 1992; Hbk 26.2)

874 C The 1987 Michigan lawsuit resulted in an award of 850000 dollars, and prompted the AAO-funded studies of 1992. (Companion Ch 2; Hbk 26.1)

875 E Conventional incisor retraction does not produce distal condylar displacement, which was the mechanism alleged by the functional orthodontists. (Luecke and Johnstone 1992; Hbk 26.2)

876 A Approximately 75 per cent have at least one sign and approximately 33 per cent at least one symptom, but only a small minority seek treatment. (Companion Ch 2; Hbk 26.2)

877 C Symptoms commonly begin in late adolescence, exactly when orthodontic treatment is provided, which is the principal confounder in this literature. (Behrents and White 1992; Hbk 26.2)

878 E MRI images the disc, the retrodiscal tissues and effusion. CBCT images bone but cannot image the disc or muscle. (Mosby ch. on TMD; Hbk 26.3)

879 B Maximum opening of at least 40 mm is regarded as normal, and deviation or deflection on opening should be recorded. (Companion Ch 2; Hbk 26.3)

880 D Conservative and reversible care comes first. Occlusal adjustment has little beneficial value and is irreversible. (Kandasamy 2022; Hbk 26.3)

881 C Juvenile idiopathic arthritis can destroy the condyle and produce a progressive Class II with an anterior open bite, and should be recognised. (Mosby ch. on TMD; Hbk 26.3)

882 A Conservative functional management is preferred in children. Prolonged immobilisation risks ankylosis, which is the outcome to avoid at all costs. (Companion Ch 15; Hbk 26.4)

883 E Because the condyle is the mandible's principal growth site, a childhood fracture is the commonest cause of significant mandibular asymmetry. (Companion Ch 15; Hbk 26.4)

884 D Deviation to the affected side on opening, reduced ramus height and an occlusal cant develop as growth proceeds. (Companion Ch 15; Hbk 26.4)

885 B Chronic nasal obstruction is associated with mouth breathing, low tongue posture, lips apart and head extension, producing a narrow maxilla, increased face height and backward rotation. (Companion Ch 7; Hbk 26.5)

886 E The association should be stated without asserting the cause. Experimental work in primates is not human evidence of causation. (Companion Ch 7; Hbk 26.5)

887 B Adenoid tissue peaks at approximately 3 to 5 years and involutes through adolescence, so many children grow out of the obstruction. (Mosby ch. on airway; Hbk 26.5)

888 D Paediatric obstructive sleep apnoea affects approximately 2 per cent of children, and adenotonsillectomy is first-line treatment in most. (Companion Ch 2; Hbk 26.5)

889 A Do not diagnose or treat obstructive sleep apnoea orthodontically without a sleep study and appropriate referral. (Companion Ch 2; Hbk 26.5)

890 C Mandibular advancement appliances are effective in mild to moderate obstructive sleep apnoea; CPAP remains the gold standard in moderate to severe disease. (Koretsi 2018; Hbk 26.5)

891 D Maxillomandibular advancement is the most effective surgical option for severe adult obstructive sleep apnoea. (Mosby ch. on airway; Hbk 26.5)

892 B These are the symptoms of sleep-disordered breathing and require proper assessment, not orthodontic treatment in the first instance. (Companion Ch 2; Hbk 26.5)

893 E Occlusal adjustment has little beneficial value and is irreversible, and should not be undertaken to treat temporomandibular disorder. (Mosby ch. on TMD; Hbk 26.3)

894 A Manfredini found a lack of evidence implicating occlusion in the pathophysiology of temporomandibular disorder. (Manfredini 2017; Hbk 26.2)

895 C Prevalence has not changed and symptoms remit spontaneously, which no occlusal explanation accounts for. (Proffit Ch 1; Hbk 26.2)

896 B Screening and documenting joint status before treatment is the recognised standard, and protects both patient and clinician. (Kandasamy 2022; Hbk 26.3)

897 D Kandasamy's guidance is to suspend active mechanics if symptoms appear and manage with conservative, reversible therapies. (Kandasamy 2022; Hbk 26.3)

898 A Habit interruption before the permanent incisors are established has reasonable evidence; exercises alone do not, and the stronger claims made for proprietary systems are unsupported. (Companion Ch 12; Hbk 26.6)

899 C A lower lip trap proclines the uppers and retroclines the lowers, perpetuating the malocclusion. Its elimination is one of the soft tissue effects of a functional appliance. (Companion Ch 12; Hbk 26.6)

900 E Broad classification into myofascial pain, internal derangement, degenerative disease and inflammatory arthropathy guides management and referral. (Mosby ch. on TMD; Hbk 26.3)

901 A Do not promise a cure, and do not let a click drive an irreversible occlusal change you would not otherwise make. (Macfarlane 2009; Luther 2010; Hbk 26.2)

Scored under 70% on this block? Re-read Revision Handbook §26, then sit it again in a fortnight.

Block AA — Adult orthodontics and orthognathic surgery (Q902–931 · Handbook §27)

902 C The centre of resistance migrates apically, so the same force applied at the bracket produces a larger moment and more tipping. Lighter forces and higher moment-to-force ratios are required. (Companion Ch 6; Hbk 27.1)

903 E Probing every adult and recording pre-existing recession is essential clinically and medico-legally. (Nanda; Hbk 27.1)

904 B Adjunctive treatment serves a larger goal, such as uprighting a tipped molar before a bridge, and is often appropriate in general practice. (Proffit Ch 18; Hbk 27.1)

905 D Intrusion during molar uprighting requires 10 to 15 g per tooth, and the uprighted tooth should be retained until the lamina dura reappears. (Mosby ch. on molar uprighting; Hbk 27.1)

906 C Dolichocephalic, high-angle patients are poor candidates because uprighting mechanics tend to be extrusive and open the bite further. (Mosby ch. on molar uprighting; Hbk 27.1)

907 A The uprighted molar should be retained until the lamina dura reappears radiographically. (Mosby ch. on molar uprighting; Hbk 27.1)

908 E Proffit's alveolar bone argument is that bone exists only where a tooth erupted, and can be developed by erupting a tooth through the site. (Proffit Ch 13; Hbk 27.1)

909 D The decision must be made before treatment begins, because pre-surgical orthodontics decompensates in the opposite direction from camouflage. (Proffit Ch 19; Hbk 27.2)

910 B Pre-surgical orthodontics typically takes 12 to 18 months, with a further 4 to 6 months of post-surgical detailing. (Proffit Ch 19; Hbk 27.2)

911 E Calling pre-surgical orthodontics aligning the arches is one of the twelve costly slips. Decompensation deliberately worsens the bite. (Proffit Ch 19; Hbk 27.2)

912 B Le Fort I impaction corrects vertical maxillary excess and a gummy smile, and produces autorotation of the mandible. (Proffit Ch 19; Hbk 27.3)

913 D The inferior alveolar nerve is at risk, with altered sensation common transiently and occasionally permanent. It must be consented for specifically. (Proffit Ch 19; Hbk 27.3)

914 A The vertical subsigmoid osteotomy, performed extra-orally or intra-orally, is used for mandibular setback. (Proffit Ch 19; Hbk 27.3)

915 C You cannot orthodontically produce a chin. A genioplasty is required, and it is a very stable movement in any direction. (Mosby ch. on treatment planning; Hbk 27.3)

916 D Maxillary impaction, mandibular advancement in the low- and average-angle patient, and genioplasty in any direction are the most stable movements. (Proffit Ch 19; Hbk 27.4)

917 B Maxillary down-grafting and surgical expansion are the least stable movements. (Proffit Ch 19; Hbk 27.4)

918 E Stability depends on direction, then fixation, then technique, in that order. Very stable means about a 90 per cent chance of no significant change. (Proffit Ch 19; Hbk 27.4)

919 A Condylar resorption is a recognised risk particularly in young women with a high-angle pattern and pre-existing joint disease, and must be consented for. (Proffit Ch 19; Hbk 27.5)

920 C The sequence is corticotomy, latency of about 5 to 7 days, distraction at approximately 1 mm per day, then consolidation. (Proffit Ch 19; Hbk 27.3)

921 B Joint assessment with the patient's own concern recorded first is the correct first step, and the decision to operate is made at the planning stage. (Proffit Ch 19; Hbk 27.2)

922 D Body dysmorphic disorder affects up to 17 per cent of those seeking orthognathic surgery and must be screened for. (Dons 2022; Hbk 27.5)

923 A Comprehensive treatment corrects the malocclusion and requires the full workup. Adjunctive treatment serves a larger restorative or periodontal goal. (Proffit Ch 18; Hbk 27.1)

924 C Movement is slower in adults, hyalinisation is more prolonged, and treatment takes longer than in an adolescent. (Companion Ch 13; Hbk 27.1)

925 E The restorative endpoint should be agreed first so that the tooth movement is designed to reach it. (Proffit Ch 18; Hbk 27.1)

926 A Porcelain requires surface preparation and silane coupling. Bands are an alternative where bonding is unreliable. (Proffit Ch 18; Hbk 27.1)

927 C Camouflage should be offered honestly with its limitations stated, and the discussion recorded. Decompensation must not be started for a plan the patient has declined. (GDC Standards; Hbk 27.2)

928 E Surgically assisted rapid palatal expansion, or bone-borne expansion, is indicated beyond approximately 5 mm in the non-growing patient. (Companion Ch 13; Hbk 24.2 and 27.3)

929 B The lower centre line is held on the chin during pre-surgical orthodontics, since the mandible will be moved as a unit at surgery. (Cobourne cases; Hbk 27.2)

930 D Post-surgical orthodontics typically takes 4 to 6 months to settle the occlusion, detail and finish. (Proffit Ch 19; Hbk 27.2)

931 C Anterior maxillary segmental osteotomy is described by Wunderer and the mandibular equivalent by Kole; both are used for bimaxillary protrusion and open bite correction. (Proffit Ch 19; Hbk 27.3)

Scored under 70% on this block? Re-read Revision Handbook §27, then sit it again in a fortnight.

Block AB — Cleft lip and palate and craniofacial syndromes (Q932–961 · Handbook §28)

932 A Lip repair is performed at 3 to 6 months, and palate repair at 6 to 12 months, before speech develops. (Companion Ch 16; Hbk 28.2)

933 E The rule of tens is 10 weeks of age, 10 pounds in weight and a haemoglobin of 10 grams per decilitre. (Companion Ch 16; Hbk 28.2)

934 D Alveolar bone grafting is performed at 8 to 10 years, before the canine erupts, when about two thirds of its root is formed. (Companion Ch 16; Hbk 28.2)

935 B The graft is placed before the canine erupts, when approximately two thirds of its root is formed, so that the canine erupts into grafted bone. (Companion Ch 16; Hbk 28.2)

936 E Expand before grafting: the graft needs access and the arch needs form. (Mosby ch. on clefts; Hbk 28.2)

937 B Cancellous bone, usually harvested from the iliac crest, is the standard graft material. (Companion Ch 16; Hbk 28.2)

938 D The graft does not correct a skeletal Class III relationship; that requires orthognathic surgery once growth is complete. (Companion Ch 16; Hbk 28.2)

939 A The missing lateral incisor on the cleft side is the commonest anomaly, alongside supernumerary and malformed teeth, often on both sides of the cleft. (Companion Ch 16; Hbk 28.3)

940 C The central incisor adjacent to the cleft is characteristically rotated and tipped towards the cleft, and is often hypoplastic. (Companion Ch 16; Hbk 28.3)

941 D Dental anomalies are commoner in bilateral cases. (Cobourne cases; Hbk 28.3)

942 B Hearing must be monitored, since middle ear effusion is near-universal in cleft palate, and grommets are inserted as required. (Companion Ch 16; Hbk 28.2)

943 E Velopharyngeal incompetence may require a pharyngoplasty, following speech assessment and therapy. (Companion Ch 16; Hbk 28.2)

944 A Both involve FGFR2 mutations with craniosynostosis and midface hypoplasia. Syndactyly distinguishes Apert. (Companion Ch 16; Hbk 28.4)

945 C Pierre Robin is a sequence: the small mandible causes the glossoptosis and the cleft palate, and airway obstruction is the principal early concern. (Companion Ch 16; Hbk 28.4)

946 B Treacher Collins syndrome results from TCOF1 mutation, and is autosomal dominant. (Companion Ch 16; Hbk 28.4)

947 D Downward-sloping palpebral fissures, malar and mandibular hypoplasia, ear anomalies and conductive deafness are characteristic, with cleft palate in about 30 per cent. (Companion Ch 16; Hbk 28.4)

948 A Cleidocranial dysplasia, from RUNX2 mutation, produces multiple supernumeraries and failure of eruption, along with absent or hypoplastic clavicles and delayed fontanelle closure. (Companion Ch 16; Hbk 28.4)

949 C The Pruzansky classification grades the mandibular ramus in hemifacial microsomia. (Companion Ch 16; Hbk 28.4)

950 E Epibulbar dermoid and vertebral anomalies distinguish Goldenhar from isolated hemifacial microsomia. (Companion Ch 16; Hbk 28.4)

951 A Examiners use both terms in the same paper. A Turner tooth is a localised hypoplasia; Turner syndrome is a chromosomal disorder. (Companion Ch 16; Hbk 28.4)

952 C Midface hypoplasia with relative mandibular prognathism, macroglossia, hypodontia, delayed eruption and periodontal disease are characteristic, alongside cardiac anomalies. (Companion Ch 16; Hbk 28.4)

953 E The 22q11 deletion, also known as DiGeorge or velocardiofacial syndrome, includes cleft palate or velopharyngeal incompetence, cardiac anomalies, a characteristic facies and learning difficulties. (Companion Ch 16; Hbk 28.4)

954 B Veau described four classes: soft palate only, soft and hard palate, unilateral complete lip and palate, and bilateral complete. (Companion Ch 16; Hbk Part 2 Card 5)

955 D LAHSHAL records lip, alveolus, hard palate and soft palate on both sides, giving a reproducible description of the cleft. (Companion Ch 16; Hbk Part 2 Card 5)

956 C Centralisation of cleft care with multidisciplinary team working and national audit measurably improved outcomes. (Companion Ch 16; Hbk 28.2)

957 A Maxillary hypoplasia is common in cleft patients, and distraction osteogenesis is used for large advancements. (Companion Ch 16; Hbk 28.2)

958 E Retention is prolonged and permanent retention is usual, because of scarring, the transverse deficiency and the tendency to relapse. (Companion Ch 16; Hbk 28.3)

959 D Presurgical orthopaedics is used in some centres but the evidence remains contested. (Companion Ch 16; Hbk 28.2)

960 B Recurrence risk rises with the number of affected relatives, reflecting the multifactorial aetiology. (Companion Ch 16; Hbk 28.1)

961 E Maternal smoking, alcohol, folate deficiency, anticonvulsants, retinoids, corticosteroids and maternal diabetes all increase risk; folic acid reduces it. (Companion Ch 16; Hbk 28.1)

Scored under 70% on this block? Re-read Revision Handbook §28, then sit it again in a fortnight.

Block AC — Risks of orthodontic treatment and the medical history (Q962–991 · Handbook §29)

962 B Dental aesthetics and trauma prevention have strong evidence. Periodontal health, caries, speech and mastication all require qualifiers. (Companion Ch 2; Hbk 29.1)

963 D Bollen and Papageorgiou found a lack of evidence that orthodontic treatment improves periodontal health. (Bollen 2008; Hbk 29.1)

964 A Macey found weak evidence that crowding is a caries risk factor or that treatment reduces later caries incidence. (Macey 2020; Hbk 29.1)

965 C Demineralisation, root resorption, recession and relapse are the four that Cobourne's cases specify. (Cobourne cases; Hbk 29.2)

966 D Pain peaks at 48 hours and returns to baseline by 7 days. Analgesics may be taken preemptively. (Monk 2017; Hbk 29.2)

967 B Paracetamol is generally preferred, since prostaglandin inhibitors may slow tooth movement, although the effect is small and clinically debated. (Proffit Ch 8; Hbk 29.2)

968 E Sameshima and Sinclair reported severe root resorption in 1 to 5 per cent of treated teeth, greatest in the anterior maxilla. (Sameshima and Sinclair 2004; Hbk 29.2)

969 A Previous trauma and previously identified resorption are among the factors that should be identified before treatment, alongside abnormal root morphology. (Companion Ch 6; Hbk 29.2)

970 C Gingival hyperplasia is common with poor hygiene and usually recovers after debond, so excision is reserved for persistent cases. (Companion Ch 2; Hbk 29.2)

971 B NICE advises against routine antibiotic prophylaxis solely to prevent infective endocarditis, although this conflicts with AHA and ESC guidance for the highest-risk patients. (NICE; Thornhill 2016; Hbk 29.3)

972 D Mortality from infective endocarditis is approximately 21 per cent, which is why the NICE position remains contested. (Companion Ch 7; Hbk 29.3)

973 A Removable appliances should be avoided in epilepsy because of the risk of displacement and inhalation during a seizure. (Companion Ch 7; Hbk 29.3)

974 C Gingival hyperplasia associated with phenytoin and with transplant immunosuppression makes bonded tubes preferable to bands. (Companion Ch 7; Hbk 29.3)

975 E Palate-covering removable appliances should be avoided in asthma. (Companion Ch 7; Hbk 29.3)

976 A Zymperdikas found that bisphosphonates slow or inhibit tooth movement and delay eruption. Aims should be kept simple and stopping the drug does not help. (Zymperdikas 2020; Hbk 29.3)

977 C Light forces and avoidance of extractions where possible are appropriate in sickle cell disease, thalassaemia, childhood malignancy and diabetes. (Companion Ch 7; Hbk 29.3)

978 E Elective orthodontic treatment should be deferred for at least 6 months after transplantation, and until malignancy is in remission. (Companion Ch 7; Hbk 29.3)

979 B Nickel titanium should be avoided. Stainless steel is safe because the nickel is bound within the austenitic lattice. (Chakravarthi 2012; Hbk 29.3)

980 D Elective treatment is deferred. Where urgent care is required, the second trimester is preferred, and radiographs are taken only where justified. (Companion Ch 7; Hbk 29.3)

981 C Patient selection and deferral until hygiene is adequate is the single most effective measure, since demineralisation in elective treatment is foreseeable. (Companion Ch 2; Hbk 29.2)

982 A Daily high-fluoride toothpaste plus professional high-concentration fluoride every 6 to 8 weeks is the recommended regimen. (Benson 2019; Hbk 29.2)

983 E Joss-Vassalli, Aziz and Tepedino associate recession more with retroclination in high-angle Class III cases than with proclination. (Joss-Vassalli 2010; Hbk 29.2)

984 D Many early lesions arrest and remineralise superficially. Explanation, recording, intensive prevention and review precede any restorative intervention. (Companion Ch 2; Hbk 29.2)

985 B Klaus reported high discontinuation rates in children and adolescents, especially with removable appliances. (Klaus 2017; Hbk 29.2)

986 E The warning, the two mandatory safety mechanisms and the instruction to stop and attend if the bow disengages must all be recorded. (Samuels 1996; Hbk 29.2)

987 B A small but real thickness of enamel is lost, which is minimised by using a tungsten carbide bur at low speed and polishing. (Companion Ch 10; Hbk 29.2)

988 **D** Failure to achieve the aim is a real and consentable risk, and the discussion should include what would be done in that event. (GDC Standards; Hbk 29.2)

989 **A** Latex-free elastics, gloves and dam are required. Latex allergy affects approximately 4.3 per cent of the general population. (Wu 2016; Hbk 29.3)

990 **C** Elective treatment is deferred while the disorder is active, with liaison, preventive care and management of erosion in the interim. (Companion Ch 7; Hbk 29.3)

991 **D** Javidi reported improvement in the emotional and social domains of OHRQoL, while Kenealy found little objective evidence of long-term psychological benefit. (Javidi 2017; Kenealy 2007; Hbk 29.1)

Scored under 70% on this block? Re-read Revision Handbook §29, then sit it again in a fortnight.

Block AD — Orthodontic practice, infection control and governance (Q992–1016 · Handbook §30)

992 **B** The sequence is clean, inspect, sterilise, store and track. Cycles must be validated and recorded daily. (Local decontamination guidance; Hbk 30.1)

993 **E** Steam sterilisation is performed at 134 to 137 degrees Celsius, with daily validation and recording of cycles. (Local decontamination guidance; Hbk 30.1)

994 **A** Single-use items must never be reprocessed. This includes prophylaxis cups and brushes, saliva ejectors, needles and cartridges. (Local decontamination guidance; Hbk 30.1)

995 **C** Bands removed from stock cannot be returned. A dedicated band-sizing kit, sterilised between patients, is used instead. (Local decontamination guidance; Hbk 30.1)

996 **B** Impressions and appliances must be rinsed, disinfected and labelled both before dispatch and on receipt, and the process documented. (Local decontamination guidance; Hbk 30.1)

997 **D** Debonding and clean-up generate aerosol, requiring high-volume aspiration and eye protection for both patient and operator. (Local decontamination guidance; Hbk 30.1)

998 **A** Wire ends are sharps. A distal end cutter that retains the fragment should be used and every wire end accounted for. (Local decontamination guidance; Hbk 30.1)

999 **C** Justification for the individual patient, optimisation of dose and limitation of the number of exposures govern every radiograph. (Companion Ch 7; Hbk 30.2)

1000 **E** Stochastic risk means there is no safe threshold. Every exposure must be individually justified. (Companion Ch 7; Hbk 30.2)

1001 **A** The practitioner justifies the exposure, the referrer requests it, the operator carries it out, and the legal person holds overall responsibility. (Ionising radiation regulations; Hbk 30.2)

1002 **C** Both the justification and the report must be recorded. A radiograph taken is a radiograph that must be reported. (Companion Ch 7; Hbk 30.2)

1003 **E** A signed form is evidence of consent, not consent itself. The content of the discussion must be recorded. (GDC Standards; Hbk 30.3)

1004 **B** Contemporaneous notes must record the clinical detail, the hygiene status, breakages, instructions and the appointment interval. (GDC Standards; Hbk 30.3)

1005 **D** PAR or ICON is used to audit an operator or service over many cases. IOTN measures need, not outcome. (Companion Ch 2; Hbk 30.3)

1006 **C** Clinical audit covers case outcomes, failure and breakage rates, decalcification, duration, discontinuation, radiograph rejects and decontamination records. (GDC Standards; Hbk 30.3)

1007 **A** Severe skeletal discrepancies, likely surgical cases, complex impactions, cleft and craniofacial cases, significant asymmetry, severe hypodontia and reduced periodontium cases should be referred. (Hbk 30.4)

1008 **E** Volunteering the boundary before being asked is the single most reliable way of demonstrating judgement. (Hbk 30.4 and Part 5)

1009 **D** The complaint in the patient's own words, the history, the examination in three planes, IOTN, the provisional problem list and the records justified and taken must all be recorded. (GDC Standards; Hbk 30.3)

1010 **B** A material change of plan requires the consent discussion to be repeated and both the reason and the discussion recorded. (GDC Standards; Hbk 30.3)

1011 **E** Dental unit water lines require flushing, treatment and monitoring in line with local guidance, with a Legionella risk assessment. (Local decontamination guidance; Hbk 30.1)

1012 **B** Clinical, sharps, amalgam and domestic waste must be segregated, with consignment notes retained. (Local decontamination guidance; Hbk 30.1)

1013 **D** A disposable or sterilisable tip sleeve is used and the wand wiped between patients, in accordance with the manufacturer's instructions. (Local decontamination guidance; Hbk 30.1)

1014 **A** Final records, retention issued, instructions given and review arranged must all be recorded at debond. (GDC Standards; Hbk 30.3)

1015 **C** A quality assurance programme with reject audit, staff training and dose monitoring is required. (Ionising radiation regulations; Hbk 30.2)

1016 **D** Recognising the limit, explaining it, recording the discussion and referring is the required response, and it is what the examiners are testing. (GDC Standards; Hbk 30.4)

Scored under 70% on this block? Re-read Revision Handbook §30, then sit it again in a fortnight.

Answer key grid

Ten answers per row. Mark a block, then read the explanations.

	1	2	3	4	5	6	7	8	9	10
1	A	C	E	B	D	C	A	E	D	B
11	E	B	D	A	C	D	B	E	A	C
21	B	D	A	C	E	A	C	E	B	D
31	C	A	E	D	B	E	B	D	A	C
41	D	B	E	A	C	B	D	A	C	E
51	A	C	E	B	D	C	A	E	D	B
61	E	B	D	A	C	D	B	E	A	C
71	B	D	A	C	E	A	C	E	B	D
81	C	A	E	D	B	E	B	D	A	C
91	D	B	E	A	C	B	D	A	C	E
101	A	C	E	B	D	C	A	E	D	B
111	E	B	D	A	C	D	B	E	A	C
121	B	D	A	C	E	A	C	E	B	D
131	C	A	E	D	B	E	B	D	A	C
141	D	B	E	A	C	B	D	A	C	E
151	A	C	E	B	D	C	A	E	D	B
161	E	B	D	A	C	D	B	E	A	C
171	B	D	A	C	E	A	C	E	B	D
181	C	A	E	D	B	E	B	D	A	C
191	D	B	E	A	C	B	D	A	C	E
201	A	C	E	B	D	C	A	E	D	B
211	E	B	D	A	C	D	B	E	A	C
221	B	D	A	C	E	A	C	E	B	D
231	C	A	E	D	B	E	B	D	A	C
241	D	B	E	A	C	B	D	A	C	E
251	A	C	E	B	D	C	A	E	D	B
261	E	B	D	A	C	D	B	E	A	C
271	B	D	A	C	E	A	C	E	B	D
281	C	A	E	D	B	E	B	D	A	C
291	D	B	E	A	C	B	D	A	C	E
301	A	C	E	B	D	C	A	E	D	B
311	E	B	D	A	C	D	B	E	A	C
321	B	D	A	C	E	A	C	E	B	D
331	C	A	E	D	B	E	B	D	A	C
341	D	B	E	A	C	B	D	A	C	E
351	A	C	E	B	D	C	A	E	D	B
361	E	B	D	A	C	D	B	E	A	C
371	B	D	A	C	E	A	C	E	B	D
381	C	A	E	D	B	E	B	D	A	C
391	D	B	E	A	C	B	D	A	C	E
401	A	C	E	B	D	C	A	E	D	B
411	E	B	D	A	C	D	B	E	A	C
421	B	D	A	C	E	A	C	E	B	D
431	C	A	E	D	B	E	B	D	A	C
441	D	B	E	A	C	B	D	A	C	E
451	A	C	E	B	D	C	A	E	D	B

	1	2	3	4	5	6	7	8	9	10
461	E	B	D	A	C	D	B	E	A	C
471	B	D	A	C	E	A	C	E	B	D
481	C	A	E	D	B	E	B	D	A	C
491	D	B	E	A	C	B	D	A	C	E
501	A	C	E	B	D	C	A	E	D	B
511	E	B	D	A	C	D	B	E	A	C
521	B	D	A	C	E	A	C	E	B	D
531	C	A	E	D	B	E	B	D	A	C
541	D	B	E	A	C	B	D	A	C	E
551	A	C	E	B	D	C	A	E	D	B
561	E	B	D	A	C	D	B	E	A	C
571	B	D	A	C	E	A	C	E	B	D
581	C	A	E	D	B	E	B	D	A	C
591	D	B	E	A	C	B	D	A	C	E
601	A	C	E	B	D	C	A	E	D	B
611	E	B	D	A	C	D	B	E	A	C
621	B	D	A	C	E	A	C	E	B	D
631	C	A	E	D	B	E	B	D	A	C
641	D	B	E	A	C	B	D	A	C	E
651	A	C	E	B	D	C	A	E	D	B
661	E	B	D	A	C	D	B	E	A	C
671	B	D	A	C	E	A	C	E	B	D
681	C	A	E	D	B	E	B	D	A	C
691	D	B	E	A	C	B	D	A	C	E
701	A	C	E	B	D	C	A	E	D	B
711	E	B	D	A	C	D	B	E	A	C
721	B	D	A	C	E	A	C	E	B	D
731	C	A	E	D	B	E	B	D	A	C
741	D	B	E	A	C	B	D	A	C	E
751	A	C	E	B	D	C	A	E	D	B
761	E	B	D	A	C	D	B	E	A	C
771	B	D	A	C	E	A	C	E	B	D
781	C	A	E	D	B	E	B	D	A	C
791	D	B	E	A	C	B	D	A	C	E
801	A	C	E	B	D	C	A	E	D	B
811	E	B	D	A	C	D	B	E	A	C
821	B	D	A	C	E	A	C	E	B	D
831	C	A	E	D	B	E	B	D	A	C
841	D	B	E	A	C	B	D	A	C	E
851	A	C	E	B	D	C	A	E	D	B
861	E	B	D	A	C	D	B	E	A	C
871	B	D	A	C	E	A	C	E	B	D
881	C	A	E	D	B	E	B	D	A	C
891	D	B	E	A	C	B	D	A	C	E
901	A	C	E	B	D	C	A	E	D	B
911	E	B	D	A	C	D	B	E	A	C
921	B	D	A	C	E	A	C	E	B	D
931	C	A	E	D	B	E	B	D	A	C
941	D	B	E	A	C	B	D	A	C	E
951	A	C	E	B	D	C	A	E	D	B
961	E	B	D	A	C	D	B	E	A	C
971	B	D	A	C	E	A	C	E	B	D

	1	2	3	4	5	6	7	8	9	10
981	C	A	E	D	B	E	B	D	A	C
991	D	B	E	A	C	B	D	A	C	E
1001	A	C	E	B	D	C	A	E	D	B
1011	E	B	D	A	C	D				

Mark sheet

One row per block. Record the date and the score out of the block total. Anything under 70 per cent sends you back to the Handbook section named in the last column before you sit it again.

Block	Subject	Total	1st: date / score	2nd	3rd	Hbk
A	Concepts of growth and development	25				§1
B	Craniofacial growth, prenatal and postnatal	30				§2
C	Development of the dentition	40				§3
D	The biology of tooth movement, eruption and root resorption	45				§4
E	Occlusion, malocclusion and aetiology	30				§5
F	Treatment need, indices and epidemiology	25				§6
G	Evidence-based orthodontics	30				§7
H	Ethics, consent, communication and the psychosocial patient	30				§8
I	Clinical examination and records	30				§9
J	Imaging and cephalometrics	40				§10
K	Study models, space analysis and the digital set-up	25				§11
L	Diagnosis, the problem list and treatment planning	40				§12
M	Extraction decisions and space management	40				§13
N	Biomechanics, force control and anchorage	51				§14
O	Orthodontic materials, wires, bonding and instruments	44				§15
P	Removable and functional appliances	41				§16
Q	Fixed appliances — labial, lingual and aligners	40				§17
R	Treatment mechanics: Stage I, II and III	35				§18
S	Retention, relapse and stability	35				§19
T	Interceptive orthodontics and the developing dentition	45				§20
U	Class I malocclusion and bimaxillary protrusion	25				§21
V	Class II malocclusion	40				§22
W	Class III malocclusion	30				§23
X	Transverse problems and facial asymmetry	25				§24
Y	Vertical problems: deep bite and open bite	30				§25
Z	TMD, condylar trauma, the airway and myofunctional therapy	30				§26
AA	Adult orthodontics and orthognathic surgery	30				§27
AB	Cleft lip and palate and craniofacial syndromes	30				§28

Block	Subject	Total	1st: date / score	2nd	3rd	Hbk
AC	Risks of orthodontic treatment and the medical history	30	_____	_____	_____	§29
AD	Orthodontic practice, infection control and governance	25	_____	_____	_____	§30

Part 3 — 207 viva questions with model answers

How to use this bank. Read the question, close the book, and answer it *out loud* to the end before reading the model. Answering in your head is not practice; the oral examination tests whether you can produce a structured spoken answer under pressure, and that is a separate skill from knowing the material.

The model answers are written in the first person, at the length and register of a good spoken answer — roughly twenty to forty seconds. They are not the only correct answers, and an examiner would accept a good deal of variation. What they are meant to demonstrate is *shape*: a direct opening sentence, the reasoning, the number where a number exists, and a close that says what you would actually do. Where a model answer names a study, name it or say nothing; a mis-attributed reference is worse than none.

The italic note beneath each answer is the examiner's point — what the question is really testing, or the trap it is set to catch.

The sections

Questions	Handbook	Section, and where it sits in *Notes In Orthodontics*
V1–V10	§1–§2	Growth, development and the craniofacial complex Chapters 3 and 4; Appendix A, Topic 1 background
V11–V20	§3	Development of the dentition Chapter 5; crash notebook *The arithmetic of the developing arch*
V21–V30	§4	The biology of tooth movement and root resorption Topic 1; Chapter 6; Appendix A, Topic 1
V31–V38	§5	Occlusion, aetiology and epidemiology Chapter 2
V39–V48	§6–§7	Treatment need, indices and the evidence base Chapters 1 and 2; Appendix D; crash notebook *Ten things the evidence overturned*
V49–V58	§8	Ethics, consent and communication Part V, unit V.1; Topic 9 in part; Chapter 7
V59–V68	§9–§10	Clinical examination and records Topic 2; Chapter 7; Appendix A, Topic 2
V69–V80	§11–§12	Space analysis, diagnosis and treatment planning Topic 3; Chapter 8; Appendix A, Topic 3
V81–V90	§13	Extraction decisions and space management Topic 4; Chapter 8; Appendix A, Topic 4
V91–V102	§14	Biomechanics and anchorage Topics 5 and 6; Chapters 6 and 11; Appendix A, Topic 5
V103–V111	§15	Materials, wires and bonding Topic 7; Chapter 10; Appendix A, Topic 7
V112–V121	§16–§17	Removable, functional and fixed appliances Topics 6 and 8; Chapters 9, 10 and 14; Appendix A, Topics 6 and 8
V122–V129	§18	Treatment mechanics and finishing

		Chapters 10 and 13; Supplements E and F
V130–V139	§19	Retention and stability
		Topic 20; Chapter 13; Appendix A, Topic 20
V140–V151	§20	Interceptive orthodontics and impacted teeth
		Topics 13, 14 and 15; Chapter 12; Appendix A, Topics 13–15
V152–V161	§22–§23	Class II and Class III management
		Topics 12 and 17; Chapters 9, 12 and 13; Appendix A, Topics 12 and 17
V162–V173	§24–§25	Transverse and vertical problems
		Topics 11, 16 and 19; Chapters 9, 12, 13 and 15
V174–V183	§26, §29	TMD, the airway and the medically complex patient
		Topics 10 and 18; Chapter 2, pp. 102–105; Chapter 7
V184–V193	§27, §28, §30	Adults, orthognathic surgery, clefts and governance
		Chapters 15 and 16; Part V, units V.2–V.4 for governance
V194–V207	Handbook Part 5	The questions that test judgement
		Crash notebook *The last hour*; the twelve traps

Growth, development and the craniofacial complex (V1–V10)

Notes In Orthodontics: Chapters 3 and 4; Appendix A, Topic 1 background
Supplements: Revision Handbook **§1–§2**
Practise this: 10 questions · answer each one aloud before reading the model

V1 Why does it matter whether this patient is still growing?

"Because it determines which of the three treatment routes is available to me. If the patient is growing I can attempt growth modification, overbite reduction by buccal segment extrusion is stable because condylar growth compensates, sutural expansion and facemask protraction are possible, and I can expect spontaneous improvement after well-timed extraction. If growth has finished, every one of those becomes either a dentoalveolar compromise or a surgical problem. So growth status is the first thing I establish and the last thing I return to.

The single most powerful discriminator in orthodontic planning. Say it in the first sentence of any planning answer.

V2 How would you assess whether a 12-year-old girl is at her growth spurt?

"I would use several indices rather than one. Clinically, serial standing height gives me height velocity, which is the direct measure, and I would ask about menarche, because menarche follows peak height velocity by about a year. If I already had a lateral cephalogram taken for another reason, I would stage the cervical vertebrae: CS3, at a mean age of 12, means the peak occurs during this year. What I would not do is take a radiograph purely to stage growth, and I would add that none of these indices is universally accepted.

Tests whether you will justify a radiograph. The examiner is listening for the caveat as much as for the staging.

V3 Why is chronological age of so little use in orthodontics?

"Because it correlates poorly with everything that matters. Two 11-year-olds can be two years apart in dental age and three years apart in skeletal age. Every clinically useful index of readiness is a maturational one: dental age from tooth formation, skeletal age from cervical vertebral maturation or a hand-wrist film, sexual maturation, and morphological age from height velocity.

A short answer. The mark is for naming the five ages, not for elaborating.

V4 Is the mandible's growth modifiable?

"Within limits, yes. The condyle is a secondary, adaptive cartilage rather than a primary growth centre, which is precisely why the functional environment can influence it. But the randomised evidence is that the effect is a few millimetres at best and largely transient. So my honest position is Petrovic's: growth is modifiable within genetically determined limits, and those limits are individual.

A deliberately open question. The examiner wants the qualification, not enthusiasm.

V5 What is the difference between a site and a centre of growth?

"A site is simply a location where growth occurs. A centre is a location where growth occurs independently, under its own genetic control. The distinction matters because the whole nasal-septum-versus-sutures argument turns on it: Scott held that cartilage was a true centre and the sutures merely responded, whereas Moss held that neither was a centre and that both responded to the functional matrix.

Precision of language. It is a two-sentence answer if you know it and a rambling one if you do not.

V6 Describe Bjork's three mandibular rotations.

"Total, or true, rotation is rotation of the mandibular body relative to the cranial base, and it can only be measured on implants. Matrix, or apparent, rotation is what a cephalogram shows without implants, which is rotation of the soft tissue matrix relative to the cranial base. Intramatrix rotation is the difference between them, and it represents angular remodelling of the lower border. Total equals matrix plus intramatrix.

Frequently asked, rarely answered cleanly. Learn the equation.

V7 What are the clinical consequences of a backward-rotating growth pattern?

"It is the pattern that will defeat me. Anterior face height grows more than posterior, so I get an increased lower face height and a tendency to anterior open bite. Overbite reduction by extrusion is unstable, anchorage is harder, and any open bite recurs. Practically it means no cervical headgear, no Class II elastics if I can avoid them, and no extrusive mechanics at all. I would want that pattern identified before I chose the mechanics, not after.

A grading question. Recognising the high-angle face and changing the mechanics is what separates a pass from a good pass.

V8 Where do clefts of the lip and of the palate arise developmentally?

"They are different failures at different times. Cleft lip, with or without the primary palate, is a failure of the maxillary process to fuse with the medial nasal process, and it happens by week 8. Isolated cleft palate is a failure of the palatine shelves to elevate above the tongue or to fuse, between weeks 6 and 10. The clearest evidence that they are distinct is that the sex ratios are opposite: cleft lip is male 2 to 1 and isolated cleft palate female 4 to 1.

The opposite sex ratios are the point. Give them unprompted.

V9 What continues to change in the face after the age of eighteen?

"Rather a lot, and slowly. Behrents showed continued rotational growth throughout adult life, in the individual's original pattern, and described the cumulative effect as surprisingly large. Practically it means continued uprighting of the lower labial segment, which is why late lower incisor crowding happens in untreated people too, continued vertical growth, and soft tissue change: the nose and upper lip lengthen and incisor show at rest reduces. It is also why an implant placed in a young adult can end infra-occluded.

Connects growth to retention, to late crowding and to implant timing in one answer. Examiners like that.

V10 Which theory of growth control would you defend, and why?

> *"The modern synthesis rather than any single classical theory. Moss's functional matrix explains the environmental influence and is the intellectual basis of functional appliances, but on its own it overstates the case. Petrovic's servosystem model is the most defensible: growth as a cybernetic system with local and systemic feedback, in which the condylar cartilage is a secondary and adaptive tissue. That formulation lets me say that growth is modifiable and that the modification is limited, which is what the evidence actually shows.*

Do not simply list the theories. Take a position and justify it.

Development of the dentition (V11–V20)

Notes In Orthodontics: Chapter 5; crash notebook *The arithmetic of the developing arch*
Supplements: Revision Handbook **§3**
Practise this: 10 questions · answer each one aloud before reading the model

V11 What are incisor liability and leeway space, and which is larger where?

"Incisor liability is the fact that the permanent incisors are wider than their predecessors, so there is a deficit to be met: about 7 mm in the maxilla and 5 mm in the mandible. Leeway space is the fact that the primary canine and molars are wider than the permanent canine and premolars, so there is an excess to be used: about 1.5 mm per quadrant in the maxilla and 2.5 mm in the mandible. Note the asymmetry: the larger liability is maxillary and the larger leeway is mandibular.

Two numbers do most of the work in this chapter. Get the asymmetry the right way round.

V12 How is the incisor liability met?

"Three ways. First, the primate and developmental spacing already present in the primary dentition, which is why generalised spacing there is normal and necessary. Second, an increase in intercanine width, about 3 mm in the mandible and 4.5 mm in the maxilla. Third, the permanent incisors erupt more labially than their predecessors. If the primary dentition had no spacing at all, I would expect the permanent dentition to be crowded.

The question behind the question is whether you will reassure a parent about spaced primary teeth.

V13 A parent is worried that the lower permanent incisors are crowded in an 8-year-old. What do you say?

"That this is normal and I expect it to improve. Space for the mandibular incisors is negative for about two years after they erupt, so transient early crowding is the rule rather than the exception, and the leeway space becomes available as the primary molars are shed. I would reassure, review at a defined date rather than vaguely, and do nothing at this stage. What I would not do is expand the lower arch, because the mandibular arch is the template and lower expansion relapses.

Tests restraint. The wrong answer is an appliance.

V14 Why does the second premolar end up crowded in the mandible and the canine in the maxilla?

"Because the last tooth to erupt in each arch is the one that ends up short of space. In the mandible the sequence is canine, then first premolar, then second premolar, so the second premolar is left. In the maxilla it is first premolar, second premolar, then canine, so the canine is left. That is one reason the maxillary canine is the tooth most often displaced.

A simple mechanism that explains two clinical observations. Give the mechanism, not just the fact.

V15 What does a distal step terminal plane in the primary dentition tell you?

"That the molar relationship will be Class II, and that it will not self-correct. A mesial step becomes Class I. A flush terminal plane, which is the commonest at around 60 to 75 per cent, is a coin toss: it may correct to Class I using the leeway space or remain end to end. So the distal step is the one that gives me advance warning.

Examiners like this because it rewards understanding rather than recall of a percentage.

V16 You find a peg-shaped upper lateral incisor in a 10-year-old. What does that make you do?

"Look for the others. Dental anomalies co-occur far more often than chance, which reflects a shared genetic field effect. So I would palpate both canines in the buccal sulcus, check for hypodontia elsewhere, particularly the lower second premolars, look for infra-occluded primary molars, and consider a radiograph if the clinical picture warranted it. Anomalies travel in packs.

A one-line answer that demonstrates a habit of mind. It scores well.

V17 When would you extract a first permanent molar of poor prognosis?

"Ideally when the bifurcation of the second permanent molar is just calcifying, which is about 8 to 10 years of dental age, so that the second molar erupts into the space. Before I committed, I would confirm the presence and position of all the permanent teeth including the third molars, and I would consider whether a compensating extraction of the opposing molar was needed to prevent over-eruption. Spontaneous closure is more reliable in the maxilla, so timing matters most in the lower arch.

Three elements: the timing, the radiographic check, and the compensating extraction. Miss any one and the answer is incomplete.

V18 What is the difference between a balancing and a compensating extraction?

"A balancing extraction is removal of the contralateral tooth in the same arch, to prevent a centre line shift, and it is most relevant for canines. A compensating extraction is removal of the opposing tooth in the same quadrant, to preserve the buccal occlusion and prevent over-eruption, and it is considered mainly for first permanent molars. Neither is automatic; both are decisions I would justify from the centre line and the occlusion in the individual case.

Definitions plus the refusal to apply them mechanically. That last sentence is the mark.

V19 How does a tooth erupt?

"The dental follicle is the essential structure. It coordinates the osteoclastic resorption that creates the eruption pathway through bone, and the tooth follows. The evidence is elegant: remove the follicle and the tooth does not erupt, but remove the tooth and leave the follicle and a pathway still forms. Eruption begins at about two thirds to three quarters of root formation and continues to occlusal contact, then slowly thereafter to compensate for wear.

The follicle experiment is the detail that separates a good answer from an average one.

V20 A lower second molar has failed to erupt and does not respond to traction. What is your differential?

"Two possibilities that must be separated. Mechanical obstruction, usually from the third molar, does respond to intervention, and I would work up the escalating ladder from observation, through disimpaction and third molar removal, to surgical uprighting. Primary failure of eruption, which is a loss-of-function mutation in PTH1R, does not respond, and orthodontic traction on such a tooth causes ankylosis. So I would not apply traction until I was reasonably confident which I was dealing with.

The trap is answering with mechanics. The examiner is testing whether you know when not to pull.

The biology of tooth movement and root resorption (V21–V30)

Notes In Orthodontics: Topic 1; Chapter 6; Appendix A, Topic 1
Supplements: Revision Handbook **§4**
Practise this: 10 questions · answer each one aloud before reading the model

V21 Why does an ankylosed tooth not move, and what does that tell you?

"Because orthodontic tooth movement is mediated by the periodontal ligament, and an ankylosed tooth has none. The same is true of an osseointegrated implant. That single observation is the cleanest available proof that the site of the response is the ligament and not bone alone, and it is why I would open any answer on this topic with biology rather than mechanics: I apply force, but the movement is produced by remodelling of the ligament and the alveolar bone.

The classic opening question. Lead with the PDL and the rest of the viva is easier.

V22 What is hyalinisation, and why does it delay tooth movement?

"It is a cell-free, glass-like zone of necrotic tissue that forms where the ligament is compressed and the blood supply is cut off. Osteoclasts cannot resorb bone beneath it, so direct frontal resorption stops and the tooth cannot move until the necrotic tissue is removed by undermining resorption from the adjacent marrow spaces. That process takes 20 to 30 days, which is the lag phase. Reitan showed it occurs even at 30 grams, and more with tipping than with bodily movement, so it is not simply a complication of heavy force.

The trap is calling hyalinisation a complication of excessive force. It happens at 30 g.

V23 If lighter forces are biologically preferable, why not use the lightest possible force?

"Because below a threshold nothing happens: the response is a dose-response curve with a plateau, as Quinn and Yoshikawa showed, not a straight line. And what matters is not the absolute force but the differential stress generated in the ligament, which depends on the root surface area loaded and on the type of movement. That is why intrusion needs 10 to 20 grams per tooth, since the force is dissipated through the small apical area, while bodily movement needs 50 to 100 centinewtons. The honest position is that no universal optimum exists.

Tests whether you will over-simplify. The examinable position is that there is no single number.

V24 Describe the RANKL and osteoprotegerin axis.

"RANKL is expressed by osteoblasts, osteocytes and periodontal ligament cells, and it binds RANK on pre-osteoclasts. It is essential for osteoclastic resorption: delete it in transgenic mice and tooth movement stops. Osteoprotegerin is a soluble decoy receptor that binds RANKL and prevents it reaching RANK, and gene transfer of osteoprotegerin reduces tooth movement. Clinically, RANKL rises on the compression side and osteoprotegerin on the tension side. RANKL peaks at three days and returns to baseline between days six and 58, which implies other pathways sustain the post-lag phase.

Reversing RANKL and OPG is one of the twelve costly slips. Say compression and tension explicitly.

V25 How would you explain the risk of root resorption to a patient at consent?

> *"I would say that some root shortening happens in most people treated with fixed appliances, that it is usually a fraction of a millimetre and of no consequence, but that in about one to five per cent of teeth it exceeds a third of the root, and that it is commonest in the upper front teeth. I would then say what I do about it: light forces, keeping treatment as short as I can, and a radiograph part-way through so that I can pause treatment for three to six months if I see anything developing. And I would record that I had said it.*

Quantified, honest, and closes with the mitigation and the record. That is the shape of a good consent answer.

V26 Which risk factors for root resorption are actually supported by evidence?

> *"Only heavy force has substantive evidence, from Weltman's work. Everything else on the usual list is suggested rather than established: continuous rather than intermittent force, intrusion, the total amount of movement, Class II elastics, abnormal root morphology, treatment duration, roots driven into cortical bone, allergy and asthma, and teeth without occlusal contact. I would mention previous trauma and previously identified resorption as the two I would look for before starting.*

Separating what is proved from what is asserted. That distinction is itself the mark.

V27 What happens if you apply a force greater than the anchorage unit's threshold?

> *"The anchorage unit moves. That is Newton's third law and there is no way around it: the reaction has to go somewhere, and if the force exceeds what the anchor unit will resist, the space closes from the wrong end. In practice I would see the molars coming forward, the overbite deepening and the centre line shifting. It is also why heavier forces are not simply faster: they hyalinise the ligament on the active side and they move the anchorage on the reactive side.*

Links force levels to anchorage. A complete answer joins the two rather than treating them separately.

V28 Do acceleration techniques work?

> *"Not in the sense patients mean. Corticotomy, micro-osteoperforation and piezocision can increase the local rate through the regional acceleratory phenomenon, but the effect appears short-lived and there is sparse evidence that any of them shortens overall treatment time. El-Angbawi found the same for photobiomodulation: it may accelerate discrete phases but does not influence overall treatment time. What actually shortens treatment is good bracket positioning, an efficient wire sequence, light continuous force, avoiding breakages, and the patient attending.*

A commercially attractive claim with weak evidence. The examiner wants the last sentence.

V29 How does periodontal bone loss change the mechanics?

> *"It moves the centre of resistance apically. The bracket is where it always was, so the moment arm is now longer, and the same force produces a bigger moment and more tipping. So I would use lighter forces and aim for a higher moment-to-force ratio to get controlled movement, and I would want the periodontal disease treated and stable before I placed anything. The reduced periodontium changes the mechanics, not merely the risk.*

That final sentence is the one examiners are listening for.

V30 What is the difference between frontal and undermining resorption, and which do you want?

> *"Frontal resorption is direct resorption of the bone surface adjacent to the compressed ligament by osteoclasts recruited from the ligament itself. It is what happens with appropriate force, and it is what I want. Undermining resorption proceeds from the adjacent marrow spaces because a hyalinised zone prevents osteoclasts acting at the surface. It is slower, it produces the lag phase, and it is what heavy force forces me to accept. Proffit frames the whole mechanism as the choice between the two.*

A clean conceptual answer that shows you understand why light force is preferred.

Occlusion, aetiology and epidemiology (V31–V38)

Notes In Orthodontics: Chapter 2
Supplements: Revision Handbook §5
Practise this: 8 questions · answer each one aloud before reading the model

V31 What are the limitations of Angle's classification?

"It assumes the first permanent molar is in a stable and correct anteroposterior position, which is often untrue, and it cannot be used at all if that molar is missing or displaced. It describes only the anteroposterior plane, so it ignores the vertical and the transverse. And it says nothing about the skeletal pattern. That is why the British Standards incisor classification is used alongside it, and why I would describe a malocclusion in all three planes rather than by class alone.

Tests whether you will describe a malocclusion properly rather than label it.

V32 What proportion of malocclusion has a specific identifiable cause?

"Five per cent or fewer. That short list is the syndromes and single-gene disorders, embryological defects such as clefting, trauma including condylar fracture and incisor avulsion, pathology such as cysts and tumours, functional deviations, and local dental causes such as early tooth loss. Everything else is multifactorial, the product of a genetically determined skeletal pattern and tooth size interacting with the soft tissue environment.

A number and a list. Both are examinable.

V33 Explain equilibrium theory and why it matters clinically.

"Teeth occupy the position where the resting soft tissue forces balance. The critical point is that it is light, sustained forces that move teeth, not heavy brief ones, so resting lip and tongue posture matters and chewing does not. Clinically it explains why a lower lip trap perpetuates a Class II division 1, why a tooth moved outside the neutral zone returns towards it, and therefore why lower incisor advance of more than about two millimetres very probably needs permanent retention.

Links aetiology to retention. The best answers in orthodontics usually connect two chapters.

V34 How would you describe an overbite properly?

"In three respects. Whether it is increased, average or reduced, remembering that normal is 2 to 4 millimetres or about a third to a half of the lower incisor crown. Whether it is complete, to tooth or to tissue, or incomplete. And whether it is traumatic, which means looking specifically at the palatal mucosa behind the upper incisors and at the labial gingiva of the lower incisors. All three go in the notes.

A pure examination-technique answer. Say all three or lose the mark.

V35 What is a pseudo-Class III?

"A case that looks Class III in intercuspation but manipulates back to an edge-to-edge incisor relationship in centric relation, because there is an anterior displacement on closing. The true relationship is Class I or mild Class III with a displacement, and the prognosis is entirely different from a true skeletal Class III. It is the reason I seat every patient into centric relation before I diagnose anything, and the reason the path of closure is one of the most important findings I can record.

If you do not mention centric relation unprompted in a Class III question, expect to be asked this.

V36 Does the airway cause long face syndrome?

"The association is real but weak, and causation is not established. Experimental nasal obstruction in primates produced facial change, but the human evidence is observational and confounded. So what I would say is that chronic nasal obstruction is associated with mouth breathing, a low tongue posture and lips apart, and with a narrower maxilla and increased face height; but I would not assert that it causes it, and I would not offer rapid maxillary expansion as a treatment for breathing, because that is not supported.

Stating an association without asserting a cause is a mark of a careful candidate.

V37 How common is Class III, and does that vary?

"In United States adolescent samples it is about 5 to 6 per cent, and the worldwide figure is about 6.8 per cent, but the regional variation is striking: it is considerably more common in East Asian populations, and in those populations maxillary retrusion is the more frequent component rather than mandibular prognathism. That matters practically, because it changes the balance between protraction and setback.

Epidemiology with a clinical consequence attached. That is what earns the mark.

V38 What are Andrews' six keys, and what were they derived from?

"Correct molar relationship, correct crown angulation, correct crown inclination, no rotations, tight contacts with no spaces, and a flat or minimal curve of Spee. He derived them from 120 non-orthodontic normal occlusions, which is the point: they describe what an ideal occlusion actually looks like in nature, and that is what made the pre-adjusted prescription possible seven years later. A seventh key covering crown size and shape is often added.

The sample size and the link to the prescription are the details that mark a prepared candidate.

Treatment need, indices and the evidence base (V39–V48)

Notes In Orthodontics: Chapters 1 and 2; Appendix D; crash notebook *Ten things the evidence overturned*
Supplements: Revision Handbook **§6–§7**
Practise this: 10 questions · answer each one aloud before reading the model

V39 How does IOTN work, and what is the commonest error in applying it?

"It has two independent components. The Dental Health Component grades 1 to 5 and is hierarchical: you work down the MOCDO list, which is missing teeth, overjet, crossbite, displacement of contact points and overbite, and record only the single worst trait. The commonest error is to treat it as cumulative and add traits together. The Aesthetic Component is a separate ten-photograph scale, and the photograph is chosen for the degree of aesthetic impairment rather than for morphological similarity.

Hierarchical, not cumulative. That is the whole question.

V40 What is the difference between IOTN and PAR?

"IOTN measures need for treatment, before treatment, and it is hierarchical. PAR measures the outcome of treatment, is scored from casts before and after, is accumulative and has no maximum. A reduction of more than 30 per cent counts as improved and 22 points or more as greatly improved. PAR is a coarse instrument best used to audit an operator or a service over many cases, rather than to judge a single case.

Confusing the two is a common and avoidable error.

V41 Which index detected the difference between aligners and fixed appliances?

"The ABO Objective Grading System, not PAR. PAR shows no difference between them, whereas ABO-OGS shows aligners significantly worse, particularly for buccolingual inclination, occlusal contacts and relationship, overjet and root angulation. PAR is the coarser measure and the two do not correlate, which is itself worth knowing.

A favourite question because it rewards understanding of what an index actually measures.

V42 Name three orthodontic beliefs that the evidence has overturned.

"Early Class II treatment does not improve the final result: O'Brien's Manchester trial in 2009 found no long-term difference, with the late group finishing slightly better. Self-ligation is not faster: Scott and Fleming found no difference in rate, treatment time, visits or occlusal outcome. And orthodontics neither causes nor cures temporomandibular disorder: the American Association of Orthodontists studies of 1992. I would add that third molars are not a primary cause of late lower incisor crowding, and that interceptive canine extraction has very weak evidence.

Naming the study is the fastest way to demonstrate scholarship. Cite accurately or not at all.

V43 What does an I squared of 80 per cent tell you about a meta-analysis?

> *"That there is high heterogeneity: 80 per cent of the variability between studies is attributable to real differences between them rather than to chance. The conventional thresholds are 25 per cent low, 50 per cent moderate and 75 per cent high. Practically it makes me cautious about the pooled estimate, and it makes me want to know what differed between the trials — population, intervention, outcome measure or follow-up.*

Do not just recite the thresholds. Say what you would do with the information.

V44 A colleague tells you a new bracket system reduces treatment time by four months. How do you respond?

> *"By asking what the evidence is. Penning's randomised trial of a customised appliance found comparable outcomes with no reduction in treatment time, more breakages, more complaints and more planning time, and found that treatment time was dictated by case difficulty and by the treating orthodontist rather than by the appliance. I would want to see a randomised comparison with a similar case mix before I accepted the claim, and I would ask who funded it.*

Tests scepticism without cynicism. The named trial does the work.

V45 What is intention-to-treat analysis and why does it matter?

> *"It means analysing participants in the group to which they were randomised, whatever treatment they actually received. It matters because it preserves the benefit of randomisation: if you analyse only those who completed the treatment, you have reintroduced selection bias through the back door, and in orthodontics the people who drop out are systematically different from those who do not. It gives the more conservative and the more realistic estimate of effect.*

The orthodontic angle — dropouts are not random — is what lifts this above a textbook definition.

V46 Why is blinding difficult in orthodontic trials, and what can be done?

> *"The operator obviously cannot be blinded to whether they have placed a functional appliance, and neither can the patient. What can and should be blinded is the outcome assessor, and that is the blinding that matters most, because it protects against detection bias in the measurement of the outcome. That is why orthodontic Cochrane reviews so often downgrade for risk of bias and for imprecision from small unblinded trials.*

Knowing which blinding is achievable, and why it matters, is more useful than lamenting that trials are unblinded.

V47 Where does a case series sit in the hierarchy of evidence?

> *"Below cross-sectional and cohort studies and above expert opinion. The critical limitation is that without a control group it cannot establish causation, only describe. That matters in orthodontics because a great deal of what is asserted about new appliances rests on case series, and a case series of consecutively treated successful cases tells you almost nothing about the failures.*

A short answer with one sharp observation is better than a long recitation of the hierarchy.

V48 How would you explain to a patient that the evidence for something is weak?

> *"Plainly, and without pretending otherwise. I would say that this is something we do because it seems reasonable and because experience supports it, but that it has not been tested properly, and*

> *then I would say what the alternatives are and which of them is easiest to reverse if it does not work. Patients handle uncertainty better than they handle being told something confidently that later turns out not to be true.*

The examiner is testing honesty and communication together. The reversibility point is the clinical one.

Ethics, consent and communication (V49–V58)

Notes In Orthodontics: Part V, unit V.1; Topic 9 in part; Chapter 7
Supplements: Revision Handbook **§8**
Practise this: 10 questions · answer each one aloud before reading the model

V49 What would you discuss when consenting a 13-year-old and her mother for fixed appliances?

"I would cover what I propose and why; all reasonable alternatives, including doing nothing and what happens if we do nothing; and the material risks. Those are demineralisation, which affects up to half of fixed appliance patients and can be visible within six months; root resorption, which exceeds a third of the root in one to five per cent of teeth; recession; relapse, and that retention is lifelong if the result is to be held; and pain, which peaks at 48 hours. I would cover the time commitment and the cost including replacement retainers, and the possibility that I do not achieve the aim. I would give it in writing, check understanding by asking her to tell me back what we had agreed, and record it. And I would satisfy myself that the child assents, not only that her mother consents.

A set-piece. Rehearse it until it is automatic and the rest of the viva is easier.

V50 What is the Montgomery standard?

"That the risks to be disclosed are those to which a reasonable patient in this patient's position would attach significance, rather than those a responsible body of clinicians would disclose. It shifted the test from a professional standard to a patient-centred one. Practically it means I have to think about what matters to this particular patient: a professional singer will attach significance to a transient speech disturbance that most patients would not.

The example is what shows you have understood it rather than memorised it.

V51 A parent consents but the child clearly does not want treatment. What do you do?

"I would not start. Legally the parent's consent may be sufficient, but clinically the child's assent is decisive, because a technically ideal plan the child will not wear is not a plan. I would explore why: often it is fear, or a friend's account, or simply that the parent wants this more than the child does. I would offer to see the child again, defer the decision, and record the discussion. Elective treatment on an unwilling child is bad orthodontics before it is anything else.

Tests the distinction between legal sufficiency and clinical wisdom.

V52 How do you assess capacity?

"Capacity is decision-specific and time-specific, so I assess it for this decision at this time rather than as a general attribute. The functional test is whether the person can understand the information, retain it long enough to decide, weigh it in the balance, and communicate the decision. In a child under 16 I would additionally consider Gillick competence, which asks whether they have sufficient understanding and intelligence to understand fully what is proposed.

Two frameworks, correctly distinguished. Do not conflate them.

V53 An adult self-refers for aligners, dissatisfied with two previous courses of treatment, and shows you extensive photographs of a discrepancy you cannot see. What do you do?

> *"I would not commit to treatment at that visit. Serial dissatisfaction with a defect that is invisible or trivial to me is a red flag for body dysmorphic disorder, which affects 5 to 13 per cent of orthodontic patients and up to 17 per cent of those seeking orthognathic surgery. I would explore what she expects treatment to achieve, ask what success would look like to her, and if I remained concerned I would say honestly that I did not think I could deliver what she was hoping for, and consider whether onward referral for psychological assessment was appropriate.*

The mark is for not treating. Recognising the red flags and declining is the safe answer.

V54 Is orthodontic treatment worth having? What are the benefits, honestly stated?

> *"The benefits are narrower than they are often claimed to be. Dental aesthetics and reduction in incisor trauma have strong evidence. Improvement in the emotional and social domains of oral health related quality of life has moderate evidence. Caries reduction, periodontal health, masticatory efficiency and speech all have weak evidence or none, and long-term psychological benefit has little objective evidence. I would say that plainly rather than overclaim, because overclaiming is both dishonest and, when the patient discovers it, corrosive.*

Overclaiming the benefits is one of the twelve traps. State the confidence level for each.

V55 A 14-year-old wants treatment but her plaque control is poor. What do you do?

> *"I would not start. Demineralisation affects up to half of fixed appliance patients, appears within six months, and in elective treatment I chose to begin, it is my responsibility. I would put a documented oral hygiene programme in place with a defined review, explain in writing that appliances will not be placed until plaque control is adequate, and re-assess. That is not a punishment; it is the single most effective preventive measure available to me.*

The wrong answer is to start and reinforce hygiene at each visit. Say why.

V56 How would you handle a patient who wants a treatment you think is wrong for them?

> *"I would find out first why they want it, because the reason is usually reasonable even when the request is not. Then I would explain plainly what I think it would and would not achieve in their case, offer what I would recommend and why, and give them the alternatives including doing nothing. If they still wanted it and I could deliver it safely and defensibly, I would; if I could not, I would say so, offer a second opinion, and record the whole discussion.*

Neither capitulation nor refusal. The examiner wants a process.

V57 When does consent need to be revisited during treatment?

> *"Whenever the plan changes materially. A decision to extract that was not part of the original plan, a decision to proceed to surgery, a change of appliance, or a decision to accept a compromise rather than the original aim all require the discussion to be repeated and recorded. Consent is a process rather than an event, and a form signed at the assessment visit does not cover a decision made eighteen months later.*

A frequently missed point, and a common source of complaint.

V58 What would make you decline to treat a patient?

> *"Inadequate plaque control that has not improved despite a documented programme; expectations I cannot meet, including the body dysmorphic picture; a case beyond my competence that I should refer rather than attempt; an active eating disorder or a medical situation where treatment should be deferred; and a patient who will not agree to the retention plan, because without retention the result is temporary and the treatment is not worth its risks.*

Five reasons, one of them about your own limits. That last is the one examiners are waiting for.

Clinical examination and records (V59–V68)

Notes In Orthodontics: Topic 2; Chapter 7; Appendix A, Topic 2
Supplements: Revision Handbook **§9–§10**
Practise this: 10 questions · answer each one aloud before reading the model

V59 Take me through your orthodontic examination.

"In a fixed order. The presenting complaint in the patient's own words and what they expect treatment to achieve, then the dental, medical, social and family history and any habits. Then extra-oral, with the patient upright in natural head position, teeth in occlusion and lips at rest: the three planes of space, lips and interlabial gap, incisor show at rest and on smiling, the smile arc, the nasolabial angle and the joints. Then intra-oral: oral health first, then each arch with the crowding quantified in millimetres, then the occlusion in three planes, then the path of closure. Then a provisional problem list, and only then do I decide which records I need.

The last clause is the one that scores: the examination decides the records, not the other way round.

V60 Why does the path of closure matter so much?

"Because it can change the diagnosis, not merely the detail. An anterior displacement from retruded contact position to intercuspal position may mean this is a pseudo-Class III rather than a true skeletal Class III, which is an entirely different prognosis. A lateral displacement means an apparently unilateral crossbite is really a bilateral constriction, which changes both the appliance and the urgency. And a displacement is associated with attrition, with recession on the loaded tooth and with asymmetric growth. Posselt showed that a discrepancy of nought to two millimetres is normal and that only about ten per cent of people have the two positions coincident.

One finding, three consequences. That structure is worth borrowing.

V61 From what age do you palpate for the maxillary canines, and why?

"From nine to ten years, at every examination. The canine has the longest eruptive path in the dentition, over twenty millimetres, and it fails to erupt in about two to three per cent of children with up to 85 per cent of those palatal. If I cannot palpate it buccally, I assume it is palatal and investigate. The real threat is not the impaction but resorption of the adjacent incisor roots, which is silent and which I will only find if I look.

A screening habit with a stated rationale. Give both.

V62 What radiographs would you take for a 12-year-old with crowding and an unerupted upper canine?

"A panoramic radiograph as the survey film, and periapicals with a parallax shift to localise the canine, using the rule that the object moving with the tube is palatal. I would consider a CBCT only if conventional imaging had not established the relationship of the canine to the lateral incisor root, or whether that root was resorbing, and only because the answer would change whether I retained the lateral incisor. I would not take a lateral cephalogram unless I intended significant incisor movement, growth modification or surgery.

Justification for each film, and refusal of the one that is not needed.

V63 Justify a lateral cephalogram to me.

> *"I would take one if I intended to move the incisors significantly, to modify growth, or if surgery was a possibility, because in each of those the skeletal pattern and incisor inclination change the plan. I would not take one routinely, and specifically not if only minor tooth movement is planned. I would not take one to predict growth, because cephalometric growth prediction has been shown to be at best unreliable. And I would record the justification and report the film.*

Three indications, two refusals, and the record. Radiation justification is easy marks and routinely omitted.

V64 What are the four sources of error in cephalometric analysis?

> *"Reproducibility, from landmark identification error, where each landmark has its own non-circular envelope of error and where the error is cumulative because landmarks are reused across an analysis. Validity, because points A and B are convenient rather than anatomical and both remodel as the incisors move. Projection, because a three-dimensional object is rendered in two dimensions and every machine has its own enlargement factor. And superimposition, because it assumes stable reference structures and no method has been proved completely accurate. Houston's summary is that cephalometrics is definitely not an exact science.*

Four headings and a quotation. Weight the limitations equally with the uses.

V65 What is the Eastman correction and when would you apply it?

> *"It corrects ANB for the fact that ANB is distorted by the anteroposterior position of nasion and by the inclination of the sella-nasion plane. If the sella-nasion to maxillary plane angle differs from eight degrees, or if SNA differs from 81 degrees, ANB is adjusted by half a degree for each degree of difference. I would apply it whenever SNA is markedly outside the normal range, because otherwise I may call a Class I pattern Class II purely because of where nasion happens to sit.*

A correction with a reason attached, and a clinical consequence. Do not just recite the arithmetic.

V66 When would a CBCT be justified in orthodontics?

> *"Where there is a specific question that conventional imaging has not answered and where the answer will change management. In practice that is impacted or ectopic teeth where the relationship to adjacent roots is unclear, suspected resorption of an adjacent incisor root, clefts and alveolar bone graft planning, complex asymmetry, orthognathic planning, and occasionally assessment of a temporary anchorage device site. The dose is far higher than a panoramic, incidental findings are reported in a quarter to nearly all scans, and it cannot image the disc or muscle, which needs MRI.*

Indications, dose, incidental findings and the limitation. All four should appear.

V67 Why can you not apply two-dimensional cephalometric norms to a three-dimensional analysis?

> *"Because the landmark definitions are not the same. A landmark defined on a projected two-dimensional image is not the same point as the equivalent three-dimensional anatomical landmark, and there is no agreed normative database for three-dimensional measurements. Three-dimensional imaging removes projection and superimposition error in principle, but the dose is higher and the*

norms are not transferable, so it does not simply replace the lateral cephalogram.

A question that separates candidates who have thought about imaging from those who have memorised norms.

V68 How would you record incisor display?

"At rest and on smiling, photographed both times. For rest position I ask the patient to say Emma or Mississippi and then relax, and I measure the incisor show, which is about two to four millimetres in a young adult and reduces with age. For the posed smile I ask them to bite together and say cheese, and I record how much of the incisor crown shows, which is 75 to 100 per cent in an average smile, together with the gingival display, the smile arc and the buccal corridors. I would do this before planning anything vertical.

The instructions are the detail. Anyone can say measure the incisor show; few say how.

Space analysis, diagnosis and treatment planning (V69–V80)

Notes In Orthodontics: Topic 3; Chapter 8; Appendix A, Topic 3
Supplements: Revision Handbook **§11–§12**
Practise this: 12 questions · answer each one aloud before reading the model

V69 Take me through your treatment planning sequence.

"Nine stages, each constraining the next. Diagnosis, from the examination, the records and the patient's own concerns. A ranked problem list. Aims, both facial and occlusal, reconciled against the skeletal pattern. Patient factors: age and growth, dental and periodontal health, medical history, compliance and expectations. The space analysis. The anchorage requirement, which follows directly from it. The appliance. The phasing. And retention, planned and agreed before treatment begins. The point is that it is a sequence rather than a list: the problem list sets the aims, the aims set the space requirement, the space requirement sets the extraction pattern, and the extraction pattern sets the anchorage demand.

The last sentence converts a list into an argument. That is the difference between a pass and a good pass.

V70 Where does treatment planning begin?

"With the position of the dentition, and particularly the maxillary incisors, within the face. That is a deliberate shift from the classical approach of planning around mandibular incisor position and isolated static occlusal goals. The soft tissues define the limits of orthodontic treatment in each individual, so I start from the face and work inwards, rather than achieving a Class I occlusion and hoping the face tolerates it.

Facial aims are routinely forgotten. Volunteer them.

V71 Your space analysis shows 6 mm of crowding in the lower arch. Take me through how that changes your plan.

"Six millimetres is moderate crowding, so it is a judgement rather than an automatic extraction. First I would ask what else needs space: levelling the curve of Spee consumes roughly a millimetre of arch length per millimetre of depth, and any change in incisor position costs two millimetres of space per millimetre of movement. Then I would ask where the space can come from: expansion yields only about half its width and the lower arch is the template, so I would not expand it; interproximal reduction offers a modest amount. If the total requirement approached eight millimetres I would extract, and I would then compare the lower requirement with the upper, because the difference between the arches determines the molar relationship I finish to and therefore the extraction pattern.

The examiner is testing whether you can convert a number into a decision.

V72 How do you decide between growth modification, camouflage and surgery?

"Growth first. If the patient is growing and the discrepancy is skeletal, growth modification is available, though I would be honest that the skeletal gain is a few millimetres at best and largely transient, and that it is far more effective in Class II than Class III. If the patient is not growing and the discrepancy is mild to moderate, camouflage, limited by the alveolar envelope, by the soft tissue

response of about 0.7 millimetres of lip retraction per millimetre of incisor retraction, and by how much dental compensation is already present. If the discrepancy is severe or the facial aims cannot be met by moving teeth, surgery. And that decision has to be made before treatment starts, because pre-surgical orthodontics decompensates in the opposite direction from camouflage.

A set-piece. Prepare it, because it comes up in every oral in one form or another.

V73 What are Proffit's thresholds for Class II surgery?

"An overjet greater than ten millimetres, pogonion to the nasion perpendicular of eighteen millimetres or more, a mandibular body length under seventy millimetres, or an anterior face height over 125 millimetres. I would add that these are indicators rather than rules: a patient at one threshold whose chief concern is the occlusion rather than the face may still be a reasonable camouflage case, and one below all four with a severe aesthetic concern may not be.

Quote the numbers, then refuse to apply them mechanically.

V74 What are the thresholds that suggest surgery in Class III?

"A Wits appraisal of minus eleven millimetres or worse, an ANB of minus four degrees or worse, a reverse overjet of minus five millimetres or worse, a Holdaway angle of 3.5 degrees or less, a gonial angle of 120 degrees or less, a lower anterior face height of 66 per cent or more, and a lower incisor inclination of eighty degrees or less. Discriminant analysis on those variables predicts the need for surgery with over ninety per cent accuracy. In practice the one that decides it for me is usually the lower incisor inclination, because it tells me how much decompensation has already happened and therefore how little room is left.

Seven numbers, then the one that matters clinically. That last step is the judgement.

V75 What is the envelope of discrepancy?

"It is a description of how far teeth can be moved in each direction by orthodontics alone, by orthodontics with growth modification, and by surgery. Three things about it are worth saying. It is asymmetric: maxillary incisors can be retracted considerably further than they can be advanced. Growth modification extends it only modestly, and only in a grower. And the soft tissue limits are frequently tighter than the hard tissue limits, so the fact that a tooth can be moved somewhere does not mean the lip will tolerate it.

Three observations rather than a diagram. The third is the clinically useful one.

V76 Which parts of your plan would change if this patient were 32 rather than 12?

"Growth modification disappears, so a skeletal discrepancy becomes camouflage or surgery. Overbite reduction reverses: in a child I can level by relative intrusion and let the buccal segments extrude, whereas in an adult the same manoeuvre rotates the mandible down and back and relapses, so I would need true intrusion. Movement is slower, hyalinisation is more prolonged and treatment takes longer. I would probe every tooth and record pre-existing recession. And I would plan the restorative endpoint with a restorative colleague first, rather than moving teeth and consulting afterwards.

Four or five specific differences beat a general statement that adults are different.

V77 How do you assess motivation and compliance at the first visit?

> *"There is no reliable predictor, so I use proxies. Whether the patient can articulate their own concern in their own words, or whether the concern is entirely the parent's. Their attendance history. Their plaque control at the assessment visit, which is a behaviour rather than a promise. And whether they can describe what they would be willing to do — wearing something at night, avoiding certain foods. I would say plainly that compliance is a planning factor rather than an afterthought, because a technically ideal plan the patient will not wear is not a plan.*

Honesty about the absence of predictors, then the proxies you actually use.

V78 A colleague's diagnostic case lists twelve findings and no argument. What is missing?

> *"A ranking, and a decision. A problem list has to be ranked in order of importance to the patient, with the chief concern high, because that is what determines which problems I treat if I cannot treat all of them. Then I would want to see aims stated so that a colleague could measure me against them, rather than a rubber-stamp objective such as achieving a good functional occlusion, and I would want at least three options with reasons for rejecting two of them.*

The commonest failure in the diagnostic case component. Rank, and choose.

V79 How does the space analysis determine the extraction pattern?

> *"Through the difference between the arches. If the requirement is equal in both, the case finishes to Class I molars and I extract corresponding premolars. If the maxillary requirement is greater, the case finishes to Class II molars and I extract further back in the mandible: upper first premolars with lower second premolars. If the mandibular requirement is greater, as in a Class III camouflage, I extract further forward in the mandible. The principle is simply to put the space where the movement is needed.*

Reversing the Class II and III patterns is one of the twelve costly slips. Give the principle so you can reconstruct it under pressure.

V80 What would make you refer rather than treat?

> *"A severe skeletal discrepancy, or any case where surgery is a realistic possibility. An impacted tooth needing surgical exposure and complex traction. Cleft and craniofacial cases. Significant asymmetry, particularly if it is progressive. Severe hypodontia. A reduced periodontium needing complex mechanics. A suspected primary failure of eruption. And in general, any case where the aims I have set cannot be delivered by the mechanics I am competent to use. I would rather say that at the assessment visit than discover it eighteen months in.*

Volunteering the boundary before you are asked is the strongest single move in an oral examination.

Extraction decisions and space management (V81–V90)

Notes In Orthodontics: Topic 4; Chapter 8; Appendix A, Topic 4
Supplements: Revision Handbook **§13**
Practise this: 10 questions · answer each one aloud before reading the model

V81 What factors govern your choice of extraction?

"Four. The condition of the teeth, which comes first: a compromised tooth should always be removed in preference to a healthy one, even if it sits in a less convenient place. The site and degree of crowding, because space is most useful where the movement is needed. The type of malocclusion, which determines the differential requirement between the arches. And the age, because a well-timed extraction in the mixed dentition permits spontaneous improvement whereas in the permanent dentition appliances become obligatory.

Prognosis first. Say it first.

V82 Why do you extract further back in the mandible in a Class II case?

"Because I need more space in the maxilla than in the mandible. Extracting upper first premolars and lower second premolars gives a large anterior space in the upper arch to retract the labial segment, while the smaller and more posteriorly sited lower space allows the lower molars to come forward, so the case finishes to a Class I canine relationship on a Class II molar relationship. The general principle is to put the space where the movement is needed, and in Class III it is exactly reversed.

Give the principle as well as the pattern, so you can reconstruct it if you blank.

V83 When would you extract a second premolar rather than a first?

"When the crowding is mild to moderate and sited more posteriorly, when I want the anchorage balance to favour the anterior segment, or when the second premolar itself is absent, hypoplastic or displaced. In a Class II case, second premolar extraction in the lower arch also eases the mechanics. Anchorage loss, incidentally, is the same for first and second premolar extraction, so that is not the discriminator people sometimes think it is.

The Haque finding is a useful corrective to a common assumption.

V84 When would you extract a lower incisor?

"Rarely. Severe lower labial segment crowding, a lower incisor of genuinely poor prognosis, or occasionally mild Class III camouflage in a non-growing patient. The costs are real: a tooth-size discrepancy that shows up on a Bolton analysis, residual spacing that may need restorative closure, and a deepened overbite. I would want a diagnostic set-up or at least a careful Bolton calculation before committing.

Knowing the costs is what makes this a safe answer rather than a reckless one.

V85 Do extractions damage the face?

"The comparative evidence says not, in the way the claim is usually made. Rushing, Paquette and Iared found minimal soft tissue differences between comparable cases treated with and without extraction, with both groups perceived to benefit. Bowman and Johnston found that lip retraction must be considerable before a profile is judged unattractive. Meyer and Christou found that extraction does not widen the buccal corridors or produce a less attractive smile. What decides it for me is the space requirement in millimetres and the nasolabial angle, not a general position for or against extraction.

Three named findings and then a clinical criterion. That is the shape.

V86 What about the claim that extractions narrow the airway?

"There is no evidence for it. Larsen and Orabi looked specifically at this and found no evidence that extraction-based treatment reduces the oropharyngeal airway or increases sleep apnoea, and the claim has been described in the literature as scaremongering and marketing. If a parent raises it I would say that plainly, and offer to discuss the actual evidence rather than dismiss the concern.

Answer the claim and address the person raising it. Both are marked.

V87 How much space does expansion actually give you?

"About half the amount of expansion. That is the O'Higgins and Lee figure, and it is why expansion is a poor substitute for extraction when the crowding is significant. It is also why I would not expand the lower arch at all: the mandibular arch is the template, lower intercanine width is among the least modifiable dimensions, and lower expansion relapses, so it buys me a small amount of space at the cost of permanent retention.

The number, then the reason it does not solve the problem.

V88 When would you extract a second permanent molar?

"Where I need space at the back of the arch, to ease Class II mechanics, or to allow a crowded third molar to erupt. Before committing I would want the third molar present, of good size, correctly angulated at thirty degrees or less to the second molar long axis, and at an appropriate stage of development. It is a decision that depends entirely on a tooth I cannot yet see erupt, so I would be cautious and I would explain the uncertainty.

The conditions are the answer. Anyone can name the extraction.

V89 How much interproximal reduction is available across an arch?

"A modest amount. Roughly half a millimetre per anterior contact and up to a millimetre per contact in the buccal segments, which across a full arch comes to somewhere in the order of four to eight millimetres, and rather less than that if I want to stay comfortable. Enamel is finite, the technique matters, and I would want good fluoride protection afterwards. It is a useful adjunct rather than a substitute for extraction in significant crowding.

Quantify it and then say what it cannot do.

V90 A first permanent molar is heavily restored and hypomineralised, and there is 5 mm of lower crowding. Which tooth comes out?

> *"The first permanent molar. Prognosis beats convenience of position, and a compromised molar should be removed in preference to a healthy premolar. I would time it against the development of the second permanent molar, ideally when its bifurcation is just calcifying at about eight to ten years of dental age, confirm all the permanent teeth including third molars are present on a radiograph, and consider a compensating extraction of the opposing molar to prevent over-eruption. Spontaneous closure is less reliable in the mandible, so I would plan for appliance treatment.*

Prognosis, timing, radiographic check, compensating extraction. Four elements.

Biomechanics and anchorage (V91–V102)

Notes In Orthodontics: Topics 5 and 6; Chapters 6 and 11; Appendix A, Topic 5
Supplements: Revision Handbook **§14**
Practise this: 12 questions · answer each one aloud before reading the model

V91 What is the moment-to-force ratio and why does it matter?

"It is the single number that decides what kind of movement I get. The centre of resistance sits inside bone and is inaccessible, so I have to apply force at the crown, which means a moment is always produced. A ratio of zero gives uncontrolled tipping, with crown and root moving in opposite directions. About seven to one gives controlled tipping, with the centre of rotation near the apex. About ten to one gives translation, with the centre of rotation at infinity. Twelve to one and above gives root movement.

The four ratios and the four movements. Learn them as a set.

V92 Why does a removable appliance only tip teeth?

"Because it achieves only two-point contact on a tooth, so it cannot generate a couple. Without a couple the moment-to-force ratio is effectively zero, and the tooth tips about a centre of rotation near its centre of resistance. That is why bodily movement, rotation and torque are beyond a removable appliance, and it is one of its two fatal limitations. The other is compliance.

A mechanical explanation of a clinical limitation. Give the mechanism.

V93 What is a couple?

"Two equal, opposite, parallel and non-collinear forces. The important properties are that it produces no net translational force, and that it produces pure rotation about the centre of resistance regardless of where on the tooth it is applied. That is why a rectangular wire engaged in a rectangular slot can torque a root without translating the crown, and it is the whole basis of third order control.

The two properties are the answer; the clinical application shows you understand them.

V94 What is the difference between a determinate and an indeterminate force system?

"A one-couple system, such as a cantilever or a utility arch engaged in one bracket and tied as a point contact at the other end, is statically determinate: the force system can be calculated and measured. A two-couple system, which is what a continuous archwire engaged in several brackets is, is statically indeterminate: the force system can only be estimated. Most of clinical orthodontics is indeterminate, which is precisely why clinical experience matters and why segmental mechanics are used when the force system needs to be known.

A distinction with a reason for caring about it.

V95 How do you plan anchorage?

"I start from the space analysis, because the anchorage requirement follows from it rather than the other way round. I classify the demand as minimum, moderate or maximum. Then I recognise that anchorage is spent or saved during levelling and aligning, before any active mechanics begin, so I use lacebacks, I bend the wire back distal to the molar, and in a maximum anchorage case I keep the anchor teeth out of the initial aligning wire. Then I reinforce: second molars in the appliance, a transpalatal or lingual arch, headgear, interarch elastics, or skeletal anchorage. And the number I would quote is that unsupported, about two-thirds of the space created is lost.

Sequence, classification, timing, reinforcement, number. Five elements.

V96 How much anchorage does a transpalatal arch actually give you?

"Very little in the sagittal plane. Diar-Bakirly found it provides very little additional anteroposterior anchorage, and Rebellato found much the same for a lower lingual arch, which additionally can procline the lower incisors. They are useful for molar derotation, for transverse control and, in the case of the lingual arch, for preserving leeway space; but if I need real anchorage I would not rely on them.

A useful corrective. Examiners like candidates who know which received practices have weak evidence.

V97 What is Proffit's anchorage ladder?

"A way of quantifying the trade-off as a ratio of anterior retraction to posterior protraction. Reciprocal space closure without any reinforcement is about one to one, and that is determined by periodontal ligament area rather than by tooth number. A stabilising lingual or palatal arch shifts it to about two to one. Headgear combined with interarch elastics reaches three or four to one. And skeletal anchorage is effectively absolute. Putting numbers on it makes the conversation with the patient about compliance a good deal easier.

The ratios convert a vague concept into a plan.

V98 How does a temporary anchorage device work?

"It does not change the force system at all. Newton's third law still applies and the reaction still has to go somewhere; the screw simply places the reaction in bone, where nothing can move. So what changes is the biological response, not the mechanics. In direct anchorage the line of force must pass through the screw. In indirect anchorage I ligate the anchor teeth to the screw with a large rectangular stainless steel wire, rigid enough not to flex.

The opening sentence is the one that demonstrates understanding rather than familiarity.

V99 Would you place a miniscrew yourself?

"In some sites, with appropriate training. The maxillary buccal interradicular region has a pooled failure rate of about 7.5 per cent, and the palate is the most reliable site at about 1.3 per cent in the median palate, because placement there is not restricted by the dentition. The mandibular buccal shelf fails in about 65 per cent of cases, and that is not somewhere I would be placing screws. More generally, if a case depends on a screw in a difficult site, I would refer the whole case rather than the screw.

The site-specific failure rates, then the judgement about referral.

V100 What insertion torque do you aim for, and why does it matter?

> *"Five to ten newton centimetres. Below that the screw has poor primary stability and is likely to be mobile from the outset. Above it, I risk microfracture of the surrounding bone and therefore poor secondary stability. I would also avoid screws shorter than eight millimetres or narrower than 1.2 millimetres, because failure rates rise below those dimensions, and the commonest single cause of failure is root contact.*

A number with a mechanism at each end of the range.

V101 Your miniscrew is mobile at four weeks. What do you do?

> *"Remove it. A mobile screw will not become stable and it perpetuates inflammation in the tissue around it. Then I would work out why: root contact is the commonest cause, so I would look again at the radiograph and at where I placed it. I would re-site at least a few millimetres away, or change the anchorage plan altogether. And I would reflect that I should not have designed a case in which the failure of one screw leaves me with nothing to fall back on.*

The last sentence is about planning rather than about screws, and it is the one that scores.

V102 What force and duration would you prescribe for headgear, and what safety precautions?

> *"It depends on the purpose. For anchorage support, 250 to 350 grams per side for 10 to 12 hours a day. For distalisation or an orthopaedic effect, 450 to 500 grams per side for 14 hours a day. Two independent safety mechanisms are mandatory: a self-releasing snap-away neck strap and a locking facebow. I would warn the patient in writing about the two injury mechanisms Samuels described, the catapult injury and nocturnal disengagement, tell them to stop wearing it and attend if the bow ever comes out, and record that I had done so.*

Quoting one headgear figure is one of the twelve traps. Give both, and the safety devices unprompted.

Materials, wires and bonding (V103–V111)

Notes In Orthodontics: Topic 7; Chapter 10; Appendix A, Topic 7
Supplements: Revision Handbook **§15**
Practise this: 9 questions · answer each one aloud before reading the model

V103 What are the properties of an ideal archwire?

"High strength, low stiffness, high range and high formability, plus biocompatibility, environmental stability, the ability to be welded or soldered, low friction and reasonable cost. No single wire has all of them, which is exactly why a wire sequence exists. The useful identity is Proffit's: strength equals stiffness multiplied by range, so I cannot raise one without paying somewhere else.

The identity is what turns a list into an understanding.

V104 Compare stainless steel and nickel titanium.

"Stainless steel is the 18-8 austenitic alloy, about 70 per cent iron with 17 to 20 per cent chromium and 8 to 12 per cent nickel. It is stiff and strong, has excellent formability, low friction and low cost, and it can be welded and soldered, which makes it the working and finishing wire and the wire that carries arch form. Nickel titanium has a modulus about 0.26 that of steel, giving very low stiffness and a very large range, with shape memory and superelasticity, which makes it the alignment wire. But it cannot be bent, welded or soldered, so its role ends when alignment ends.

A comparison organised by what each is used for, not just by properties.

V105 What is superelasticity?

"A real, measurable phenomenon rather than an advertising term. It arises from a stress-induced transformation from the austenitic to the martensitic phase, which produces a plateau in the load-deflection curve so that the wire delivers a roughly constant force over a range of deflection, with hysteresis between loading and unloading. Clinically that means a relatively constant light force as the tooth moves, which is the property I want during alignment.

Mechanism, curve, clinical consequence. Three sentences.

V106 Which archwire would you choose for initial alignment, and does the choice matter?

"A light, flexible, large-range wire: 0.012 to 0.018 inch round nickel titanium or a multistrand steel. As to whether it matters, the honest answer is less than the marketing suggests. The Cochrane evidence found insufficient evidence that any archwire material is superior, with the single exception that coaxial superelastic nickel titanium produced greater movement over twelve weeks than single strand. Papageorgiou found copper nickel titanium slightly slower to reach a working wire and more painful.

A choice, then the evidence that the choice matters less than people think.

V107 Describe your bonding procedure.

"Clean with pumice and no fluoride, isolate, etch with 37 per cent phosphoric acid for 15 to 30 seconds, rinse and dry to a frosted appearance, prime, place the adhesive and the bracket, remove excess, and cure. A self-etching primer combines the etch and prime steps, gives a shallower etch pattern and has comparable clinical bond failure rates. For molars carrying headgear or a palatal arch, or for heavily restored teeth, I would band and cement with glass ionomer for the fluoride release.

A sequence with the numbers in it, and the decision about banding.

V108 Why do ceramic brackets cause problems at debond?

"Because they bond too well. They bond by silane coupling, which is a chemical bond rather than a purely mechanical one, and the ceramic is stiffer and more brittle than the enamel. So at debond the weakest point can be the enamel rather than the adhesive, and enamel fracture is a recognised risk. They also have higher friction and they abrade opposing enamel. I would use the manufacturer's debonding instrument, work slowly, and warn the patient in advance.

The mechanism explains the risk. Give both.

V109 How do you minimise demineralisation during fixed appliance treatment?

"Patient selection first: I do not start until plaque control is demonstrably adequate, because demineralisation affects up to half of fixed appliance patients and can be visible within six months, and in elective treatment that makes it my responsibility. Then daily high-fluoride toothpaste at 2800 or 5000 parts per million, professional high-concentration fluoride every six to eight weeks, glass ionomer for band cementation, steel ligatures rather than elastomeric modules in a poor brusher, diet advice, hygiene reinforcement recorded at every visit, and a willingness to debond early if it is not improving.

Selection first, then the regimen, then the willingness to stop. All three.

V110 A patient reports a nickel allergy. What do you do?

"I would not regard it as a contraindication to orthodontic treatment. Nickel allergy affects about 10 to 30 per cent of females and 1 to 3 per cent of males, and it is a type IV delayed hypersensitivity. Stainless steel is safe, because the nickel is bound within the austenitic lattice, and intra-oral reactions are in fact rare. What I would avoid is nickel titanium, and I would avoid extra-oral nickel-containing components such as headgear, where reactions are commoner. I would record the discussion.

Saying nickel allergy contraindicates orthodontics is one of the twelve traps.

V111 What is the diode laser used for in orthodontics, and what are its limits?

"It operates at 810 to 980 nanometres, is absorbed by melanin and haemoglobin, and cuts soft tissue while sparing enamel and metal. I would use it for exposing a partially erupted tooth, for gingival recontouring, for a frenectomy or an operculectomy, with the advantages of no sutures, no dressings and controlled bleeding. Its limit is that it is a finishing instrument: I would diagnose the cause of any gingival disproportion — altered passive eruption, hyperplasia from poor hygiene, or simply an unerupted tooth — before reaching for it.

The last sentence is the clinical judgement the question is really asking about.

Removable, functional and fixed appliances (V112–V121)

Notes In Orthodontics: Topics 6 and 8; Chapters 9, 10 and 14; Appendix A, Topics 6 and 8
Supplements: Revision Handbook **§16–§17**
Practise this: 10 questions · answer each one aloud before reading the model

V112 What does a functional appliance actually do?

"Mostly dentoalveolar change. The upper incisors retrocline, the lower incisors procline, buccal segment eruption is guided, and the lip trap is eliminated so lip competence is gained. There is a skeletal effect but it is a few millimetres at best and essentially transient, and there is usually a headgear effect restraining the maxilla. The more sophisticated explanation of the mechanism is that a Class II relationship is partly locked by intercuspation, and the appliance unlocks it and lets normal mandibular growth express itself alongside the dentoalveolar change.

Saying the appliance grows the mandible is overclaiming. The unlocking formulation reads far better.

V113 When would you start functional appliance therapy?

"At the circumpubertal period, around cervical vertebral maturation stage CS3, which has a mean age of about twelve years. Starting earlier wastes compliance on a phase that will not deliver a lasting skeletal effect; starting later wastes the growth. The exception is a child at high risk of incisor trauma with a very large overjet and incompetent lips, where I might start earlier, accepting that the benefit is trauma reduction rather than a better final result.

Timing, the reason, and the exception.

V114 Twin block or Herbst?

"It depends principally on compliance. Pacha's randomised trial found 37 per cent failure with the twin block against 17 per cent with the Herbst, and a larger mean overjet reduction with the Herbst at 7.1 against 5.8 millimetres, with the twin block taking about a month and a half longer. Against that, the Herbst required more visits and produced more emergencies. So for a compliant patient I would use a twin block; where compliance is the limiting factor I would consider a fixed functional.

One trial with four numbers, then a decision rule.

V115 A patient in a twin block develops lateral open bites. What do you do?

"Nothing alarming, because I expect them. Overjet reduction outpaces overbite reduction, so posterior open bites are the rule during twin block treatment. I would trim the bite blocks progressively to allow the posterior teeth to erupt, and I would make sure the case is settled before the appliance is abandoned. What I would not do is stop treatment or tell the patient something has gone wrong.

Recognising a normal finding as normal is itself a mark.

V116 Was early Class II treatment worth it?

"Not for the reasons it was traditionally offered. The trials — Florida, North Carolina and the Manchester twin block trial — showed that early treatment reduces the overjet dentoalveolarly and gives a transient rise in self-esteem, but does not improve the final skeletal or occlusal outcome, does not shorten overall treatment, and results in more attendances. O'Brien found the late-treated group finished slightly better. The one durable benefit is a reduction in incisor trauma, with a relative risk of 0.58. So I would treat early only for trauma risk or significant teasing.

The trial names and the number. Then the policy.

V117 Do self-ligating brackets treat faster?

"No. Scott, Fleming and others found no difference in rate of movement, treatment time, number of visits or occlusal outcome, the pain benefit was marginal, and the expansion claims are unsupported. The distinction between active clips that store energy against the wire and passive slides that do not is a design difference rather than a demonstrated clinical advantage. What decides treatment time is case difficulty, bracket positioning and the operator, which is also what Penning found for customised appliances.

A short, well-referenced no. Do not hedge.

V118 What can aligners not do well?

"Root torque, extrusion, rotation of round-sectioned teeth such as canines and premolars, bodily space closure at extraction sites, molar uprighting and large anteroposterior correction. The reason is that a viscoelastic plastic shell persuades whereas a wire in a slot overpowers, and the reaction forces are accepted rather than designed because the appliance acts on every tooth at once. Attachments help with extrusion and rotation, but they do not change the underlying limitation.

A list with a mechanism. The plastic persuades formulation is memorable and accurate.

V119 What would you tell a patient asking for aligners?

"That they are less visible, less painful in the first few days and easier to keep clean, and that in a mild to moderate Class I case they can produce a good result. Then the two numbers: only about six per cent of cases finish without a refinement, and over seventeen per cent convert to fixed appliances to complete treatment. I would say that fixed appliances outperform aligners on every occlusal measure on the ABO grading system, that they need to be worn 20 to 22 hours a day, and that compliance is the weakest element in the system.

The two numbers make the consent honest and are what the examiner is listening for.

V120 Why did the pre-adjusted appliance not solve everything?

"Because it moved the error rather than removing it. Angle's edgewise appliance required all three orders to be bent into the wire tooth by tooth, which is why results were traditionally left ten to twenty per cent undertreated. Andrews built the prescription into the bracket, but the prescription can only express itself if the bracket is in the right place, so the error moved from wire bending to bracket positioning. That is why bracket position, not prescription, decides the result, and why bracket repositioning at the working wire stage is the commonest thing that rescues a case.

A historical answer with a present-day clinical consequence.

V121 When would you consider a lingual appliance?

> *"For an adult whose sole objection to treatment is visibility and who will accept the trade-offs. Those are speech disturbance, usually transient over one to three weeks, tongue discomfort and ulceration, more difficult access and hygiene, longer chairside time, cost and a steep learning curve for me. Biomechanically the bracket sits palatal to the centre of resistance and closer to it vertically, so the moment systems differ, and the anterior interbracket span is shorter, which raises effective stiffness. The comparative evidence is limited, so I would not claim a better outcome, only a less visible one.*

Indication, trade-offs, mechanics, and an honest statement about the evidence.

Treatment mechanics and finishing (V122–V129)

Notes In Orthodontics: Chapters 10 and 13; Supplements E and F
Supplements: Revision Handbook **§18**
Practise this: 8 questions · answer each one aloud before reading the model

V122 When is anchorage actually spent in a case?

"Earlier than most people think — during levelling and aligning, before any active mechanics begin. As the initial wire straightens it pushes the incisors labially and the molars mesially, so by the time I reach a working wire the anchorage may already be gone. That is why I use lacebacks, which distalise the canine without tipping it, why I bend the wire back distal to the molar tube, and why in a maximum anchorage case I keep the anchor teeth out of the initial aligning wire. Without lacebacks the lower incisor advances about 1.4 millimetres; with them it retracts about 1.0.

The timing is the insight. The numbers make it concrete.

V123 Take me through your wire sequence in a 0.022 slot.

"Alignment on a light round nickel titanium, 0.012 to 0.018, with second molars bonded as early as I can because they are needed for bite opening and anchorage. Then levelling on round stainless steel, 0.016 to 0.020. Then a 0.019 by 0.025 stainless steel working wire, left passive for about a month so the slots line up, because level slot line-up is what makes sliding mechanics possible. Then space closure on that wire with tiebacks reactivated every four to six weeks at about 150 grams. Then finishing on a wire that fills or nearly fills the slot, 0.021 by 0.025, varying stiffness by changing material rather than dimension.

A sequence with the reasons attached. The passive month is the detail few candidates give.

V124 Why 150 grams for space closure?

"Because more is not faster. The MBT work found that attempts to transfer traditional edgewise force levels of 500 to 600 grams to a pre-adjusted appliance produced the roller coaster effect, and that 200 grams gave no advantage over 150. Biologically it makes sense: heavier force hyalinises the ligament on the active side, lengthening the lag phase, and exceeds the anchorage threshold on the reactive side, so the space closes from the wrong end.

A number with both a clinical and a biological justification.

V125 What is the commonest reason a case does not finish well?

"Bracket positioning. The pre-adjusted prescription can only express itself if the bracket is where it should be, so pre-adjustment moved the error from wire bending to bracket placement. Practically I reassess every bracket at the working wire stage, against marginal ridge levels and the gingival contour, and reposition before I go into finishing wires. It is far more efficient than putting a compensating bend in a wire.

Honest and specific. It is also true, which examiners appreciate.

V126 How do you finish a case?

"Deliberately rather than by luck. I reposition brackets at the working wire stage, confirm root parallelism on a panoramic radiograph particularly at extraction sites, then finish in a wire that fills the slot with first, second and third order detailing bends. I check the occlusion in five planes — horizontal, vertical, transverse, dynamic and cephalometric or aesthetic — against the ABO criteria. Then I settle in light round wires with vertical elastics for about six weeks, keeping every space closed, and only then go to retainers. I never go straight from rectangular wires to retention.

Six steps. The settling period is the one candidates omit.

V127 What aesthetic criteria would you check before debonding?

"The midline against the face, the smile arc following the lower lip, incisor show at rest and on smiling, gingival levels with the central and canine zeniths on a line and the lateral half a millimetre below it, incisal embrasures increasing in size from the midline distally, and connector areas following the fifty, forty, thirty proportion. Also the buccal corridors and the vertical position and torque of the canines, which are the teeth most often left slightly extruded or under-torqued.

The appearance of the teeth, not just the occlusion, is the target. Nanda's criteria are all checkable.

V128 What arch form would you use?

"The patient's own. The mandibular arch is the template, and intercanine width is among the least modifiable dimensions, so I preserve the original form rather than imposing a standard one. Changing arch form results in relapse in up to seventy per cent of cases. I would also be aware that preformed superelastic nickel titanium archwires can expand the lower intercanine width by up to 5.9 millimetres if allowed to express fully, so arch form has to be a deliberate decision rather than something the wire does to me.

The last sentence is the practical warning that turns a principle into an action.

V129 When would you use loop mechanics rather than sliding mechanics?

"Where friction is the limiting factor and I need a force system I can predict — for instance, a segmented approach in a case with reduced periodontal support, or where I want differential space closure. The trade-off is that a closing loop avoids friction but produces a force system that is harder to control and requires accurate wire bending, whereas sliding mechanics are simpler and more forgiving but depend on level slot line-up and on the friction being manageable. For most cases in general practice, sliding mechanics on a 0.019 by 0.025 steel wire is the safer choice.

The trade-off and then an honest statement of what you would actually do.

Retention and stability (V130–V139)

Notes In Orthodontics: Topic 20; Chapter 13; Appendix A, Topic 20
Supplements: Revision Handbook **§19**
Practise this: 10 questions · answer each one aloud before reading the model

V130 Why do teeth relapse?

"Three causes on three different timescales. Elastic recoil of the gingival and supracrestal fibres, which takes six to twelve months to reorganise and which is why rotations relapse most readily. Soft tissue pressures, which act continuously and for life, so any tooth moved outside the zone of equilibrium returns towards it. And continued growth, which is slow, lifelong and in the individual's original pattern. Only the first is time-limited, which is why retention has no natural endpoint.

Three causes, three timescales, and the conclusion that follows.

V131 What do you tell a patient about how long they need to wear a retainer?

"The honest sentence: if you want the teeth to stay where they are, you wear a retainer for as long as you want that, and there is no endpoint. I would say it before treatment starts rather than at debond, because it is part of the decision to have treatment at all. I would also tell them that by four to five years after treatment most patients have stopped wearing their retainers, and that relapse is the commonest reason people return.

Honest, specific and delivered at the right time. All three are marked.

V132 Which retainer would you choose for the lower arch?

"In most cases a bonded retainer from three to three, with a vacuum-formed retainer over the top for night wear. The evidence is that fixed retainers are marginally more effective in the lower arch for the first twelve months, particularly for rotations, and equivalent thereafter, and that vacuum-formed retainers outperform Hawleys in the lower labial segment. The trade-off with a bonded retainer is that it needs lifelong monitoring: bond failure, wire fracture, unwanted torque or so-called wire syndrome, calculus and caries.

A choice with the evidence and the trade-off. The monitoring point is essential.

V133 What is Houston's rule about overbite stability?

"That overbite reduction relapses unless the interincisal angle is corrected and an occlusal stop is established. It is the reason a Class II division 2 case cannot simply have the upper incisors tipped labially: without genuine palatal root torque the interincisal angle remains increased and the overbite comes back. And it is why, whichever of the three mechanisms I use to reduce the overbite, I have to end with the lower incisors contacting something.

One rule with two clinical consequences.

V134 Is late lower incisor crowding relapse?

> *"No — it is ageing. It occurs in untreated people as well as treated ones. Sinclair and Little showed mandibular arch width and length decreasing with age in untreated subjects, with increasing anterior crowding, and Thilander found the same. The major driver is continued mandibular growth rotation uprighting the lower labial segment. No extraction pattern and no archwire sequence prevents it, and third molars are not a primary cause — NICE advises against removing them for this indication. The honest answer is long-term bonded retention, and saying so at consent.*

Confusing relapse with age-related change is one of the twelve traps. Say ageing, and say it early.

V135 How far can you advance lower incisors and still expect stability?

> *"About two millimetres, and beyond that permanent retention is very likely to be needed. The reason is equilibrium: the tooth has been moved out of the zone where the resting soft tissue forces balance, and it will return towards it. The same logic applies to lower intercanine expansion, which relapses reliably, and it is one of the reasons the mandibular arch is treated as the template rather than as something to be developed.*

A number and the principle behind it.

V136 What is circumferential supracrestal fiberotomy, and when would you do it?

> *"A surgical division of the supracrestal gingival fibres around a rotated tooth, performed before appliance removal, to reduce the elastic recoil that drives rotational relapse. The evidence for it is weak but it is not harmful. I would consider it for a severely rotated tooth, alongside a bonded retainer rather than instead of one, and I would perform it before debond rather than after, as with a frenectomy for a diastema.*

The timing and the fact that it supplements rather than replaces retention.

V137 What retention would you plan after closing a midline diastema?

> *"A permanent bonded retainer. A closed diastema relapses reliably, and this is one of the situations where I would say to the patient at the outset that the retention is permanent rather than time-limited. If a frenectomy is indicated, I would perform it after the space is closed and before the appliance is removed, because operating first leaves scar tissue that resists closure.*

Two specifics: permanent retention, and the sequence with the frenectomy.

V138 A patient returns after a week without wearing their retainer and says it feels tight. What do you say?

> *"Not to force it, and to come in. Tightness means the teeth have moved, and forcing an ill-fitting retainer can fracture it or move teeth unpredictably. I would check the fit, assess how much movement has occurred, and if it is minor consider an active retainer or a short course of aligners to recover it. Then I would revisit the conversation about why retention has no endpoint, without making the patient feel blamed for a very common lapse.*

Clinical management plus the communication. Both are marked.

V139 When can retention be stopped?

> *"Strictly speaking it cannot, because the three causes of relapse do not all resolve. Fibre recoil settles over six to twelve months, but soft tissue pressures act for life and growth continues slowly throughout adult life in the original pattern. So the answer I give patients is that retention becomes less demanding rather than unnecessary, and that a night-time retainer indefinitely is the price of a stable result. What I would not do is tell a sixteen-year-old that two years of retention will be enough.*

The refusal to give a comforting but false answer is what scores.

Interceptive orthodontics and impacted teeth (V140–V151)

Notes In Orthodontics: Topics 13, 14 and 15; Chapter 12; Appendix A, Topics 13–15
Supplements: Revision Handbook **§20**
Practise this: 12 questions · answer each one aloud before reading the model

V140 Which interceptive treatments have good evidence?

> *"Two clearly. Correction of a posterior crossbite with a displacement at ages seven to eleven, where the Cochrane evidence is high certainty that a quadhelix or expansion plate beats observation, and that the quadhelix beats a plate and is about three months faster. And maintenance of the leeway space with a lower lingual arch where the space is needed. Beyond those the evidence weakens quickly: arch expansion purely to relieve crowding is weak, and two-phase Class II treatment buys nothing but cost and time except where trauma risk or teasing justify it.*

Grading interventions by evidence is exactly what this question is for.

V141 A seven-year-old has a unilateral crossbite with a 2 mm displacement. What do you do?

> *"Treat it on diagnosis. The displacement tells me this is really a bilateral maxillary constriction that the child is closing out of, which is why it looks unilateral in intercuspation. It is one of the few interceptive interventions with high-certainty evidence, and untreated it causes attrition, is associated with recession on the loaded tooth and with asymmetric mandibular growth. I would use a quadhelix, because it beats a plate and compliance is not a variable in a seven-year-old, overcorrect, and retain for three to six months.*

Diagnosis, evidence, consequence of not treating, appliance, overcorrection, retention. Six elements in one answer.

V142 When would you extract a primary canine to intercept a displaced permanent canine?

> *"Between ten and eleven years, or ten to thirteen on the Royal College guidance, provided there is space in the arch, the canine crown lies distal to the midline of the lateral incisor root and its angulation to the midsagittal plane is twenty to thirty degrees. I would review at twelve months with a defined endpoint and abandon it if there is no improvement. I would also be honest about the evidence: the 2021 Cochrane review found only very weak evidence that it increases eruption and no evidence at all that it reduces the need for surgery.*

The criteria, the review interval, and the honest statement about the evidence.

V143 What actually worries you about a palatally displaced canine?

> *"Not the impaction — resorption of the lateral and sometimes the central incisor root. It is silent, it can be rapid, and by the time it is obvious the incisor may be lost. That is why I palpate from nine or ten years, why I localise carefully with parallax, and why the one justification I would readily accept for a CBCT here is establishing the relationship of the canine to the incisor root and whether that root is resorbing, because the answer changes whether I keep the lateral incisor.*

Naming the real threat rather than the obvious one.

V144 What determines the prognosis for aligning an impacted canine?

"Three positional factors. Height: prognosis worsens above the apical third of the incisor roots. Horizontal position: worse at or across the midline of the lateral incisor. And angulation: worse above about 55 degrees to the midsagittal plane. Ericson and Kurol's split is useful — overlap of less than half the lateral incisor root gives about 91 per cent normalisation after primary canine extraction, and more than half about 64 per cent. Age matters too: the older the patient, the poorer the prognosis.

Three factors, each with a threshold, plus the Ericson and Kurol numbers.

V145 Open or closed exposure?

"The Cochrane review by Parkin found no difference in success or other outcomes, on low to very low quality evidence. Cassina subsequently reported that open exposure was associated with just over two months less time to alignment and lower odds of ankylosis. For a buccal canine lying below the mucogingival junction I would use an apically repositioned flap, following Kokich, so the tooth erupts through attached tissue. In practice the decision is usually made with the surgeon and depends on the depth and position of the tooth.

Evidence, then the one situation where the technique is determined by the anatomy.

V146 What are the alternatives to exposing and aligning an impacted canine?

"Leaving it and monitoring, if it is asymptomatic, there is no resorption of adjacent roots, and the patient accepts retaining the primary canine — with a defined review, because monitoring is not the same as ignoring. Extracting the canine and either closing the space or restoring it. Autotransplantation, with a success rate of around 88 per cent, splinted semi-rigidly for no more than three weeks and then root treated. And of course accepting the primary canine while it lasts, with the honest statement that it will not last for ever.

Four options with the conditions attached to each.

V147 Take me through your management of an impacted lower second molar.

"A ladder, with a defined endpoint at each step. Observe for six to twelve months; no progress means an established impaction. Disimpact with a separator, brass wire or an open coil placed below the contact point. Remove the third molar if it is contributing to the obstruction. Surgically upright, exposing, removing the third molar, luxating and bonding a bracket at surgery. And extract the second molar as a last resort, which is the least successful outcome because it commits the patient to unpredictable third molar eruption. Throughout I would keep primary failure of eruption in mind, because traction on such a tooth causes ankylosis.

Five steps and the differential diagnosis. The last is what stops you doing harm.

V148 A nine-year-old has lost an upper central incisor by avulsion. What do you do?

"Attempt re-implantation, always, even if the long-term prognosis is poor, because it maintains both the space and the alveolar bone and buys years. Then the decision is between maintaining the space with a denture or a retainer with a pontic, which risks alveolar resorption, and allowing the space to close and reopening it later, which preserves bone but commits the patient to further orthodontics. Definitively, a resin-retained bridge or eventually an implant — and implants must wait until

growth is complete, because continued vertical alveolar growth leaves the crown in infra-occlusion.

Re-implantation first. The alveolar bone argument runs through the whole answer.

V149 When can orthodontic force be applied to a previously traumatised tooth?

"It depends on the injury. Three months after an uncomplicated crown fracture, a subluxation or a minor lateral luxation. Twelve months after a moderate or severe lateral luxation, an intrusion, an extrusion or a re-implantation, and usually twelve months after endodontic treatment following trauma. After a complicated crown fracture, until a hard tissue barrier is visible. A root fracture healing with dentine and cementum needs twelve to 24 months, and one healing with connective tissue should be treated as a short-rooted tooth with light forces.

A table learned as a list. It is frequently examined and easily forgotten.

V150 How do you monitor a traumatised tooth during orthodontic treatment?

"Before: vitality testing and periapical radiographs, recorded as a baseline. During: light forces, avoiding excessive movement, vitality testing every three months and a repeat radiograph six to nine months after starting movement. If resorption appears, I pause for three months, which is shorter than the three to six months quoted for orthodontically induced resorption generally. And I would have consented specifically for the three complications of moving such a tooth: loss of vitality, pulpal calcification and resorption where resorption has already occurred.

Baseline, monitoring interval, action, and the three complications named at consent.

V151 When would you use serial extraction?

"Rarely, and only in severe crowding with a Class I skeletal pattern and no vertical problem. The sequence is primary canines at about eight or nine, first primary molars at nine or ten, then the first premolars as they erupt. The disadvantages are substantial: a long period of unattractive spacing, a deepening overbite, retroclination of the lower incisors, a fixed appliance usually still needed to finish, and an irreversible commitment made at a young age. It is contraindicated in Class II and Class III, in deep or open bite, and in hypodontia.

The conditions and the disadvantages together. Anyone can recite the sequence.

Class II and Class III management (V152–V161)

Notes In Orthodontics: Topics 12 and 17; Chapters 9, 12 and 13; Appendix A, Topics 12 and 17
Supplements: Revision Handbook **§22–§23**
Practise this: 10 questions · answer each one aloud before reading the model

V152 How would you treat a 12-year-old with an 11 mm overjet and a skeletal Class II?

"I would establish first that he is growing, and where he is in the growth curve. Given the size of the overjet and the trauma risk I would not wait long. A twin block worn full time, timed to about CS3, expecting the correction to be largely dentoalveolar, with lateral open bites appearing and managed by trimming the blocks, over about six to nine months. Overcorrect, then transition into a fixed appliance to align, level and detail. Retain with a vacuum-formed retainer and, if the skeletal correction is at risk, night-time functional appliance wear for at least a year. At consent I would cover the trauma risk while the overjet persists, that the skeletal gain is small and largely transient, and the possibility of surgery if growth is unfavourable.

A full case answer. Growth first, appliance, expected findings, transition, retention, consent.

V153 Which direction of headgear pull would you choose, and why?

"From the vertical dimension rather than from the anteroposterior discrepancy. High pull or occipital for a hyperdivergent patient with increased vertical proportions, because it is intrusive on the molar. Cervical or low pull for a hypodivergent patient with reduced vertical proportions and a deep bite, because it is extrusive. Straight or combination pull for average proportions where I want pure distalisation. The single commonest error is to choose the headgear from the Class II and ignore the face.

The rule is one sentence. The error it prevents is the reason for the question.

V154 Why is the high-angle Class II the difficult case?

"Because everything that corrects the anteroposterior discrepancy tends to open the bite. Buccal segment extrusion, cervical headgear and Class II elastics all rotate the mandible down and back, which increases face height, worsens the Class II and relapses. So I have to correct the sagittal relationship without any extrusion: high-pull headgear or skeletal anchorage, possibly molar intrusion, and consideration of extraction to allow the mandible to rotate forwards. Anchorage is also at greater risk in these patients.

Naming the mechanism, then listing what you would do instead.

V155 What is the mechanical problem in Class II division 2?

"The increased interincisal angle combined with a deep, often complete, overbite. Retroclined upper central incisors have to be torqued labially at the root rather than merely tipped, or the interincisal angle stays increased and the overbite relapses — that is Houston's rule. So the sequence is to procline the upper labial segment to convert the case to a division 1 pattern, reduce the overbite, then retract if extraction space has been created. And because there is often little or no overjet, many division 2 cases are treated non-extraction in the upper arch, since extraction space there has

nowhere to go.

Problem, sequence, and the extraction consequence.

V156 How would you retain a corrected deep overbite?

"A Hawley retainer with an anterior bite plane, worn long-term and often into the early twenties. The reason is that overbite reduction is easy and holding it is difficult: the correction relapses unless the interincisal angle is genuinely corrected and an occlusal stop established, and the growth pattern that produced the deep bite continues. A vacuum-formed retainer holds alignment but does not provide the bite plane, so I would use both.

The appliance and the reason it is not a vacuum-formed retainer.

V157 Why is Class III harder than Class II?

"Because the growth works against me. In Class II residual mandibular growth helps to correct the discrepancy; in Class III it worsens it, and mandibular growth in Class III patients persists beyond adolescence and later in males. So an early correction can be undone, monitoring is often the right answer, and surgical decisions have to wait for serial superimposition to confirm growth has ceased.

One sentence of mechanism that explains the whole chapter.

V158 When would you treat a Class III child early?

"When the case is favourable: a Class I or mild Class III skeletal base, average or reduced lower face height, an average or increased overbite, relatively uncompensated incisors, and an anterior displacement on closing. In those, a facemask with or without expansion at 12 to 14 hours a day before age ten is worth attempting, and Mandall found over sixty per cent avoided surgery against 36 per cent untreated. In an unfavourable case — moderate to severe base, increased face height, reduced overbite, heavily compensated incisors, no displacement — I would monitor into the late teens and avoid mandibular extractions in case surgery is needed.

The favourable and unfavourable lists, then the two policies. This is a set-piece.

V159 What do you tell the parents of a Class III seven-year-old who want treatment now?

"I would separate two things. If there is an anterior crossbite with a displacement, that is treated on diagnosis, because it causes attrition and recession on the lower incisor and can drive asymmetric growth, and it corrects in three or four weeks with an incline plane. The skeletal Class III is a different question: early protraction improves the chance of avoiding surgery, sixty per cent against 36, but most of the skeletal effect is lost with growth and Proffit reports a 25 to 33 per cent recurrence of the crossbite at eight years. So I would treat the displacement now, and be honest that the skeletal picture may still declare itself in the teens.

Separating the two problems is the whole answer.

V160 Does a chin cup work?

"Not in the way it is often described. There is no agreement that it inhibits mandibular growth. What it largely does is redirect the mandible downwards and backwards, which improves the incisor relationship in the short term at the cost of increasing face height, and the short-term gain is not

maintained after growth. So I would not offer it as a means of controlling mandibular prognathism.

A short, evidence-based no with the mechanism attached.

V161 How would you camouflage a mild Class III?

"Extract further forward in the mandible — lower first premolars with upper second premolars — procline the uppers and retrocline the lowers. Occasionally a single lower incisor extraction. I would be cautious with Class III elastics, and would avoid them altogether in a high-angle case because they extrude the lower molars and rotate the mandible down and back. And I would check first how much compensation is already present: lower incisors at eighty degrees or less tell me there is very little room left, which usually means camouflage is not the answer.

The pattern, the caution, and the check that decides whether camouflage is possible at all.

Transverse and vertical problems (V162–V173)

Notes In Orthodontics: Topics 11, 16 and 19; Chapters 9, 12, 13 and 15
Supplements: Revision Handbook **§24–§25**
Practise this: 12 questions · answer each one aloud before reading the model

V162 How do you classify a crossbite?

"By site, cause and displacement. Site: anterior or posterior, unilateral or bilateral, buccal or lingual. Cause: dental, where individual teeth are displaced; skeletal, where there is a genuine transverse discrepancy of the bases; or functional, where a bilateral discrepancy produces a lateral displacement into an apparently unilateral crossbite. And with or without a mandibular displacement on closing, which is the single most important of the three because it determines urgency and prognosis.

Three axes, with a statement of which matters most.

V163 Which expansion appliance would you choose?

"For a dental or mild skeletal discrepancy in the mixed dentition, a quadhelix: it beats an expansion plate on the Cochrane evidence, is about three months faster, and takes compliance out of the equation. For a true transverse skeletal deficiency, rapid maxillary expansion at one to four turns a day over two to four weeks, generating forces up to ten kilograms, achieving up to four millimetres at the suture which is about half the screw expansion, and retained in situ for three to six months. Beyond about five millimetres in a non-growing patient, surgically assisted expansion.

Three scenarios, each with the appliance and the numbers.

V164 How much of rapid maxillary expansion is skeletal, and how much relapses?

"Up to four millimetres at the suture, about fifty per cent of the total screw expansion, with roughly ten per cent of the skeletal change relapsing at four to eight months. Dental expansion is better retained with slow expansion at about 95 per cent than with rapid at about 85 per cent. I would add Proffit's dissent: at four months rapid and slow expansion have produced about the same skeletal to dental split, so rapid activation is not actually an effective way of minimising tooth movement.

The numbers, then the dissenting position. Naming a disagreement shows you have read more than one source.

V165 Would you use expansion to treat a child's breathing?

"No. Rapid maxillary expansion is not recommended as a treatment for nasal obstruction or sleep-disordered breathing on current evidence. If the parents are worried about the breathing — and snoring, witnessed apnoeas, restless sleep or daytime behavioural problems are real concerns — I would refer to ENT or paediatrics for proper assessment including a sleep study, rather than treat it with an orthodontic appliance. I would expand if there were an orthodontic indication, and say plainly that any airway benefit is not the reason.

Refusing an attractive but unsupported indication, while taking the underlying concern seriously.

V166 In what order do you correct crossbites?

"Posterior before anterior, always. If I correct the anterior crossbite first and then expand, I lose the correction. And the anterior correction is self-retaining only if a positive overbite results, which is why I check that there is room and that the overbite will be positive before proclining. An anterior crossbite corrected with a positive overbite is usually stable and needs no retention.

A rule, a reason, and the condition for stability.

V167 How do you reduce an overbite, and which method is stable?

"Three mechanisms: intrude the incisors, extrude the buccal segments, or procline the labial segments. Which is stable depends entirely on growth. In a grower, buccal segment extrusion gives relative incisor intrusion and is stable, because condylar growth compensates for the increased posterior vertical dimension. In a non-grower the same manoeuvre rotates the mandible down and back, worsens the Class II, increases face height and relapses, so I need true intrusion — a utility or bypass arch, segmental mechanics, or miniscrews, at about sixty grams total or fifteen per tooth. Proclination gives about 0.1 to 0.2 millimetres of overbite change per degree, and is unstable unless the interincisal angle is genuinely corrected.

Three mechanisms and the growth-dependent stability. This is one of the twelve traps.

V168 How much intrusion can you actually achieve?

"Modest amounts. Segmental mechanics achieve about 1.5 millimetres in the maxilla and 1.9 in the mandible in non-growing patients, on Ng's figures. Miniscrews give about 0.67 to 0.95 millimetres more than segmental mechanics with less molar extrusion, though on low-quality evidence. Segmental techniques give greater true intrusion than a continuous reverse curve of Spee, and a Ricketts utility arch produces more lower incisor proclination. So I would be realistic with the patient about how much of the overbite reduction will come from intrusion.

Quantified expectations. Being realistic with the patient is the clinical point.

V169 What is the key sign of a skeletal anterior open bite?

"An interlabial gap of more than three millimetres at rest, in a face with an increased lower face height, a high mandibular plane angle and a backward mandibular rotation. That distinguishes it from a habit-related open bite, which is typically asymmetric, often localised to the digit, and associated with a posterior crossbite, proclined uppers and retroclined lowers. The distinction matters because a habit-related open bite usually closes spontaneously if the habit stops before the permanent incisors are established, and a skeletal one will not.

One sign, the differential, and the consequence for management.

V170 How would you correct a skeletal anterior open bite?

"Vertical control throughout, and no extrusive mechanics. High-pull headgear, posterior bite blocks, and above all molar intrusion with temporary anchorage devices, because one millimetre of molar intrusion gives about three millimetres of open bite reduction, which is the most efficient lever available. Extraction may help by allowing the mandible to rotate forwards. If that is not enough, posterior maxillary impaction with mandibular autorotation once growth is complete. And I would say at consent that about twenty per cent of open bite corrections relapse, surgical and non-surgical

alike.

The lever, the alternatives, and the relapse number.

V171 Is the tongue the cause of an anterior open bite?

"Usually not. Tongue posture is generally adaptive to the open bite rather than causative — the tongue fills the space that exists. A genuinely endogenous tongue thrust is rare, and those cases are unstable and need permanent retention. So I would not blame the tongue reflexively, I would not promise that exercises will close the bite, and I would look for the actual aetiology: habit, skeletal pattern, or iatrogenic extrusion in a high-angle patient.

Reversing the usual assumption. The iatrogenic cause is the one to volunteer.

V172 How would you assess a facial asymmetry in a growing child?

"Clinically first: from in front, from above and behind, and with the patient biting on a spatula to reveal a true occlusal cant. Then the path of closure, because a deviation from retruded contact position means a functional shift rather than a true skeletal asymmetry. Mild asymmetry is normal — clinically significant asymmetric jaw development affects fewer than 0.5 per cent, and the mandible is involved in 85 to 90 per cent of cases, most often after a childhood condylar fracture. In a grower with mild asymmetry the answer is usually to watch, but with a defined review interval and records I can superimpose.

Clinical assessment, the prevalence, and the restraint.

V173 What would make you refer an asymmetry?

"Progression. Accelerating deviation of the chin, a new or worsening occlusal cant, or new temporomandibular symptoms appearing in the teens. Those suggest an active condylar hyperplasia, and operating on an active process invites relapse, so growth has to be monitored before committing to surgery. Scintigraphy is sensitive but non-specific with false positives; serial three-dimensional facial superimposition, using the forehead, mid-nose and infraorbital regions, is a reasonable non-invasive alternative.

Progression is the trigger. The imaging caveat is the detail.

TMD, the airway and the medically complex patient (V174–V183)

Notes In Orthodontics: Topics 10 and 18; Chapter 2, pp. 102–105; Chapter 7
Supplements: Revision Handbook **§26, §29**
Practise this: 10 questions · answer each one aloud before reading the model

V174 What is the relationship between orthodontics and temporomandibular disorder?

"Orthodontics is TMJ-neutral. Weak statistical associations exist between certain occlusal traits and TMD signs, but there is no evidence of causation in either direction: orthodontics neither causes nor prevents temporomandibular disorder. The studies I would name are the American Association of Orthodontists funded work of 1992, following the Brimm versus Malloy award of 850,000 dollars in 1987. The principal confounder is that symptoms commonly begin in late adolescence, which is exactly when we treat.

State it, defend it with the named studies and the confounder, and stop there.

V175 A patient develops jaw pain and clicking during fixed appliance treatment. What do you do?

"Stop active mechanics and manage conservatively. I would take a proper history, examine the joints and muscles, record range of opening, deviation, sounds and tenderness, and reassure the patient about the natural history, because most symptoms remit. Soft diet, jaw rest, avoiding wide opening and parafunction, analgesia, and physiotherapy or a stabilisation splint if needed. What I would not do is reach for an occlusal explanation, perform an occlusal adjustment, or promise that completing the treatment will resolve it.

Suspend, assess, manage conservatively, and resist the occlusal explanation.

V176 Which imaging would you use for the temporomandibular joint?

"MRI for the disc, the retrodiscal tissues and any effusion — it is the investigation of choice for internal derangement. CT or CBCT for bone, including condylar morphology and degenerative change. A panoramic radiograph gives a screening view of condylar form only. The point to make is that CBCT cannot image the disc or muscle, so a CBCT ordered to investigate clicking will not answer the question.

The last sentence is the one that shows you understand what each modality does.

V177 What do you tell a parent worried about their child's snoring?

"That it is worth taking seriously, and that it is not an orthodontic problem in the first instance. Snoring, witnessed apnoeas, restless sleep, mouth breathing, daytime somnolence or behavioural problems all warrant referral to ENT or paediatrics, with a sleep study if indicated. Paediatric obstructive sleep apnoea affects about two per cent of children, and adenotonsillectomy is the first-line treatment in most. I would not diagnose or treat it orthodontically, and I would not offer expansion as a treatment for it.

Taking the concern seriously while declining the orthodontic role.

V178 What is the relationship between the airway and facial form?

"An association, not an established cause. Chronic nasal obstruction is associated with mouth breathing, a low tongue posture and lips apart, and with a narrower maxilla, increased face height and backward mandibular rotation — the so-called adenoid facies. Experimental nasal obstruction in primates produced facial change, but human evidence is observational and confounded. Adenoid tissue peaks at three to five years and involutes through adolescence, so many children grow out of the obstruction. I would state the association and refuse to assert the cause.

Precision about the strength of a claim is itself examinable.

V179 How does a childhood condylar fracture affect orthodontic management?

"Considerably, because the condyle is the mandible's principal growth site and a fracture there is the commonest cause of significant mandibular asymmetry. Acutely, the management is conservative and functional — early mobilisation, soft diet, analgesia, physiotherapy — because prolonged immobilisation risks ankylosis, which is the outcome to avoid at all costs. Orthodontically, I would monitor growth with serial records I can superimpose, consider a functional appliance to encourage condylar growth and maintain symmetry, and expect that definitive correction of an established asymmetry will be orthognathic.

Acute management, the ankylosis risk, and the orthodontic role.

V180 Which medical conditions change your force levels?

"Sickle cell disease, thalassaemia, a history of childhood malignancy and its treatment, diabetes, previous radiotherapy, and any patient with short roots. In all of those I would use lighter forces, keep the aims simple, and keep treatment short. I would also keep in mind that bisphosphonates slow or inhibit tooth movement and delay eruption, and that stopping the drug does not help, so in those patients the aims have to be modest from the outset.

Group them by what they change — force, extraction, appliance, timing — rather than listing conditions.

V181 Which conditions change your appliance choice?

"Epilepsy, where I would avoid removable appliances because of the risk of displacement during a seizure. Phenytoin therapy or transplant immunosuppression, where gingival hyperplasia makes bonded tubes preferable to bands. Asthma, where I would avoid a palate-covering removable. Nickel allergy, where I would avoid nickel titanium but use stainless steel freely. And a patient likely to need MRI, where ceramic brackets or aligners may be preferable.

Five conditions, each with the appliance consequence.

V182 Does a patient with a heart murmur need antibiotic cover for band placement?

"Not under NICE guidance, which advises that antibiotic prophylaxis solely to prevent infective endocarditis should not routinely be provided for dental procedures. I would add that this conflicts with the American Heart Association and European Society of Cardiology positions, which still recommend cover for the highest-risk patients, and that mortality from infective endocarditis is around 21 per cent, which is why the position remains contested. In an individual high-risk patient I would liaise with the cardiologist rather than decide alone.

Give the guidance, acknowledge the conflict, and say what you would actually do.

V183 When would you defer orthodontic treatment on medical grounds?

> *"At least six months after an organ transplant. Until a malignancy is in remission. While an eating disorder is active, with liaison, prevention and management of erosion in the interim. During pregnancy for anything elective, providing urgent care ideally in the second trimester. And where a medical condition is uncontrolled and its control would change the risk — poorly controlled diabetes being the obvious example. In each case I would explain the reason and set a review rather than simply refusing.*

Five situations and the principle that deferral is explained and reviewed rather than imposed.

Adults, orthognathic surgery, clefts and governance (V184–V193)

Notes In Orthodontics: Chapters 15 and 16; Part V, units V.2–V.4 for governance
Supplements: Revision Handbook **§27, §28, §30**
Practise this: 10 questions · answer each one aloud before reading the model

V184 What is different about treating an adult?

"No growth, so growth modification is unavailable and every millimetre comes from tooth movement or from surgery. Overbite mechanics reverse: buccal segment extrusion, which is stable in a child, rotates the mandible down and back in an adult. The periodontium is often reduced, which moves the centre of resistance apically and changes the mechanics rather than just the risk. Movement is slower. There are restorations to bond to and missing teeth to plan around. Aesthetic expectations are higher and more specific. And I would probe every tooth and record pre-existing recession before I start.

Seven differences. Growth, mechanics, periodontium and records are the four that matter most.

V185 What is decompensation, and why does it matter?

"It is the deliberate removal of the dental compensation that has developed around a skeletal discrepancy, and it is what pre-surgical orthodontics is for. In a Class III that means retroclining the uppers and proclining the lowers, so the reverse overjet gets worse before surgery. It matters because it is the opposite of camouflage: I cannot start camouflage and change my mind, which is why the camouflage-or-surgery decision has to be made before treatment begins. And it matters at consent, because the patient has to understand that the bite will look worse for eighteen months.

Calling pre-surgical orthodontics aligning the arches is one of the twelve traps.

V186 What is the hierarchy of stability in orthognathic surgery?

"Direction of movement matters more than fixation, which matters more than surgical technique. Most stable are maxillary impaction, mandibular advancement in a low- or average-angle patient, and genioplasty in any direction. Then maxillary advancement, and maxilla up with mandible forward. Mandibular setback is problematic. Least stable are maxillary down-grafting and surgical maxillary expansion. Very stable means about a ninety per cent chance of no clinically significant change.

The ordering principle first, then the list. The definition of very stable is a nice detail.

V187 What would you consent a patient for before orthognathic surgery?

"Nerve injury, particularly the inferior alveolar nerve in a sagittal split, with altered sensation common transiently and occasionally permanent. Infection, bleeding, unfavourable split, relapse, and the possibility of further orthodontics afterwards. Temporomandibular symptoms, and condylar resorption, which is a particular risk in young women with a high-angle pattern and pre-existing joint disease. Plate removal. The eighteen months of pre-surgical orthodontics during which the bite gets worse. The psychological adjustment to a changed face. And the possibility of dissatisfaction even with a technically excellent result, which is why I would screen for body dysmorphic disorder.

A long list, and the last item is the one that distinguishes a thoughtful answer.

V188 Take me through the cleft care timeline.

> *"At birth, diagnosis, feeding advice, hearing assessment and parental support. Lip repair at three to six months, by the rule of tens. Palate repair at six to twelve months, before speech develops, with hearing monitored because glue ear is near-universal. Prevention and fissure sealing through the mixed dentition, with arch expansion. Alveolar bone graft at eight to ten years, before the canine erupts and when about two thirds of its root is formed, using cancellous bone usually from the iliac crest. Definitive orthodontics in adolescence, and orthognathic surgery in the late teens if the maxillary hypoplasia warrants it. Throughout, multidisciplinary team care with national audit.*

A timeline learned as a sequence of ages. Expand before you graft is the detail examiners listen for.

V189 Why is the alveolar bone graft timed to the canine?

> *"Because the purpose is to provide bone for the canine to erupt into. Grafting before eruption, at about two thirds of root formation, means the canine erupts through the graft and maintains it, whereas grafting after eruption leaves a graft with no functional stimulus. The graft also stabilises the maxillary segments, closes the oronasal fistula, supports the alar base and makes later implant placement possible. And I would expand first, because the graft needs surgical access and the arch needs its form established.*

Purpose first, then the other four benefits, then the sequencing point.

V190 What dental problems would you expect in a unilateral cleft?

> *"A missing lateral incisor on the cleft side, most commonly, together with supernumerary and malformed teeth often on both sides of the cleft. The central incisor adjacent to the cleft is characteristically rotated and tipped towards it and often hypoplastic. Anterior and posterior crossbites from transverse and sagittal deficiency and from surgical scarring. Higher caries risk and enamel defects, with difficult hygiene. Dental anomalies are commoner in bilateral cases. And retention is prolonged, usually permanent.*

The list, plus the two generalisations about bilateral cases and retention.

V191 How would you distinguish Apert from Crouzon syndrome?

> *"By the hands and feet. Both involve FGFR2 mutations and both produce craniosynostosis with midface hypoplasia, exophthalmos and a relative mandibular prognathism. Apert additionally has syndactyly, and its midface hypoplasia is generally more severe. Crouzon has no limb anomalies. I would also be careful to distinguish a Turner tooth, which is a localised enamel hypoplasia caused by infection or trauma to the primary predecessor, from Turner syndrome, which is 45,X — examiners do use both terms in the same paper.*

One discriminating feature, plus the terminological trap.

V192 What would you record in the notes at an orthodontic assessment?

> *"The complaint in the patient's own words and what they expect treatment to achieve. The history. The examination in three planes with the crowding quantified. The IOTN. A provisional problem list. Which records I justified and took, with the justification for any radiograph and a report on it.*

Then the plan: facial and occlusal aims, the options discussed including no treatment, the extraction pattern, the anchorage plan, the appliance, the estimated duration and the agreed retention plan. And the consent discussion: what was said, what was warned about, what was given in writing, who was present, and that the patient agreed.

A structured list. It is also the answer to any complaint that arises three years later.

V193 What would you audit in your own orthodontic practice?

"Case outcomes using PAR or ICON, because those are designed for auditing an operator over many cases. Failure and breakage rates. Decalcification rates at debond, which is the one that tells me most about my patient selection. Treatment duration. Discontinuation rates. Radiograph reject rates and dose. And decontamination cycle records. If I found a decalcification rate I was unhappy with, I would change my selection criteria rather than my fluoride regimen first.

An audit list with one worked inference. That last sentence is the mark.

The questions that test judgement (V194–V207)

Notes In Orthodontics: Crash notebook *The last hour*; the twelve traps
Supplements: Revision Handbook **Handbook Part 5**
Practise this: 14 questions · answer each one aloud before reading the model

V194 Present this case to me.

"A 13-year-old growing girl with a mild skeletal Class II pattern, increased vertical proportions and a symmetrical face. Her lips are incompetent with an interlabial gap of about five millimetres and a lower lip trap. Intra-orally she is caries-free with fair hygiene, all teeth present, with six millimetres of upper crowding and four of lower. She has a Class II division 1 incisor relationship with an overjet of eight millimetres and an incomplete overbite, on a half-unit Class II molar relationship, with no displacement on closing. My working diagnosis is a Class II division 1 on a mild skeletal Class II base with a vertical tendency, and the decision in this case turns on whether the vertical pattern will tolerate the mechanics needed to reduce that overjet.

Sixty seconds, in a fixed order, ending with what the decision turns on. Rehearse it until it is automatic.

V195 What is the first thing you want to know about any orthodontic patient?

"Two things, and I would give them in this order. What the patient themselves is concerned about, because that becomes the outcome against which the treatment will be judged and it ranks highly in the problem list. And whether they are still growing, because that determines which of the three treatment routes is available and it changes the mechanics for overbite reduction, expansion and protraction.

The two opening moves. Both should be automatic.

V196 You do not know the answer to a question. What do you do?

"Say so, and then say how I would find out. Something like: I am not certain of that figure, and I would rather not guess; what I would do is look it up in the Handbook before I saw the patient again, and in the meantime the principle I would work from is this. That scores; a guess defended does not, and a mis-attributed reference is worse than no reference at all.

Examiners are testing safety, not omniscience. This answer is worth practising out loud.

V197 An examiner pushes you towards a plan you would not carry out. How do you respond?

"Politely, and by saying where my limit is and why. Something like: I can see the argument for that, but it is beyond what I would attempt myself, and what I would actually do is refer the case at this point. Being led step by step into a plan I would not deliver is a worse outcome than disagreeing with an examiner, because the examination is testing whether I know my own limits.

Being led is a recognised failure mode. Practise the sentence that stops it.

V198 You realise mid-answer that you have said something wrong. What do you do?

> *"Correct it immediately and move on. Something like: I said 150 grams for intrusion, and that is wrong — intrusion needs ten to twenty grams per tooth, because the force is concentrated at the apex. Correcting myself costs one mark; leaving it costs the examiner's confidence for the rest of the viva.*

A recoverable error, handled well, can leave a better impression than no error at all.

V199 How would you structure an answer to a broad question?

> *"By signposting. Something like: there are three things I would want to say about that — first, second, third — and then finishing in the clinic with what I would actually do. Structure earns breadth, numbers earn depth, and weighing the evidence earns judgement. What loses marks is a stream of associations with no shape.*

The single most useful technique in the oral. It also buys thinking time.

V200 What would you do if you disagreed with a colleague's treatment plan for a shared patient?

> *"Speak to the colleague first, not to the patient. I would ask what their reasoning was, because there is usually something I do not know, and I would put my concern in terms of the specific finding rather than the general plan. If I still disagreed and thought the patient was at risk of harm, I would say so clearly, document the discussion, and if necessary suggest a second opinion. What I would not do is undermine a colleague in front of a patient.*

Professionalism questions appear in every oral. Colleague first, patient's interests throughout, record it.

V201 A patient is unhappy with their result. How do you handle it?

> *"Listen first, and find out specifically what they are unhappy about, because it is often not what I would have predicted. Then look at it with them against the records and the aims we agreed at the start. If they are right and something is correctable, I would say so and offer to correct it. If it is a limitation we discussed and agreed at consent, I would show them that discussion in the notes and explain it again. Either way I would record the conversation, and I would treat it as audit data about my own practice.*

Listening, records, honesty, and the audit loop. Four elements.

V202 How do you decide whether a case is within your competence?

> *"By asking whether the aims I have set can be achieved by the mechanics I am competent to deliver, and whether I would be comfortable managing the foreseeable complications. If the answer to either is no, the case goes. In practice that means referring severe skeletal discrepancies and likely surgical cases, complex impactions needing exposure and traction, cleft and craniofacial cases, significant or progressive asymmetry, severe hypodontia, reduced periodontium cases needing complex mechanics, and suspected primary failure of eruption. And I would rather decide that at the assessment visit than eighteen months in.*

A test rather than a list, followed by the list. The test is what generalises.

V203 What are the commonest reasons candidates lose marks in an oral like this?

> *"Not saying whether the patient is growing. Quoting a number confidently and wrongly. Presenting tooth movement as mechanics rather than biology. Reversing the Class II and Class III extraction patterns, or RANKL and OPG. Forgetting the facial aims. Calling pre-surgical orthodontics aligning the arches. Overclaiming the benefits of treatment. Omitting radiation justification. And leaving retention out of the plan, or mentioning it as a sentence in the conclusion rather than as a paragraph in the plan.*

The twelve traps, condensed. Read them on the morning of the examination.

V204 Tell me about a case that went wrong.

> *"I would choose one honestly and describe it in the same order I would describe any case: what I planned, what happened, when I noticed, what I did, and what I would do differently. For example, a case where I underestimated the anchorage demand, saw the molars coming forward at the working wire stage, and had to add reinforcement mid-treatment rather than having planned it at the start. What I learned was to do the space analysis formally rather than by eye, because the anchorage requirement follows from it.*

Every candidate has one. Preparing it turns a difficult question into an easy one.

V205 What is the single most important thing you have learned about orthodontics?

> *"That growth status decides almost everything, and that most of what is presented as skeletal correction is dentoalveolar and largely transient. The corollary is that the profession has often been more confident than the evidence justified — early Class II treatment, self-ligation, third molars and crowding, orthodontics and TMD — and that being able to say which is which is more useful than knowing any individual technique.*

An open question. A prepared, specific answer beats a general one.

V206 Why should a general dentist with an interest in orthodontics treat orthodontic cases at all?

> *"Because a great deal of orthodontic need is straightforward, because access to specialist care is limited, and because interceptive treatment in particular is most effectively delivered by the clinician who sees the child regularly — palpating for canines from nine or ten years, spotting a crossbite with a displacement, timing a first molar extraction. What makes that safe is knowing the boundary: treating what I can treat well, referring what I cannot, and recognising the difference early.*

The question behind the question is whether you understand the level the Diploma is pitched at.

V207 If you had one sentence to give a colleague sitting this examination, what would it be?

> *"Say whether the patient is growing in your first sentence, say where you would refer before you are asked, and finish every answer with what you would actually do.*

The three moves that carry an oral examination. Everything else is detail.

Viva tracker

One row per section. Tick a box each time you have answered every question in that section aloud, *to the end, before reading the model. Three ticks is the target. The last column is the Handbook section to re-read if the words would not come.*

Questions	Section	No.	1st	2nd	3rd	Hbk
V1–V10	Growth, development and the craniofacial complex	10	☐	☐	☐	§1–§2
V11–V20	Development of the dentition	10	☐	☐	☐	§3
V21–V30	The biology of tooth movement and root resorption	10	☐	☐	☐	§4
V31–V38	Occlusion, aetiology and epidemiology	8	☐	☐	☐	§5
V39–V48	Treatment need, indices and the evidence base	10	☐	☐	☐	§6–§7
V49–V58	Ethics, consent and communication	10	☐	☐	☐	§8
V59–V68	Clinical examination and records	10	☐	☐	☐	§9–§10
V69–V80	Space analysis, diagnosis and treatment planning	12	☐	☐	☐	§11–§12
V81–V90	Extraction decisions and space management	10	☐	☐	☐	§13
V91–V102	Biomechanics and anchorage	12	☐	☐	☐	§14
V103–V111	Materials, wires and bonding	9	☐	☐	☐	§15
V112–V121	Removable, functional and fixed appliances	10	☐	☐	☐	§16–§17
V122–V129	Treatment mechanics and finishing	8	☐	☐	☐	§18
V130–V139	Retention and stability	10	☐	☐	☐	§19
V140–V151	Interceptive orthodontics and impacted teeth	12	☐	☐	☐	§20
V152–V161	Class II and Class III management	10	☐	☐	☐	§22–§23
V162–V173	Transverse and vertical problems	12	☐	☐	☐	§24–§25
V174–V183	TMD, the airway and the medically complex patient	10	☐	☐	☐	§26, §29
V184–V193	Adults, orthognathic surgery, clefts and governance	10	☐	☐	☐	§27, §28, §30
V194–V207	The questions that test judgement	14	☐	☐	☐	Handbook Part 5

SPEAK IT, TIME IT, SHORTEN IT

A good spoken answer runs twenty to forty seconds. On your first pass most answers will run to ninety, because you will say everything you know. On the second pass, say the same answer in half the time by leading with the conclusion and keeping only the reasoning that supports it. The third pass is for the sections you still dread — and those are the ones the examiner will find.